The Complete Book of

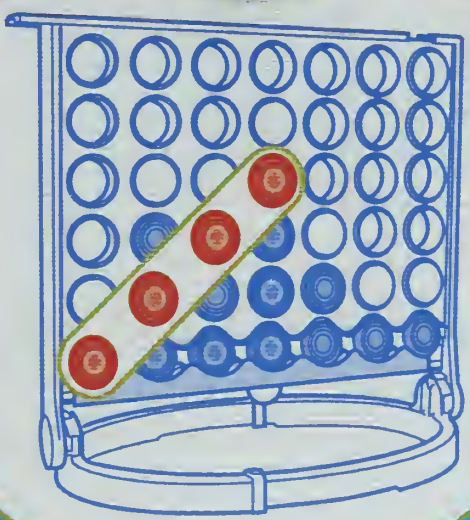
The original game of

Connect 4

History

Strategy

Puzzles



JAMES DOW ALLEN

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The Complete Book of



The Complete Book of Connect 4

HISTORY

STRATEGY

PUZZLES

JAMES DOW ALLEN

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For Lily and Scott

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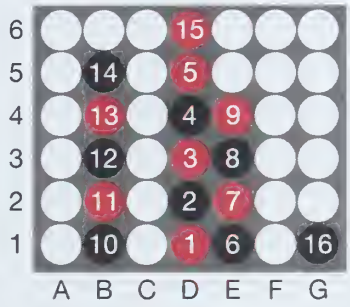
Preface

The Complete Book of CONNECT 4 outlines the strategic and tactical principles of CONNECT 4 and allows you to assess your skills with any of several problem sets. Please do not get discouraged if the problems seem quite difficult; even experts will find many of them to be *very* difficult.

As you read this book, take a break from time to time and try one of the easy or medium problem sets. The solutions at the back of the book show the winning move, and some of them point you to a diagram and accompanying discussion elsewhere in the book. As you reread the book, your play should improve, and eventually you will want to try some of the harder problem sets.

How I Came to Write This Book

DIAGRAM 1
Red to play, with only one move to win



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 D6 g1 (17) __

I first started playing CONNECT 4 at a beach resort during the 1980s, where this game was a popular way to pass the time between swims. I took the game more seriously than most of the other players, but even after I figured out some basic principles about odd-numbered rows and even-numbered rows, there were still several opponents who could beat me consistently. One evening my opponent and I kept playing the same opening over and over (see Diagram 1), and I kept losing. I made it my goal to figure out this opening.

Diagram 1 may still be a rather tricky problem even after you've read this book, but perhaps you'd like to try the challenge of solving it right now! I give the solution in Chapter 1, but as a hint, here's the solution with details omitted:

Red may need to win in the upper right side of the board, but Black has that side protected by threatening a diagonal four-in-row; therefore Red needs to force Black eventually to play ___; Red can do this if he eventually plays ___, but Black can foil that idea if he sets up a future four-in-row by playing ___, so Red needs to play ___ now.

Although it's been almost 25 years, I still remember saying "Aha!" to myself when I guessed Red's winning move in Diagram 1. That was the moment I started

to become a good player. I wanted to *prove* that my move in Diagram 1 was the only winning move, so I wrote a computer program to verify the solution. By analyzing more than one billion positions, the computer program was able to confirm that First Player can always win CONNECT 4.

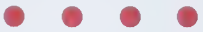
In 1988, CONNECT 4 was the most difficult game that had ever been solved by computer up to that time. It would have taken millions of dollars' worth of computer time to solve the game in 1970, but by 1988, processing speeds had improved so that a thousand hours was enough time to solve the game on an ordinary workstation. (Computers are even faster now, of course, and your home PC, properly programmed, could solve the game today in a matter of minutes.)

Perfect computer play does *not* mean that it's easy for a human to play perfectly. The game continued to fascinate me, and I continued to search for interesting positions and principles of winning play. The results are in this book.

Acknowledgments

I want to thank Alan Pruce for much helpful criticism and input, especially the "Pruce Defense." Frank Stalone was also very helpful and introduced me to the Third-Row Gambit. (Neither Alan nor Frank claims invention of these opening styles.) John Tromp, a master programmer who has made important discoveries in go and other games, helped review the early draft but also deserves special acknowledgment for another reason: While the CONNECT 4 software I wrote 22 years ago to solve the game may have historic or sentimental interest, for several years I've preferred to use John Tromp's excellent Fhourstones implementation. (Most of the analyses in this book have been verified by computer, but if there are any errors, they were probably introduced during my editing and shouldn't be blamed on Fhourstones.) A special thank you goes to Jeffrey Harris who spent many hours checking the logic in the discussions. (I hope Jeffrey has now become an expert CONNECT 4 player!) Finally, I should thank the anonymous bartender in Bang Lamung, Thailand, who 25 years ago used Diagram 1 to show me that CONNECT 4 is not an easy game!

Introduction



CONNECT 4®, by Hasbro (and earlier, Milton Bradley), is a very popular game among all age-groups and skill levels. The game is as simple to learn as tic-tac-toe yet offers ample scope for tactics and strategy. Some people who seldom play other games find CONNECT 4 to be fun and to have just the right level of challenge. Some gaming connoisseurs enjoy the several Web sites that offer online play with CONNECT 4 experts. I can guarantee that if you study this book, you will be able to thoroughly trounce most of the players who *haven't* read the book.

CONNECT 4 is an alignment, or connection, game—one strives to *align*, or *connect*, one's pieces along a row, column, or diagonal—but while other alignment and connection games are played on horizontal boards, CONNECT 4 is played on a vertical board, with the disks dropping to the lowest empty slot in whichever column they are played. As we will see, this makes the strategy of CONNECT 4 vastly different from that of other connection games. In most board games, you'd be happy to make two moves in a row if your opponent offered to allow this. In CONNECT 4, if your opponent makes this offer, it's likely that he or she is trying to trick you into exchanging a won game for a lost game!

The most famous connection game is the child's game tic-tac-toe (also called naughts and crosses or three-in-a-row), which was probably played at least as early as Egypt's Nineteenth Dynasty, 3,300 years ago, and certainly in ancient Rome (perhaps under the name *terni lapilli*). A variety of other types of connection games exist, including nine men's morris and even bingo. The most difficult among popular connection games may be the ancient Japanese game gomoku, but CONNECT 4 is also extremely difficult. There are *very* few, if any, human players who can claim to have fully mastered it.

The basic concepts underlying CONNECT 4 aren't as old as tic-tac-toe or gomoku, but they are old enough that the place and date of its original invention have been forgotten. Similar games have been played in many different countries under many different names, including force four, four across, four-in-a-line, four-in-a-row, line4, plot four, quadrow, stack4, tetralink, and the captain's mistress. Its foreign-language names include *conecta 4* (Spanish), *vier op 'n rij* (Dutch), *forza quattro* (Italian), *puissance quatre* (French) and *Vier gewinnt* (German). In this book I discuss Hasbro's version of the game, named CONNECT 4, previously a registered trademark of Milton Bradley.

Captain James Cook

Most of the different names for CONNECT 4 are self-explanatory, but the name "captain's mistress" has a special story. The captain in question was the famous British explorer James Cook (1728–79).

Captain James Cook was born to the family of a simple farm laborer but became an officer in the British navy and captained three very important exploration voyages. He was the first European to discover several islands, including the Hawaiian (or Sandwich) Islands, New Caledonia, Niue, and, of course, the Cook Islands. He searched (unsuccessfully) for the Northwest Passage from both the Pacific and the Atlantic, and mapped Newfoundland, New Zealand, and many points in between; he is most famous for being the first European to set foot on Australia, claiming that sixth continent for the British Empire. (Cook failed in an attempt to also be first to set foot on the seventh continent, Antarctica, though he was first to cross the Antarctic Circle.)

Cook's first ship was His Majesty's Bark *Endeavour*, and his first mission was to verify rumors of a great continent west of New Zealand. As a "cover story" for Cook's secret mission to claim Australia for King George III, he first visited Tahiti to observe the 1769 transit of Venus; for this reason he was accompanied by the astronomer Charles Green. Also on board Cook's ship were two naturalists, Sir Joseph Banks and Daniel Solander. (Australia's Botany Bay is named after their discoveries.)

During the long, uneventful days at sea, it is said that Cook and his scientist friends entertained themselves playing a vertical checker game that was very similar to what we now call CONNECT 4. Because of Cook's fascination with this game, the crew of the *Endeavour* (and his later ship HMS *Resolution*) called his wooden board the captain's mistress. Cook's voyages were before the era of steam engines, and vessels often listed sharply for long periods, even when the seas weren't causing yawing or rolling. We don't know whether Cook, Green, Banks, and Solander played other board games, such as chess, but I suspect they chose their vertical checker game because the pieces wouldn't be knocked over by the ship's listing or rocking, while chess pieces might. (Today, CONNECT 4 is a preferred game for travel in cars and airplanes for the same reason.)

Before he made his famous discovery voyages, James Cook wrote that he intended to go "not only farther than any man has been before me, but as far as I think it is possible for a man to go." If anyone tries to tell you that CONNECT 4 is a pointless child's game, remind them that a similar game (with the nickname "captain's mistress") was the pastime of the great Captain Cook.

Chapter 1

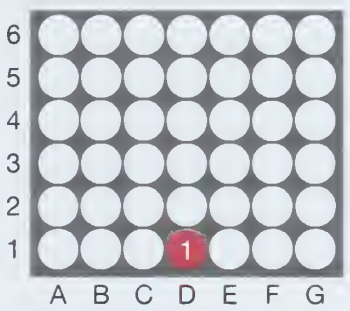
Rules and Terminology

I present the rules of CONNECT 4 by showing a game as it unfolds. This also gives us a chance to agree on some basic terminology.

The game of CONNECT 4 is played by two players on a vertical board of six rows and seven columns. One player has 21 disks of one color, say, red. The other player has 21 disks of another color, say, black. It doesn't matter what the two colors are, as long as they can be easily distinguished. In CONNECT 4, the name of the game is the same as the object of the game. You want to align (connect) four disks of your own color in a straight line. (Call that *four-in-row.*) The four disks must be contiguous but may line up horizontally, vertically, or diagonally. As soon as you've achieved this, the game is over—you've won!

It may happen that even after both players play all 21 of their disks, neither player has achieved four in row. The final position of such a game is called a *draw*; positions that will lead to draws assuming perfect future play may also be referred to as draws, and the games leading to those positions may be called *drawn games.* There is surprising variation in CONNECT 4, depending on a player's style. Some players may find 20 percent of their games ending in draws. Others will find drawn games to be rare.

DIAGRAM 2
Black to play (will lose)



Moves: D1! (2) __

Unlike chess (in which White always goes first), in CONNECT 4 each player retains the same color throughout a series of games, but usually the players alternate going first. However, it simplifies things in this book to have the Red player play first in every game; Black always plays second.

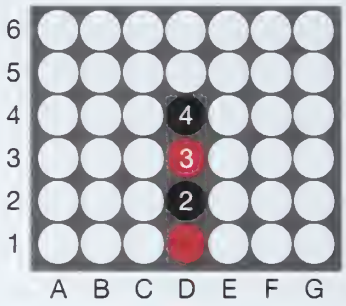
I label the columns A through G, as shown in Diagram 2. Red can start in any column, but it's best by far for Red to start in the central column (D), and you will rarely see any player deviate from this, except whimsically. Gravity makes the disk drop down to the first (bottommost) row; the cell it lands on is called *D1.* Hence, we say this game starts "1 D1." Instead of "D1," I could have written just "D"—once a player chooses which column to drop his or her disk into, there's only one row it can end up on.

At the top of every diagram I indicate what the outcome of the game will be, *assuming perfect future play*. Since Black has no chance against a perfect Red player, I indicate that Black "will lose." (Don't despair if you're Black and Red starts with the winning move D1! *Very* few humans play *perfectly*.)

In chess notation, particularly good moves are shown with an exclamation mark ("!"). I adopt that idea here but change the meaning somewhat. When a move is the *only* move that leads to victory (assuming perfect play), I mark that move with an exclamation mark. I've done that here because D1 is Red's *only* winning first move. I also use the exclamation mark when a move draws and all other moves lose.

DIAGRAM 3

Red to play, with only one move to win



Moves: D1! (2) d2 D3! d4 (5) __

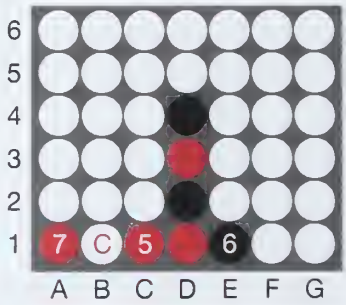
The game then continues with players alternating turns. At the bottom of Diagram 3, I show a record of all the plays leading up to the diagrammed position. Since the "1" label on Red's first disk was already shown in a previous diagram, I reduce clutter by omitting it in this diagram.

I always show Red's moves in uppercase and Black's moves in lowercase. This adds clarity when I discuss moves or move sequences. Black's 4 d4 is probably the player's best move, but I do *not* show it with an exclamation mark; that mark is reserved for a move that gives a better result than any other move *against perfect computer play*, but here a "perfect" Red player will win whether Black plays well or poorly.

As shown, I give each disk a different number. This means I won't speak of Red's first, second, third moves, and so on, but rather Red's disk 1, disk 3, disk 5, etc. Black plays the even-numbered disks. It works out nicely that Red plays the odd-numbered disks because in Chapter 4 when I discuss major threat analysis, I advise you to commit to memory the following mantra: "Red (First Player) likes odd numbers."

DIAGRAM 4

Black to play, with only one move to win



Moves: D1! d2 D3! d4 (5) C1?? e1 A1 (8) __

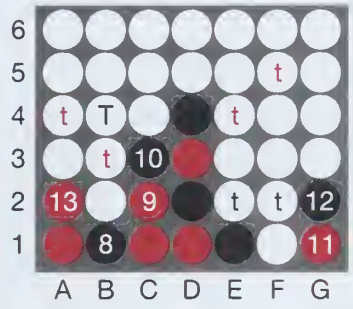
I write a question mark ("?") after a move that is a *blunder*. A blunder is a move that causes a won game to be drawn or a drawn game to be lost. I write a double question mark ("??") after a *severe blunder*, which causes a won game to be lost. In Diagram 4, Red's move 5 C1 loses, even though a winning move was available, thus it is marked with a double question mark.

As the game continues in Diagram 4, Red threatens to achieve four-in-row and win the game. While Red *played* in the cell A1, the empty cell B1 is the important point. I will use *threat* to refer to any empty cell a player hopes to occupy in the future. By analogy with chess, Red might say "check" as a courtesy when he or she plays 7 A1 (warning the Black player, since it isn't much fun to win through the opponent's simple oversight). In this book, I use the term *check* to describe a cell available for immediate play whose occupation would complete four-in-row for one of the players, like B1 in Diagram 4. I've marked the check with a red *C* (for "check"). (By the same analogy with chess, completion of a game-winning four-in-row can be called *mate*.)

The game continues as shown in Diagram 5. Black plays 8 b1, of course; despite being obvious, by our convention the move is shown with an exclamation mark. Black 10 c3 is also shown with an exclamation mark; that it is a mandatory play for Black will be clear to you after reading this book. For Black, b4 is a threat cell but won't become a check (immediate threat) until cells B2 and B3 have been occupied. An important threat that is not check (usually because the cell under it is empty) is called a *strategic threat*. Black's threat at b4 is shown with an uppercase *T* and called a *major threat* because Black already has three of the four disks needed for four-in-row. Black of course hopes Red is eventually forced to play B3; if Black is the one who plays b3, Red will simply answer at B4.

DIAGRAM 5

Black to play and win



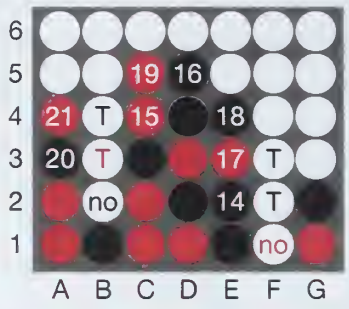
Moves: D1! d2 D3! d4 C1?? e1 A1 (8) b1! C2 c3! G1 g2 A2 (14) __

Red hasn't achieved a major threat yet, but I use lowercase red *t*'s to show *minor threat pairs* for Red at A4-B3 and at E4-F5. A *minor threat* is one of two cells needed to convert two aligned disks to four-in-row. I also show a minor threat pair for Black at e2-f2 since capturing both those cells would give Black a win. While a cell that is available for *immediate* play is never called a *major threat* (it would be called a *check*), it may be termed a *minor threat*, as e2 is here.

A threat is called an *odd-row threat* or an *even-row threat* if it is on, respectively, an odd row (either the third or fifth row) or even row (either the second, fourth, or sixth row); this attribute is called the threat's *row parity*. Remember that it is the *empty* cell needed to complete four-in-row that determines the threat's row parity. The two threat cells of a minor threat pair may have the same row parity or they may have different row parities; in the latter case they are called a *mixed threat pair*.

A game position might have many cells that meet the definition of major or minor threat, but in the diagrams in this book, I label the threats with *T* or *t* only when they are decisive or relevant to the discussion; even relevant threats may be left unlabeled to avoid cluttering the diagrams. This first example game, however, is just to introduce terminology, and threats are labeled for that purpose.

DIAGRAM 6
Black to play and win

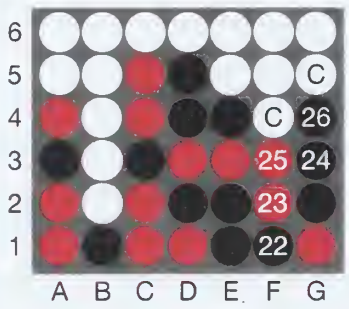


Moves: D1! d2 D3! d4 C1?? e1 A1 b1! C2 c3! G1 g2 A2 (14) e2 C4 d5 E3 e4 C5 a3 A4 (22) __

After eight more moves, the position in Diagram 6 is reached. Red establishes a major threat at B3. This *undercuts* (renders worthless) Black's major threat at b4. (This is because Black will only get a chance to play there if Red had just played B3 and won the game! However, a Black threat like b4 may not be *completely* worthless. Eventually Red will be forced to spend a turn blocking it with a disk at B4—a turn Red might have preferred to spend elsewhere.)

On the right side of the board, Black has established major threats at f2 and f3; these threats will allow Black to win the game very quickly, just by playing in column F. Two major threats of the same color in neighboring rows, such as the ones just mentioned, are called an *adjacent threat pair*. (In this book the word *adjacent* always refers to neighboring *rows*, not neighboring *columns*.)

DIAGRAM 7
Red to play (will lose)



Moves: D1! d2 D3! d4 C1?? e1 A1 b1! C2 c3! G1 g2 A2 e2 C4 d5 E3 e4 C5 a3 A4 (22) f1 F2 g3 F3 g4 (27) __

Black will lose if he or she ever plays a disk at b2, so we've marked that cell with a black "no." Similarly, Red won't play F1, so it's marked with a red "no." The game continues in Diagram 7. Black could have won by playing 24 f3, but Black chose a different way to win just so we could complete our overview of basic terminology. In the final position, Black has checks at f4 and g5. Red will play one of these cells; Black will play the other and win. I will refer to pair of checks for the same player, such as Black's checks at f4 and g5 in Diagram 7, as a *double check*.

I've been careful to mark all the moves in all the diagrams in this book with exclamation points and question marks as previously explained. This makes it possible to count question marks in the move list and to determine the perfect-play

result of any position reached during the moves in that game. For example, any move list with an odd total number of question marks will lead to a drawn position (assuming perfect further play). In Diagram 4, the move list tells us that the game should properly be won for Red at any point before Red's 5 C1 blunder and is properly won by Black at any point thereafter. It also means that you can use any move sequence in this book as a basis for studying good play, *as long as you verify that the moves aren't listed with question marks.* (I use the same meanings even in the solutions to quick victories; exclamation points are not awarded for moves that are the only way to quick victory if there are other winning moves.)

The game just described ended in double check, but in the section below called "How to Become a Master Player" I discuss a real expertly played game that ends in triple check!

Terminology

During this first example game, I introduced some special terminology, such as *odd-row threat* and *major threat.* We will need such words in order to explain CONNECT 4 strategic principles. A useful term I *never* use in this book is *double threat.* I avoid it because different people might interpret *double threat* in any of several different ways:

- A double check
- An adjacent threat pair
- A pair of non-adjacent same-colored threats in the same column
- A pair of threats by both Red and Black at the same cell
- A pair of major threats
- A minor threat pair

Terms such as *triple threat*, *quadruple threat*, and *sextuple threat* seem less likely to cause confusion than *double threat*, and I *do* use these terms, giving them *very* specific meanings.

How to Become a Master Player

This book shows you some of the tactical and strategic ideas you will need to become a master at CONNECT 4, but there is no secret formula for success. CONNECT 4 is an easier game than chess or even checkers, but you'll still need to think for yourself and plan ahead. Look back at Diagram 1; did you guess the right move? Let's deduce it step by step.

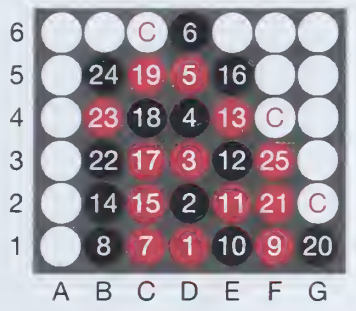
1. Red does have threats in the upper left side of the board, but they are not quite enough to win, so Red must be prepared to force victory on the right side depending on how Black defends.
2. To have any chance of building a threat on the right side, Red must somehow get past the Black major threat at F2.
3. If you've just started this book, you may not yet realize that a major threat at F2 (or at any point on the second, fourth, or sixth row) is not going to disappear by itself! Black will not play f1 unless there is a very strong compulsion to do so.
4. Red can compel Black to play f1. All Red has to do is play C4. It may not be obvious that Red will get C4, but he or she does get help from his or her threat at C5. Anyway, Red can reason, "Getting C4 is my only chance, so I may as well assume I will get it."
5. Now, when Black puts him- or herself in Red's shoes, Black will realize that C4 is Red's paramount goal. Can Black prevent that?
6. Yes; if Black plays b6, Red C4 loses immediately. Black wants to play b6, but it's Red's turn.
7. Red's best move is Black's best move. Red must play B6 so Black doesn't get it. Later Red will play C4, followed by F2. It may not be clear yet exactly how Red will win in the upper right side, but if Red doesn't play B6 right now, Black will play there and Red will have no chance whatsoever on the right side.

This book contains many example games with many expert moves—too many expert moves to explain each one in as much detail as I just did for Red's disk 17 in Diagram 1. If you don't understand the moves in a game, set them up on your own board and try to figure them out. (Or, for more rapid experimentation, use one of the fine CONNECT 4 software packages, such as John Tromp's Fhourstones.) Most of the example games are played expertly, perhaps with one or two mistakes to help make a point. (The game just reviewed in Diagrams 2–7 had many simple mistakes just to introduce terminology.)

Before continuing, I'll try to whet your appetite with another expertly-played game, shown in Diagram 8. By the exclamation points, we can infer that Black was able to find a sequence of moves that caused *each* of Red's first *thirteen* disks to be forced. This highly unusual most-forced game ends in Red victory via a rare triple check!

DIAGRAM 8

-Black to play (will lose)



Moves: D1! d2 D3! d4 D5! d6 C1! b1 F1! e1 E2! e3 E4! b2 C2! e5 C3! c4 C5! g1 F2! b3 B4! b5 F3! (26) __

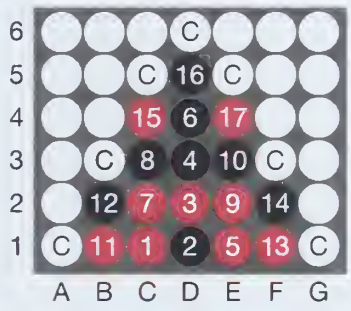
After studying this book, come back and look at this opening again. Set it up on your own board and try variations; see whether you can figure out why the players made some of the moves they did. (Black's 24 b5 allowing the triple check may seem peculiar, but experienced Black would simply resign rather than allow Red B5.) It may seem amazing, but all thirteen moves played here by Red were absolutely essential to Red's guaranteed victory. (E1 could be substituted for C1, but I treat C1 as a uniquely winning move anyway, since E1 leads to the position's mirror image and thus obviously yields the same result.) After you've studied this book from cover to cover, you should find that most of the moves in this game are evidently best.

Using example games, I develop the key concepts of CONNECT 4 play. When a problem is posed, be sure to try to guess the winning move before reading ahead. The book emphasizes *strategic principles*, but first I need to discuss *tactical sequences*.

Before leaving Chapter 1, I must mention that the triple check in Diagram 8 hardly holds the record for *most simultaneous checks*. That record is indubitably held by John Tromp's septuple check, shown in Diagram 9.

DIAGRAM 9

John Tromp's Septuple Check



Moves: C1? d1 D2 d3! E1? d4 C2 c3 E2 e3 B1 b2! F1 f2 C4 d5 E4 (18) __

Chapter 2

Tactical Ideas

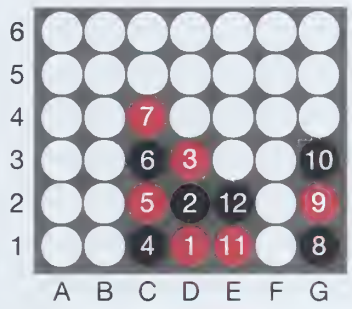
While *strategy* refers to the evaluation and selection of long-term threats, *tactics* refer to the parries and thrusts that occur move by move. Obviously, strategy and tactics work together; it is through tactical skill that a player is able to establish winning strategic threats.

Value of Opponent's Reply

An important tactic is to make your disk more valuable than your opponent's reply. In Diagram 10, any of the central cells C5, D4, or E3 might seem like a useful location for Red's disk 13, but C5 is the winning move. Black can answer 13 D4 or E3 by taking a valuable central cell (d5 or e4), but Red needn't worry about Black taking the remote cell c6. (But judge all cases on their own merits; with a slight change, D4 or E3 might be the winning move.)

DIAGRAM 10

Red to play, with only one move to win



Moves: D1! d2 D3! c1 C2 c3 C4 g1 G2! g3 E1! e2 (13) __

Even better than a move that gives up no strong cell for the opponent is one that forces the opponent to waste a move. This idea is seen in several of the problems, such as Problem 155, in which Black, already occupying f3 and f4, is delighted to play 12 f5 since Red then has to waste a turn with the otherwise almost useless blocking move at F6.

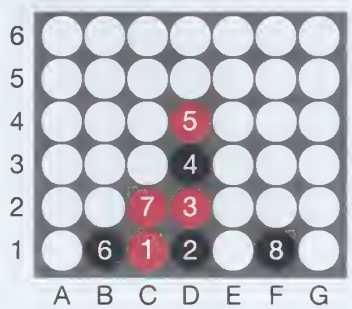
Tactical Sequences for Quick Wins

A sequence of forcing moves is called a *tactical sequence*, or *forcing combination*. Where should Red play in Diagram 11 to win quickly?

Red plays a sequence of forcing moves and wins quickly, as shown in Diagram 12. Since Red completes four-in-row by playing just *three* Red disks after Diagram 11, we

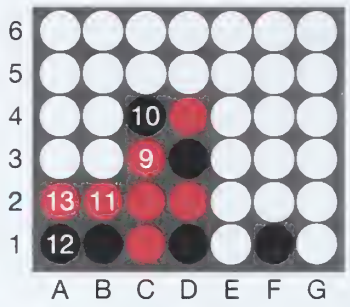
DIAGRAM 11

Red to play, and mate in three



Moves: C1? d1 D2 d3! D4 b1? C2? f1? (9) __

DIAGRAM 12
Red has won

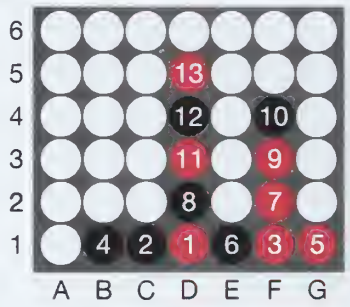


Moves: C1? d1 D2 d3! D4 b1? C2? f1? (9) C3 c4 B2! a1 A2

can say that from that diagram, Red has a guaranteed *four-in-row* in three moves, or, for brevity, *mate in three.*

It turns out that Red (the player who moves first) can win every game if he or she plays *perfectly*, but that doesn't mean Black (the player who moves second) has no chance. There are many difficult variations, and even the very best human player in the world isn't fully confident of victory when he or she moves first. When two experts play each other, Red almost always wins, but nonexpert play varies. (In casual play against opponents of medium skill, the author usually wins almost effortlessly moving second but has to concentrate when playing as the first player.)

DIAGRAM 13
Black to play and win

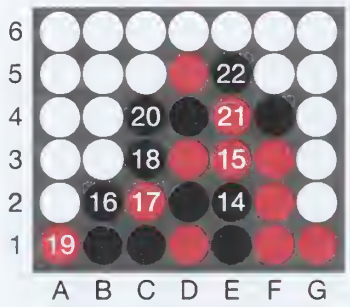


Moves: D1! c1 F1 b1 G1 e1 F2 d2 F3 f4 D3! d4 D5?? (14) __

Unlike Diagram 11 (a nonstandard opening with several bad moves), Diagram 13 comes from an expert opening, with Red 13 D5 being a not-so-obvious blunder. It is Black who can now win with a tactical sequence, specifically mate in five. Remember to find it yourself before reading ahead!

The solution is shown in Diagram 14. Take full credit if you started with 14 b2 instead of e2—that play also leads to mate in five. (The other five possibilities for Black's play 14 all lead to Red victory.)

DIAGRAM 14
Black has won



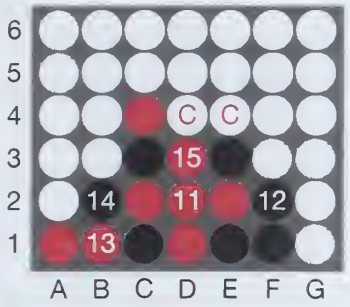
Moves: D1! c1 F1 b1 G1 e1 F2 d2 F3 f4 D3! d4 D5?? (14) e2 E3 b2! C2 c3! A1 c4 E4 e5!

Diagram 15 (related to the opening in Diagram 157; see Chapter 10) shows another winning combination. In Diagram 16, Black has "pushed" in one column with b1-b2-b3, forcing a B4 reply, and then seized b5; such a maneuver is called a *push-up play.*

In the previous examples, the forcing moves threatened a quick win. More often, a forcing move will force the opponent to prevent a strategic long-term threat. There are many examples of this strategy in this book; for example, in Diagram 95 (see Chapter 8).

DIAGRAM 15

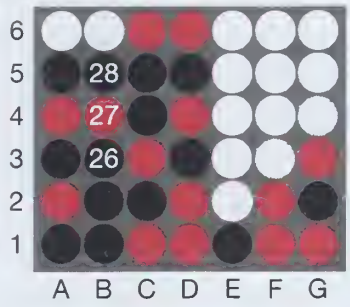
Black to play (will lose)



Moves: D1! e1 A1 c1 E2? e3! C2 c3 C4 f1? (11) D2! f2 B1! b2 D3! (16) __

DIAGRAM 16

Black has won



Moves: D1! b1 D2? d3! C1? c2 C3 c4 D4 a1? A2? c5 F1 e1! F2 d5?? G1 g2 D6 a3 A4? b2? C6? a5! G3? (26) b3! B4 b5!

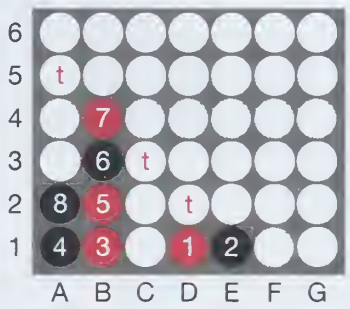
Preventing a Tactical Ploy

If your opponent has a successful tactical ploy, it may be that you should have seen it coming and prevented it. Often the ordinary strong-looking move will serve this purpose, but sometimes you will have to make an unusual move to stop a ploy. Diagram 17 shows an interesting example. Where should Red play to win in Diagram 17? (These opening moves may look quite peculiar, but this is actually an expertly played opening.)

I've indicated a possible future diagonal four-in-row that may lead to Red's victory; the threats at A5 and C3 are strong for Red because they are on odd rows. With practice you will eventually be able to recognize a pattern like this quickly. Red's plan will be to play D2 when appropriate to complete the minor odd-row threats at A5 and C3, but he or she must watch out lest Black foil this with a push-up play. An expert Black will play 10 a3 A4 a5 in response to 9 B5, or play as shown in Diagram 18 in response to 9 E2. Thus, Red's winning play is 9 A3 to prevent this push-up play. With the center still largely unplayed, A3 will strike most players as an unlikely best move. (Some Red players might try 9 D2 d3 first and then play 11 A3, but this allows Black 10 d3, 12 e2, followed by 14 d4 or e4.)

DIAGRAM 17

Red to play, with only one move to win



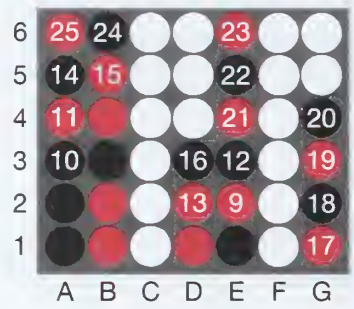
Moves: D1! e1 B1 a1 B2! b3 B4 a2 (9) __

In Diagram 18, after Red's mistaken 9 E2, most of the moves are shown with exclamation marks; Black must play very precisely to get a draw, and Red must defend very precisely to ensure that Black gets no better than a draw. If instead Red

plays correctly with 9 A3, the game is likely to continue as shown in Diagram 19. One way to understand why A3 is a better play than E2 is to look at the cells *above* these two moves, where Black would then have the opportunity to play. A4 is an almost worthless cell, while E3 is a cell in a prime central location, which Black would surely occupy if Red gave him or her the chance.

DIAGRAM 18

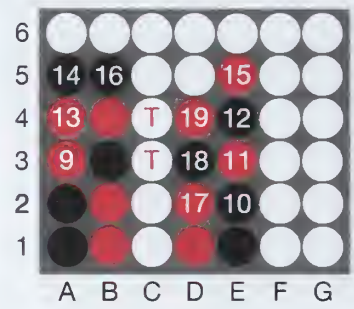
Black to play and draw



Moves: D1! e1 B1 a1 B2! b3 B4 a2 (9) E2? a3! A4! e3! D2 a5! B5! d3! G1 g2! G3 g4! E4! e5 E6! b6 A6! (26) __

DIAGRAM 19

Black to play (will lose)



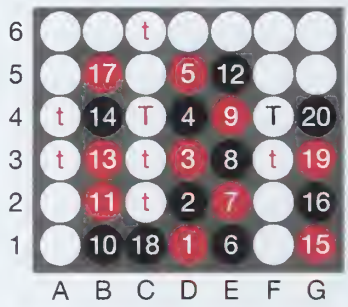
Moves: D1! e1 B1 a1 B2! b3 B4 a2 (9) A3! e2 E3! e4 A4 a5 E5! b5 D2! d3 D4! (20) __

Races

If your tactical plan depends on playing up one column, while your opponent has a tactical play in a different column, the phrase *the race is to the swift* may apply. Diagram 207, in Chapter 12, discusses a very interesting (and difficult) example of such a race; the position is repeated here as Diagram 20. Give yourself a big pat on the back if you can guess Red's next move!

DIAGRAM 20

Red to play, with only one move to win



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 e5 B3 b4 G1! g2 B5 c1 G3 g4 (21) __

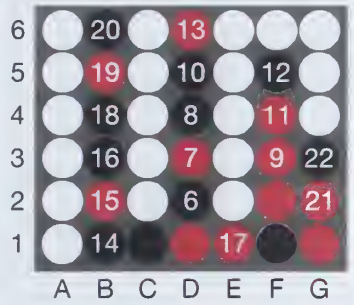
More Examples of Quick Victory

It's always more fun to win quickly than slowly, and sometimes a tactical sequence will lead to four-in-row in just a few moves. Often a player forces an adjacent threat pair and converts it to quick victory. This applies in Diagram 21. Do you see Red's quick win?

Red's winning move is shown in Diagram 22. With 23 E2, Red gets an adjacent threat pair and wins quickly. In fact, 23 E2 is the only move for Red that wins at all! If instead Red plays as shown in Diagram 23, Black builds a major odd-row threat and gets a draw.

DIAGRAM 21

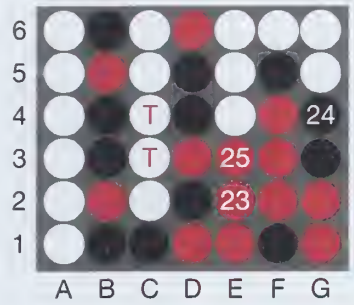
Red to play, with only one move to win



Moves: D1! c1 G1 f1 F2! (6) d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! b3 E1! b4 B5! b6 G2 g3 (23) __

DIAGRAM 22

Black to play (will lose)



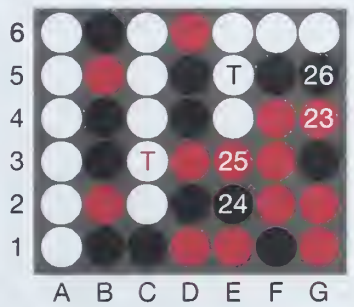
Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! b3 E1! b4 B5! b6 G2 g3 (23) E2! g4 E3! (26) __

Often the purpose of a tactical sequence won't be to get a four-in-row quickly but just to establish a winning strategic threat. Diagram 24 shows an example of this. Where should Red play next?

Diagram 24 is of historic interest. It appears in the master's thesis of Victor Allis, a computer scientist famous for his studies of CONNECT 4 and gomoku.

DIAGRAM 23

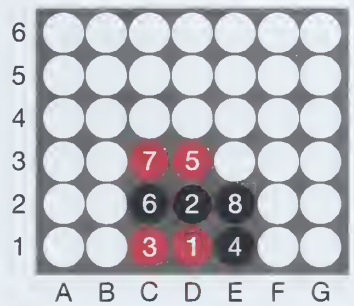
Red to play and draw



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! b3 E1! b4 B5! b6 G2 g3 (23) G4? e2! E3! g5! (27) __

DIAGRAM 24

Red to play, with only one move to win



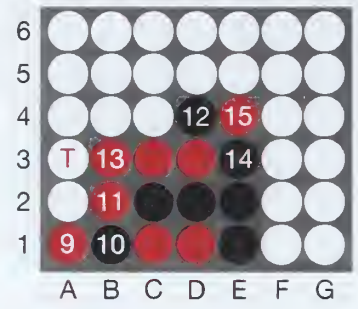
Moves: D1! d2 C1?? e1 D3 c2?? C3 e2 (9) __

Diagram 25 shows how a precise tactical sequence yields a winning strategic threat for Red.

In Diagram 26, Red can achieve a winning adjacent threat pair in three moves. As shown in Diagram 27, Red may need four additional moves to complete a four-in-row. (Black 16 g1 was essential since if Red plays there, E1-E2 would be a quick-winning adjacent threat pair.)

DIAGRAM 25

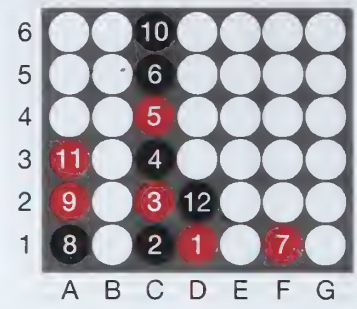
Black to play (will lose)



Moves: D1! d2 C1?? e1 D3 c2?? C3 e2 (9) A1! b1 B2! d4 B3! e3 E4! (16) __

DIAGRAM 26

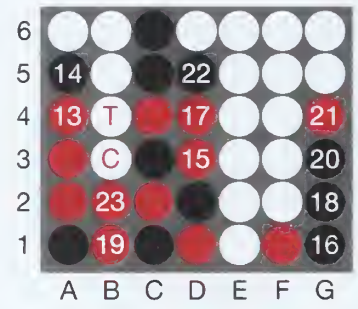
Red to play, and mate in seven



Moves: D1! c1 C2? c3! C4 c5? F1? a1? A2? c6? A3 d2 (13) __

DIAGRAM 27

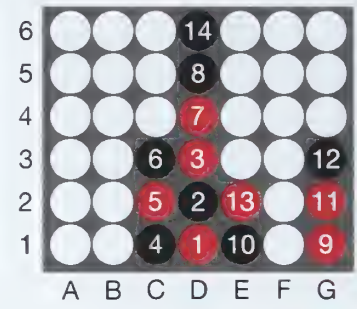
Black to play (will lose)



Moves: D1! c1 C2? c3! C4 c5? F1? a1? A2? c6? A3 d2 (13) A4 a5 D3 g1 D4 g2 B1 g3 G4! d5 B2 (24) __

DIAGRAM 28

Red to play, and mate in three

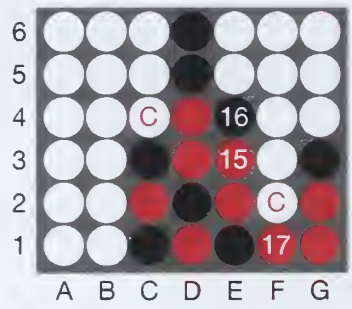


Moves: D1! d2 D3! c1 C2 c3 D4 d5 G1! e1 G2! g3 E2! d6 (15) __

Here's a final example of quick victory. (There are more in the problem sets.) In Diagram 28, Red will win no matter where he or she plays, but only one move assures Red of mate in three. Diagram 29 shows the solution.

DIAGRAM 29

Black to play (will lose)



Moves: D1! d2 D3! c1 C2 c3 D4 d5 G1! e1 G2! g3 E2! d6 (15) E3 e4 F1 (18) __

Problem Sets

This is the most important part of this book! As you study the game, work one of these problem sets from time to time and score yourself, using the "Answers to Problem Sets" section. Be aware that most of the problems are *quite difficult*. In most cases you won't be able to "see ahead" far enough to be sure what the correct move is; instead, apply your *intuition* and *guess* the correct move, by visualizing where the key threats are or will come.

Most of the problems are taken from example games discussed elsewhere in the book. (Sometimes the problem will be a mirror image of the discussion, with column A replacing column G, for example.) Problem diagrams are shown without move annotations or indications of the game result (although the moves leading up to a problem are shown, the exclamation points and question marks that would normally appear have been omitted). In each case, find the appropriate move for the player whose turn it is to move. In 46 of the problems, the correct move will be good enough only to draw (all other moves would lose), but most of these problems aren't identified as such. With very few exceptions, there is only one correct answer. The quick victory (mate) problems may have multiple winning moves, but you're looking for the one that leads to four-in-row most quickly. Some of the problems in the easier sets come with hints.

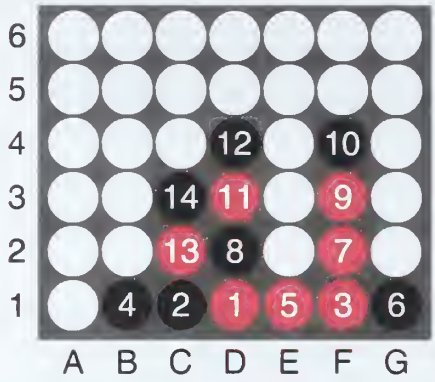
Difficulty rankings are only approximate and, anyway, perceived difficulty will vary from player to player. Also, since when you're playing a real game there will be no "little birdie" telling you whether your next move is hard to find or not, I've tried to keep you on your toes by including some hard problems in the easy sets and vice versa. To make this book useful for even the most advanced CONNECT 4 players, I've included some very hard problem sets. Some of these will stump all but the very greatest CONNECT 4 geniuses.

If you're a beginner or intermediate player, read the book cover to cover *twice*, and then, I think, you'll be an advanced player yourself. Again, please do *not* become discouraged if these problems seem very difficult.

Problem Set A (Easy)

PROBLEM 1

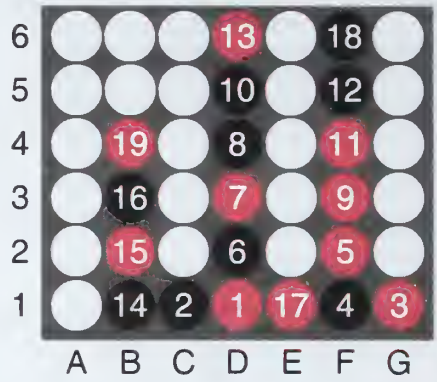
Red to play (Hint: Black threatens quick win)



Moves: D1 c1 F1 b1 E1 g1 F2 d2
F3 f4 D3 d4 C2 c3 (15) __

PROBLEM 2

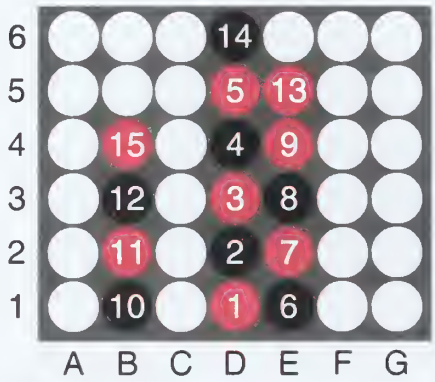
Black to play



Moves: D1 c1 G1 f1 F2 d2 D3 d4 F3 d5
F4 f5 D6 b1 B2 b3 E1 f6 B4 (20) __

PROBLEM 3

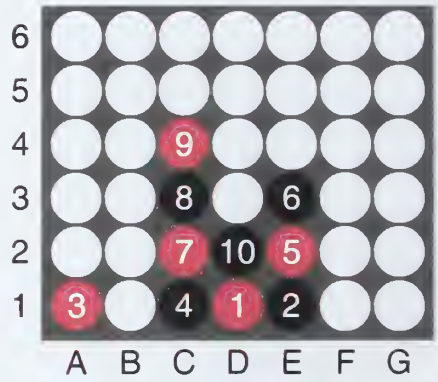
Black to play (Hint: Play where Red wants to play)



Moves: D1 d2 D3 d4 D5 e1 E2 e3
E4 b1 B2 b3 E5 d6 B4 (16) __

PROBLEM 4

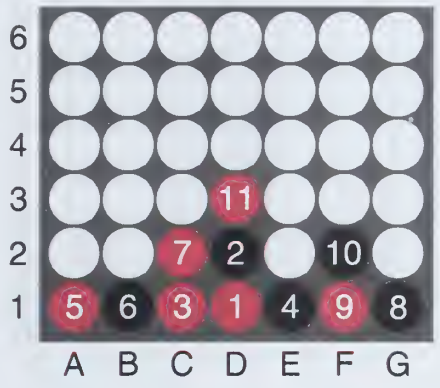
Red to play



Moves: D1 e1 A1 c1 E2 e3
C2 c3 C4 d2 (11) __

PROBLEM 5

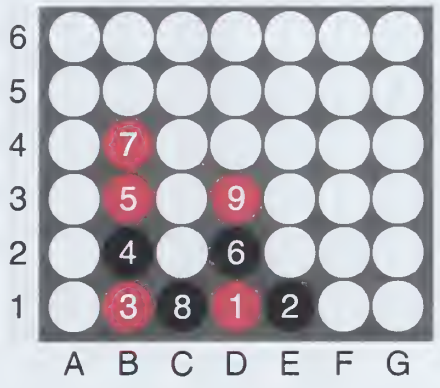
Black to play, and mate in three



Moves: D1 d2 C1 e1 A1 b1
C2 g1 F1 f2 D3 (12) __

PROBLEM 6

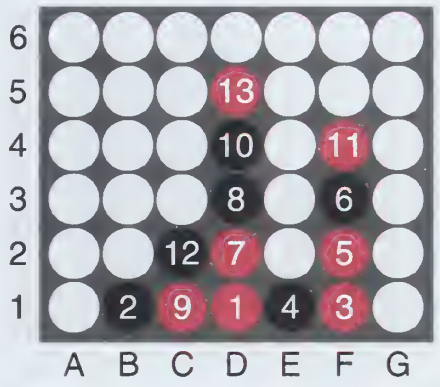
Black to play, and mate in four



Moves: D1 e1 B1 b2 B3 d2 B4 c1 D3 (10) __

PROBLEM 7

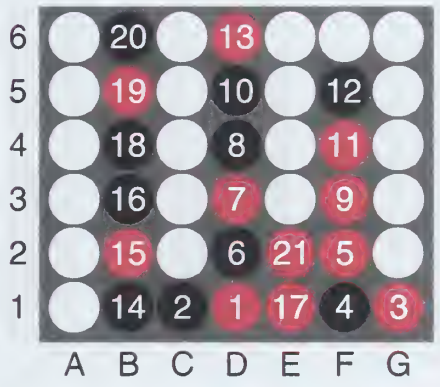
Black to play



Moves: D1 b1 F1 e1 F2 f3 D2
d3 C1 d4 F4 c2 D5 (14) __

PROBLEM 8

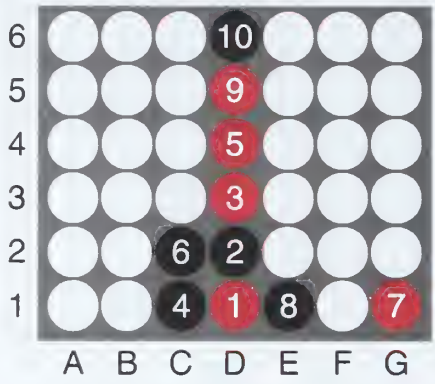
Black to play



Moves: D1 c1 G1 f1 F2 d2 D3 d4 F3 d5 F4
f5 D6 b1 B2 b3 E1 b4 B5 b6 E2 (22) __

PROBLEM 9

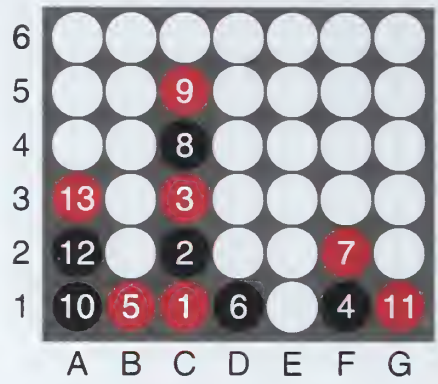
Red to play



Moves: D1 d2 D3 c1 D4 c2
G1 e1 D5 d6 (11) __

PROBLEM 10

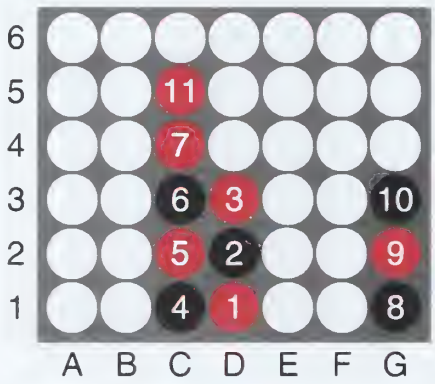
Black to play, and mate in four



Moves: C1 c2 C3 f1 B1 d1 F2
c4 C5 a1 G1 a2 A3 (14) __

PROBLEM 11

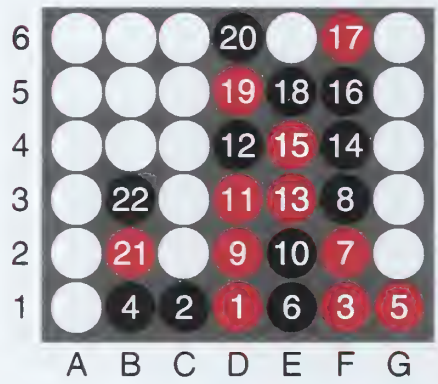
Black to play



Moves: D1 d2 D3 c1 C2 c3
C4 g1 G2 g3 C5 (12) __

PROBLEM 12

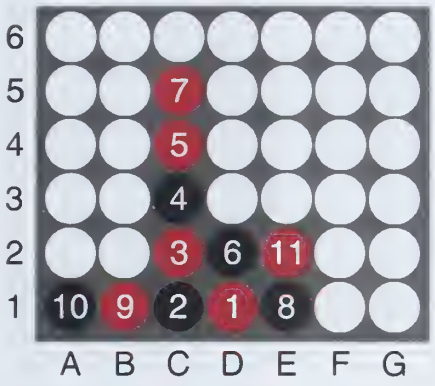
Red to play



Moves: D1 c1 F1 b1 G1 e1 F2 f3 D2 e2 D3
d4 E3 f4 E4 f5 F6 e5 D5 d6 B2 b3 (23) __

PROBLEM 13

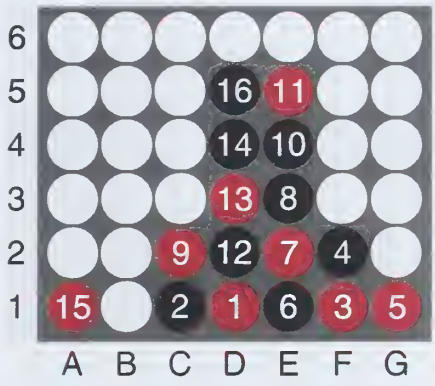
Black to play, and mate in four



Moves: D1 c1 C2 c3 C4 d2
C5 e1 B1 a1 E2 (12) __

PROBLEM 14

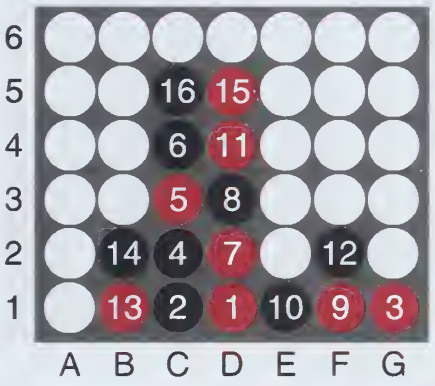
Red to play



Moves: D1 c1 F1 f2 G1 e1 E2 e3 C2
e4 E5 d2 D3 d4 A1 d5 (17) __

PROBLEM 15

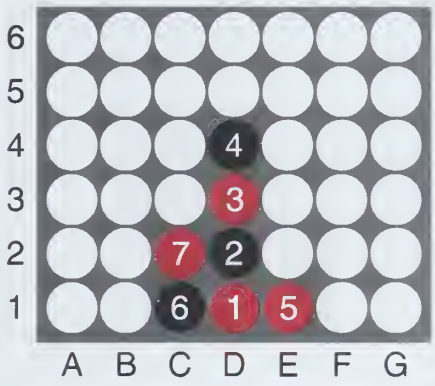
Red to play



Moves: D1 c1 G1 c2 C3 c4 D2 d3
F1 e1 D4 f2 B1 b2 D5 c5 (17) __

PROBLEM 16

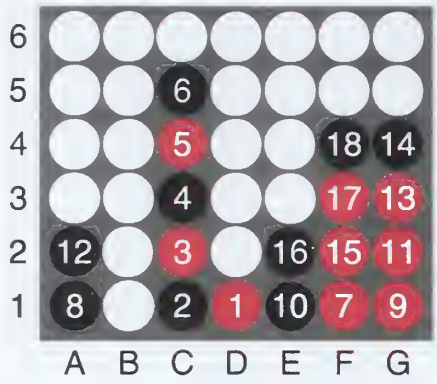
Black to play (Hint: Where does Red want to play?)



Moves: D1 d2 D3 d4 E1 c1 C2 (8) __

PROBLEM 17

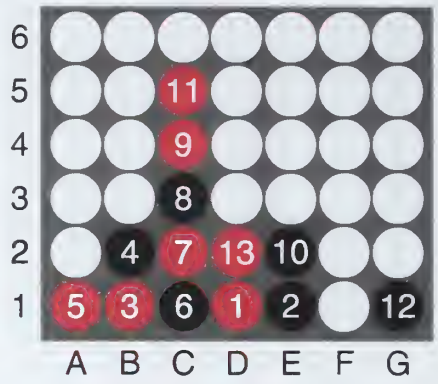
Red to play



Moves: D1 c1 C2 c3 C4 c5 F1 a1 G1
e1 G2 a2 G3 g4 F2 e2 F3 f4 (19) __

PROBLEM 18

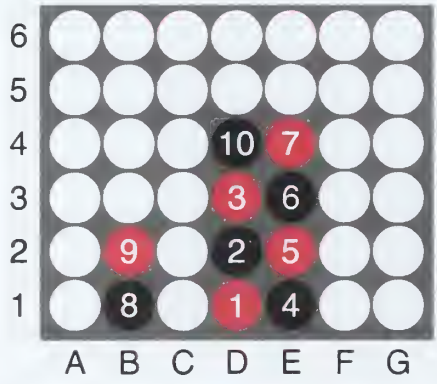
Black to play



Moves: D1 e1 B1 b2 A1 c1 C2
c3 C4 e2 C5 g1 D2 (14) __

PROBLEM 19

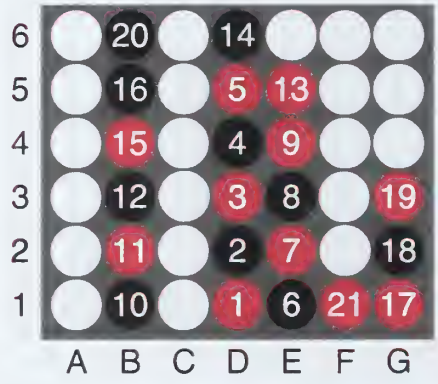
Red to play



Moves: D1 d2 D3 e1 E2 e3
E4 b1 B2 d4 (11) __

PROBLEM 20

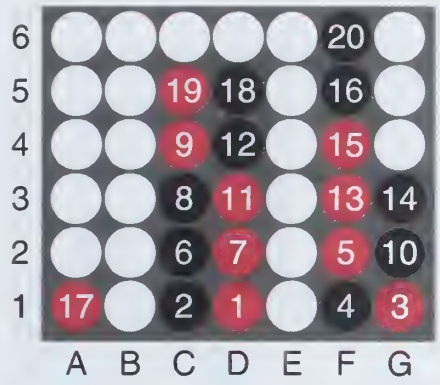
Black to play



Moves: D1 d2 D3 d4 D5 e1 E2 e3 E4 b1 B2
b3 E5 d6 B4 b5 G1 g2 G3 b6 F1 (22) __

PROBLEM 21

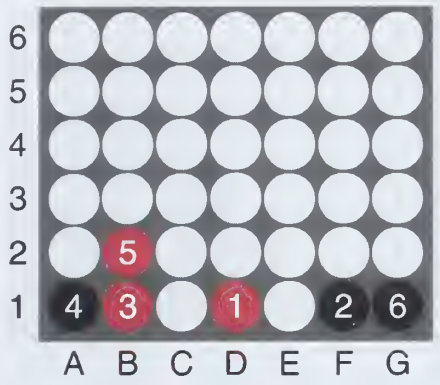
Red to play



Moves: D1 c1 G1 f1 F2 c2 D2 c3 C4 g2 D3 d4 F3 g3 F4 f5 A1 d5 C5 f6 (21) __

PROBLEM 22

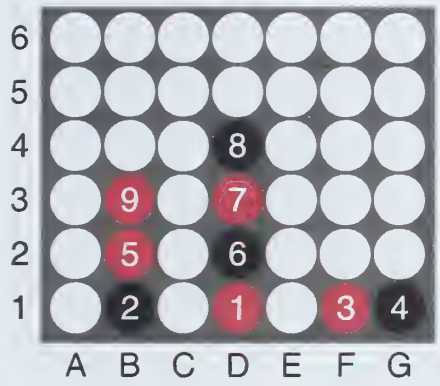
Red to play, and mate in seven



Moves: D1 f1 B1 a1 B2 g1 (7) __

PROBLEM 23

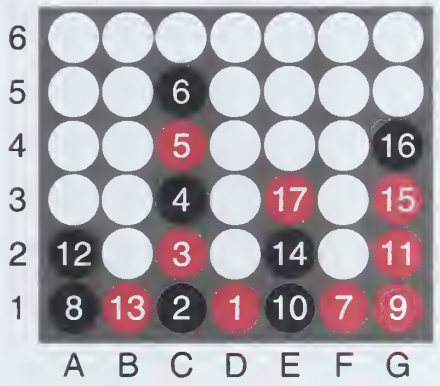
Black to play



Moves: D1 b1 F1 g1 B2 d2 D3 d4 B3 (10) __

PROBLEM 24

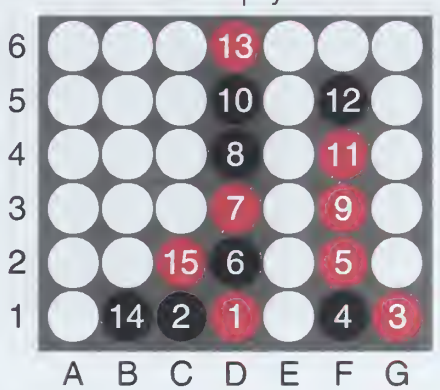
Black to play (Hint: Prevent key Red threat)



Moves: D1 c1 C2 c3 C4 c5 F1 a1 G1 e1 G2 a2 B1 e2 G3 g4 E3 (18) __

PROBLEM 25

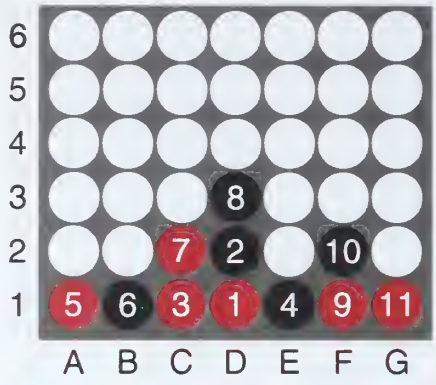
Black to play



Moves: D1 c1 G1 f1 F2 d2 D3 d4 F3 d5 F4 f5 D6 b1 C2 (16) __

PROBLEM 26

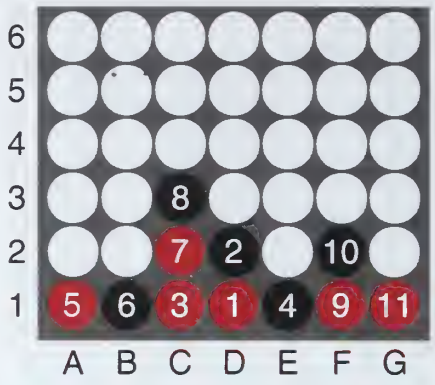
Black to play, and mate in four



Moves: D1 d2 C1 e1 A1 b1
C2 d3 F1 f2 G1 (12) __

PROBLEM 27

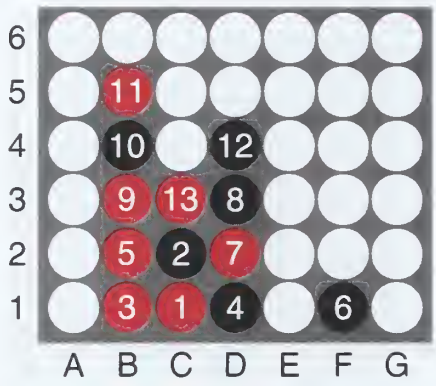
Black to play, and mate in five



Moves: D1 d2 C1 e1 A1 b1
C2 c3 F1 f2 G1 (12) __

PROBLEM 28

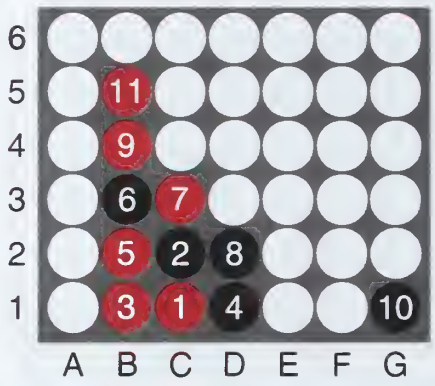
Black to play, and mate in four



Moves: C1 c2 B1 d1 B2 f1 D2
d3 B3 b4 B5 d4 C3 (14) __

PROBLEM 29

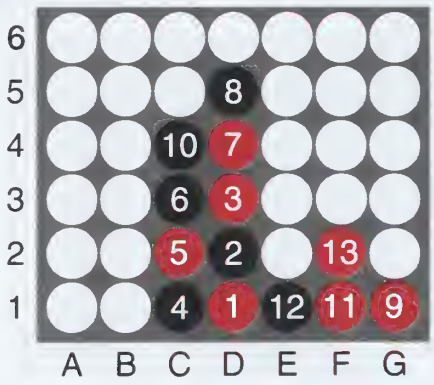
Black to play, and mate in four



Moves: C1 c2 B1 d1 B2 b3
C3 d2 B4 g1 B5 (12) __

PROBLEM 30

Black to play

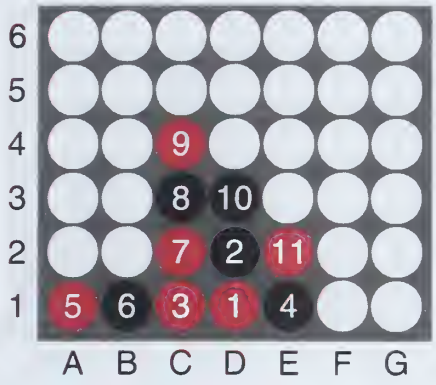


Moves: D1 d2 D3 c1 C2 c3 D4
d5 G1 c4 F1 e1 F2 (14) __

Problem Set B (Easy)

PROBLEM 31

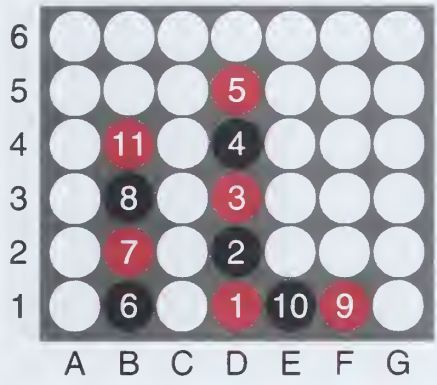
Black to play, and mate in three



Moves: D1 d2 C1 e1 A1 b1
C2 c3 C4 d3 E2 (12) __

PROBLEM 32

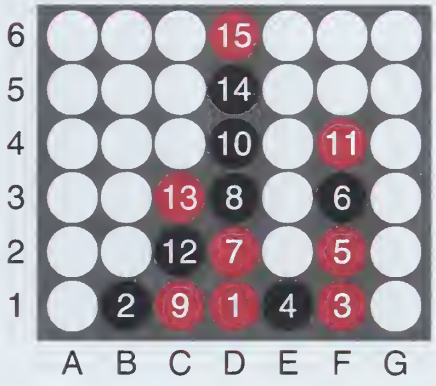
Black to play (Hint: Kill Red's third row)



Moves: D1 d2 D3 d4 D5 b1
B2 b3 F1 e1 B4 (12) __

PROBLEM 33

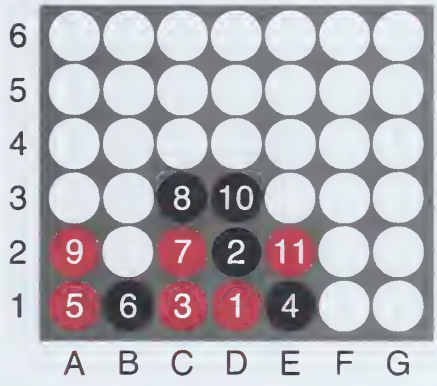
Black to play



Moves: D1 b1 F1 e1 F2 f3 D2 d3
C1 d4 F4 c2 C3 d5 D6 (16) __

PROBLEM 34

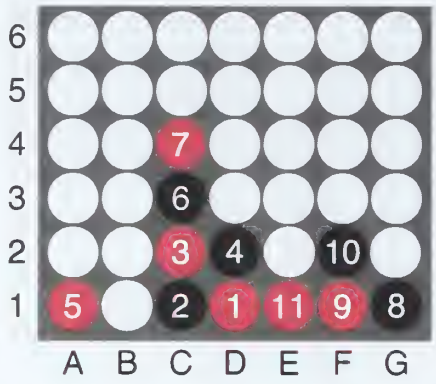
Black to play, and mate in four



Moves: D1 d2 C1 e1 A1 b1
C2 c3 A2 d3 E2 (12) __

PROBLEM 35

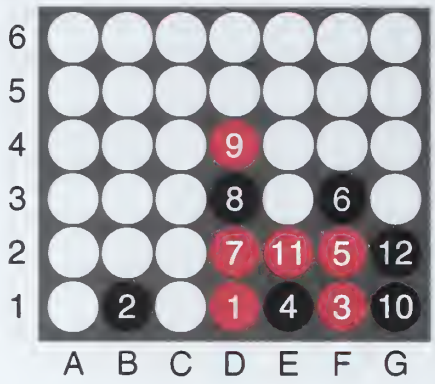
Black to play, and mate in four



Moves: D1 c1 C2 d2 A1 c3
C4 g1 F1 f2 E1 (12) __

PROBLEM 36

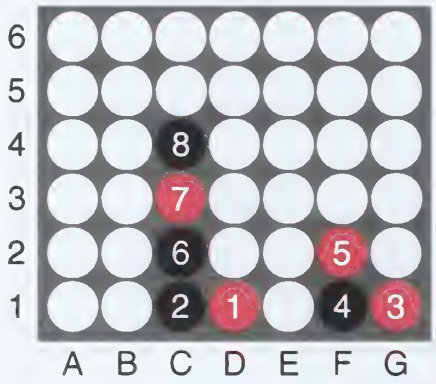
Red to play



Moves: D1 b1 F1 e1 F2 f3 D2
d3 D4 g1 E2 g2 (13) __

PROBLEM 37

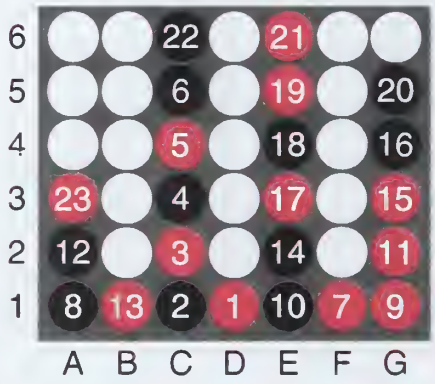
Red to play



Moves: D1 c1 G1 f1 F2 c2 C3 c4 (9) __

PROBLEM 38

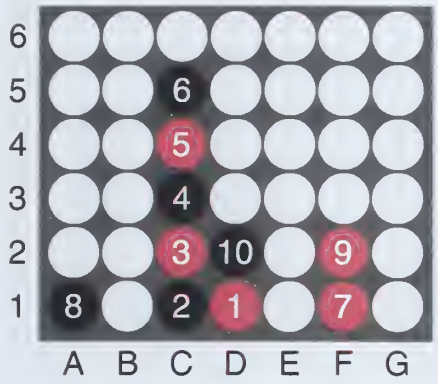
Black to play



Moves: D1 c1 C2 c3 C4 c5 F1 a1 G1 e1 G2
a2 B1 e2 G3 g4 E3 e4 E5 g5 E6 c6 A3 (24) __

PROBLEM 39

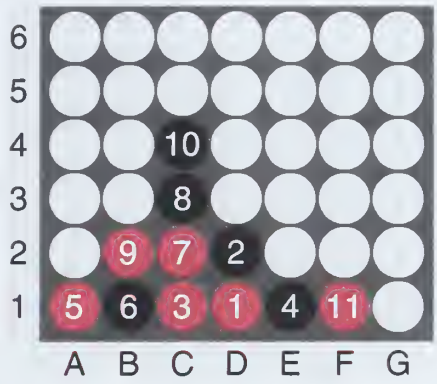
Red to play, and mate in six



Moves: D1 c1 C2 c3 C4 c5
F1 a1 F2 d2 (11) __

PROBLEM 40

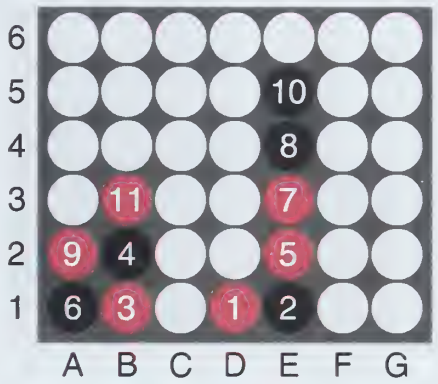
Black to play, and mate in four



Moves: D1 d2 C1 e1 A1 b1
C2 c3 B2 c4 F1 (12) __

PROBLEM 41

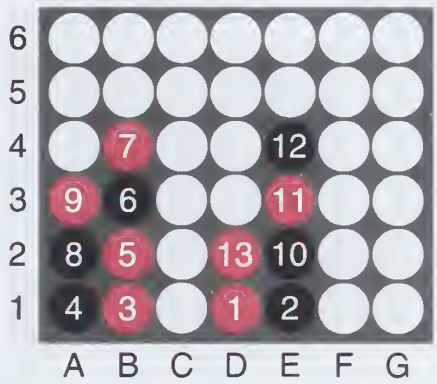
Black to play



Moves: D1 e1 B1 b2 E2 a1
E3 e4 A2 e5 B3 (12) __

PROBLEM 42

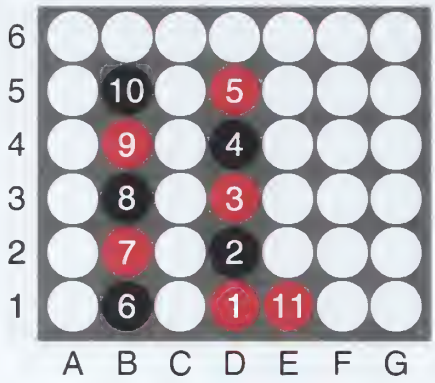
Black to play (Hint: Play where Red wants to play)



Moves: D1 e1 B1 a1 B2 b3 B4
a2 A3 e2 E3 e4 D2 (14) __

PROBLEM 43

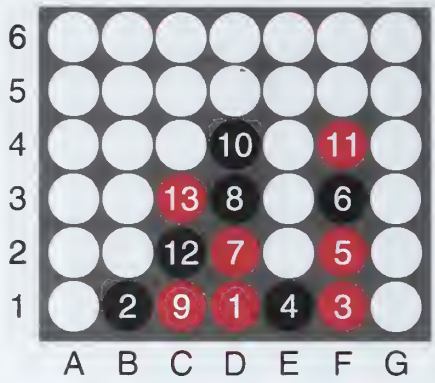
Black to play



Moves: D1 d2 D3 d4 D5 b1
B2 b3 B4 b5 E1 (12) __

PROBLEM 44

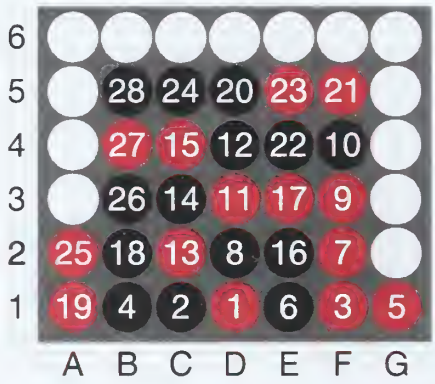
Black to play



Moves: D1 b1 F1 e1 F2 f3 D2
d3 C1 d4 F4 c2 C3 (14) __

PROBLEM 45

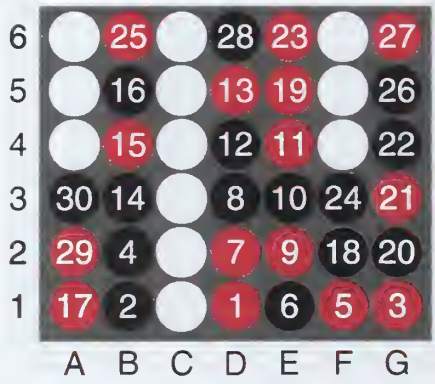
Red to play (Hint: Kill Black's threat)



Moves: D1 c1 F1 b1 G1 e1 F2 d2 F3
f4 D3 d4 C2 c3 C4 e2 E3 b2 A1 d5
F5 e4 E5 c5 A2 b3 B4 b5 (29) __

PROBLEM 46

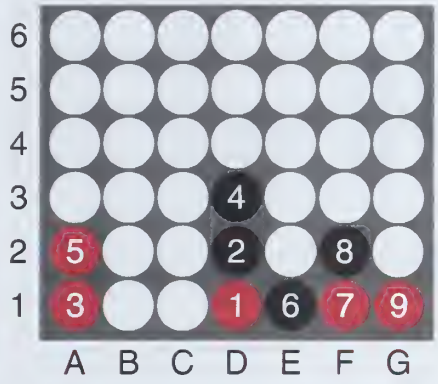
Red to play



Moves: D1 b1 G1 b2 F1 e1 D2 d3 E2
e3 E4 d4 D5 b3 B4 b5 A1 f2 E5 g2 G3
g4 E6 f3 B6 g5 G6 d6 A2 a3 (31) __

PROBLEM 47

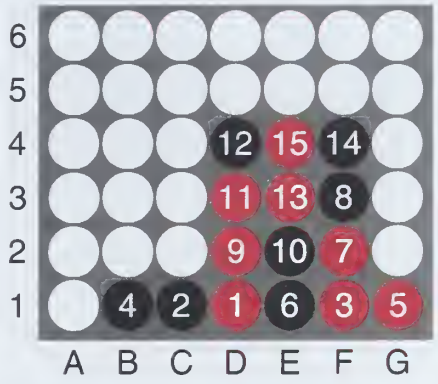
Black to play, and mate in five



Moves: D1 d2 A1 d3 A2 e1 F1 f2 G1 (10) __

PROBLEM 48

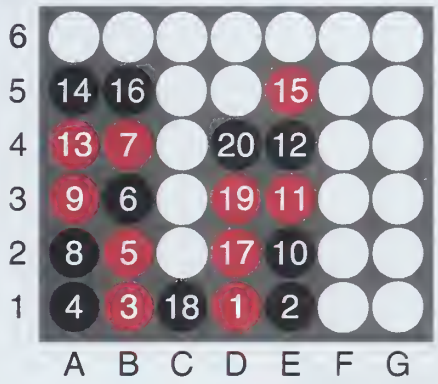
Black to play



Moves: D1 c1 F1 b1 G1 e1 F2 f3
D2 e2 D3 d4 E3 f4 E4 (16) __

PROBLEM 49

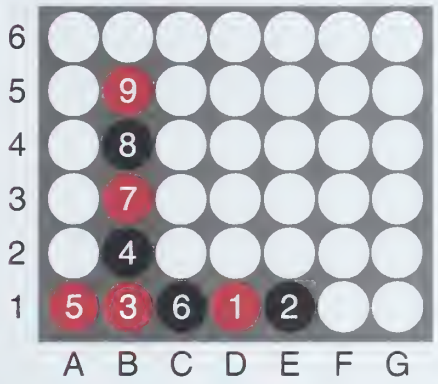
Red to play (Hint: Watch out for Black!)



Moves: D1 e1 B1 a1 B2 b3 B4 a2 A3 e2
E3 e4 A4 a5 E5 b5 D2 c1 D3 d4 (21) __

PROBLEM 50

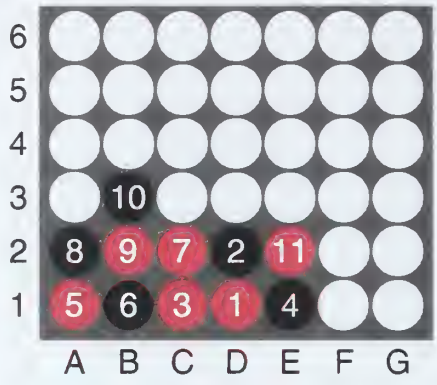
Black to play, and mate in three



Moves: D1 e1 B1 b2 A1 c1 B3 b4 B5 (10) __

PROBLEM 51

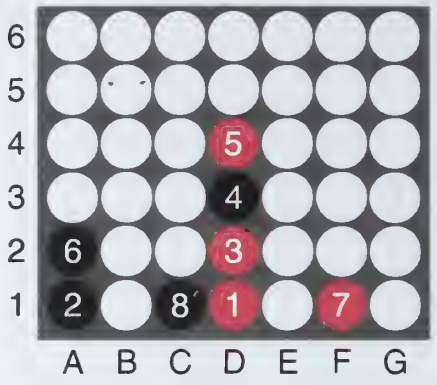
Black to play, and mate in three



Moves: D1 d2 C1 e1 A1 b1
C2 a2 B2 b3 E2 (12) __

PROBLEM 52

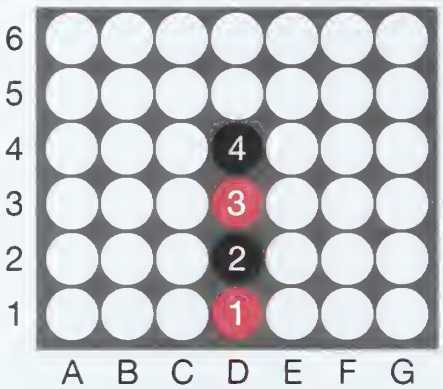
Red to play, and mate in five



Moves: D1 a1 D2 d3 D4 a2 F1 c1 (9) __

PROBLEM 53

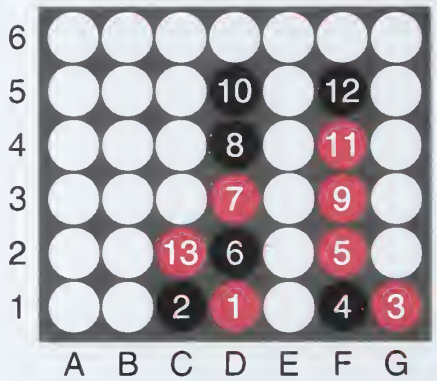
Red to play, and mate in 19



Moves: D1 d2 D3 d4 (5) __

PROBLEM 54

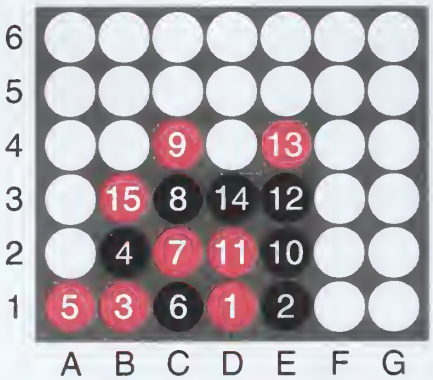
Black to play



Moves: D1 c1 G1 f1 F2 d2 D3
d4 F3 d5 F4 f5 C2 (14) __

PROBLEM 55

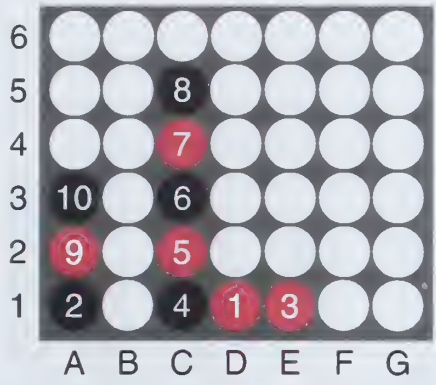
Black to play



Moves: D1 e1 B1 b2 A1 c1 C2 c3
C4 e2 D2 e3 E4 d3 B3 (16) __

PROBLEM 56

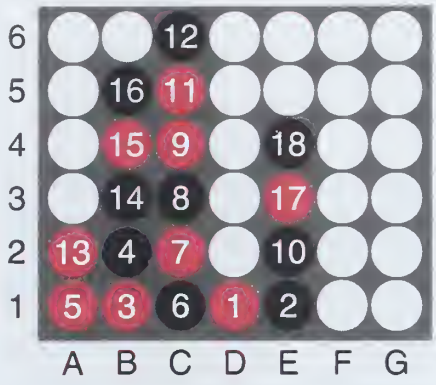
Red to play, and mate in five



Moves: D1 a1 E1 c1 C2 c3
C4 c5 A2 a3 (11) __

PROBLEM 57

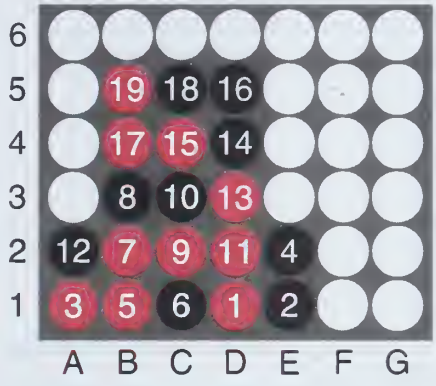
Red to play



Moves: D1 e1 B1 b2 A1 c1 C2 c3 C4
e2 C5 c6 A2 b3 B4 b5 E3 e4 (19) __

PROBLEM 58

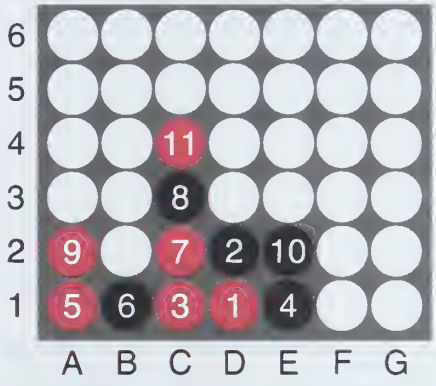
Black to play (Hint: Kill Red's right-side defense)



Moves: D1 e1 A1 e2 B1 c1 B2 b3 C2 c3
D2 a2 D3 d4 C4 d5 B4 c5 B5 (20) __

PROBLEM 59

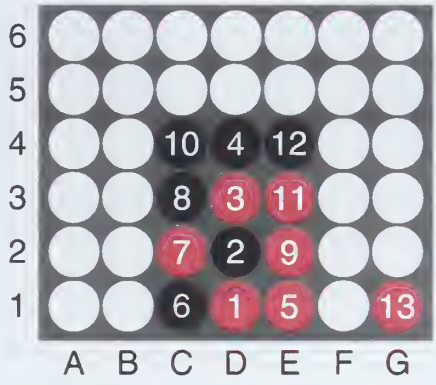
Black to play, and mate in five



Moves: D1 d2 C1 e1 A1 b1
C2 c3 A2 e2 C4 (12) __

PROBLEM 60

Black to play

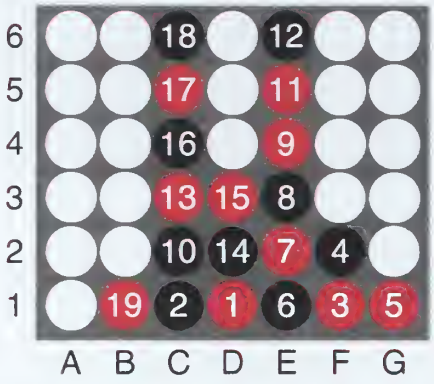


Moves: D1 d2 D3 d4 E1 c1 C2
c3 E2 c4 E3 e4 G1 (14) __

Problem Set C (Medium)

PROBLEM 61

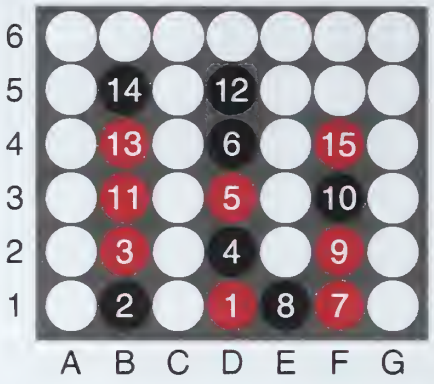
Black to play



Moves: D1 c1 F1 f2 G1 e1 E2 e3 E4 c2 E5 e6 C3 d2 D3 c4 C5 c6 B1 (20) __

PROBLEM 62

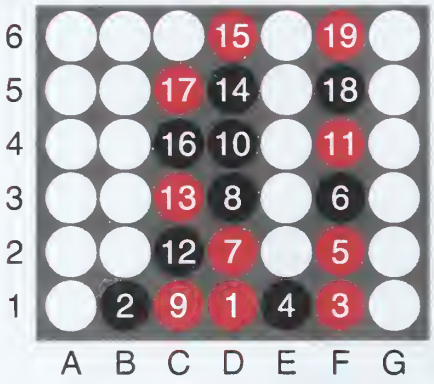
Black to play



Moves: D1 b1 B2 d2 D3 d4 F1 e1 F2 f3 B3 d5 B4 b5 F4 (16) __

PROBLEM 63

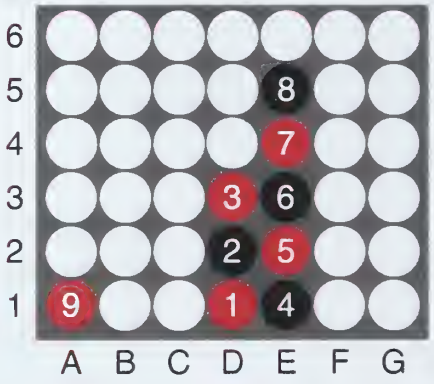
Black to play



Moves: D1 b1 F1 e1 F2 f3 D2 d3 C1 d4 F4 c2 C3 d5 D6 c4 C5 f5 F6 (20) __

PROBLEM 64

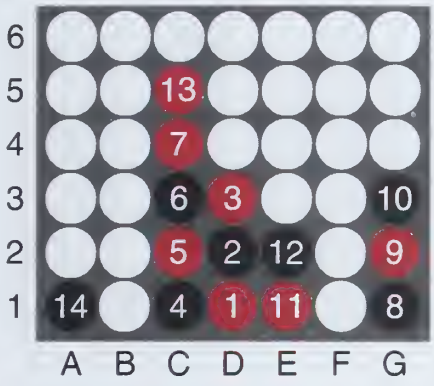
Black to play



Moves: D1 d2 D3 e1 E2 e3 E4 e5 A1 (10) __

PROBLEM 65

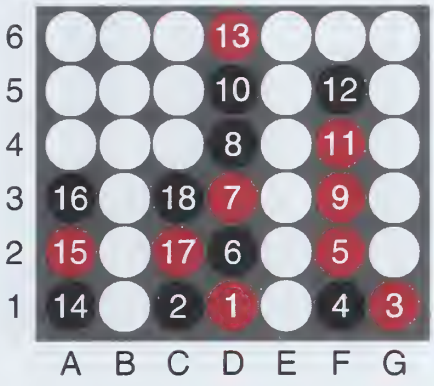
Red to play (Hint: Beware of Black d4)



Moves: D1 d2 D3 c1 C2 c3 C4 g1 G2 g3 E1 e2 C5 a1 (15) __

PROBLEM 66

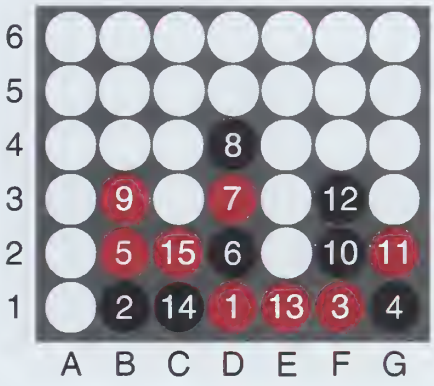
Red to play



Moves: D1 c1 G1 f1 F2 d2 D3 d4 F3 d5 F4 f5 D6 a1 A2 a3 C2 c3 (19) __

PROBLEM 67

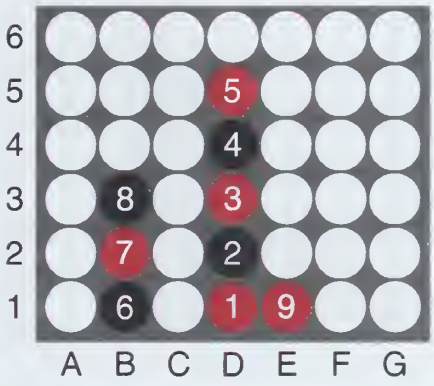
Black to play



Moves: D1 b1 F1 g1 B2 d2 D3 d4 B3 f2 G2 f3 E1 c1 C2 (16) __

PROBLEM 68

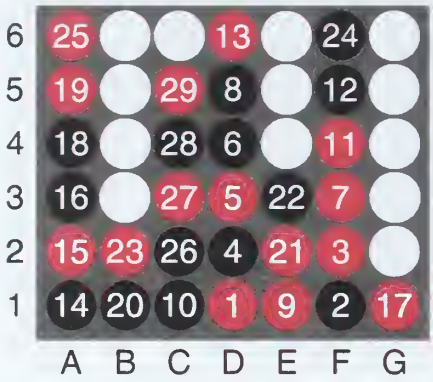
Black to play



Moves: D1 d2 D3 d4 D5 b1 B2 b3 E1 (10) __

PROBLEM 69

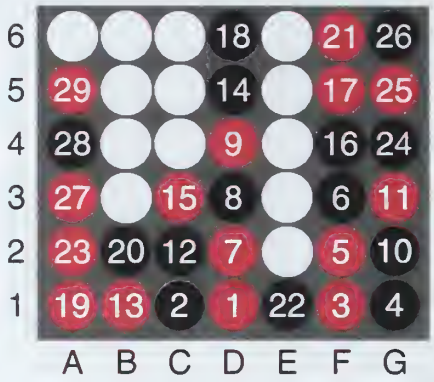
Black to play



Moves: D1 f1 F2 d2 D3 d4 F3 d5 E1 c1 F4 f5 D6 a1 A2 a3 G1 a4 A5 b1 E2 e3 B2 f6 A6 c2 C3 c4 C5 (30) __

PROBLEM 70

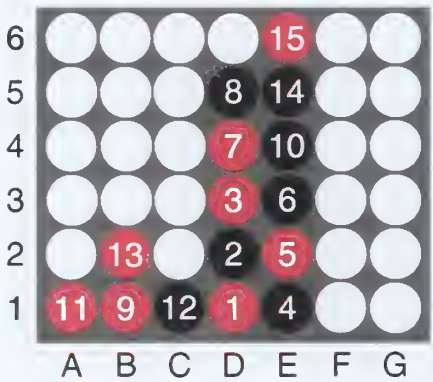
Black to play



Moves: D1 c1 F1 g1 F2 f3 D2 d3 D4 g2 G3 c2 B1 d5 C3 f4 F5 d6 A1 b2 F6 e1 A2 g4 G5 g6 A3 a4 A5 (30) __

PROBLEM 71

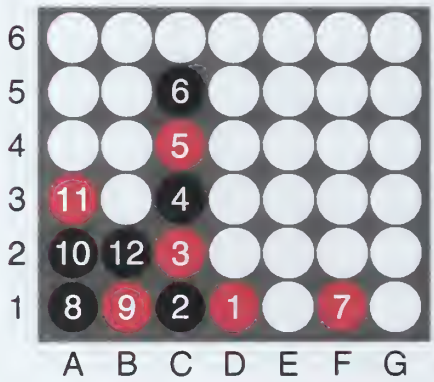
Black to play



Moves: D1 d2 D3 e1 E2 e3 D4 d5 B1 e4 A1 c1 B2 e5 E6 (16) __

PROBLEM 72

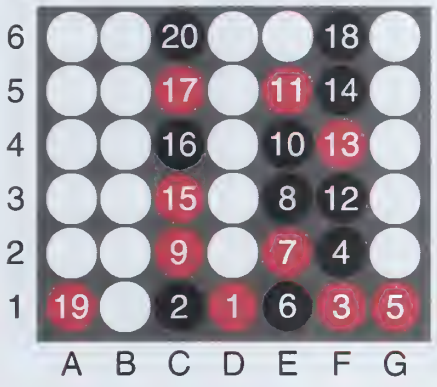
Red to play, and mate in seven



Moves: D1 c1 C2 c3 C4 c5 F1 a1 B1 a2 A3 b2 (13) __

PROBLEM 73

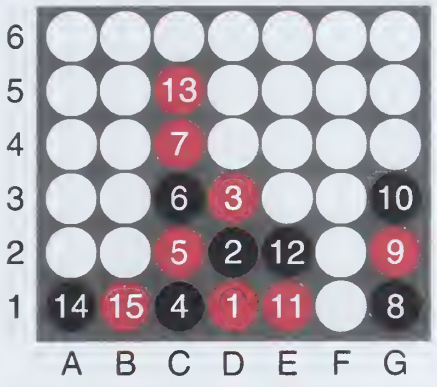
Red to play



Moves: D1 c1 F1 f2 G1 e1 E2 e3 C2 e4
E5 f3 F4 f5 C3 c4 C5 f6 A1 c6 (21) __

PROBLEM 74

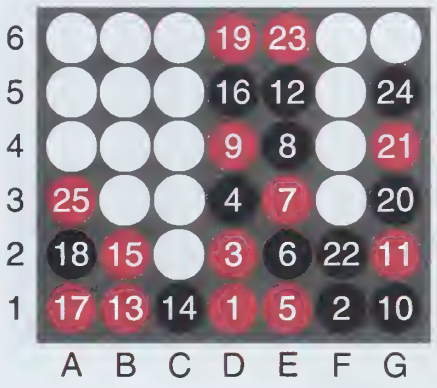
Black to play



Moves: D1 d2 D3 c1 C2 c3 C4 g1
G2 g3 E1 e2 C5 a1 B1 (16) __

PROBLEM 75

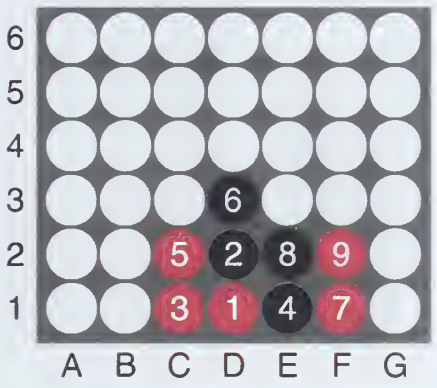
Black to play



Moves: D1 f1 D2 d3 E1 e2 E3 e4
D4 g1 G2 e5 B1 c1 B2 d5 A1 a2
D6 g3 G4 f2 E6 g5 A3 (26) __

PROBLEM 76

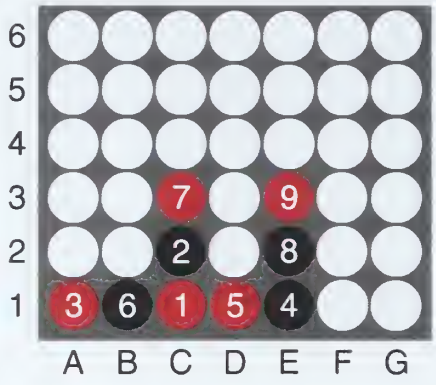
Black to play, and mate in six



Moves: D1 d2 C1 e1 C2 d3 F1 e2 F2 (10) __

PROBLEM 77

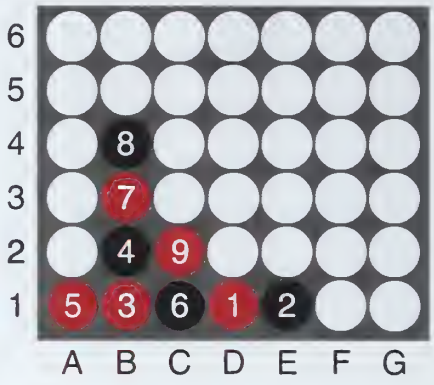
Black to play, and mate in five



Moves: C1 c2 A1 e1 D1 b1 C3 e2 E3 (10) __

PROBLEM 78

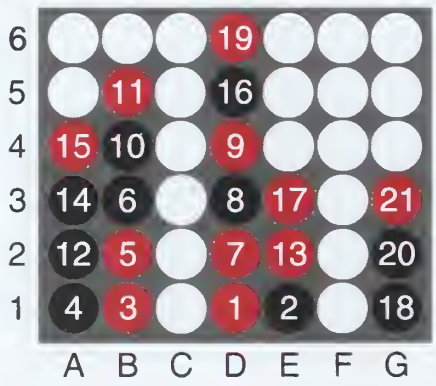
Black to play (Hint: Kill Red's third-row chances)



Moves: D1 e1 B1 b2 A1 c1 B3 b4 C2 (10) __

PROBLEM 79

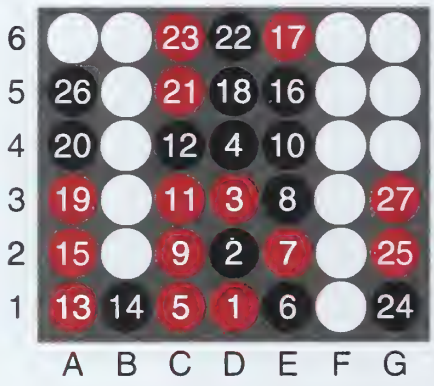
Black to play (Hint: Prevent key Red threat)



Moves: D1 e1 B1 a1 B2 b3 D2 d3 D4 b4 B5 a2 E2 a3 A4 d5 E3 g1 D6 g2 G3 (22) __

PROBLEM 80

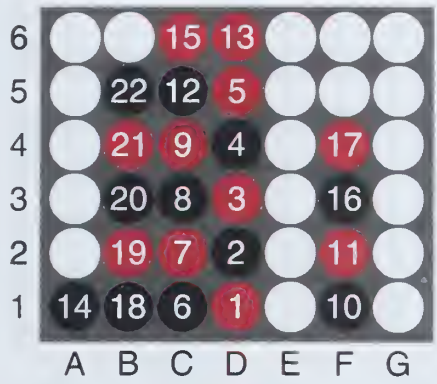
Black to play



Moves: D1 d2 D3 d4 C1 e1 E2 e3 C2 e4 C3 c4 A1 b1 A2 e5 E6 d5 A3 a4 C5 d6 C6 g1 G2 a5 G3 (28) __

PROBLEM 81

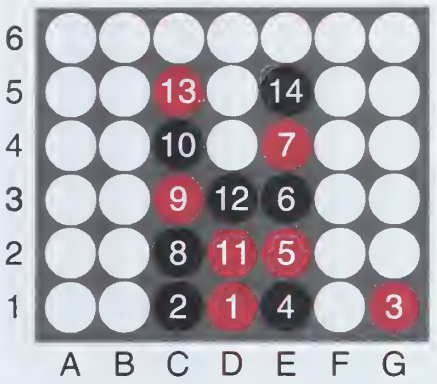
Red to play



Moves: D1 d2 D3 d4 D5 c1 C2 c3 C4 f1 F2 c5 D6 a1 C6 f3 F4 b1 B2 b3 B4 b5 (23) __

PROBLEM 82

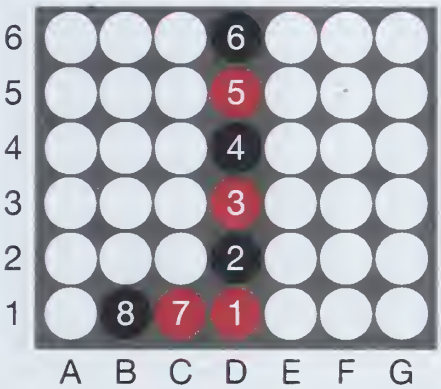
Red to play



Moves: D1 c1 G1 e1 E2 e3 E4 c2 C3 c4 D2 d3 C5 e5 (15) __

PROBLEM 83

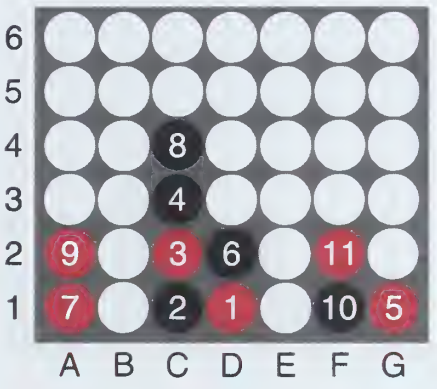
Red to play



Moves: D1 d2 D3 d4 D5 d6 C1 b1 (9) __

PROBLEM 84

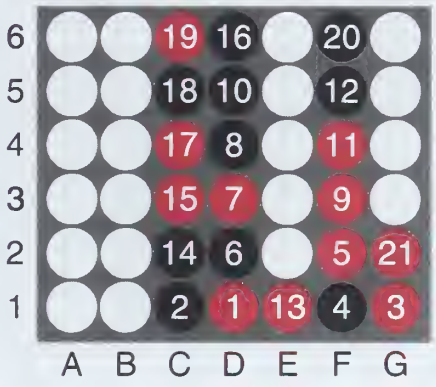
Black to play, and mate in five



Moves: D1 c1 C2 c3 G1 d2 A1 c4 A2 f1 F2 (12) __

PROBLEM 85

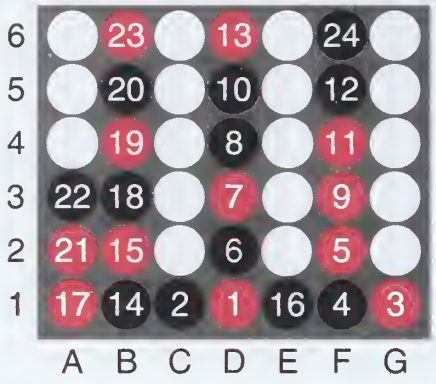
Black to play (Hint: Don't hurt yourself!)



Moves: D1 c1 G1 f1 F2 d2 D3 d4 F3 d5 F4 f5 E1 c2 C3 d6 C4 c5 C6 f6 G2 (22) __

PROBLEM 86

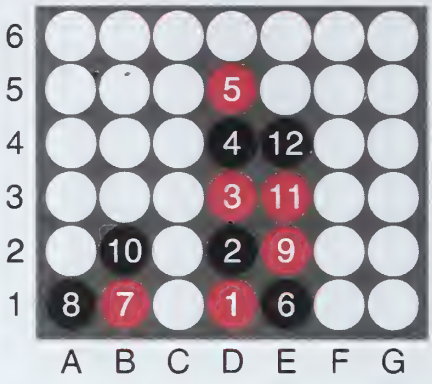
Red to play (Hint: Create a right-side win)



Moves: D1 c1 G1 f1 F2 d2 D3 d4 F3 d5 F4 f5
D6 b1 B2 e1 A1 b3 B4 b5 A2 a3 B6 f6 (25) __

PROBLEM 87

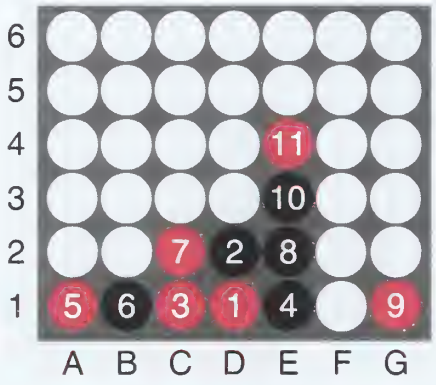
Red to play



Moves: D1 d2 D3 d4 D5 e1 B1
a1 E2 b2 E3 e4 (13) __

PROBLEM 88

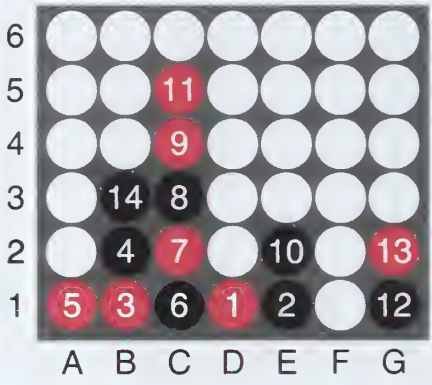
Black to play, and mate in five



Moves: D1 d2 C1 e1 A1 b1
C2 e2 G1 e3 E4 (12) __

PROBLEM 89

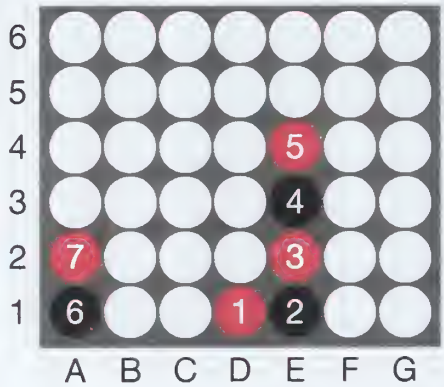
Red to play (Hint: Don't let Black win!)



Moves: D1 e1 B1 b2 A1 c1 C2 c3
C4 e2 C5 g1 G2 b3 (15) __

PROBLEM 90

Black to play

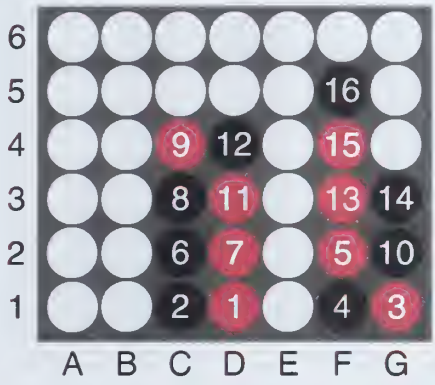


Moves: D1 e1 E2 e3 E4 a1 A2 (8) __

Problem Set D (Medium)

PROBLEM 91

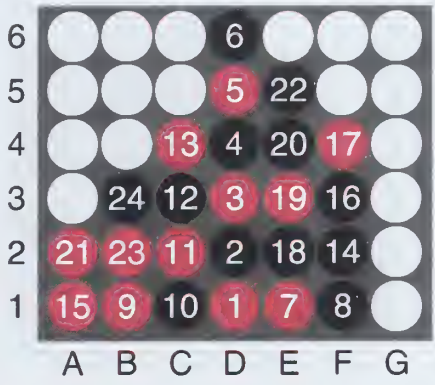
Red to play (Hint: Retain left-side chances)



Moves: D1 c1 G1 f1 F2 c2 D2 c3 C4 g2 D3 d4 F3 g3 F4 f5 (17) __

PROBLEM 92

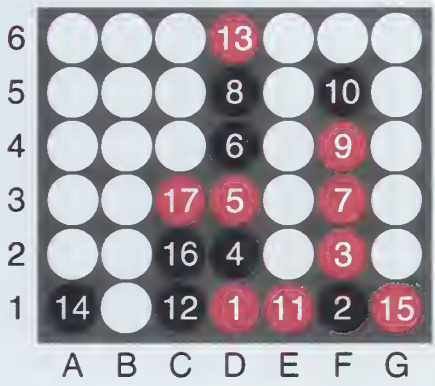
Red to play



Moves: D1 d2 D3 d4 D5 d6 E1 f1 B1 c1 C2 c3 C4 f2 A1 f3 F4 e2 E3 e4 A2 e5 B2 b3 (25) __

PROBLEM 93

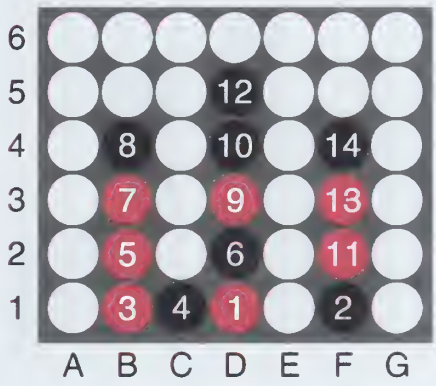
Black to play



Moves: D1 f1 F2 d2 D3 d4 F3 d5 F4 f5 E1 c1 D6 a1 G1 c2 C3 (18) __

PROBLEM 94

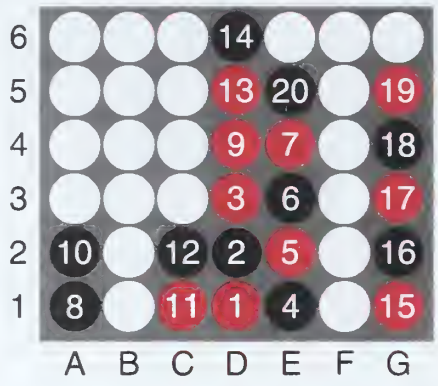
Red to play



Moves: D1 f1 B1 c1 B2 d2 B3 b4 D3 d4 F2 d5 F3 f4 (15) __

PROBLEM 95

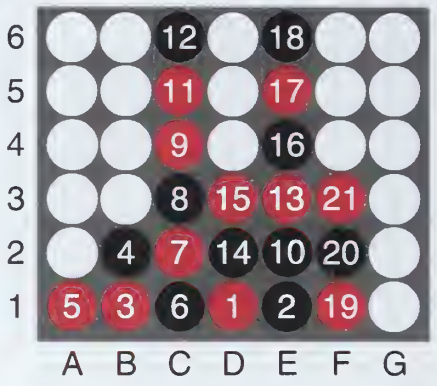
Red to play



Moves: D1 d2 D3 e1 E2 e3 E4 a1 D4 a2 C1 c2 D5 d6 G1 g2 G3 g4 G5 e5 (21) __

PROBLEM 96

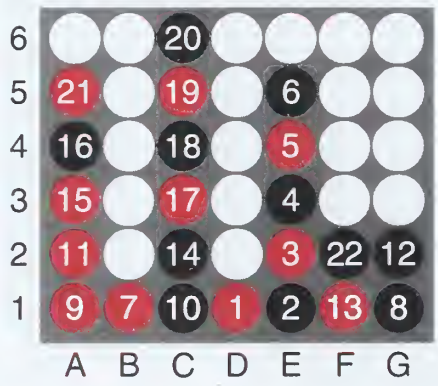
Black to play (Hint: Don't relinquish anything)



Moves: D1 e1 B1 b2 A1 c1 C2 c3 C4 e2 C5 c6 E3 d2 D3 e4 E5 e6 F1 f2 F3 (22) __

PROBLEM 97

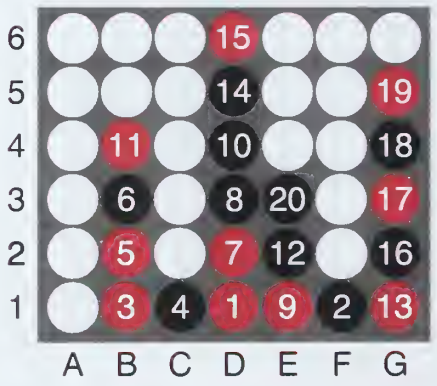
Red to play



Moves: D1 e1 E2 e3 E4 e5 B1 g1 A1 c1 A2 g2 F1 c2 A3 a4 C3 c4 C5 c6 A5 f2 (23) __

PROBLEM 98

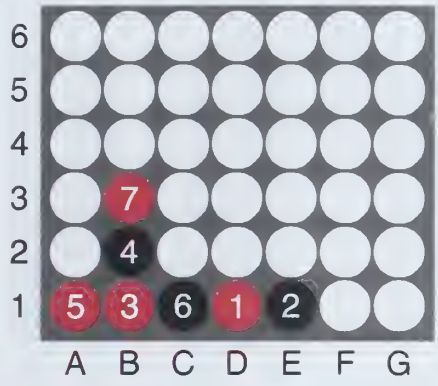
Red to play



Moves: D1 f1 B1 c1 B2 b3 D2 d3 E1 d4 B4 e2 G1 d5 D6 g2 G3 g4 G5 e3 (21) __

PROBLEM 99

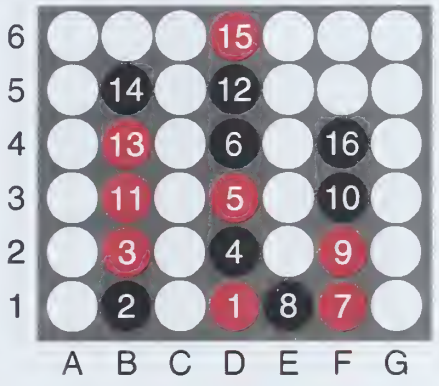
Black to play (Hint: You'd love to get d3)



Moves: D1 e1 B1 b2 A1 c1 B3 (8) __

PROBLEM 100

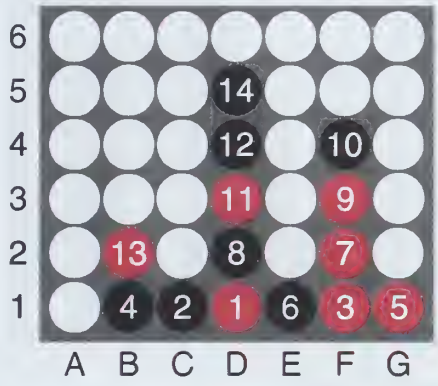
Red to play (Hint: Play where Black wants to play)



Moves: D1 b1 B2 d2 D3 d4 F1 e1
F2 f3 B3 d5 B4 b5 D6 f4 (17) __

PROBLEM 101

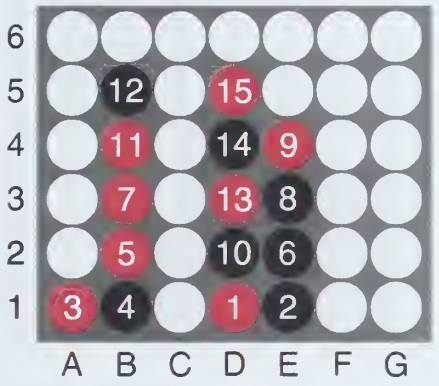
Red to play (Hint: Which cell has eventual value?)



Moves: D1 c1 F1 b1 G1 e1 F2 d2
F3 f4 D3 d4 B2 d5 (15) __

PROBLEM 102

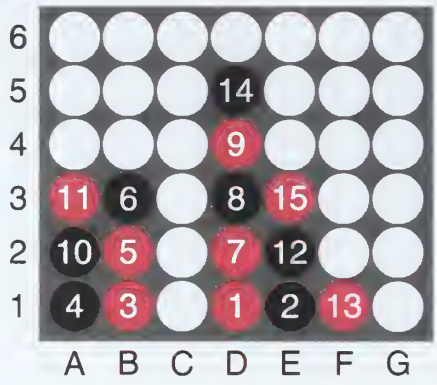
Black to play



Moves: D1 e1 A1 b1 B2 e2 B3 e3
E4 d2 B4 b5 D3 d4 D5 (16) __

PROBLEM 103

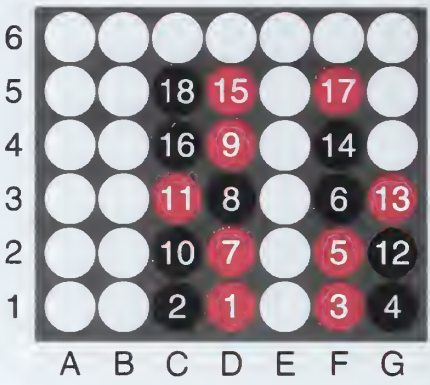
Black to play



Moves: D1 e1 B1 a1 B2 b3 D2 d3 D4 a2 A3 e2 F1 d5 E3 (16) __

PROBLEM 104

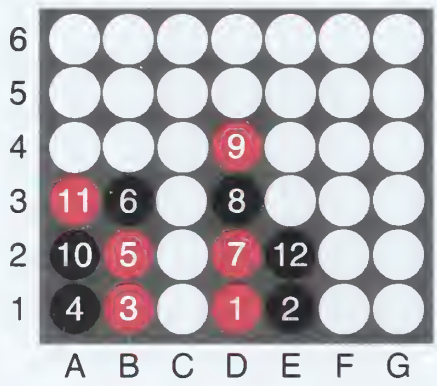
Red to play



Moves: D1 c1 F1 g1 F2 f3 D2 d3 D4 c2 C3 g2 G3 f4 D5 c4 F5 c5 (19) __

PROBLEM 105

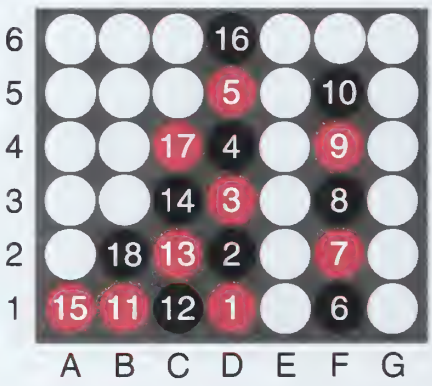
Red to play



Moves: D1 e1 B1 a1 B2 b3 D2 d3 D4 a2 A3 e2 (13) __

PROBLEM 106

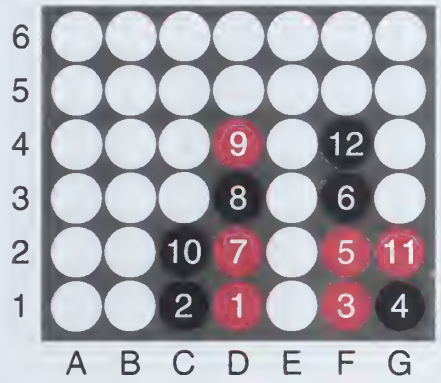
Red to play



Moves: D1 d2 D3 d4 D5 f1 F2 f3 F4 f5 B1 c1 C2 c3 A1 d6 C4 b2 (19) __

PROBLEM 107

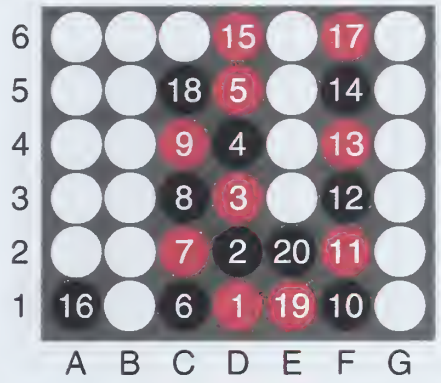
Red to play



Moves: D1 c1 F1 g1 F2 f3 D2
d3 D4 c2 G2 f4 (13) __

PROBLEM 108

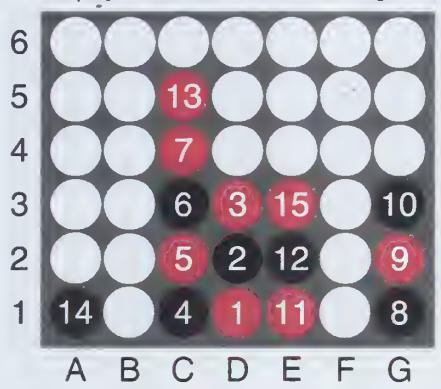
Red to play



Moves: D1 d2 D3 d4 D5 c1 C2 c3 C4 f1
F2 f3 F4 f5 D6 a1 F6 c5 E1 e2 (21) __

PROBLEM 109

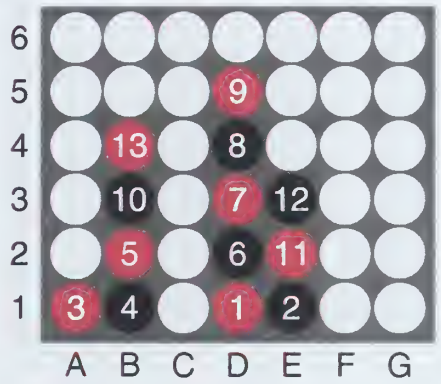
Black to play (Hint: Prevent Red's strong move)



Moves: D1 d2 D3 c1 C2 c3 C4 g1
G2 g3 E1 e2 C5 a1 E3 (16) __

PROBLEM 110

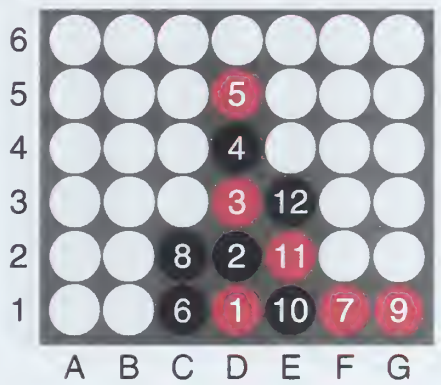
Black to play



Moves: D1 e1 A1 b1 B2 d2 D3
d4 D5 b3 E2 e3 B4 (14) __

PROBLEM 111

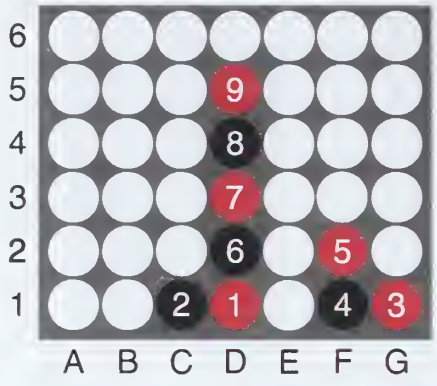
Red to play



Moves: D1 d2 D3 d4 D5 c1 F1
c2 G1 e1 E2 e3 (13) __

PROBLEM 112

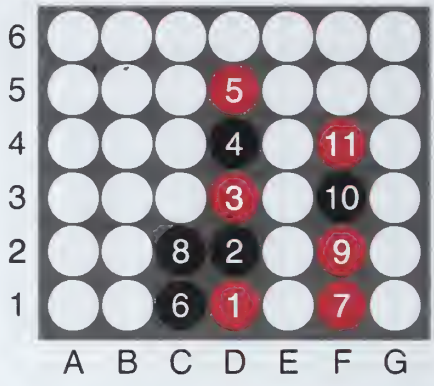
Black to play (Hint: Which is Red's key row?)



Moves: D1 c1 G1 f1 F2 d2 D3 d4 D5 (10) __

PROBLEM 113

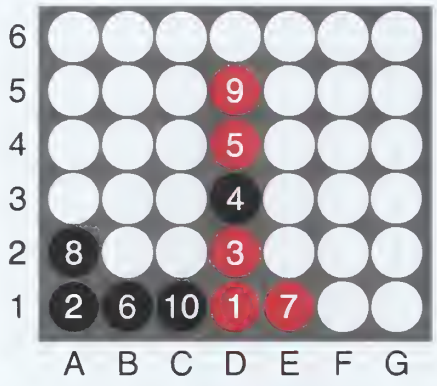
Black to play



Moves: D1 d2 D3 d4 D5 c1
F1 c2 F2 f3 F4 (12) __

PROBLEM 114

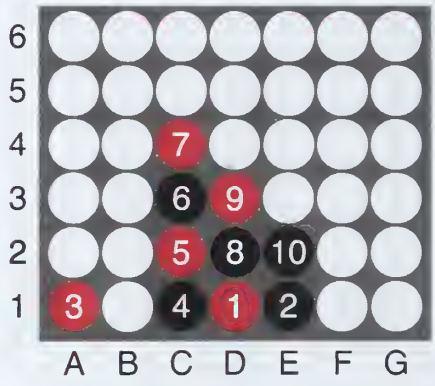
Red to play, and mate in five



Moves: D1 a1 D2 d3 D4 b1
E1 a2 D5 c1 (11) __

PROBLEM 115

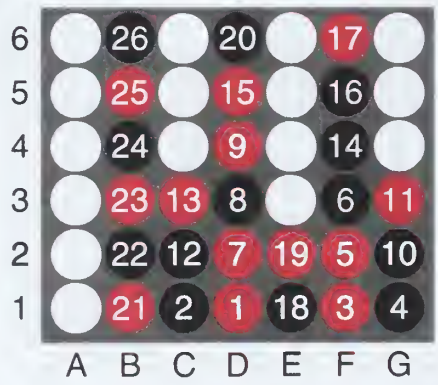
Red to play



Moves: D1 e1 A1 c1 C2 c3
C4 d2 D3 e2 (11) __

PROBLEM 116

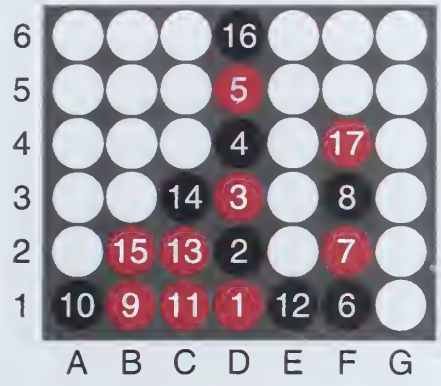
Red to play



Moves: D1 c1 F1 g1 F2 f3 D2 d3 D4 g2 G3 c2 C3
f4 D5 f5 F6 e1 E2 d6 B1 b2 B3 b4 B5 b6 (27) __

PROBLEM 117

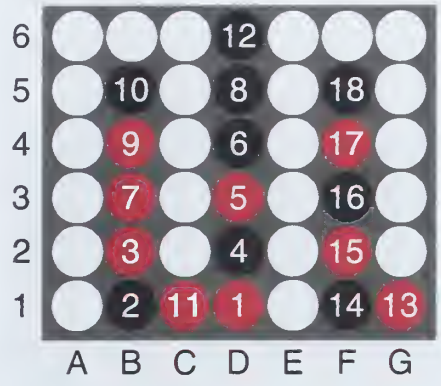
Black to play



Moves: D1 d2 D3 d4 D5 f1 F2 f3 B1 a1 C1 e1 C2 c3 B2 d6 F4 (18) __

PROBLEM 118

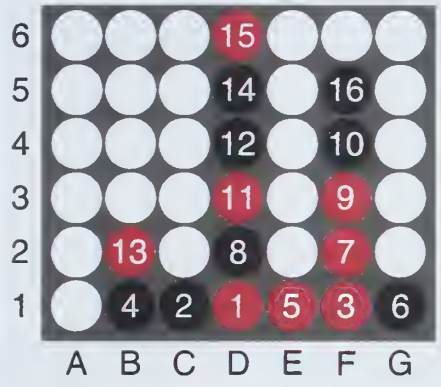
Red to play (Hint: Stop the overcutting threat)



Moves: D1 b1 B2 d2 D3 d4 B3 d5 B4 b5 C1 d6 G1 f1 F2 f3 F4 f5 (19) __

PROBLEM 119

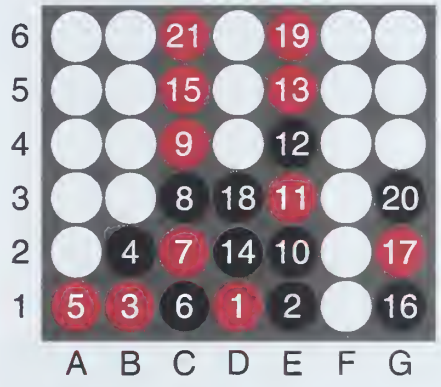
Red to play (Hint: Get a huge strategic lead)



Moves: D1 c1 F1 b1 E1 g1 F2 d2 F3 f4 D3 d4 B2 d5 D6 f5 (17) __

PROBLEM 120

Black to play (Hint: The danger is in the upper right)

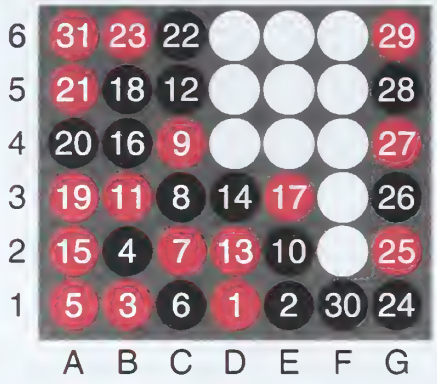


Moves: D1 e1 B1 b2 A1 c1 C2 c3 C4 e2 E3 e4 E5 d2 C5 g1 G2 d3 E6 g3 C6 (22) __

Problem Set E (Medium)

PROBLEM 121

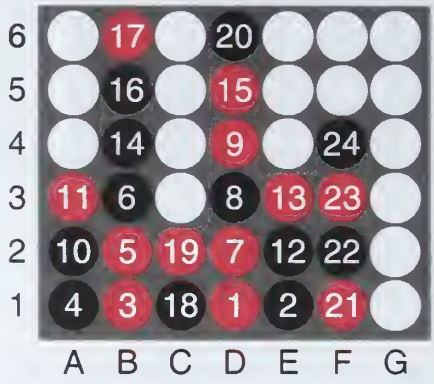
Black to play (Hint: Keep your threats intact)



Moves: D1 e1 B1 b2 A1 c1 C2 c3 C4 e2 B3 c5 D2 d3 A2 b4 E3 b5 A3 a4 A5 c6 B6 g1 G2 g3 G4 g5 G6 f1 A6 (32) __

PROBLEM 122

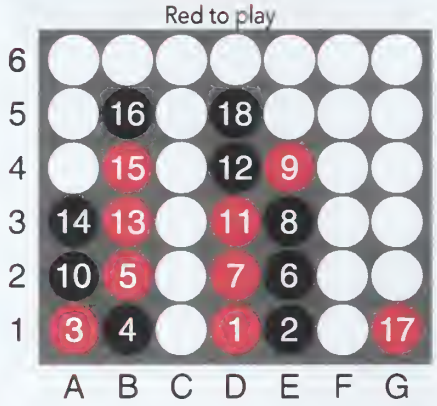
Red to play (Hint: Red needs an odd-row threat; C5 is undercut)



Moves: D1 e1 B1 a1 B2 b3 D2 d3 D4 a2 A3 e2 E3 b4 D5 b5 B6 c1 C2 d6 F1 f2 F3 f4 (25) __

PROBLEM 123

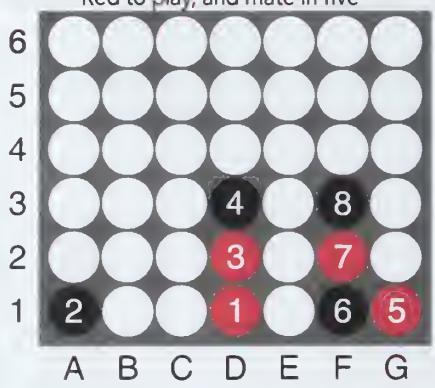
Red to play



Moves: D1 e1 A1 b1 B2 e2 D2 e3 E4 a2 D3 d4 B3 a3 B4 b5 G1 d5 (19) __

PROBLEM 124

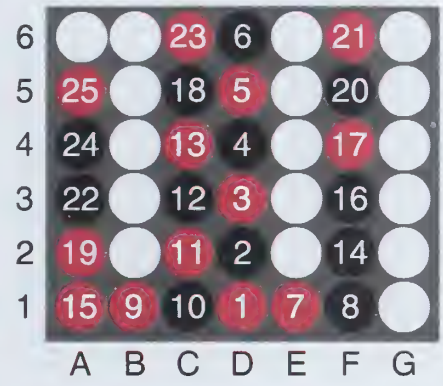
Red to play, and mate in five



Moves: D1 a1 D2 d3 G1 f1 F2 f3 (9) __

PROBLEM 125

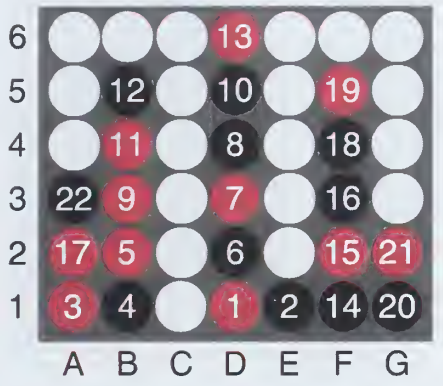
Black to play



Moves: D1 d2 D3 d4 D5 d6 E1 f1 B1 c1 C2 c3 C4 f2 A1 f3 F4 c5 A2 f5 F6 a3 C6 a4 A5 (26) __

PROBLEM 126

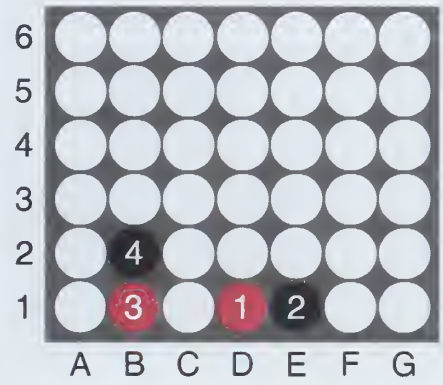
Red to play



Moves: D1 e1 A1 b1 B2 d2 D3 d4 B3 d5 B4 b5 D6 f1 F2 f3 A2 f4 F5 g1 G2 a3 (23) __

PROBLEM 127

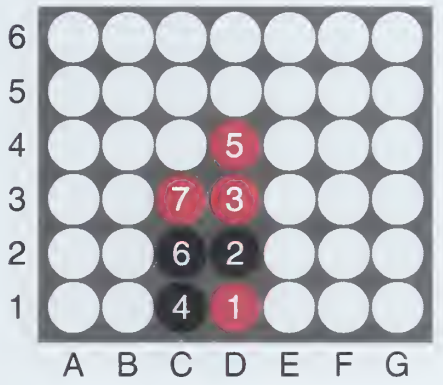
Red to play (Hint: Take the better of two cells)



Moves: D1 e1 B1 b2 (5) __

PROBLEM 128

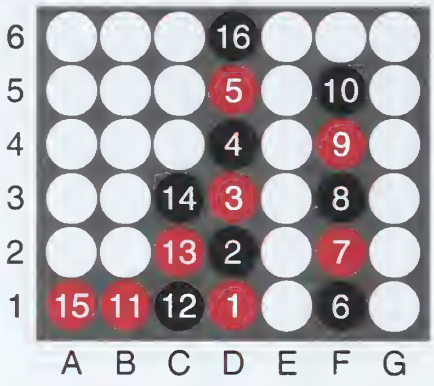
Black to play (Hint: How can Red break into your a2-f2 row?)



Moves: D1 d2 D3 c1 D4 c2 C3 (8) __

PROBLEM 129

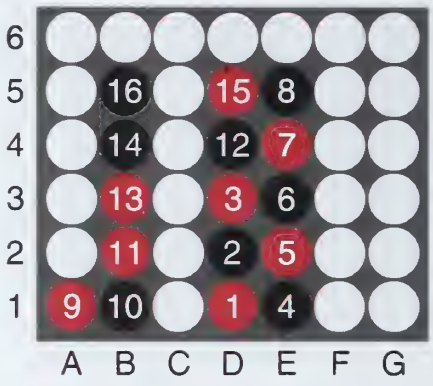
Red to play



Moves: D1 d2 D3 d4 D5 f1 F2 f3
F4 f5 B1 c1 C2 c3 A1 d6 (17) __

PROBLEM 130

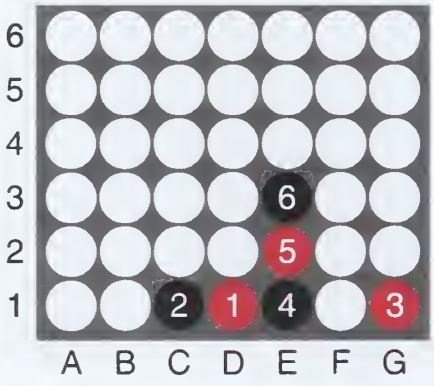
Red to play



Moves: D1 d2 D3 e1 E2 e3 E4 e5
A1 b1 B2 d4 B3 b4 D5 b5 (17) __

PROBLEM 131

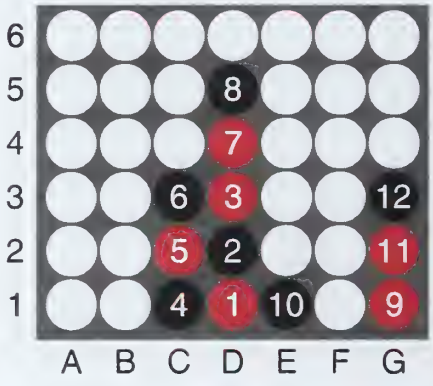
Red to play (Hint: Value near the center)



Moves: D1 c1 G1 e1 E2 e3 (7) __

PROBLEM 132

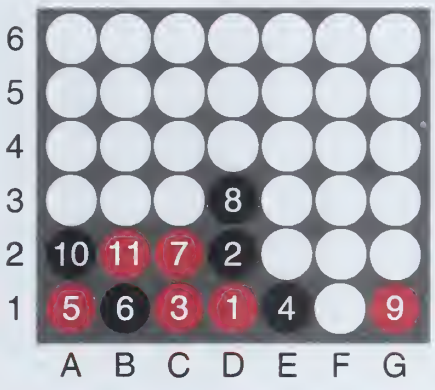
Red to play



Moves: D1 d2 D3 c1 C2 c3 D4
d5 G1 e1 G2 g3 (13) __

PROBLEM 133

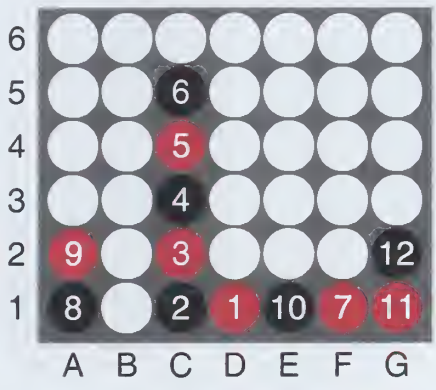
Black to play, and mate in five



Moves: D1 d2 C1 e1 A1 b1
C2 d3 G1 a2 B2 (12) __

PROBLEM 134

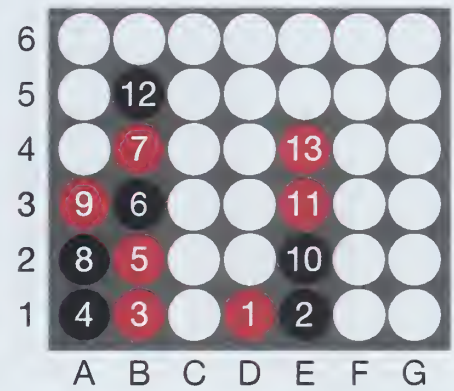
Red to play, and mate in eight



Moves: D1 c1 C2 c3 C4 c5 F1
a1 A2 e1 G1 g2 (13) __

PROBLEM 135

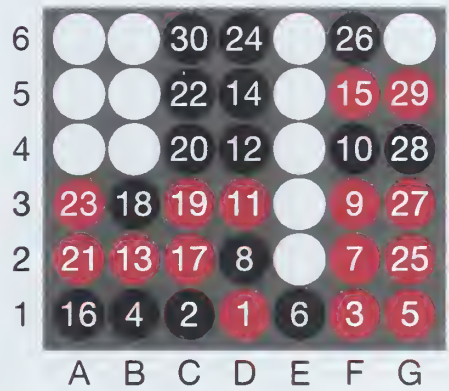
Black to play



Moves: D1 e1 B1 a1 B2 b3 B4
a2 A3 e2 E3 b5 E4 (14) __

PROBLEM 136

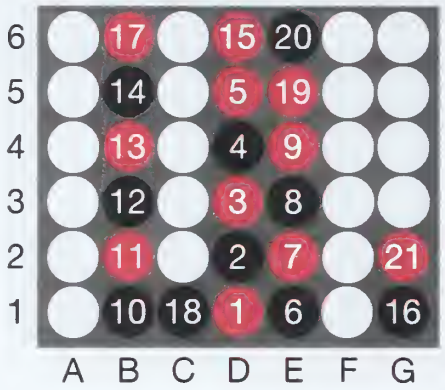
Red to play



Moves: D1 c1 F1 b1 G1 e1 F2 d2 F3 f4
D3 d4 B2 d5 F5 a1 C2 b3 C3 c4 A2 c5
A3 d6 G2 f6 G3 g4 G5 c6 (31) __

PROBLEM 137

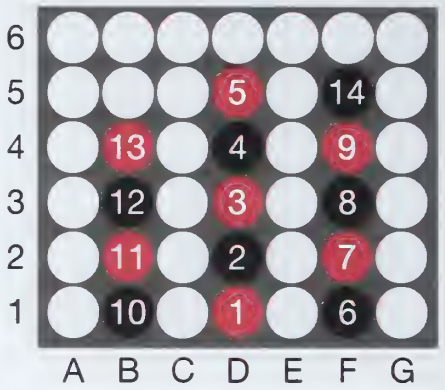
Black to play



Moves: D1 d2 D3 d4 D5 e1 E2 e3 E4 b1 B2 b3 B4 b5 D6 g1 B6 c1 E5 e6 G2 (22) __

PROBLEM 138

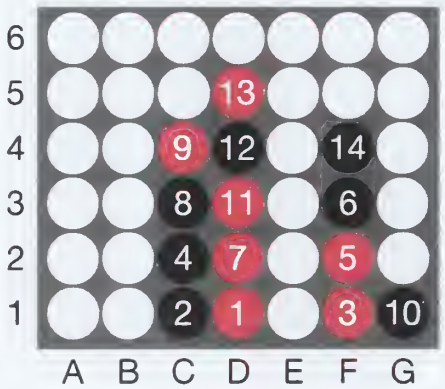
Red to play



Moves: D1 d2 D3 d4 D5 f1 F2 f3 F4 b1 B2 b3 B4 f5 (15) __

PROBLEM 139

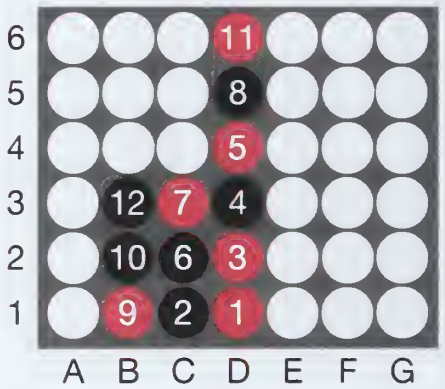
Red to play



Moves: D1 c1 F1 c2 F2 f3 D2 c3 C4 g1 D3 d4 D5 f4 (15) __

PROBLEM 140

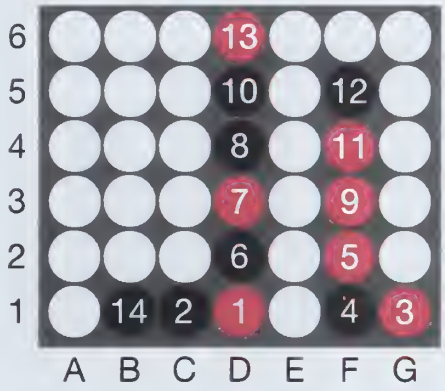
Red to play, and mate in six



Moves: D1 c1 D2 d3 D4 c2 C3 d5 B1 b2 D6 b3 (13) __

PROBLEM 141

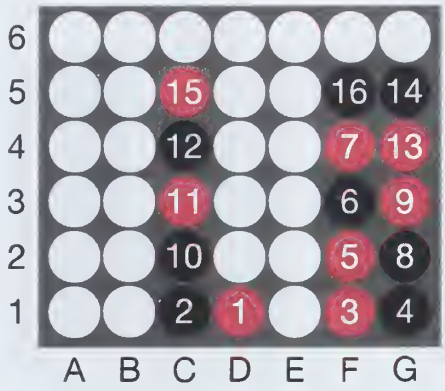
Red to play (Hint: Foil Black's left-side plans)



Moves: D1 c1 G1 f1 F2 d2 D3 d4 F3 d5 F4 f5 D6 b1 (15) __

PROBLEM 142

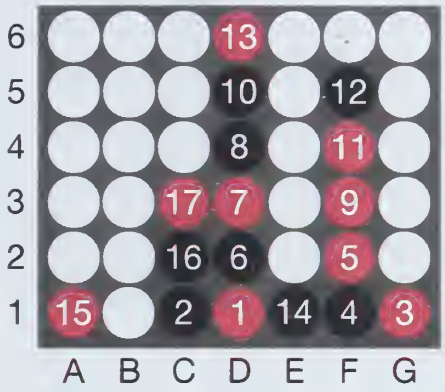
Red to play



Moves: D1 c1 F1 g1 F2 f3 F4 g2 G3 c2 C3 c4 G4 g5 C5 f5 (17) __

PROBLEM 143

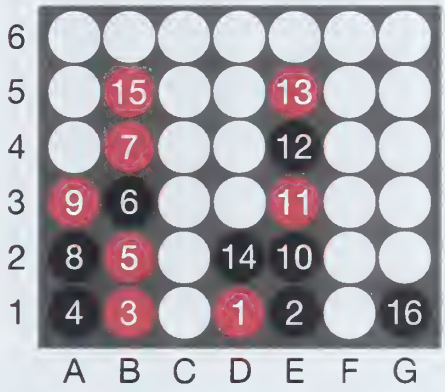
Black to play (Hint: You need to win on the left side)



Moves: D1 c1 G1 f1 F2 d2 D3 d4 F3 d5 F4 f5 D6 e1 A1 c2 C3 (18) __

PROBLEM 144

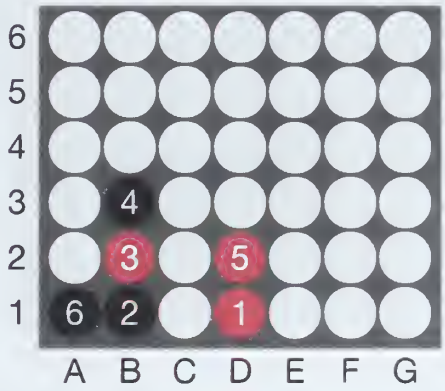
Red to play (Hint: Retain right-side chances)



Moves: D1 e1 B1 a1 B2 b3 B4 a2 A3 e2 E3 e4 E5 d2 B5 g1 (17) __

PROBLEM 145

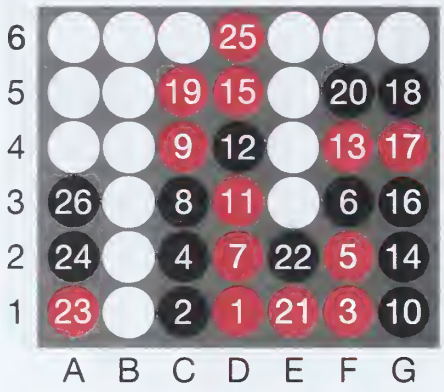
Red to play, and mate in six



Moves: D1 b1 B2 b3 D2 a1 (7) __

PROBLEM 146

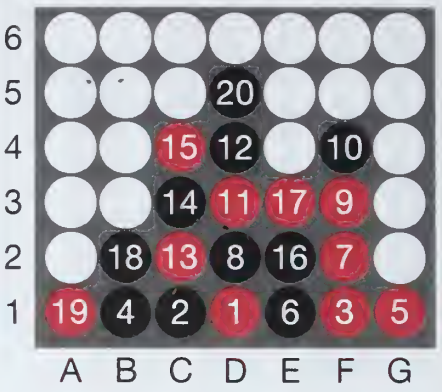
Red to play



Moves: D1 c1 F1 c2 F2 f3 D2 c3 C4 g1 D3 d4 F4 g2 D5 g3 G4 g5 C5 f5 E1 e2 A1 a2 D6 a3 (27) __

PROBLEM 147

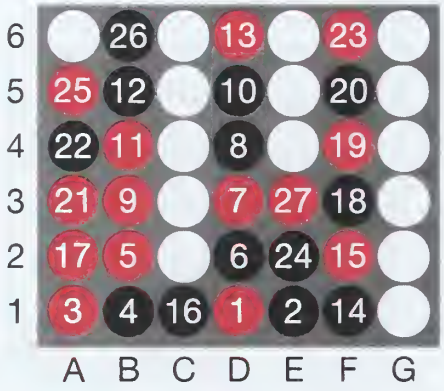
Red to play (Hint: Kill Black's threat)



Moves: D1 c1 F1 b1 G1 e1 F2 d2 F3 f4 D3 d4 C2 c3 C4 e2 E3 b2 A1 d5 (21) __

PROBLEM 148

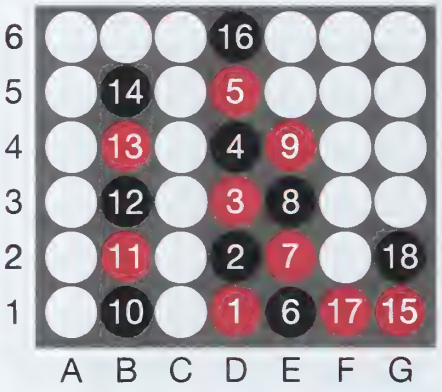
Black to play



Moves: D1 e1 A1 b1 B2 d2 D3 d4 B3 d5 B4 b5 D6 f1 F2 c1 A2 f3 F4 f5 A3 a4 F6 e2 A5 b6 E3 (28) __

PROBLEM 149

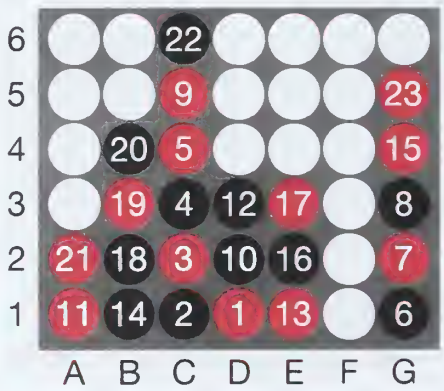
Red to play



Moves: D1 d2 D3 d4 D5 e1 E2 e3 E4 b1 B2 b3 B4 b5 G1 d6 F1 g2 (19) __

PROBLEM 150

Black to play

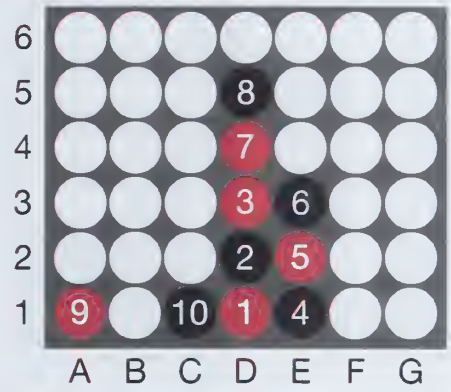


Moves: D1 c1 C2 c3 C4 g1 G2 g3 C5 d2 A1 d3 E1 b1 G4 e2 E3 b2 B3 b4 A2 c6 G5 (24) __

Problem Set F (Hard)

PROBLEM 151

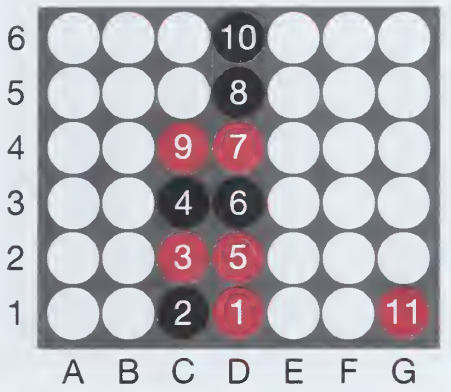
Red to play



Moves: D1 d2 D3 e1 E2 e3
D4 d5 A1 c1 (11) __

PROBLEM 152

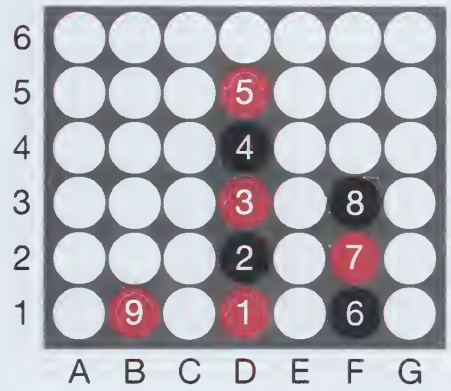
Black to play



Moves: D1 c1 C2 c3 D2 d3
D4 d5 C4 d6 G1 (12) __

PROBLEM 153

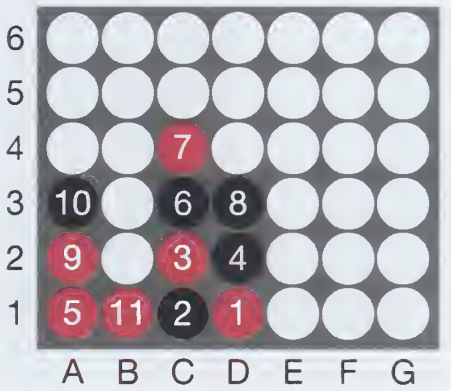
Black to play



Moves: D1 d2 D3 d4 D5 f1 F2 f3 B1 (10) __

PROBLEM 154

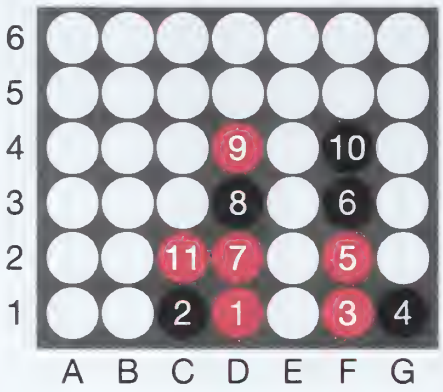
Black to play, and mate in four



Moves: D1 c1 C2 d2 A1 c3
C4 d3 A2 a3 B1 (12) __

PROBLEM 155

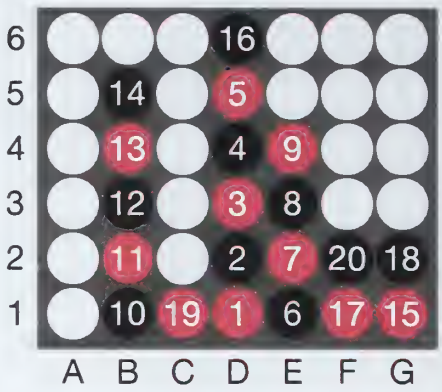
Black to play



Moves: D1 c1 F1 g1 F2 f3 D2
d3 D4 f4 C2 (12) __

PROBLEM 156

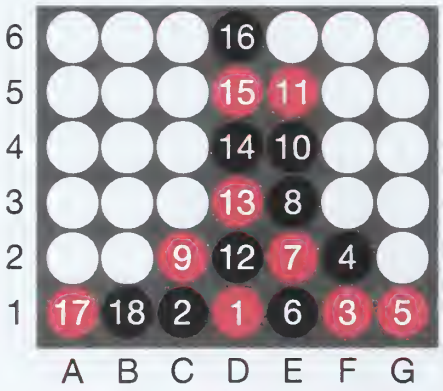
Red to play



Moves: D1 d2 D3 d4 D5 e1 E2 e3 E4 b1
B2 b3 B4 b5 G1 d6 F1 g2 C1 f2 (21) __

PROBLEM 157

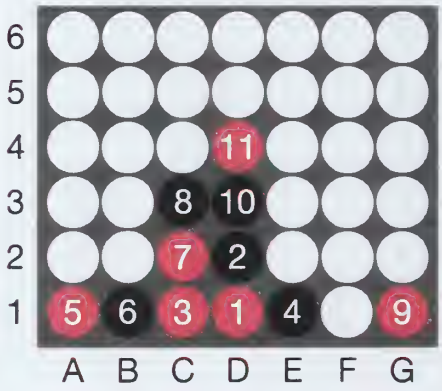
Red to play



Moves: D1 c1 F1 f2 G1 e1 E2 e3 C2 e4
E5 d2 D3 d4 D5 d6 A1 b1 (19) __

PROBLEM 158

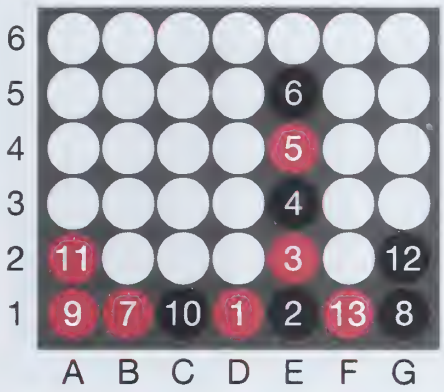
Black to play, and mate in five



Moves: D1 d2 C1 e1 A1 b1
C2 c3 G1 d3 D4 (12) __

PROBLEM 159

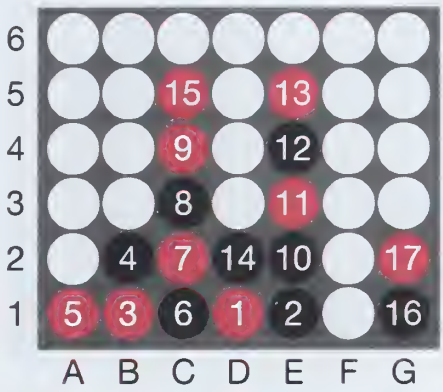
Black to play



Moves: D1 e1 E2 e3 E4 e5 B1 g1 A1 c1 A2 g2 F1 (14) __

PROBLEM 160

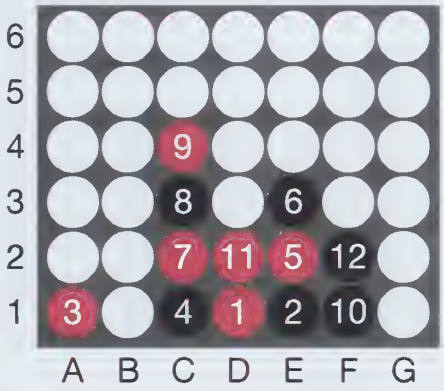
Black to play



Moves: D1 e1 B1 b2 A1 c1 C2 c3 C4 e2 E3 e4 E5 d2 C5 g1 G2 (18) __

PROBLEM 161

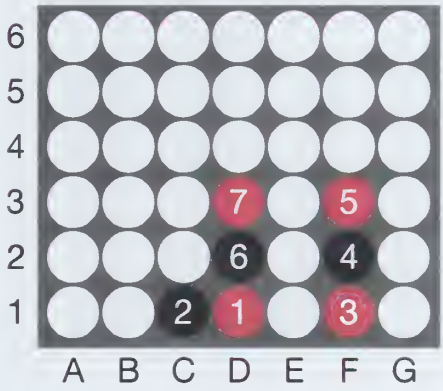
Red to play



Moves: D1 e1 A1 c1 E2 e3 C2 c3 C4 f1 D2 f2 (13) __

PROBLEM 162

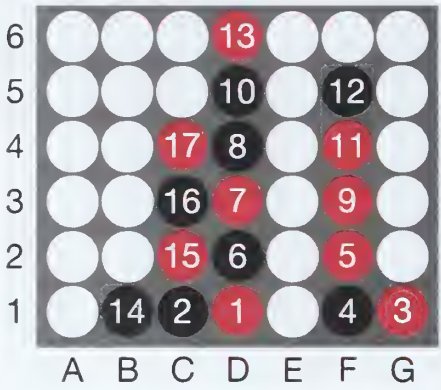
Black to play



Moves: D1 c1 F1 f2 F3 d2 D3 (8) __

PROBLEM 163

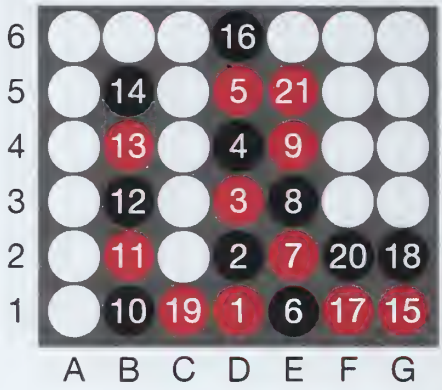
Black to play



Moves: D1 c1 G1 f1 F2 d2 D3 d4 F3 d5 F4 f5 D6 b1 C2 c3 C4 (18) __

PROBLEM 164

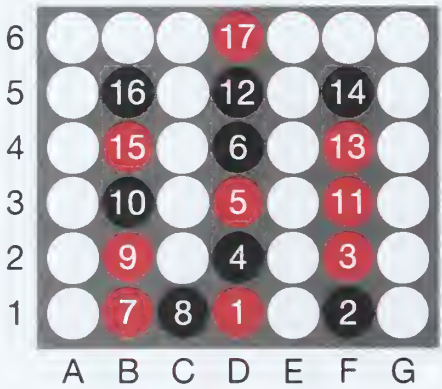
Black to play



Moves: D1 d2 D3 d4 D5 e1 E2 e3 E4 b1 B2 b3 B4 b5 G1 d6 F1 g2 C1 f2 E5 (22) __

PROBLEM 165

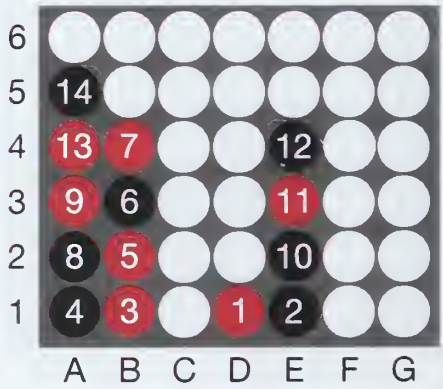
Black to play



Moves: D1 f1 F2 d2 D3 d4 B1 c1 B2 b3 F3 d5 F4 f5 B4 b5 D6 (18) __

PROBLEM 166

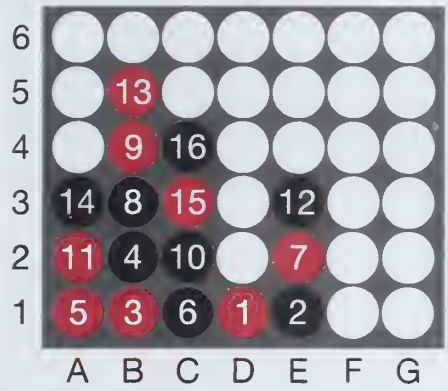
Red to play



Moves: D1 e1 B1 a1 B2 b3 B4 a2 A3 e2 E3 e4 A4 a5 (15) __

PROBLEM 167

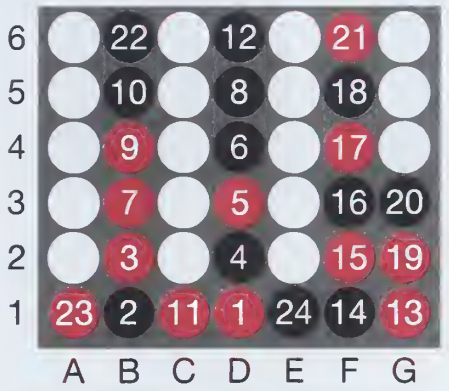
Red to play



Moves: D1 e1 B1 b2 A1 c1 E2 b3
B4 c2 A2 e3 B5 a3 C3 c4 (17) __

PROBLEM 168

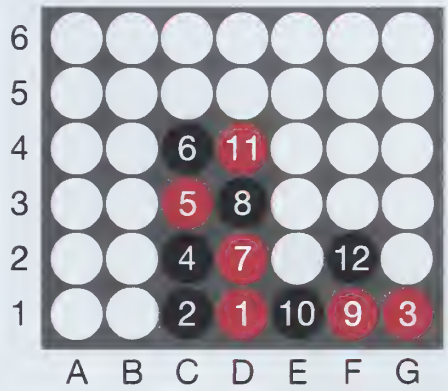
Red to play



Moves: D1 b1 B2 d2 D3 d4 B3
d5 B4 b5 C1 d6 G1 f1 F2 f3 F4 f5
G2 g3 F6 b6 A1 e1 (25) __

PROBLEM 169

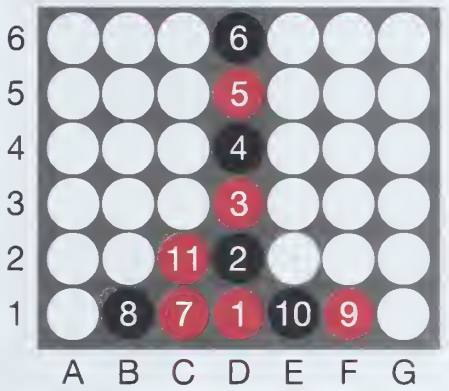
Red to play



Moves: D1 c1 G1 c2 C3 c4 D2
d3 F1 e1 D4 f2 (13) __

PROBLEM 170

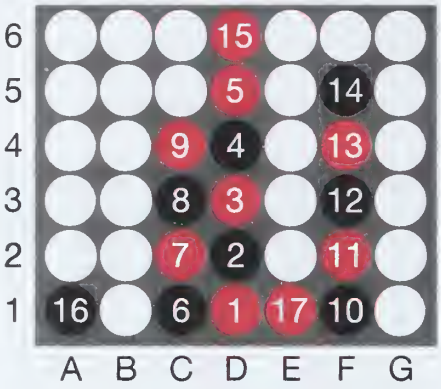
Black to play



Moves: D1 d2 D3 d4 D5 d6
C1 b1 F1 e1 C2 (12) __

PROBLEM 171

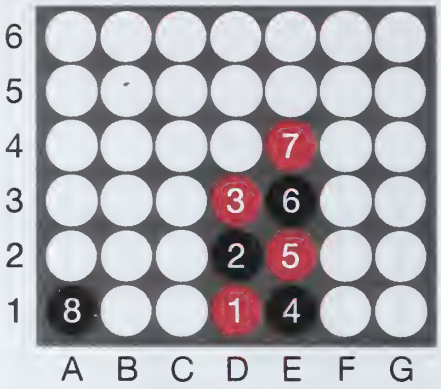
Black to play



Moves: D1 d2 D3 d4 D5 c1 C2 c3
C4 f1 F2 f3 F4 f5 D6 a1 E1 (18) __

PROBLEM 172

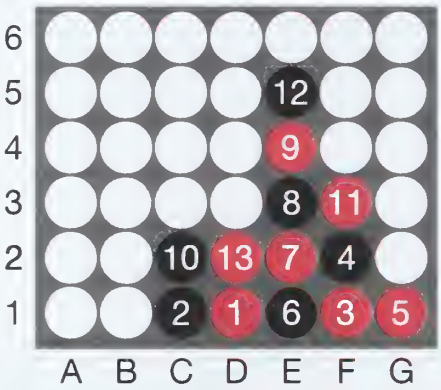
Red to play



Moves: D1 d2 D3 e1 E2 e3 E4 a1 (9) __

PROBLEM 173

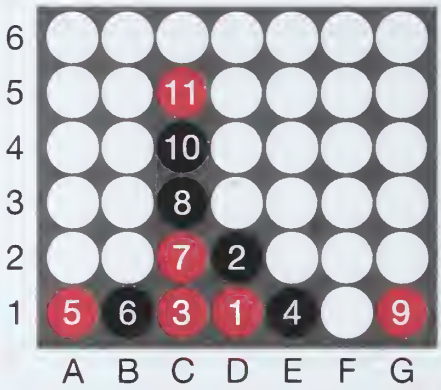
Black to play



Moves: D1 c1 F1 f2 G1 e1 E2
e3 E4 c2 F3 e5 D2 (14) __

PROBLEM 174

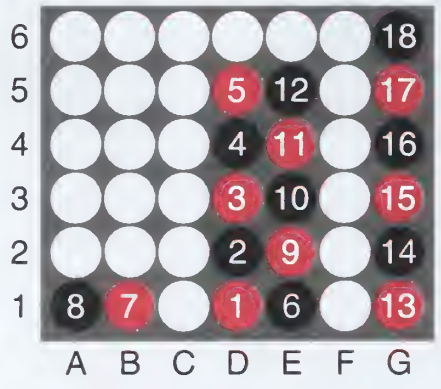
Black to play, and mate in five



Moves: D1 d2 C1 e1 A1 b1
C2 c3 G1 c4 C5 (12) __

PROBLEM 175

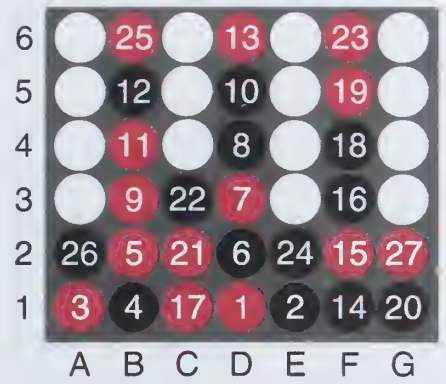
Red to play



Moves: D1 d2 D3 d4 D5 e1 B1 a1 E2
e3 E4 e5 G1 g2 G3 g4 G5 g6 (19) __

PROBLEM 176

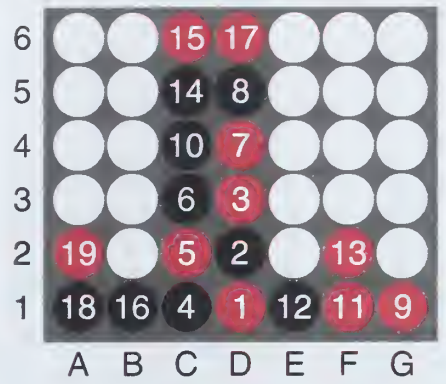
Black to play



Moves: D1 e1 A1 b1 B2 d2 D3 d4 B3 d5 B4 b5 D6 f1 F2 f3 C1 f4 F5 g1 C2 c3 F6 e2 B6 a2 G2 (28) __

PROBLEM 177

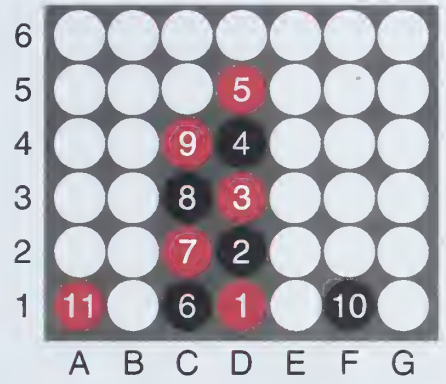
Black to play



Moves: D1 d2 D3 c1 C2 c3 D4 d5 G1 c4 F1 e1 F2 c5 C6 b1 D6 a1 A2 (20) __

PROBLEM 178

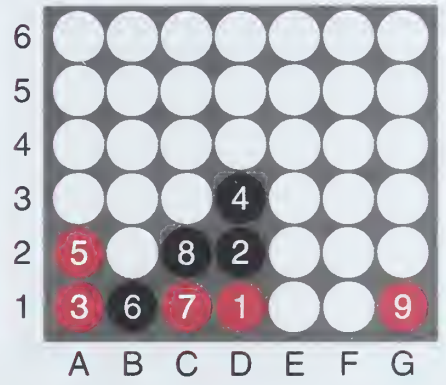
Black to play



Moves: D1 d2 D3 d4 D5 c1 C2 c3 C4 f1 A1 (12) __

PROBLEM 179

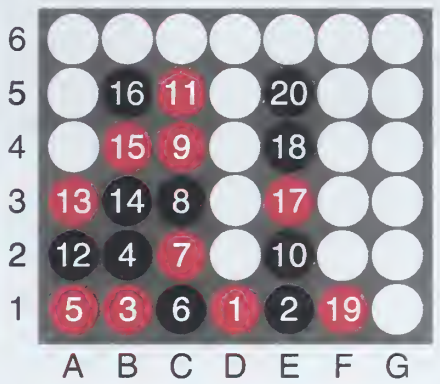
Black to play, and mate in six



Moves: D1 d2 A1 d3 A2 b1 C1 c2 G1 (10) __

PROBLEM 180

Red to play

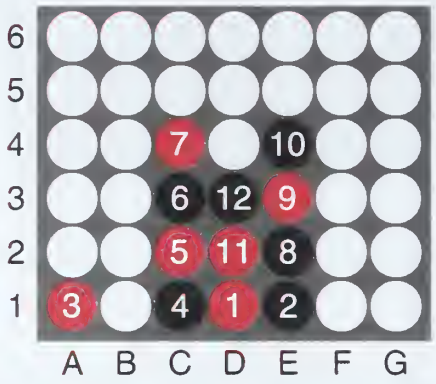


Moves: D1 e1 B1 b2 A1 c1 C2 c3 C4 e2 C5 a2 A3 b3 B4 b5 E3 e4 F1 e5 (21) __

Problem Set G (Hard)

PROBLEM 181

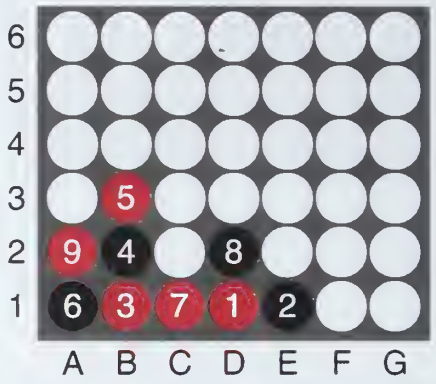
Red to play



Moves: D1 e1 A1 c1 C2 c3 C4
e2 E3 e4 D2 d3 (13) __

PROBLEM 182

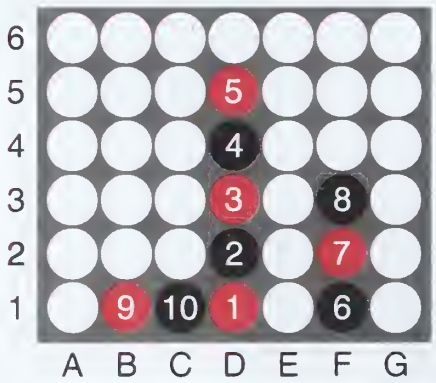
Black to play, and mate in five



Moves: D1 e1 B1 b2 B3 a1 C1 d2 A2 (10) __

PROBLEM 183

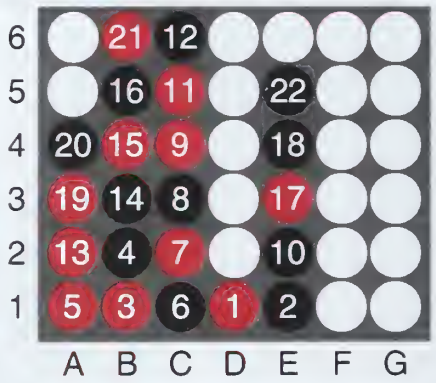
Red to play



Moves: D1 d2 D3 d4 D5 f1
F2 f3 B1 c1 (11) __

PROBLEM 184

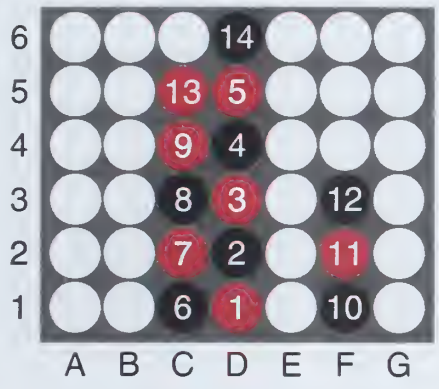
Red to play



Moves: D1 e1 B1 b2 A1 c1 C2 c3 C4 e2 C5
c6 A2 b3 B4 b5 E3 e4 A3 a4 B6 e5 (23) __

PROBLEM 185

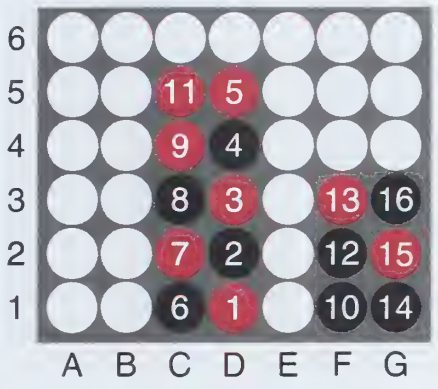
Red to play



Moves: D1 d2 D3 d4 D5 c1 C2 c3 C4 f1 F2 f3 C5 d6 (15) __

PROBLEM 186

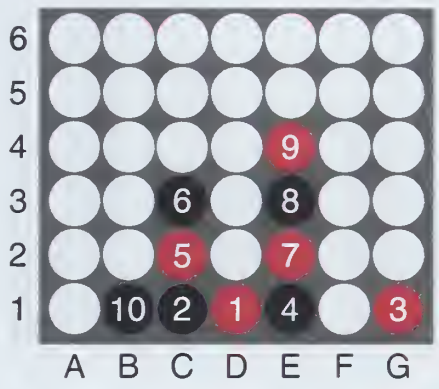
Red to play



Moves: D1 d2 D3 d4 D5 c1 C2 c3 C4 f1 C5 f2 F3 g1 G2 g3 (17) __

PROBLEM 187

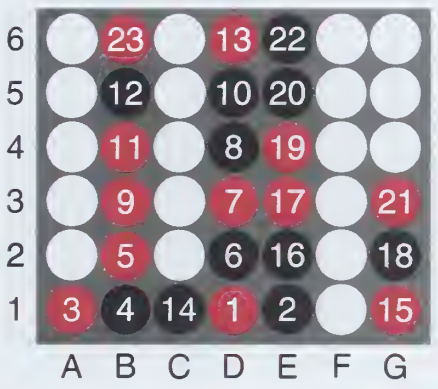
Red to play



Moves: D1 c1 G1 e1 C2 c3 E2 e3 E4 b1 (11) __

PROBLEM 188

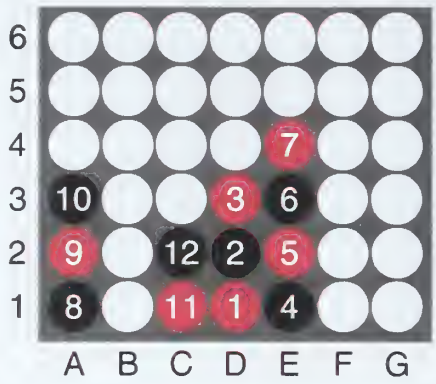
Black to play



Moves: D1 e1 A1 b1 B2 d2 D3 d4 B3 d5 B4 b5 D6 c1 G1 e2 E3 g2 E4 e5 G3 e6 B6 (24) __

PROBLEM 189

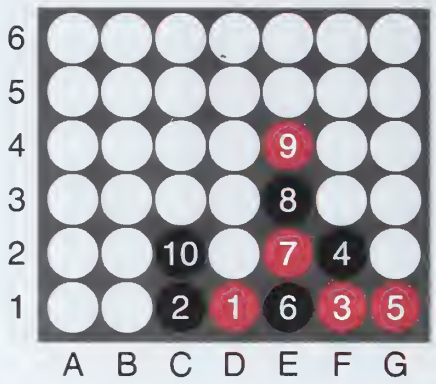
Red to play



Moves: D1 d2 D3 e1 E2 e3 E4
a1 A2 a3 C1 c2 (13) __

PROBLEM 190

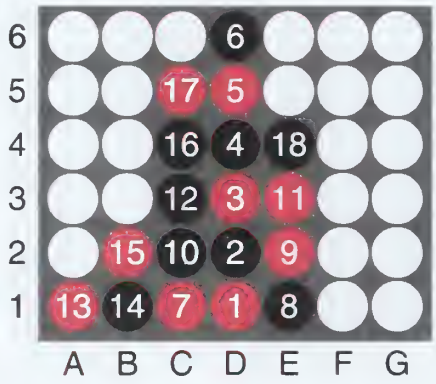
Red to play



Moves: D1 c1 F1 f2 G1 e1
E2 e3 E4 c2 (11) __

PROBLEM 191

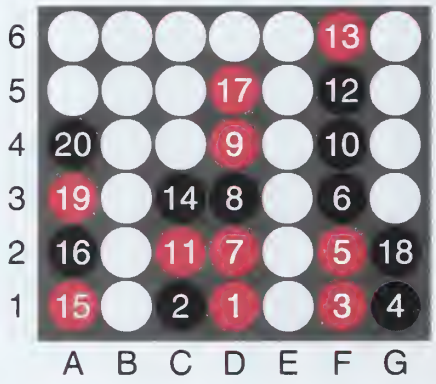
Red to play



Moves: D1 d2 D3 d4 D5 d6 C1 e1 E2
c2 E3 c3 A1 b1 B2 c4 C5 e4 (19) __

PROBLEM 192

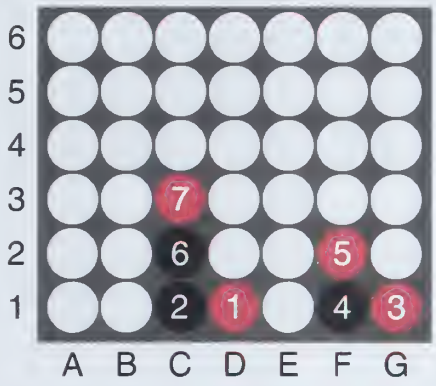
Red to play



Moves: D1 c1 F1 g1 F2 f3 D2 d3 D4 f4
C2 f5 F6 c3 A1 a2 D5 g2 A3 a4 (21) __

PROBLEM 193

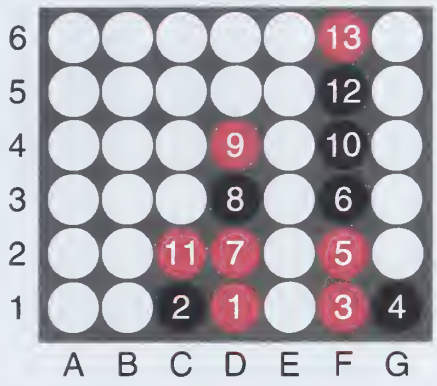
Black to play



Moves: D1 c1 G1 f1 F2 c2 C3 (8) __

PROBLEM 194

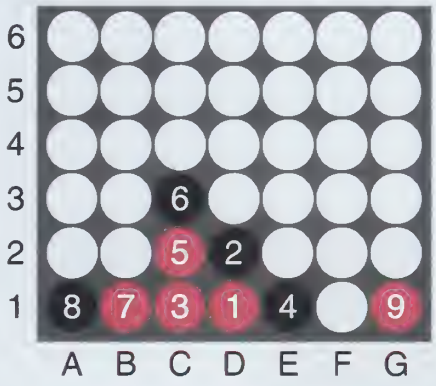
Black to play



Moves: D1 c1 F1 g1 F2 f3 D2
d3 D4 f4 C2 f5 F6 (14) __

PROBLEM 195

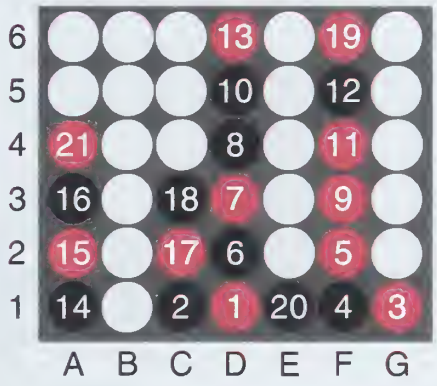
Black to play, and mate in six



Moves: D1 d2 C1 e1 C2 c3 B1 a1 G1 (10) __

PROBLEM 196

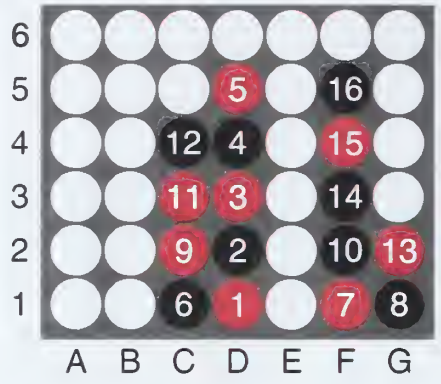
Black to play



Moves: D1 c1 G1 f1 F2 d2 D3 d4 F3 d5 F4
f5 D6 a1 A2 a3 C2 c3 F6 e1 A4 (22) __

PROBLEM 197

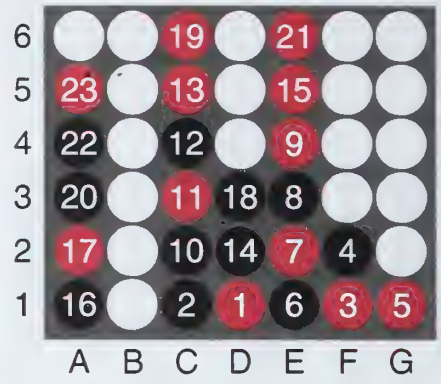
Red to play



Moves: D1 d2 D3 d4 D5 c1 F1 g1
C2 f2 C3 c4 G2 f3 F4 f5 (17) __

PROBLEM 198

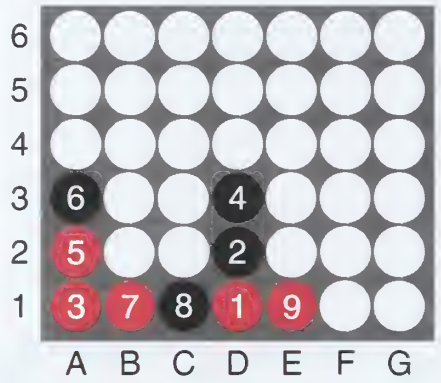
Black to play



Moves: D1 c1 F1 f2 G1 e1 E2 e3 E4 c2 C3 c4
C5 d2 E5 a1 A2 d3 C6 a3 E6 a4 A5 (24) __

PROBLEM 199

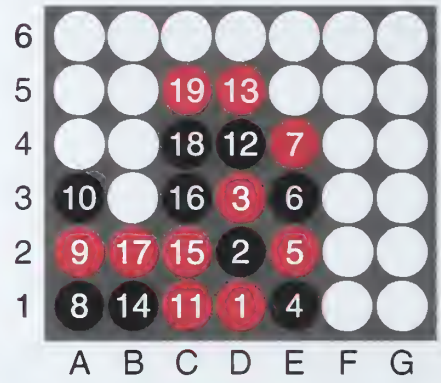
Black to play, and mate in six



Moves: D1 d2 A1 d3 A2 a3 B1 c1 E1 (10) __

PROBLEM 200

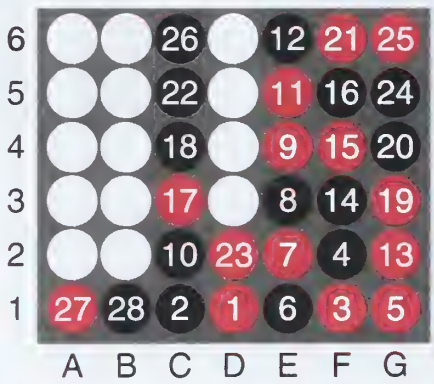
Black to play



Moves: D1 d2 D3 e1 E2 e3 E4 a1 A2 a3
C1 d4 D5 b1 C2 c3 B2 c4 C5 (20) __

PROBLEM 201

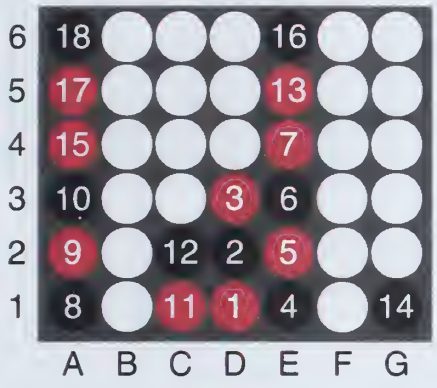
Red to play



Moves: D1 c1 F1 f2 G1 e1 E2 e3 E4 c2 E5 e6 G2 f3
F4 f5 C3 c4 G3 g4 F6 c5 D2 g5 G6 c6 A1 b1 (29) __

PROBLEM 202

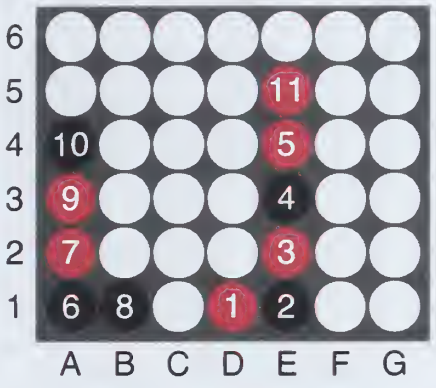
Red to play



Moves: D1 d2 D3 e1 E2 e3 E4 a1 A2 a3 C1 c2 E5 g1 A4 e6 A5 a6 (19) __

PROBLEM 203

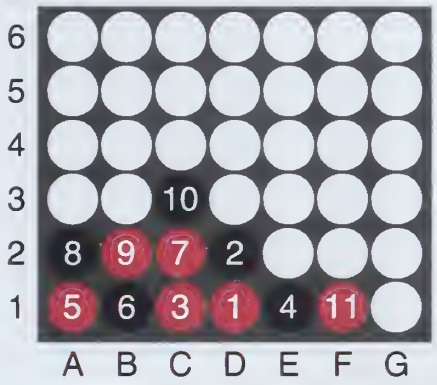
Black to play



Moves: D1 e1 E2 e3 E4 a1 A2 b1 A3 a4 E5 (12) __

PROBLEM 204

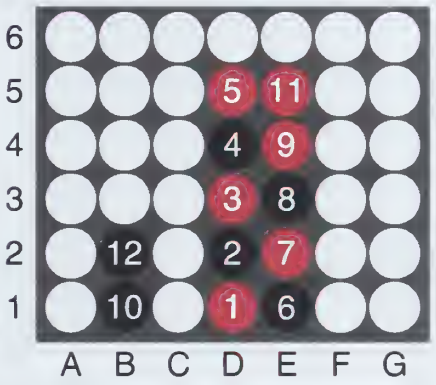
Black to play, and mate in five



Moves: D1 d2 C1 e1 A1 b1 C2 a2 B2 c3 F1 (12) __

PROBLEM 205

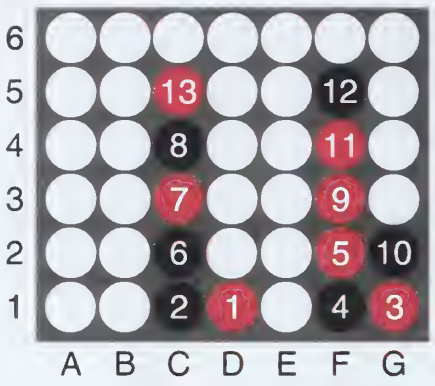
Red to play (Hint: Does your left side have a future?)



Moves: D1 d2 D3 d4 D5 e1 E2 e3 E4 b1 E5 b2 (13) __

PROBLEM 206

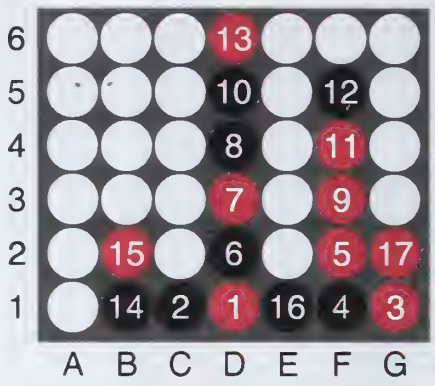
Black to play



Moves: D1 c1 G1 f1 F2 c2 C3
c4 F3 g2 F4 f5 C5 (14) __

PROBLEM 207

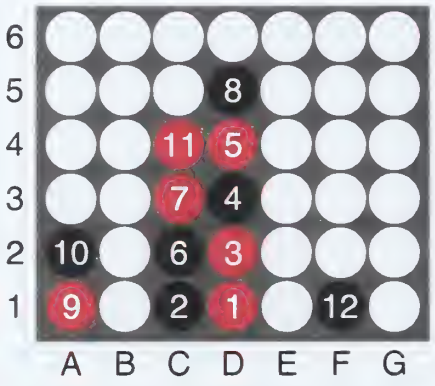
Black to play



Moves: D1 c1 G1 f1 F2 d2 D3 d4 F3
d5 F4 f5 D6 b1 B2 e1 G2 (18) __

PROBLEM 208

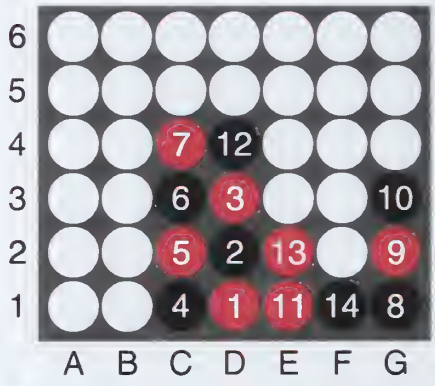
Red to play, and mate in eight



Moves: D1 c1 D2 d3 D4 c2 C3
d5 A1 a2 C4 f1 (13) __

PROBLEM 209

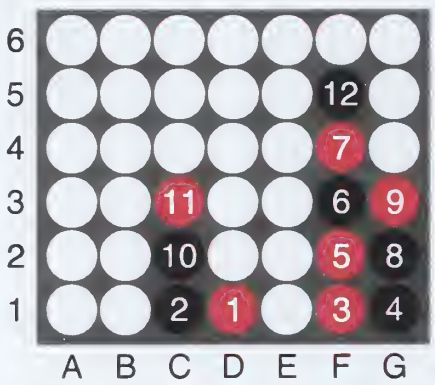
Red to play



Moves: D1 d2 D3 c1 C2 c3 C4
g1 G2 g3 E1 d4 E2 f1 (15) __

PROBLEM 210

Red to play

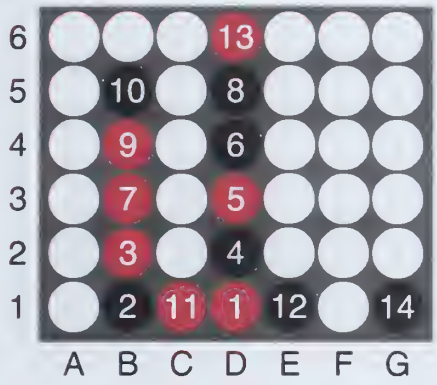


Moves: D1 c1 F1 g1 F2 f3 F4
g2 G3 c2 C3 f5 (13) __

Problem Set H (Expert)

PROBLEM 211

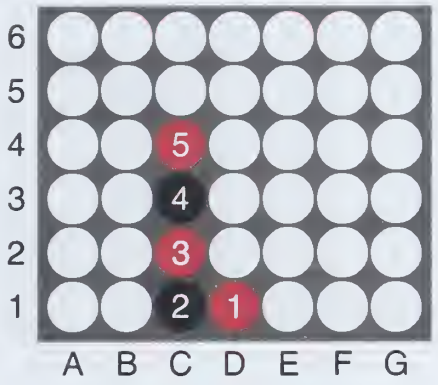
Red to play



Moves: D1 b1 B2 d2 D3 d4 B3 d5 B4 b5 C1 e1 D6 g1 (15) __

PROBLEM 212

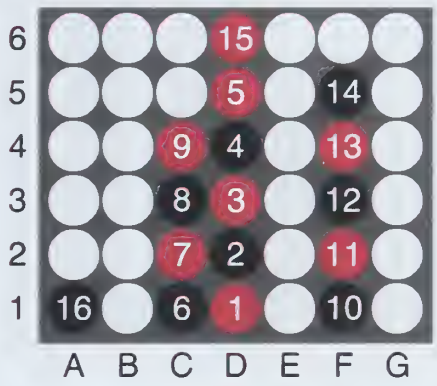
Black to play



Moves: D1 c1 C2 c3 C4 (6) __

PROBLEM 213

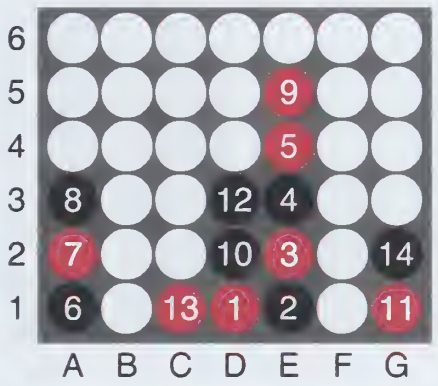
Red to play



Moves: D1 d2 D3 d4 D5 c1 C2 c3 C4 f1 F2 f3 F4 f5 D6 a1 (17) __

PROBLEM 214

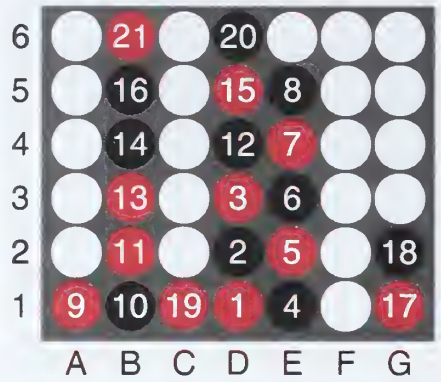
Red to play



Moves: D1 e1 E2 e3 E4 a1 A2 a3 E5 d2 G1 d3 C1 g2 (15) __

PROBLEM 215

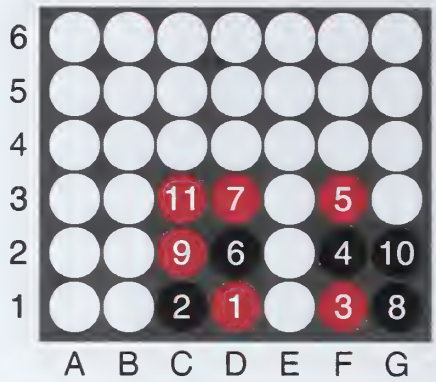
Black to play



Moves: D1 d2 D3 e1 E2 e3 E4 e5 A1 b1 B2 d4 B3 b4 D5 b5 G1 g2 C1 d6 B6 (22) __

PROBLEM 216

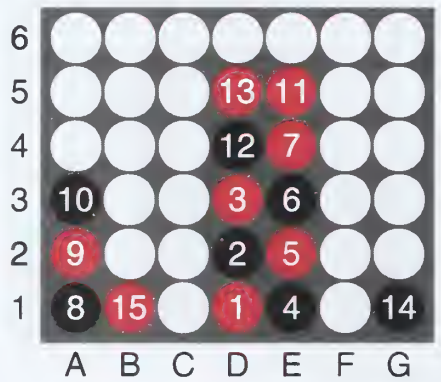
Black to play



Moves: D1 c1 F1 f2 F3 d2 D3 g1 C2 g2 C3 (12) __

PROBLEM 217

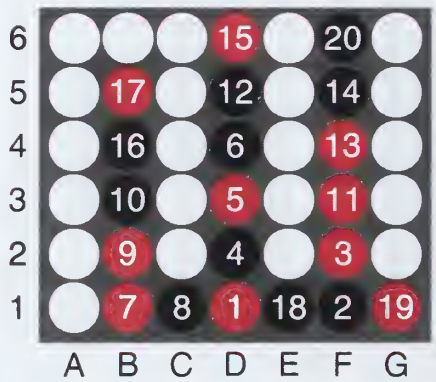
Black to play



Moves: D1 d2 D3 e1 E2 e3 E4 a1 A2 a3 E5 d4 D5 g1 B1 (16) __

PROBLEM 218

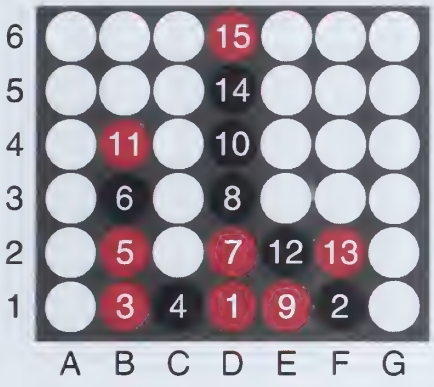
Red to play



Moves: D1 f1 F2 d2 D3 d4 B1 c1 B2 b3 F3 d5 F4 f5 D6 b4 B5 e1 G1 f6 (21) __

PROBLEM 219

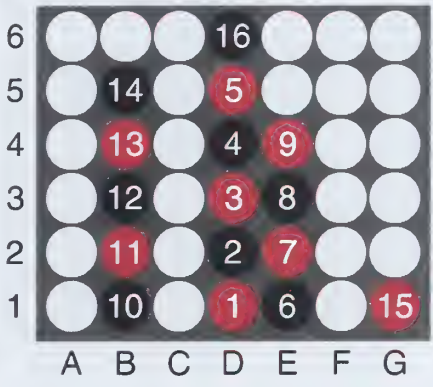
Black to play



Moves: D1 f1 B1 c1 B2 b3 D2 d3
E1 d4 B4 e2 F2 d5 D6 (16) __

PROBLEM 220

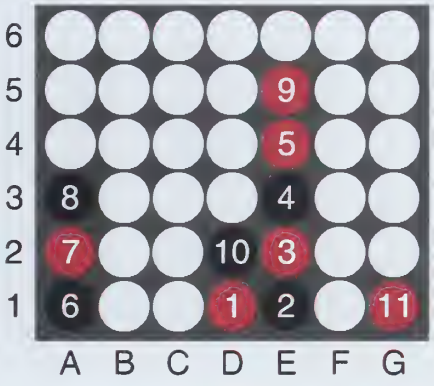
Red to play



Moves: D1 d2 D3 d4 D5 e1 E2 e3
E4 b1 B2 b3 B4 b5 G1 d6 (17) __

PROBLEM 221

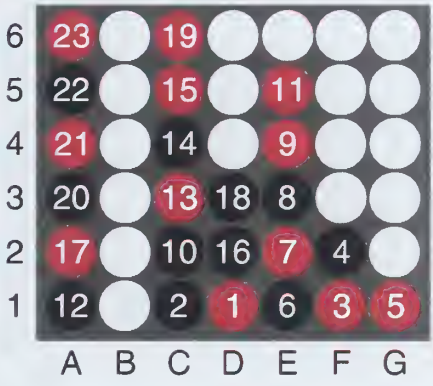
Black to play



Moves: D1 e1 E2 e3 E4 a1
A2 a3 E5 d2 G1 (12) __

PROBLEM 222

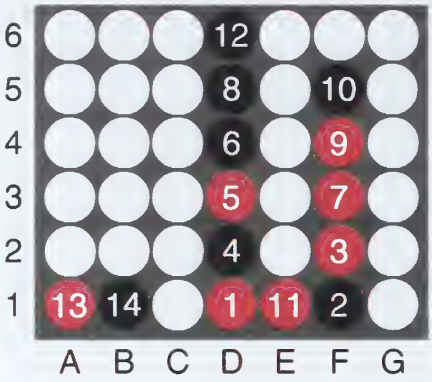
Black to play



Moves: D1 c1 F1 f2 G1 e1 E2 e3 E4 c2 E5 a1
C3 c4 C5 d2 A2 d3 C6 a3 A4 a5 A6 (24) __

PROBLEM 223

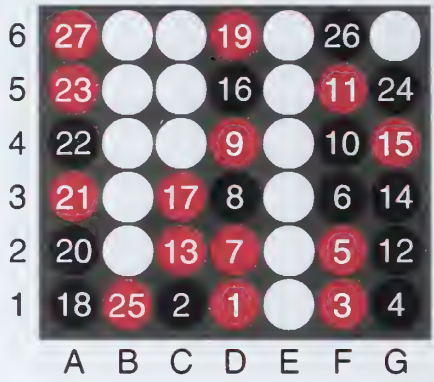
Red to play



Moves: D1 f1 F2 d2 D3 d4 F3 d5
F4 f5 E1 d6 A1 b1 (15) __

PROBLEM 224

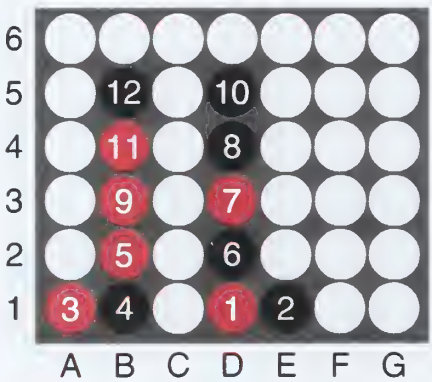
Black to play



Moves: D1 c1 F1 g1 F2 f3 D2 d3 D4
f4 F5 g2 C2 g3 G4 d5 C3 a1 D6 a2
A3 a4 A5 g5 B1 f6 A6 (28) __

PROBLEM 225

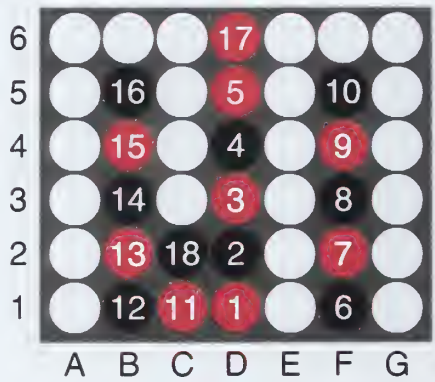
Red to play



Moves: D1 e1 A1 b1 B2 d2 D3
d4 B3 d5 B4 b5 (13) __

PROBLEM 226

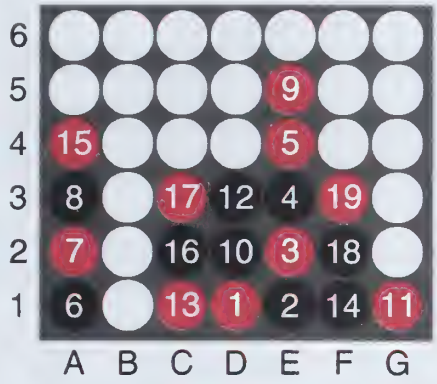
Red to play



Moves: D1 d2 D3 d4 D5 f1 F2 f3 F4
f5 C1 b1 B2 b3 B4 b5 D6 c2 (19) __

PROBLEM 227

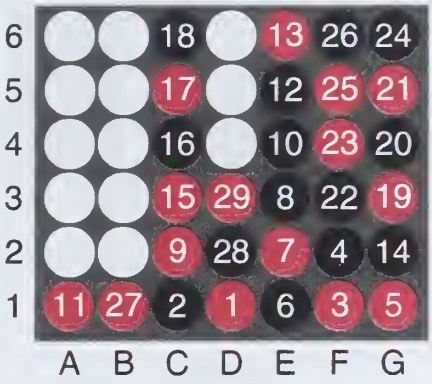
Black to play



Moves: D1 e1 E2 e3 E4 a1 A2 a3 E5 d2 G1 d3 C1 f1 A4 c2 C3 f2 F3 (20) __

PROBLEM 228

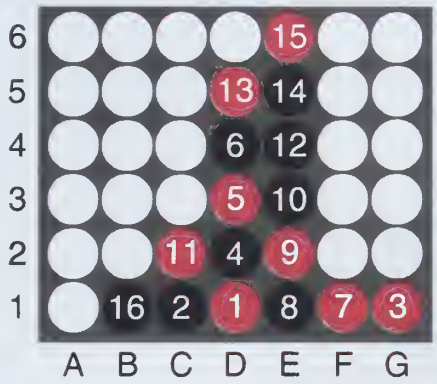
Black to play



Moves: D1 c1 F1 f2 G1 e1 E2 e3 C2 e4 A1 e5 E6 g2 C3 c4 C5 c6 G3 g4 G5 f3 F4 g6 F5 f6 B1 d2 D3 (30) __

PROBLEM 229

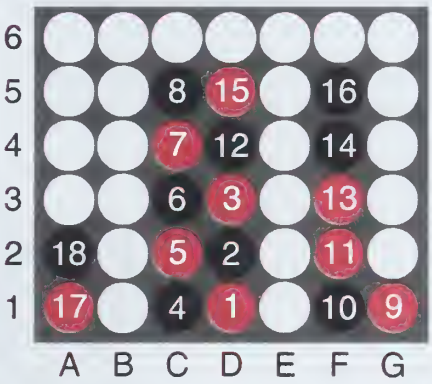
Red to play



Moves: D1 c1 G1 d2 D3 d4 F1 e1 E2 e3 C2 e4 D5 e5 E6 b1 (17) __

PROBLEM 230

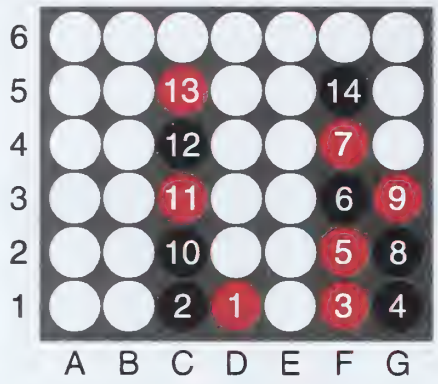
Red to play



Moves: D1 d2 D3 c1 C2 c3 C4 c5 G1 f1 F2 d4 F3 f4 D5 f5 A1 a2 (19) __

PROBLEM 231

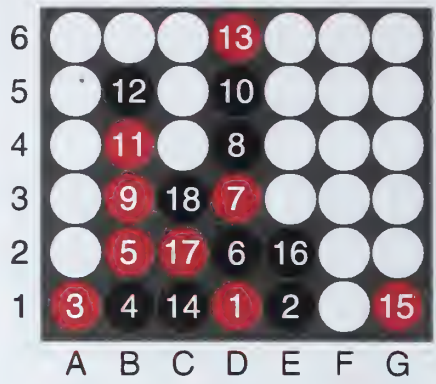
Red to play



Moves: D1 c1 F1 g1 F2 f3 F4 g2
G3 c2 C3 c4 C5 f5 (15) __

PROBLEM 232

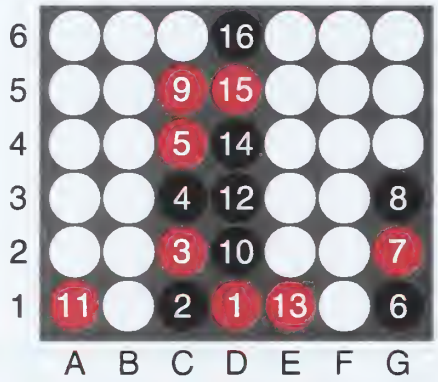
Red to play



Moves: D1 e1 A1 b1 B2 d2 D3 d4 B3
d5 B4 b5 D6 c1 G1 e2 C2 c3 (19) __

PROBLEM 233

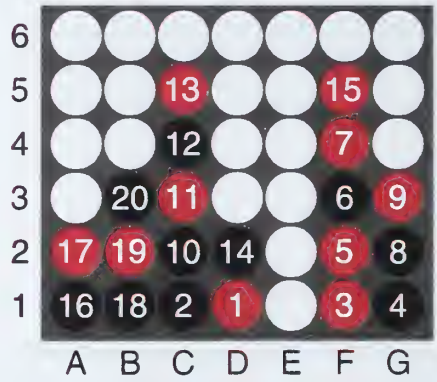
Red to play



Moves: D1 c1 C2 c3 C4 g1 G2 g3
C5 d2 A1 d3 E1 d4 D5 d6 (17) __

PROBLEM 234

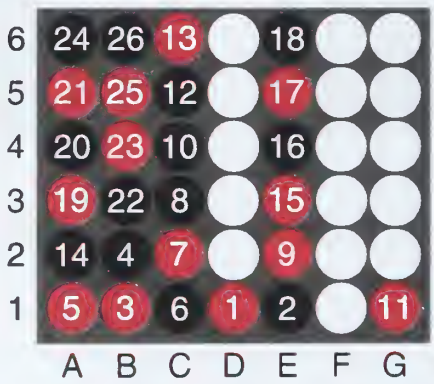
Red to play



Moves: D1 c1 F1 g1 F2 f3 F4 g2 G3 c2
C3 c4 C5 d2 F5 a1 A2 b1 B2 b3 (21) __

PROBLEM 235

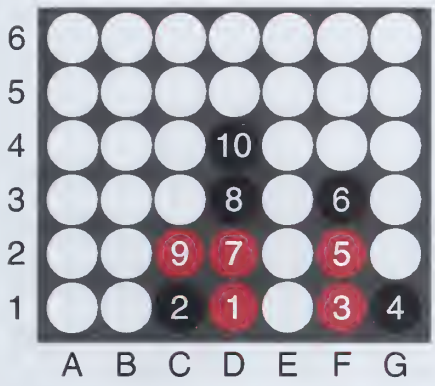
Red to play



Moves: D1 e1 B1 b2 A1 c1 C2 c3
E2 c4 G1 c5 C6 a2 E3 e4 E5 e6 A3
a4 A5 b3 B4 a6 B5 b6 (27) __

PROBLEM 236

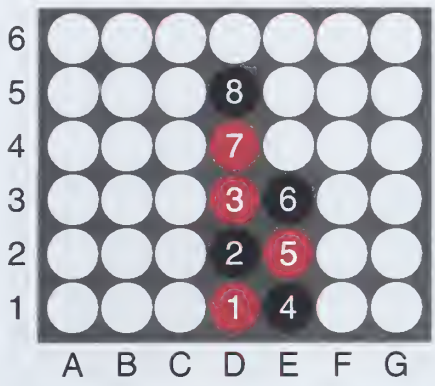
Red to play



Moves: D1 c1 F1 g1 F2 f3 D2 d3 C2 d4 (11) __

PROBLEM 237

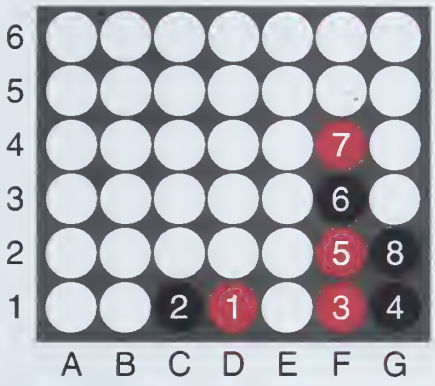
Red to play



Moves: D1 d2 D3 e1 E2 e3 D4 d5 (9) __

PROBLEM 238

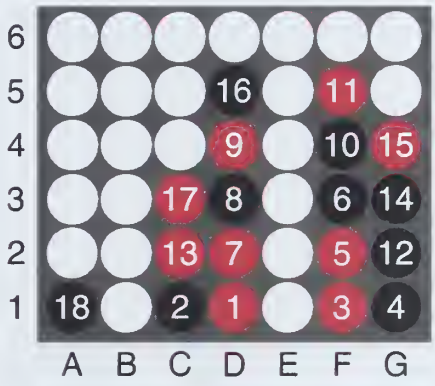
Red to play



Moves: D1 c1 F1 g1 F2 f3 F4 g2 (9) __

PROBLEM 239

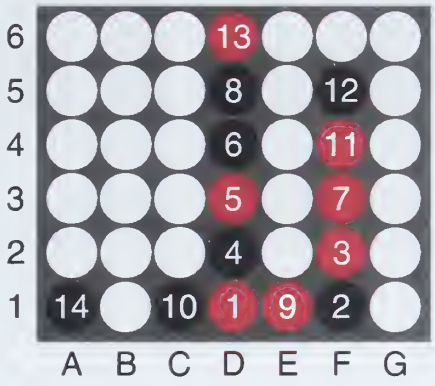
Red to play



Moves: D1 c1 F1 g1 F2 f3 D2 d3 D4 f4 F5 g2 C2 g3 G4 d5 C3 a1 (19) __

PROBLEM 240

Red to play

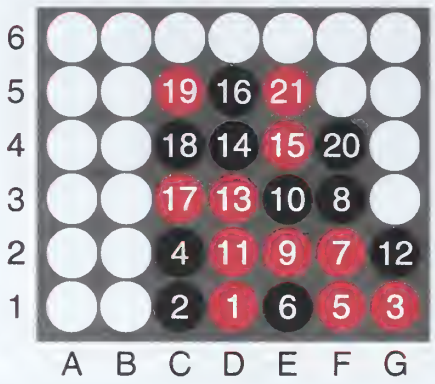


Moves: D1 f1 F2 d2 D3 d4 F3 d5 E1 c1 F4 f5 D6 a1 (15) __

Problem Set I (Expert)

PROBLEM 241

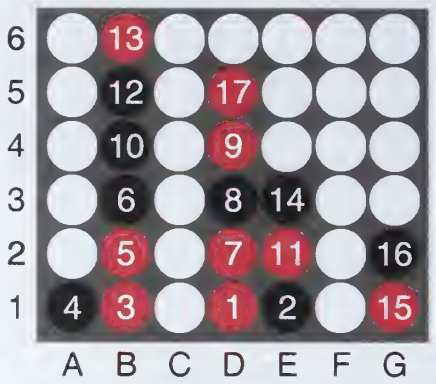
Black to play



Moves: D1 c1 G1 c2 F1 e1 F2 f3 E2 e3 D2 g2 D3 d4 E4 d5 C3 c4 C5 f4 E5 (22) __

PROBLEM 242

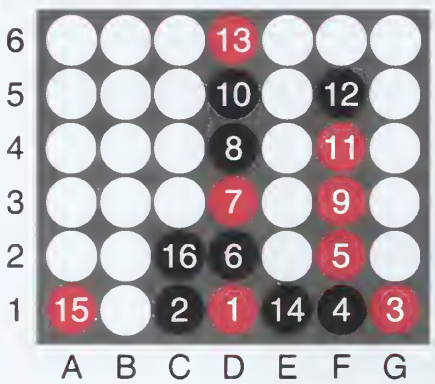
Black to play



Moves: D1 e1 B1 a1 B2 b3 D2 d3 D4 b4 E2 b5 B6 e3 G1 g2 D5 (18) __

PROBLEM 243

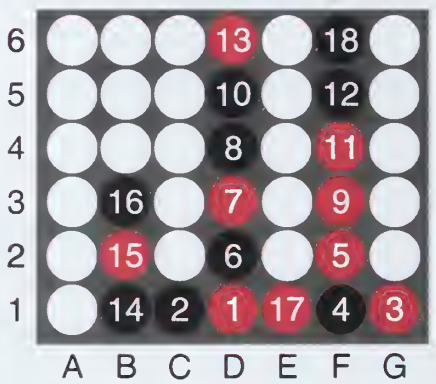
Red to play



Moves: D1 c1 G1 f1 F2 d2 D3 d4 F3 d5 F4 f5 D6 e1 A1 c2 (17) __

PROBLEM 244

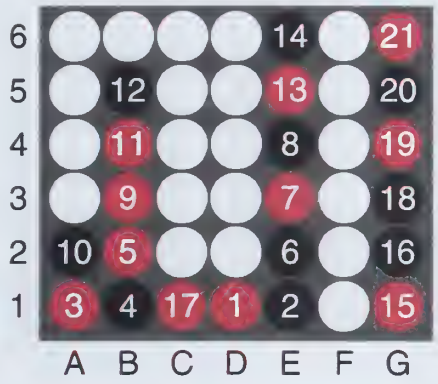
Red to play



Moves: D1 c1 G1 f1 F2 d2 D3 d4 F3 d5 F4 f5 D6 b1 B2 b3 E1 f6 (19) __

PROBLEM 245

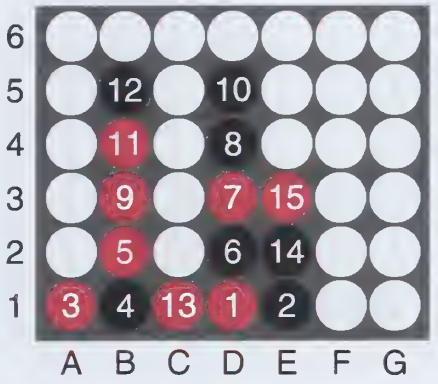
Black to play



Moves: D1 e1 A1 b1 B2 e2 E3 e4 B3 a2 B4 b5 E5 e6 G1 g2 C1 g3 G4 g5 G6 (22) __

PROBLEM 246

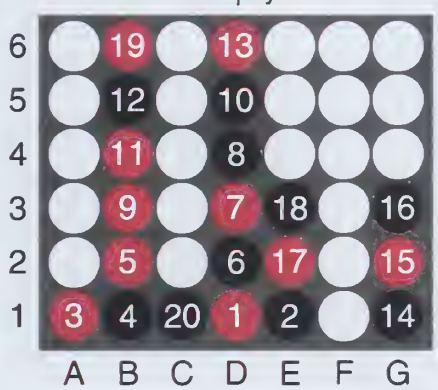
Black to play



Moves: D1 e1 A1 b1 B2 d2 D3 d4 B3 d5 B4 b5 C1 e2 E3 (16) __

PROBLEM 247

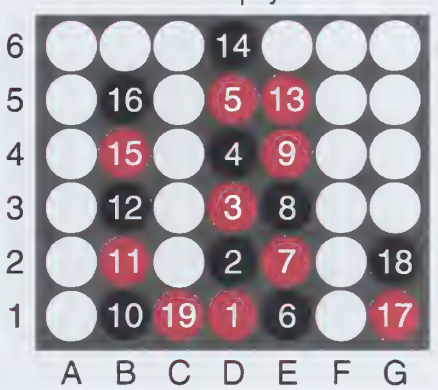
Red to play



Moves: D1 e1 A1 b1 B2 d2 D3 d4 B3 d5 B4 b5 D6 g1 G2 g3 E2 e3 B6 c1 (21) __

PROBLEM 248

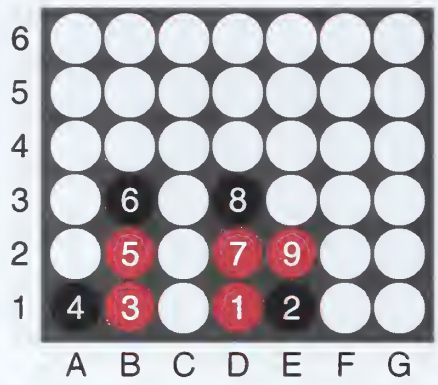
Black to play



Moves: D1 d2 D3 d4 D5 e1 E2 e3 E4 b1 B2 b3 E5 d6 B4 b5 G1 g2 C1 (20) __

PROBLEM 249

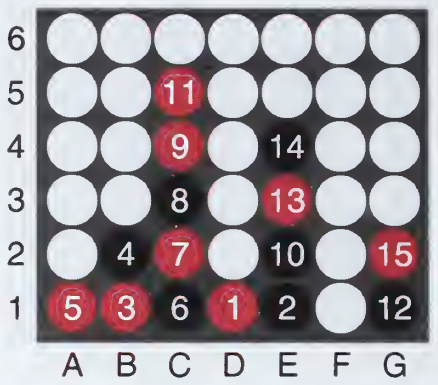
Black to play



Moves: D1 e1 B1 a1 B2 b3 D2 d3 E2 (10) __

PROBLEM 250

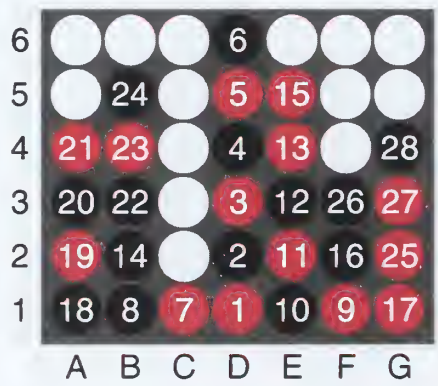
Black to play



Moves: D1 e1 B1 b2 A1 c1 C2 c3
C4 e2 C5 g1 E3 e4 G2 (16) __

PROBLEM 251

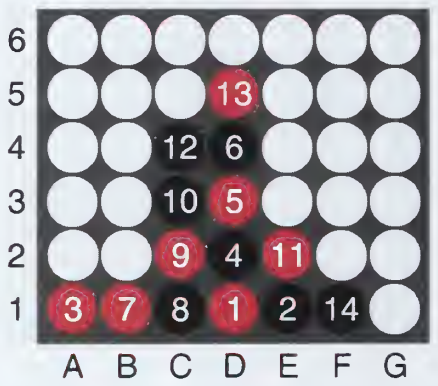
Red to play



Moves: D1 d2 D3 d4 D5 d6 C1 b1
F1 e1 E2 e3 E4 b2 E5 f2 G1 a1 A2 a3
A4 b3 B4 b5 G2 f3 G3 g4 (29) __

PROBLEM 252

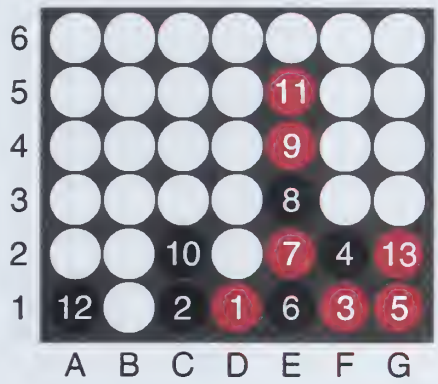
Red to play



Moves: D1 e1 A1 d2 D3 d4 B1
c1 C2 c3 E2 c4 D5 f1 (15) __

PROBLEM 253

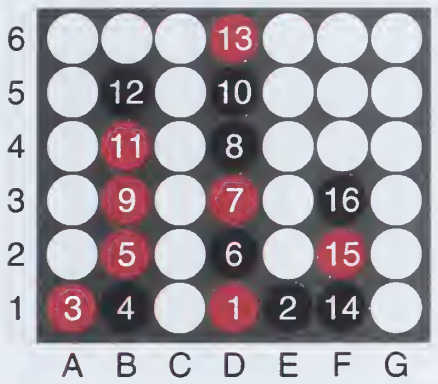
Black to play



Moves: D1 c1 F1 f2 G1 e1 E2
e3 E4 c2 E5 a1 G2 (14) __

PROBLEM 254

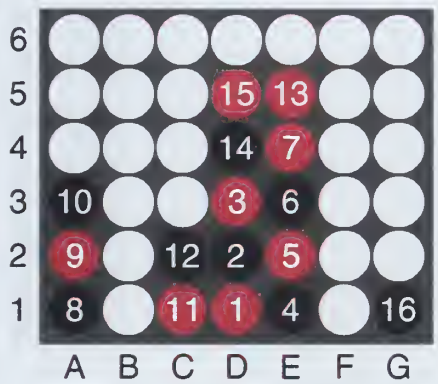
Red to play



Moves: D1 e1 A1 b1 B2 d2 D3 d4
B3 d5 B4 b5 D6 f1 F2 f3 (17) __

PROBLEM 255

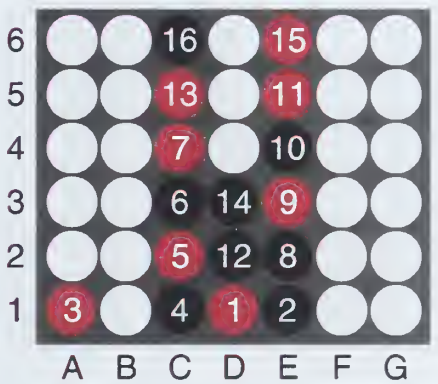
Red to play



Moves: D1 d2 D3 e1 E2 e3 E4 a1
A2 a3 C1 c2 E5 d4 D5 g1 (17) __

PROBLEM 256

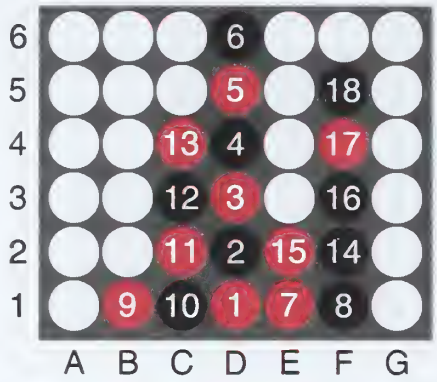
Red to play



Moves: D1 e1 A1 c1 C2 c3 C4 e2
E3 e4 E5 d2 C5 d3 E6 c6 (17) __

PROBLEM 257

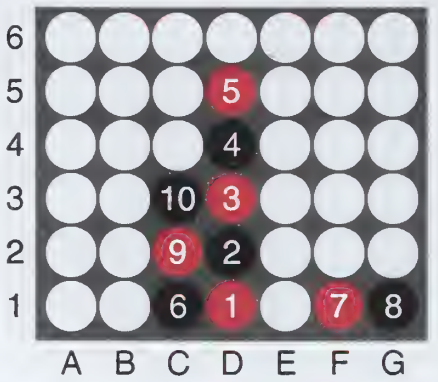
Red to play



Moves: D1 d2 D3 d4 D5 d6 E1 f1 B1 c1 C2 c3 C4 f2 E2 f3 F4 f5 (19) __

PROBLEM 258

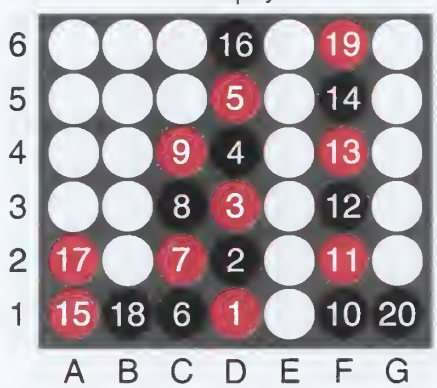
Red to play



Moves: D1 d2 D3 d4 D5 c1 F1 g1 C2 c3 (11) __

PROBLEM 259

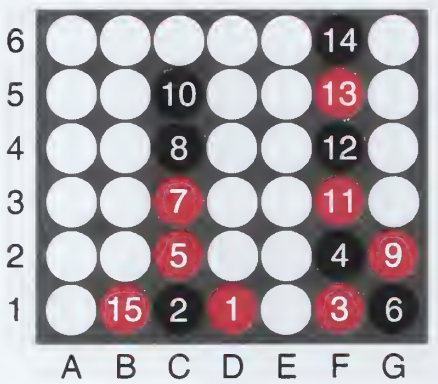
Red to play



Moves: D1 d2 D3 d4 D5 c1 C2 c3 C4 f1 F2 f3 F4 f5 A1 d6 A2 b1 F6 g1 (21) __

PROBLEM 260

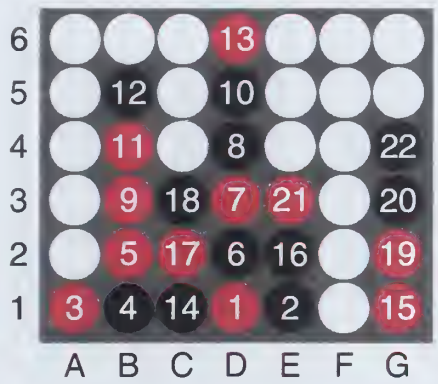
Black to play



Moves: D1 c1 F1 f2 C2 g1 C3 c4 G2 c5 F3 f4 F5 f6 B1 (16) __

PROBLEM 261

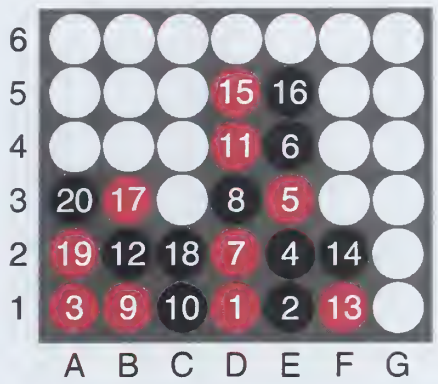
Red to play



Moves: D1 e1 A1 b1 B2 d2 D3 d4 B3 d5 B4 b5 D6 c1 G1 e2 C2 c3 G2 g3 E3 g4 (23) __

PROBLEM 262

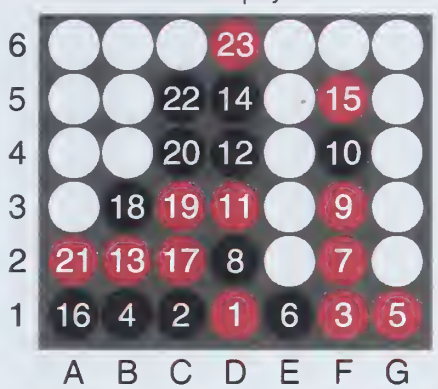
Red to play



Moves: D1 e1 A1 e2 E3 e4 D2 d3 B1 c1 D4 b2 F1 f2 D5 e5 B3 c2 A2 a3 (21) __

PROBLEM 263

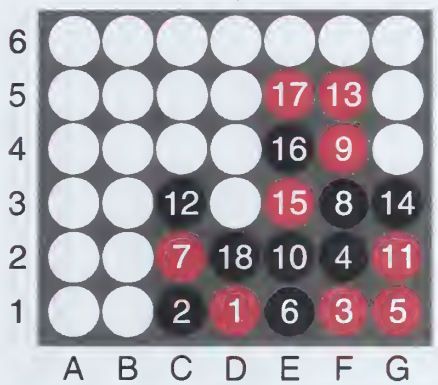
Black to play



Moves: D1 c1 F1 b1 G1 e1 F2 d2 F3 f4 D3 d4 B2 d5 F5 a1 C2 b3 C3 c4 A2 c5 D6 (24) __

PROBLEM 264

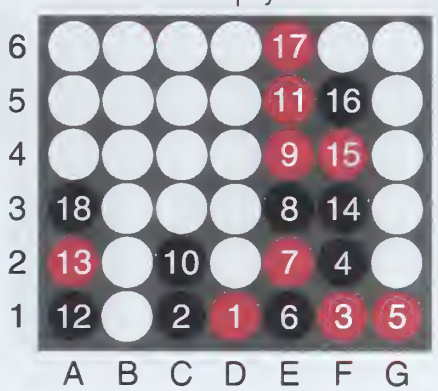
Red to play



Moves: D1 c1 F1 f2 G1 e1 C2 f3 F4 e2 G2 c3 F5 g3 E3 e4 E5 d2 (19) __

PROBLEM 265

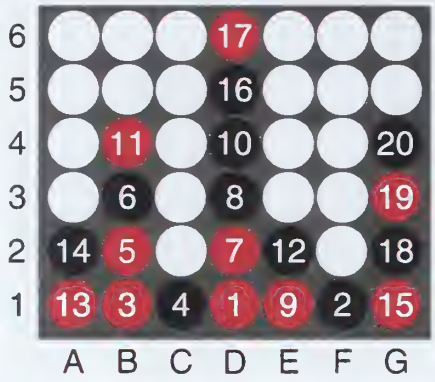
Red to play



Moves: D1 c1 F1 f2 G1 e1 E2 e3 E4 c2 E5 a1 A2 f3 F4 f5 E6 a3 (19) __

PROBLEM 266

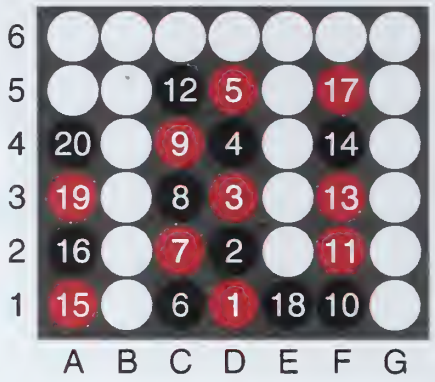
Red to play



Moves: D1 f1 B1 c1 B2 b3 D2 d3 E1 d4 B4 e2 A1 a2 G1 d5 D6 g2 G3 g4 (21) __

PROBLEM 267

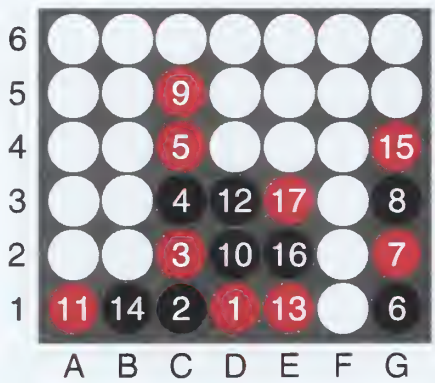
Red to play



Moves: D1 d2 D3 d4 D5 c1 C2 c3 C4 f1 F2 c5 F3 f4 A1 a2 F5 e1 A3 a4 (21) __

PROBLEM 268

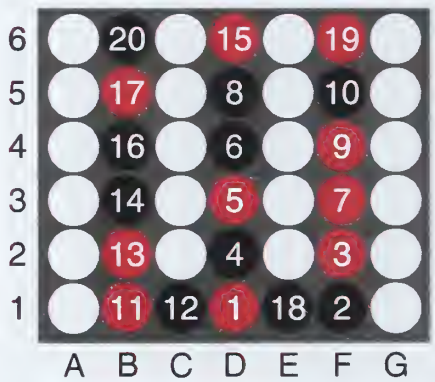
Black to play



Moves: D1 c1 C2 c3 C4 g1 G2 g3 C5 d2 A1 d3 E1 b1 G4 e2 E3 (18) __

PROBLEM 269

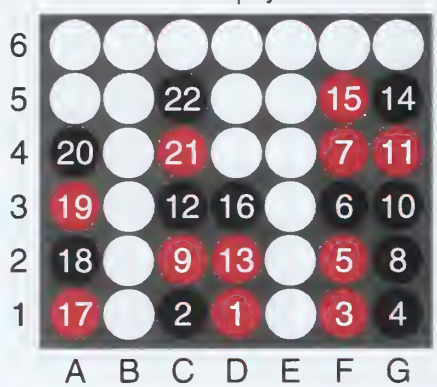
Red to play



Moves: D1 f1 F2 d2 D3 d4 F3 d5 F4 f5 B1 c1 B2 b3 D6 b4 B5 e1 F6 b6 (21) __

PROBLEM 270

Red to play

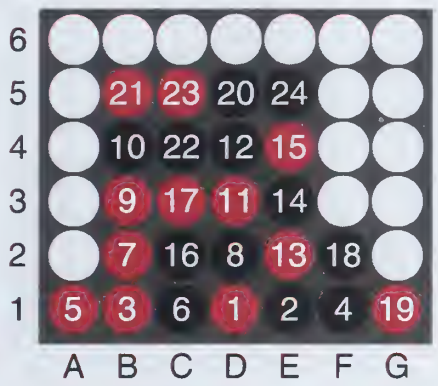


Moves: D1 c1 F1 g1 F2 f3 F4 g2 C2 g3 G4 c3 D2 g5 F5 d3 A1 a2 A3 a4 C4 c5 (23) __

Problem Set J (Challenger)

PROBLEM 271

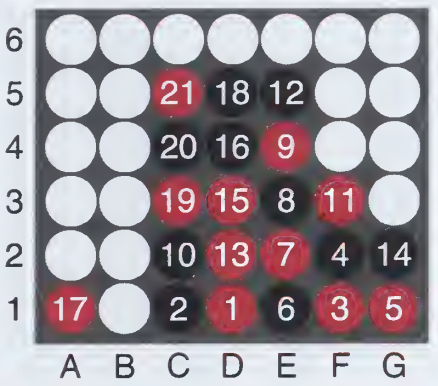
Red to play



Moves: D1 e1 B1 f1 A1 c1 B2 d2 B3 b4 D3 d4 E2 e3 E4 c2 C3 f2 G1 d5 B5 c4 C5 e5 (25) __

PROBLEM 272

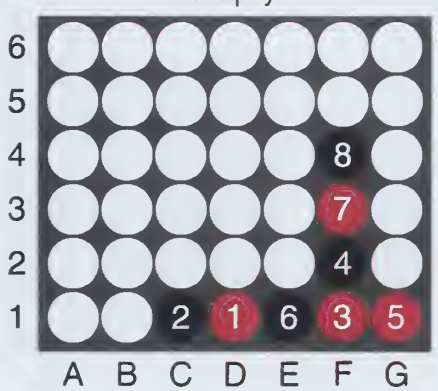
Black to play



Moves: D1 c1 F1 f2 G1 e1 E2 e3 E4 c2 F3 e5 D2 g2 D3 d4 A1 d5 C3 c4 C5 (22) __

PROBLEM 273

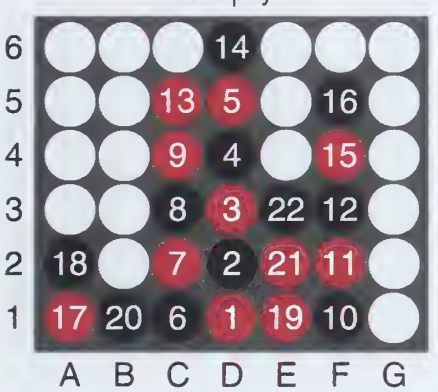
Red to play



Moves: D1 c1 F1 f2 G1 e1 F3 f4 (9) __

PROBLEM 274

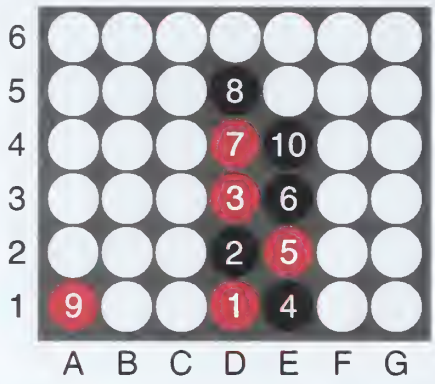
Red to play



Moves: D1 d2 D3 d4 D5 c1 C2 c3 C4 f1 F2 f3 C5 d6 F4 f5 A1 a2 E1 b1 E2 e3 (23) __

PROBLEM 275

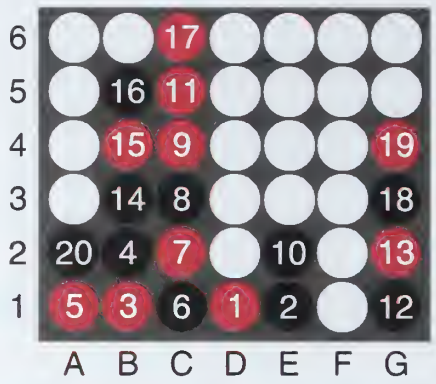
Red to play



Moves: D1 d2 D3 e1 E2 e3
D4 d5 A1 e4 (11) __

PROBLEM 276

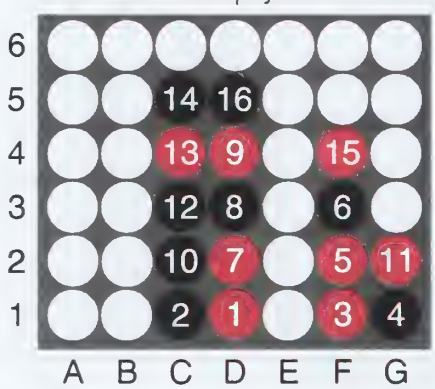
Red to play



Moves: D1 e1 B1 b2 A1 c1 C2 c3 C4 e2
C5 g1 G2 b3 B4 b5 C6 g3 G4 a2 (21) __

PROBLEM 277

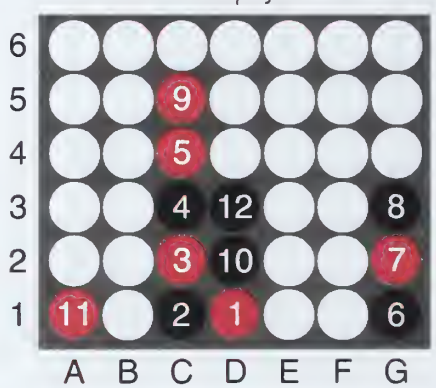
Red to play



Moves: D1 c1 F1 g1 F2 f3 D2 d3 D4
c2 G2 c3 C4 c5 F4 d5 (17) __

PROBLEM 278

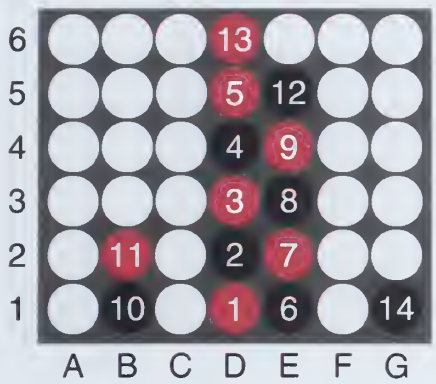
Red to play



Moves: D1 c1 C2 c3 C4 g1 G2
g3 C5 d2 A1 d3 (13) __

PROBLEM 279

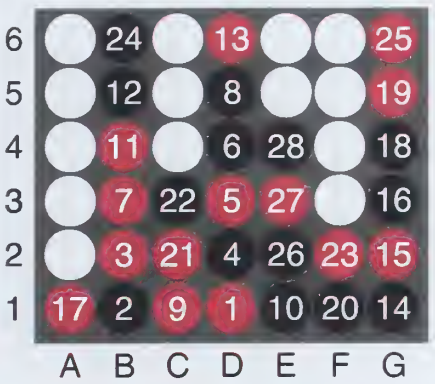
Red to play



Moves: D1 d2 D3 d4 D5 e1 E2 e3 E4 b1 B2 e5 D6 g1 (15) __

PROBLEM 280

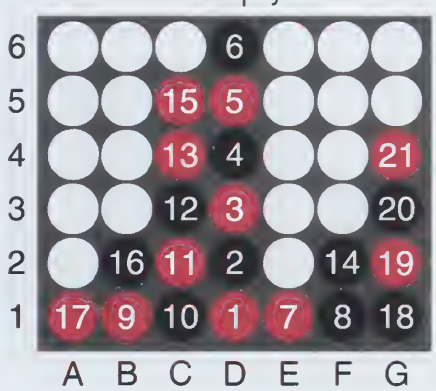
Red to play



Moves: D1 b1 B2 d2 D3 d4 B3 d5 C1 e1 B4 b5 D6 g1 G2 g3 A1 g4 G5 f1 C2 c3 F2 b6 G6 e2 E3 e4 (29) __

PROBLEM 281

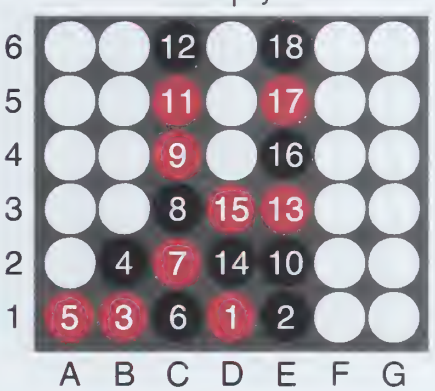
Black to play



Moves: D1 d2 D3 d4 D5 d6 E1 f1 B1 c1 C2 c3 C4 f2 C5 b2 A1 g1 G2 g3 G4 (22) __

PROBLEM 282

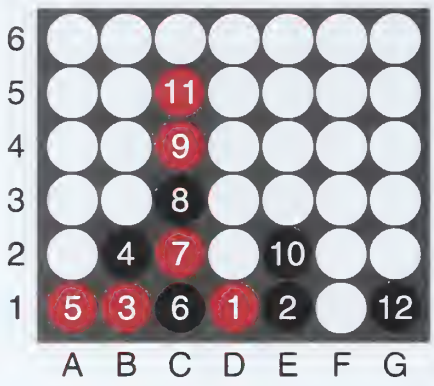
Red to play



Moves: D1 e1 B1 b2 A1 c1 C2 c3 C4 e2 C5 c6 E3 d2 D3 e4 E5 e6 (19) __

PROBLEM 283

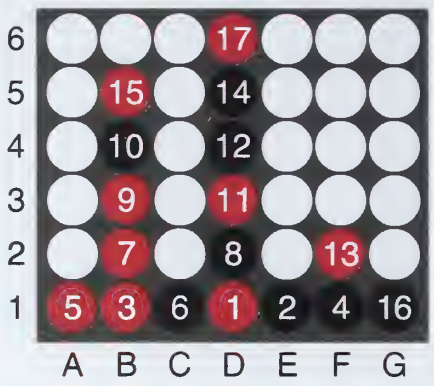
Red to play



Moves: D1 e1 B1 b2 A1 c1 C2 c3 C4 e2 C5 g1 (13) __

PROBLEM 284

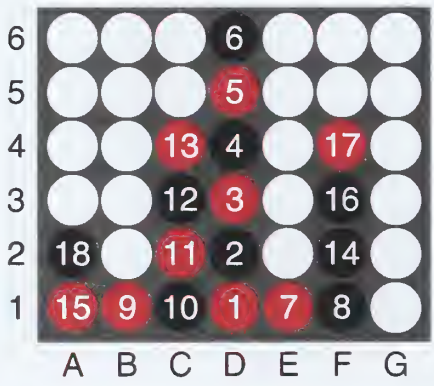
Black to play



Moves: D1 e1 B1 f1 A1 c1 B2 d2 B3 b4 D3 d4 F2 d5 B5 g1 D6 (18) __

PROBLEM 285

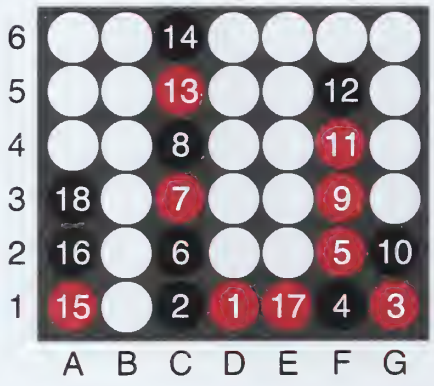
Red to play



Moves: D1 d2 D3 d4 D5 d6 E1 f1 B1 c1 C2 c3 C4 f2 A1 f3 F4 a2 (19) __

PROBLEM 286

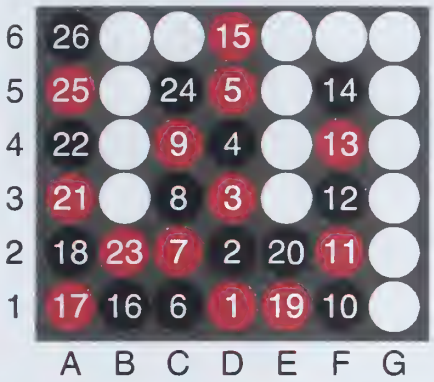
Red to play



Moves: D1 c1 G1 f1 F2 c2 C3 c4 F3 g2 F4 f5 C5 c6 A1 a2 E1 a3 (19) __

PROBLEM 287

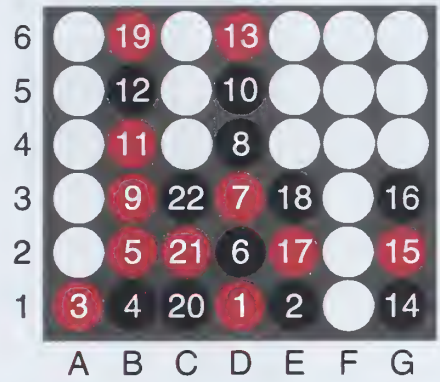
Red to play



Moves: D1 d2 D3 d4 D5 c1 C2 c3 C4 f1 F2 f3 F4 f5 D6 b1 A1 a2 E1 e2 A3 a4 B2 c5 A5 a6 (27) __

PROBLEM 288

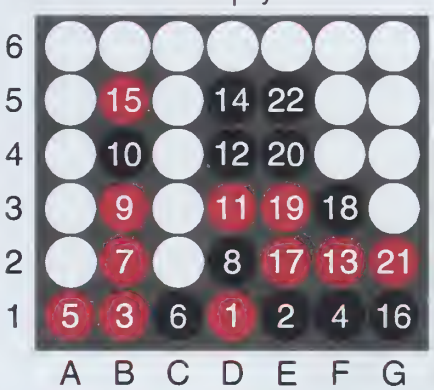
Red to play



Moves: D1 e1 A1 b1 B2 d2 D3 d4 B3 d5 B4 b5 D6 g1 G2 g3 E2 e3 B6 c1 C2 c3 (23) __

PROBLEM 289

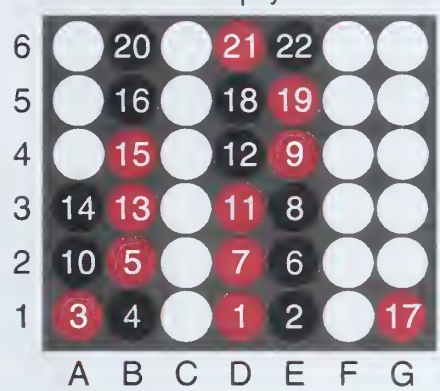
Red to play



Moves: D1 e1 B1 f1 A1 c1 B2 d2 B3 b4 D3 d4 F2 d5 B5 g1 E2 f3 E3 e4 G2 e5 (23) __

PROBLEM 290

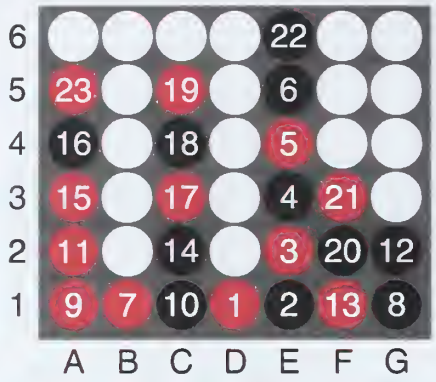
Red to play



Moves: D1 e1 A1 b1 B2 e2 D2 e3 E4 a2 D3 d4 B3 a3 B4 b5 G1 d5 E5 b6 D6 e6 (23) __

PROBLEM 291

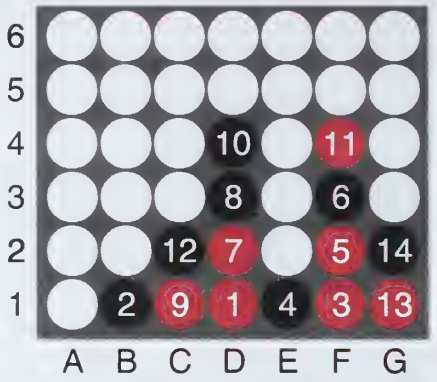
Black to play



Moves: D1 e1 E2 e3 E4 e5 B1 g1 A1 c1 A2 g2 F1 c2 A3 a4 C3 c4 C5 f2 F3 e6 A5 (24) __

PROBLEM 292

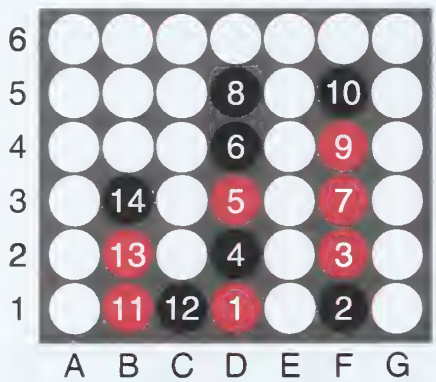
Red to play



Moves: D1 b1 F1 e1 F2 f3 D2 d3 C1 d4 F4 c2 G1 g2 (15) __

PROBLEM 293

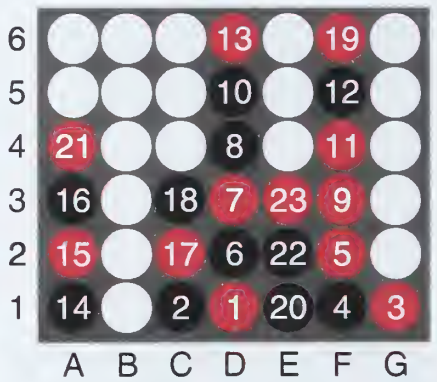
Red to play



Moves: D1 f1 F2 d2 D3 d4 F3 d5 F4 f5 B1 c1 B2 b3 (15) __

PROBLEM 294

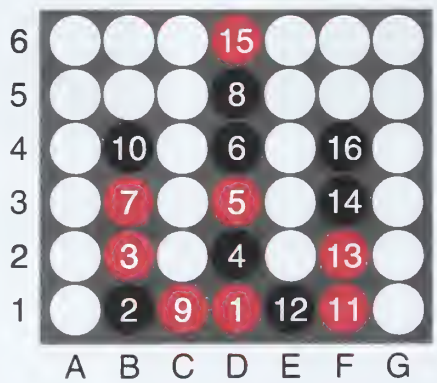
Black to play



Moves: D1 c1 G1 f1 F2 d2 D3 d4 F3 d5 F4 f5 D6 a1 A2 a3 C2 c3 F6 e1 A4 e2 E3 (24) __

PROBLEM 295

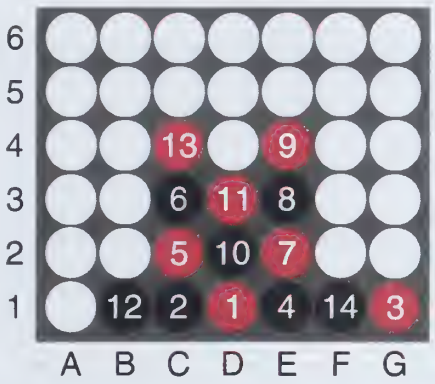
Red to play



Moves: D1 b1 B2 d2 D3 d4 B3 d5 C1 b4 F1 e1 F2 f3 D6 f4 (17) __

PROBLEM 296

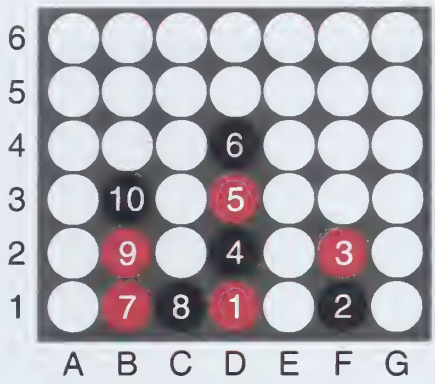
Red to play



Moves: D1 c1 G1 e1 C2 c3 E2 e3
E4 d2 D3 b1 C4 f1 (15) __

PROBLEM 297

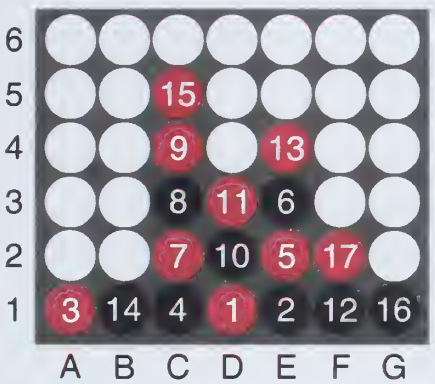
Red to play



Moves: D1 f1 F2 d2 D3 d4
B1 c1 B2 b3 (11) __

PROBLEM 298

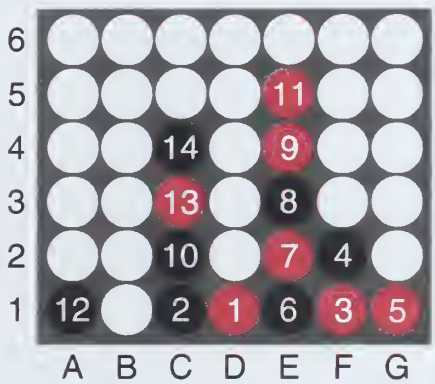
Black to play



Moves: D1 e1 A1 c1 E2 e3 C2 c3 C4
d2 D3 f1 E4 b1 C5 g1 F2 (18) __

PROBLEM 299

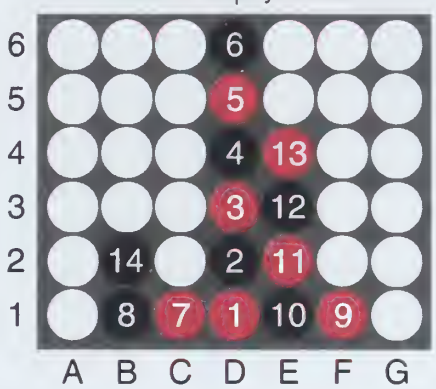
Red to play



Moves: D1 c1 F1 f2 G1 e1 E2 e3
E4 c2 E5 a1 C3 c4 (15) __

PROBLEM 300

Red to play



Moves: D1 d2 D3 d4 D5 d6 C1
b1 F1 e1 E2 e3 E4 b2 (15) __

Chapter 3

Strategic Threats (Endgame Victories)

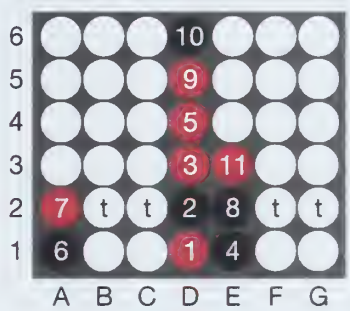
Most games between good players are won in the ending, rather than with a quickly forced four-in-row early in the game. When a player does make a key sequence of forcing moves early in the game, often he or she will do so to establish a long-term strategic threat, rather than an immediate four-in-row. Therefore, a player must understand which strategic threats will lead to a successful endgame, and which threats are worthless.

Let us begin our study of *strategic principles*. Not only will these principles tell you what threats of your own to seek, but they also will tell which of your opponent's potential threats to block. I first consider the simplest cases, and then gradually study the effects of complications.

Black Wins on the Second (or Fourth or Sixth) Row

If Red lets Black occupy two center cells in the second row (along with a third cell underneath one of them), the game may be essentially over already, although Black won't complete four-in-row until the ending. Diagram 30 shows an example of this situation.

DIAGRAM 30
Black to play and win



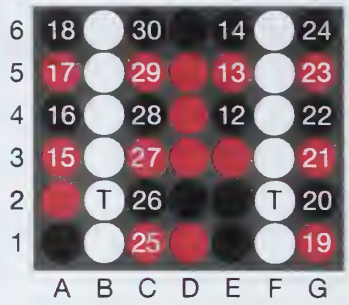
Moves: D1! d2 D3! e1 D4 a1
A2?? e2! D5 d6! E3 (12) __

In Diagram 30, the game seems to have just begun, and it is Red who may appear to be building strength in the upper center of the board. Yet Black is just as certain of victory here as he or she was with the "fireworks" in games like the one shown in Diagram 13.

Black has the entire second row locked up with minor threats and will win easily *as long he or she avoids playing in the first row.* Red will eventually be forced to play in *each* remaining first-row cell, and Black will get to play disks in *all* the second-row cells. If Black plays b1, c1, f1, or g1, Black loses, but otherwise it hardly matters where Black plays; the simplest play is for Black to take e4 now and then just play wherever Red plays. This will eventually lead to a position similar to the one in Diagram 31.

In Diagram 31, B2 and F2 have become major threats for Red as well as Black—in fact, Red has adjacent threat pairs at B2-C3 and F2-F3—but it's Red's turn to play. Whatever Red does now, Black will win on Black's next turn.

DIAGRAM 31
Red to play (will lose)

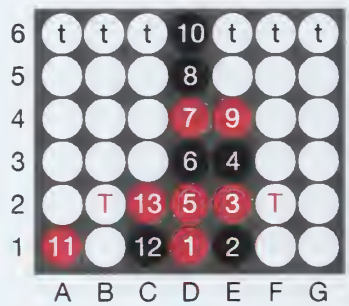


Moves: D1! d2 D3! e1 D4 a1 A2?? e2! D5 d6! E3 (12) e4 E5 e6 A3 a4! A5 a6! G1 g2! G3 g4! G5 g6! C1 c2! C3 c4! C5 c6! (31) __

Black wins easily in Diagram 30 because of Black's two disks in the second row; Red must not let Black achieve this. Notice that Red threats above the second row will be useless—the game will end before they're reached. Instead of the second row, Black can win with disks in another even-numbered row, though then he or she would have to worry about Red threats *below* the Black threats. It's essential to have a Black disk under one of the two disks on the even-numbered row. In Diagram 30, e1 is essential; otherwise, Red's first-row threats would undercut Black's second-row threats. Similarly a fourth-row winning configuration for Black might involve d4-e4-e3, and a sixth-row winning configuration might involve d6-e6-d5.

In Diagram 30, Black has only two disks aligned instead of three so hasn't yet achieved a major threat. In fact, even the second Black disk often isn't essential! Consider Diagram 32. In Diagram 32, two major second-row threats are shown for Red, while the entire sixth row is shown as minor threats for Black, even though, so far, Black has only one disk out of the four needed to complete a four-in-row.

DIAGRAM 32
Black to play and win



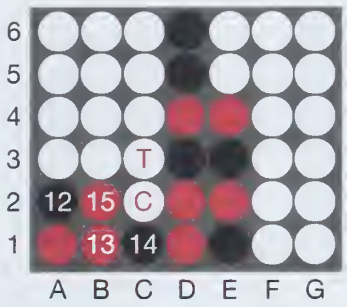
Moves: D1 e1 E2? e3! D2 d3! D4 d5 E4? d6! A1 c1! C2 (14) __

Black has an inevitable and easy win on the sixth row in Diagram 32. In fact, Black doesn't get just four-in-row; Black can get seven-in-row! Play this game out a few times, and you'll see what I mean.

The easiest way to prove that Black wins in Diagram 32 is for Black to start by playing 14 a2. This leaves every column with an even number of empty cells, and Black can then adopt the simple rule "Black plays where Red just played." Red ends up with every remaining cell on the first, third, and fifth rows (and gets no four-in-row); Black ends up with every remaining cell on the second, fourth, and sixth rows and completes not just four-in-row, but seven-in-row on the sixth row.

There are many CONNECT 4 players who haven't learned the strategic rules and will play exactly as Red did in Diagram 32. They may look at their major threats at B2 and F2 and even think they're winning! But it is Black (Second Player) who wants and needs even-row threats, not Red.

DIAGRAM 33
Black to play (will lose)



Moves: D1 e1 E2? e3! D2 d3! D4 d5 E4? d6! A1 (12) a2?? B1 c1 B2! (16) __

Just knowing this one "trick" may improve your CONNECT 4 reputation quite a bit. In fact, the ease with which Black won in these games may deceive you into thinking CONNECT 4 is an easy game. But if this is your only trick, you won't stand a chance against a good player. Read on, and become a master!

As implied by the exclamation point, Black's 12 c1 in Diagram 32 was essential. If instead Black prematurely started playing where Red just played with 12 A2, Black would lose quickly, as shown in Diagram 33.

First- and Second-Player Strategies Are Not the Same

The strategies for Red and Black are different in CONNECT 4. Black (Second Player) will usually be very happy if Red (First Player) makes the error of trying for two Red disks on an even-numbered row. Red should aim for threats on the third or fifth row.

(In the middle of a game, to know whether to strive for odd- or even-row threats, you will often need to remind yourself whether you were First Player or Second Player. Of course, it is usually very simple to figure out which player went first; First Player virtually always plays his or her first disk at D1.)

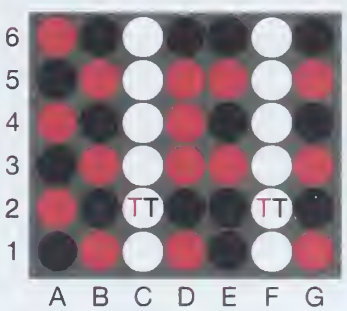
Zugzwang: It Is Not Always an Advantage to Move

I must introduce the concept of *zugzwang*, which is essential to the strategic analysis of endgames in CONNECT 4. This German word means "compulsion to move."

In Diagram 34, C2 is a major threat for both Red and Black; likewise, F2 is a major threat for both players. Whoever moves next will lose; the opposing player will get four-in-row immediately afterward.

Zugzwang is a chess term referring to a position where any move loses—your only chance would be to pass your move, but that's

DIAGRAM 34
Whoever plays next will lose



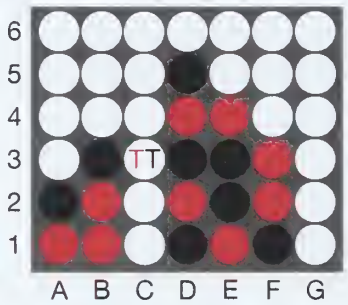
Moves: D1! d2 D3! e1 D4 e2 G1?? a1?? D5! d6 E3?? g2! G3 g4 G5 g6 A2 a3 A4 a5 A6 e4! E5 e6! B1 b2! B3 b4! B5 b6! (31) __

not allowed. Diagram 35 gives another, less obvious, example of zugzwang. If it is Red's move, then Black wins; if it is Black's move, then Red wins!

That the position shown in Diagram 35 is zugzwang is far from obvious; there's still a lot of play left in the game, and both players will need to be careful. Diagram 36 shows this game after another 14 disks have been (well) played; again,

DIAGRAM 35

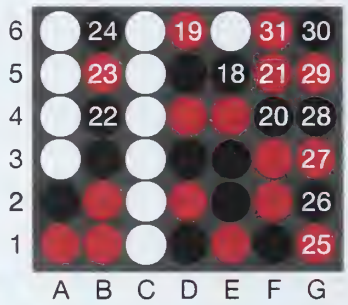
Whoever plays next will lose



Moves: B1?? d1? B2 f1? E1 e2 A1?? e3?? D2 d3 D4 d5 F2?? b3?? F3 a2 E4 (18) __

DIAGRAM 36

Whoever plays next will lose



Moves: B1?? d1? B2 f1? E1 e2 A1?? e3?? D2 d3 D4 d5 F2?? b3?? F3 a2 E4 (18) e5 D6 f4 F5! b4 B5 b6 G1 g2 G3 g4 G5 g6 F6 (32) __

whoever plays next loses. Suppose it's Black's turn and he or she temporizes by putting a disk at e6. Where must Red then play to win?

I hope you saw that Red must make the temporizing move at C1, forcing Black to play a3. If Red instead plays A3, then Black plays a4 and wins with the adjacent threat pair at C2-C3.

If it were Red's turn to play in Diagram 36, Red would be the one forced to play A3; Black would play a4 and win with the adjacent threat pair. If you've played much CONNECT 4, you already realize that zugzwang is a very common occurrence (although you may never have heard this unusual German word). In cases such as Diagram 35, where one player is in zugzwang but his or her moves do not quickly lead to a loss, I often say that the other player has *zugzwang control.*

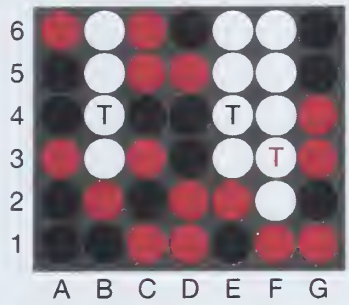
Even-Row Threats Are Not Influential

In Diagram 37, Red wins with a major odd-row threat, even though Black has two columns with major even-row threats.

Black can play f2 and lose at once, or play b3 (or e3) and sacrifice one of his or her even-row threats. Sacrificing an even-row threat will do Black no good; after b3 B4 b5 B6, it will be Black's turn to play again, and he or she will be in the same dilemma. In general, when all the safe cells have been filled in a column

with an even-row threat, there will be an even number of unfilled cells remaining in that column; those cells will be divided evenly between Red and Black. Thus columns with even-row threats have no effect on zugzwang control.

DIAGRAM 37
Black to play (will lose)



Moves: D1! e1 D2? d3! C1? c2? C3? c4? E2? d4 G1 g2 D5 b1 G3 a1 C5 a2?? G4 d6 A3 a4 B2 a5 A6 g5 C6 g6 F1! (30) __

Relative Values of Threat Types

A strategic threat can be called a *horizontal threat* or a *diagonal threat*, depending on which kind of four-in-row is threatened. (Vertical threats can be key to *tactical* sequences but only rarely fit the definition of *strategic* threat.)

All other things being equal, in order of preference Red will seek odd-row horizontal threats as most valuable, followed by diagonal threats, with even-row horizontal threats the least valuable. For Black the different threat types have roughly equal value. Both players also must seek to block their opponent's useful threats.

All other things being equal, threats are better low on the board. A major threat on the second or third row often leaves any threats above it in the same column with little or no value, while a threat on the sixth row undercuts nothing. (But be aware that the low threat usually does *not* neutralize any threats of the same row parity.)

All other things being equal, it is better to occupy cells in a central column. For example, in terms of diagonal threats, A3 participates only along A3-B4-C5-D6, so it relates diagonally to only three other cells. C3 participates along A1-B2-C3-D4-E5-F6 and E1-D2-C3-B4-A5, so it relates diagonally to nine other cells.

Some Diagonals Are Better Than Others

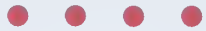
Cells on diagonals that point up toward the center of the board are better points to occupy than cells on diagonals that point up toward an edge. For example, A1-B2-C3-D4 is much more likely to develop into an important diagonal threat than D1-C2-B3-A4, and A3-B4-C5-D6 is usually better than D3-E4-F5-G6. (Similarly, B2-C3-D4-E5 may be *slightly* more likely, a priori, to develop into a good threat than C2-D3-E4-F5, and so on.) The reason for this is that, as the game develops, it is often easy to play (or threaten to play) near the bottom or center of the board. If building your threat requires instead that you capture a cell near an upper corner (e.g., A5 or B6), your opponent will often be able to direct early play elsewhere.

This general rule about "good" versus "bad" diagonals explains why corner cells such as A1, B1, and B2 are often good plays early in the game. Of course, you hope to follow up A1 or B2 with a disk at D4, but if your opponent gets D4 instead, your disk at A1 or B2 isn't wasted; at least the opponent can't use it for his or her own diagonal threat.

The remarks in this section are just tendencies and shouldn't be overemphasized. Many games are won because of a horizontal threat on the sixth row, a diagonal threat involving A5, or some other threat of "poor" a priori quality. Each case needs to be judged on its own merits.

Chapter 4

Major Threat Analysis



In this chapter I show you how to calculate victory in the very simplest case: when a game's outcome depends only on major threats. The rules in this chapter cover only straightforward major threats; I will analyze more complicated situations in Chapters 5-8.

The reader may ask: Why bother learning whether I or my opponent has a certain victory? By the time the position is stable enough to apply the rules, it will be too late! Yes, you do need to look ahead in CONNECT 4, but you'll need the strategic rules to know what to look ahead for!

As a CONNECT 4 game develops, and emerging threats come into focus, you will need to know which threats will lead to victory and which are useless. You will need to understand the effects of two threats in the same column: Does your threat neutralize the enemy's threat above it? Do two same-colored threats in the same column enhance each other? In this chapter I outline the answers to these questions and show how a game's major threats predict its outcome.

Summary

When there are two major threats in the same column, whether of the same color or different colors, four cases can be distinguished:

- Adjacent rows: Wins if same color; upper threat neutralized if different color.
- Second and fifth rows: Very strong if same color; upper threat neutralized if different color.
- Same row parity (both threats on even rows, or both threats on odd rows): Threats have no effect on each other.
- Third and sixth rows: Threats have no effect on each other.

After columns with multiple threats are resolved, the game outcome depends on the major odd-row threats. The result can often be inferred from the mantra "Red (First Player) likes odd numbers." Only when the odd-row threats imply a draw do Black's even-row threats become relevant. Red's (isolated) even-row threats are never relevant.

In the remainder of this chapter, I explain these general principles and present some transformation and victory rules. The transformation rules explain which threats can be ignored and which can be simplified; after I've simplified the threats using these transformations, the victory rules explain who will win the game.

The rules in this chapter are *not* complete, because they do not cover the following points:

- Complicated configurations of minor threats
- Columns with three or more major threats
- Tactical interactions between columns

Nevertheless, the principles will guide you to understand which potential threats have value and which don't, and if you understand these "simple" cases, you should be able to figure out more complicated cases as well.

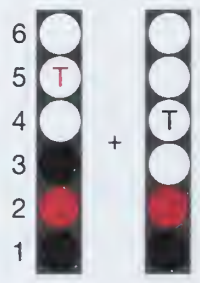
General Principles

G1: In the final column, Black gets the even-row cells, and Red gets the odd-row cells.

If the board ends up completely filled in, Black will place the final disk (disk 42) on a sixth-row cell. Working backward from there, Principle G1 becomes obvious. This principle implies that Red should seek odd-row threats, while Black should seek even-row threats.

G2: Red's odd-row threat has priority over a Black even-row threat in a separate column.

In the diagram, if Black moves in the first column, Red wins at once. It does no good for Black to play in the second column, because that column has an even number of vacant cells, and Black will be in zugzwang anyway after those vacant cells are taken. However, if it is Red's move in the diagram, he or she can play in the first column, forfeiting his or her own threat, but there are an *odd* number of vacant cells in that column and thus it will be Black's turn to move (and forfeit his or her own threat) when the second column is finally played.



In other words, even-row threats have no effect on zugzwang control, but odd-row threats lead to a reversal of zugzwang control. I refer to columns with an odd-row threat as *odd-threated columns.*

G3: Red (First Player) likes odd numbers.

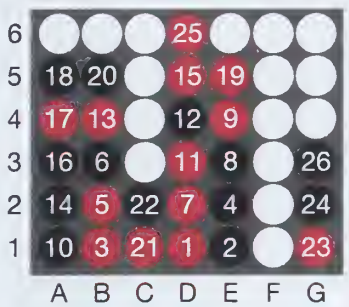
With no odd-threated columns, Black wins if he or she has an even-row threat, because then Red will be in zugzwang in the endgame. With one odd-threated column, Red wins if the odd-row threat is Red's; Red has zugzwang control. With a second odd-threated column belonging to Black, zugzwang control reverses back to Black, and so on. So Red (who can win only in an *odd* row) wants there to be an *odd* number of odd-threated columns.

Recall this principle by memorizing the mantra: "Red (First Player) likes odd numbers."

The simplest application of this principle is that if Red has one odd-row threat, he or she wants to stop Black from getting an odd-row threat in a separate column. Diagram 38 shows a fairly routine opening that has been played "perfectly"—no mistakes have been made yet. Where must Red play next to keep the game "perfect"?

Red must play 27 B6 so Black does not get an odd-row threat of his or her own at C5.

DIAGRAM 38
Red to play, with only one move to win



Moves: D1! e1 B1 e2 B2 b3 D2 e3 E4! a1 D3 d4 B4 a2 D5 a3 A4! a5 E5 b5 C1 c2 G1! g2 D6 g3 (27) __

G4: Adjacent major threats win.

If Black has major threats at two adjacent cells, such as b4 and b5, he or she can simply play up their column and win, without waiting for the endgame.

Sometimes one player will have adjacent threats on one side of the board while the other player has adjacent threats on the other side. This leads to a race; whoever plays up to his or her threats first wins.

G5: Second-row and fifth-row threats work together.

When the two threats at these two points are of two different colors, the second-row threat has maximum value; the fifth-row threat is undercut and so is worthless.

When the two threats are both one player's, he or she can force the opposing player into zugzwang, later forfeit the second-row threat, and use the fifth-row threat to force the opposing player into zugzwang *a second time.* Perhaps a simple way to think of this is to memorize that *a same-colored second- and fifth-row threat pair is equivalent to two odd-threated columns.*

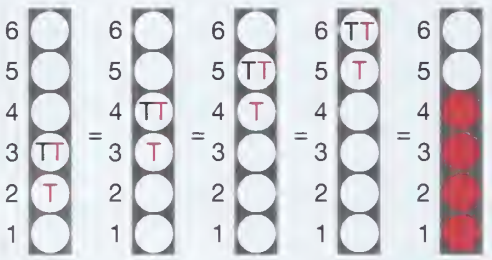
G6: When a column has major threats of both players, with the same row parity, the game will not end in a draw.

Obviously, there cannot be a draw if *both* players have a major threat at, say, F3. If one player has a threat at F3 and the other a threat at F5, a draw is theoretically possible, but only if one player makes an illogical blunder.

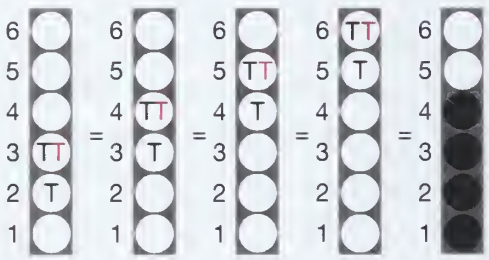
Transformation Rules

To determine the outcome of a game with only major threats, first mentally replace any complicated column with an equivalent but simpler column (or column pair). This is done with the following transformation rules.

T1: Adjacent major threats win.

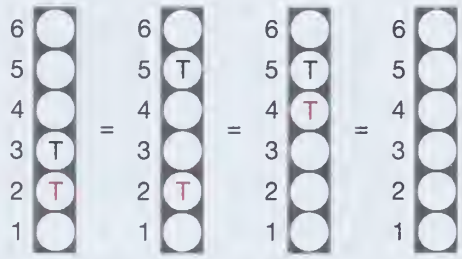


Adjacent Red major threats win.

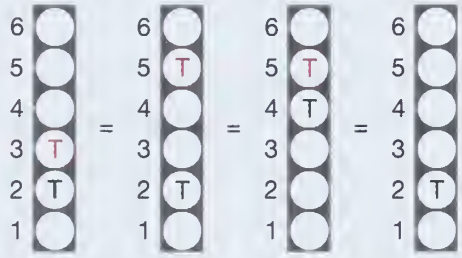


Adjacent Black major threats win.

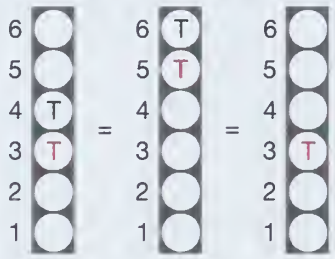
T2: Threats may neutralize (undercut) opponent's threats with opposite row parity.



A Red even-row threat neutralizes any Black odd-row threat *above* it.

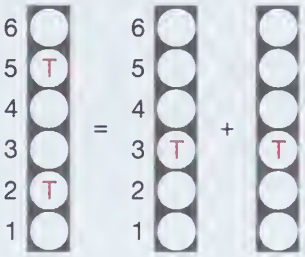


A Black even-row threat neutralizes any Red odd-row threat *above* it.

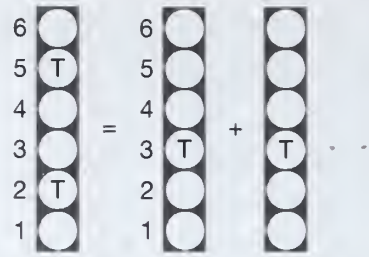


A Red odd-row threat neutralizes a Black even-row threat *immediately* above it.

T3: Second-row and fifth-row threats of the same player in the same column will exert zugzwang control *twice*.



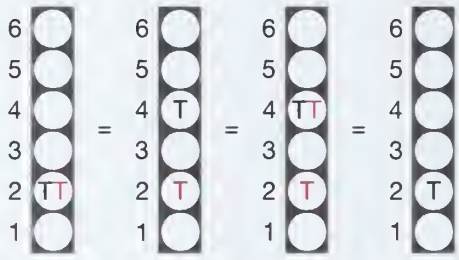
A second-row threat and a fifth-row threat in the same column are equivalent to two odd-row threats in separate columns.



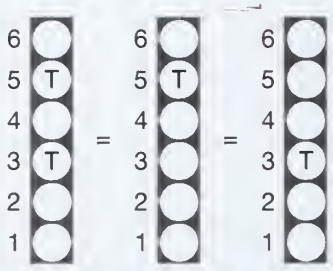
A second-row threat and a fifth-row threat in the same column are equivalent to two odd-row threats in separate columns.

(Most of the other rules are simple, but this peculiar rule, where a second-row threat can sometimes be considered equivalent to an odd-row threat, may seem surprising. This important rule also applies to minor threats.)

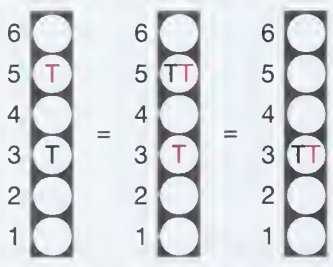
T4: Threats with the same row parity have no influence on each other.



A Red even-row threat has no effect on Black even-row threats.

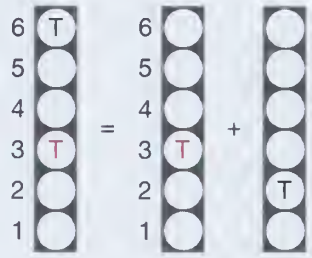


There's no bonus for two odd-row threats in the same column.

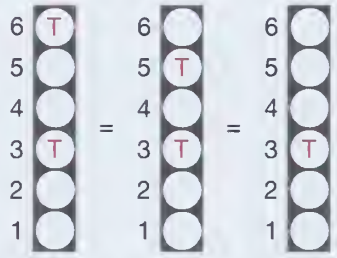


When both players have odd-row threats in the same column, the rows of the threats don't matter.

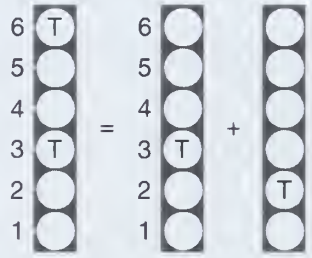
T5: Third-row and sixth-row threats do not influence each other.



A Red third-row threat has no effect on a Black sixth-row threat.

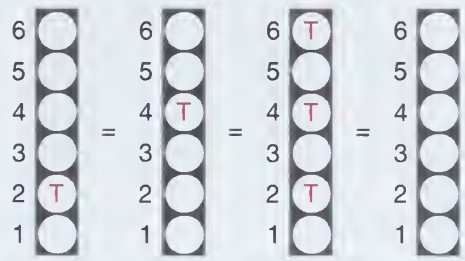


Red's sixth-row threat is always worthless (unless it is part of an adjacent pair). There's no bonus for two odd-row threats in the same column.



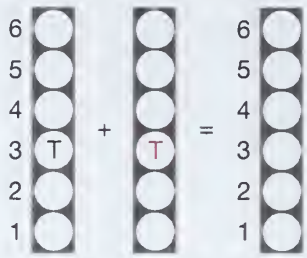
Black's sixth-row threat is independent of the third-row threat in the same column.

T6: Red's isolated even-row threats have no value.



Except as noted in other rules, Red even-row threats are worthless.

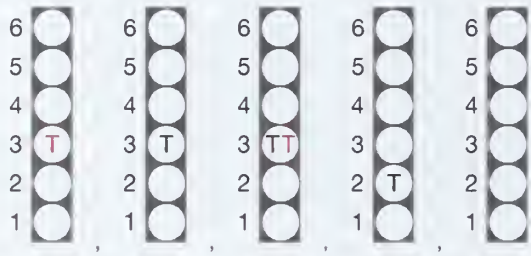
T7: Red and Black odd-row threats *in separate columns* cancel each other.



Red and Black odd-row threats *in separate columns* cancel each other.

Victory Rules

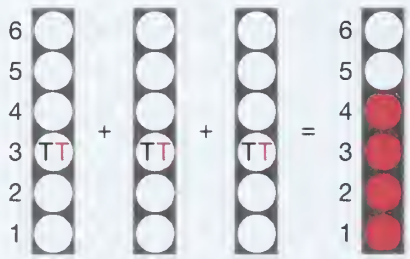
After applying the transformation rules to all the major threats in a position, you will have a simplified system with only five types of column:



The five types of fundamental column: Odd-row threat of Red, Black, or both; Black even-row threat; no threat.

Next, scan the following victory rules *in order*. The *first* rule that applies will tell you the game's outcome.

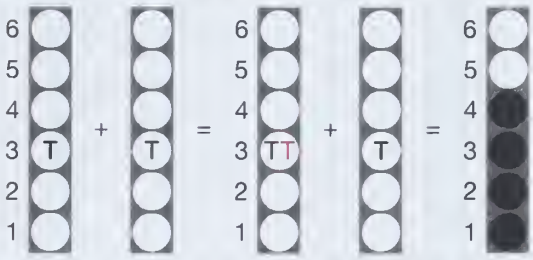
V1: Red wins with three major odd-row threats in separate columns.



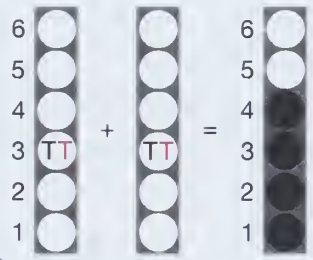
Red (First Player) likes odd numbers.

(With major odd-row threats in *four* separate columns, Black wins, but this situation will be rarer than a four-leaf clover.)

V2: Black wins with two odd-row threats in separate columns.

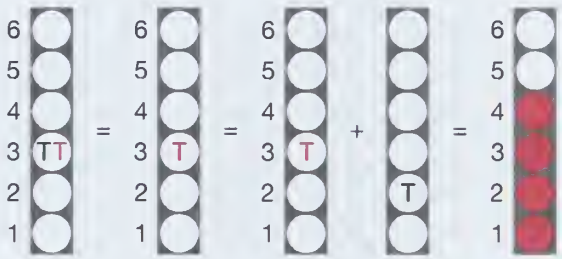


Black needs two odd-row threats (that weren't removed earlier by rule T7) to win.



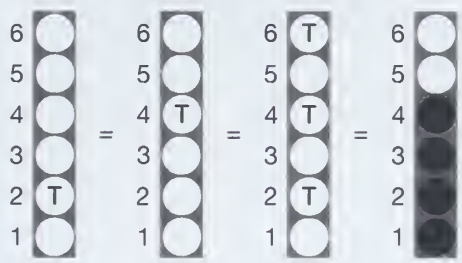
Black's two odd-row threats win despite Red's threats in the same columns. (Black likes even numbers.)

V3: Red wins with an odd-row threat.



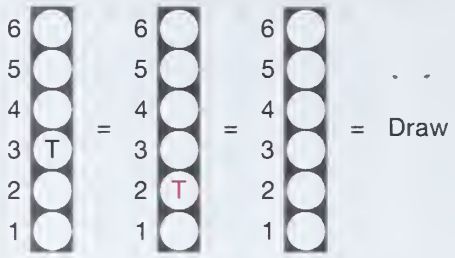
Red's odd-row threat wins.

V4: Black wins with an even-row threat.



Black's even-row threat wins *only* if neither player has a winning odd-row threat.

V5: Otherwise the game is drawn.

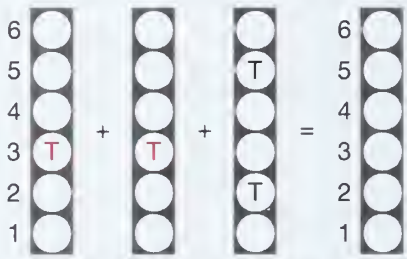


Without a winning threat, no one wins.

Columns With Two Threats

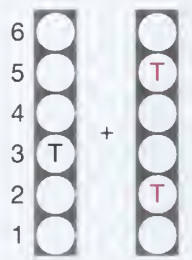
The transformation rules allow you to mentally convert any column with two major threats to a simpler form. The rules for adjacent threats (like third- and fourth-row threats) and same-parity threats (like third- and fifth-row threats) are pretty simple. The special cases (second- and fifth-row threats, or third- and sixth-row threats) are more interesting. I use some threat combinations involving these special cases to illustrate how to apply the rules.

In the following example of an endgame with four major threats in three columns, our rules correctly predict that the game will be drawn. (Zugzwang will occur four times as the ending is played out, and all four threats will be relinquished.)

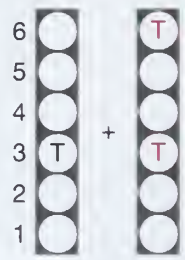


Apply T3, T7, T7, V5

The next two examples illustrate that a second- and fifth-row threat pair is always better for Red than a third- and sixth-row threat pair.

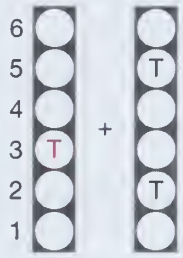


Apply T3, T7, V3: Red wins.

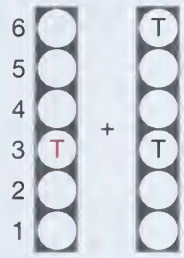


Apply T5, T7, V5: Drawn.

The next two examples illustrate that a third-/sixth-row threat pair is often better for Black than a second-/fifth-row threat pair.



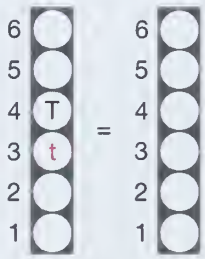
Apply T3, T7, V5: Drawn.



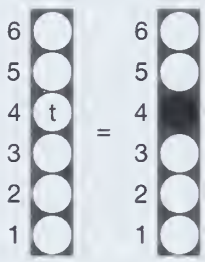
Apply T5, T7, V4: Black wins.

Minor Threats

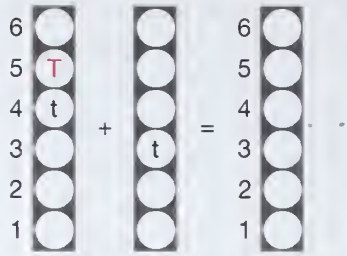
Using the same style of diagram as in the previous section, here are some of the simplest facts about minor threats (shown with a lowercase *t*). Chapter 7 discusses minor threats in more detail.



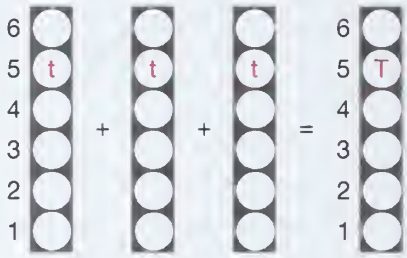
Major threats "overcut" minor threats.



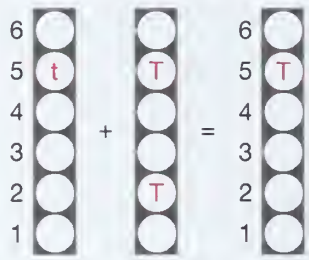
Black's isolated minor even-row threats will be fulfilled.



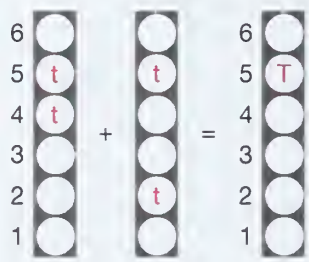
A mixed threat pair on a diagonal undercuts a major threat.



Three minor odd-row threats in separate columns are equivalent to one major odd-row threat.



An even-row threat can sometimes serve as a third threat.



Sometimes a "triple threat" is comprised of four threats.

Chapter 5

Undercutting Threats

If Red prepares a major odd-row threat on the right side, say, at F5, Black has two ways to respond. (1) Black can form his or her own odd-row threat on the left side, perhaps at a3 or a5, getting a draw—this is called *counterthreat.* (2) Black can undercut Red's F5 threat with his or her own major threat at f2 or f4. If Black is sure that Red won't get a second major odd-row threat, he or she will naturally prefer a winning undercutting threat to a counterthreat good enough only to draw. If Red does get two major odd-row threats, Black will usually seek a counterthreat *and* an undercutting threat.

If Red has a threat at C5, Black can *undercut* that threat by constructing a threat at c4. More generally, any threat is an *undercutting threat* if it lies in the same column and immediately underneath a threat of the opposing player. Unlike an ordinary Black threat at c4, whose existence would make Black reluctant to play c3, an *adjacent-row* undercutting threat can be played up to; Black would be happy to play c3 if c3 C4 c5 wins. A first-row check that destroys a second-row threat is also termed *undercutting.*

As a special case, a second-row threat undercuts an opponent's fifth-row threat, though it isn't on an adjacent row.

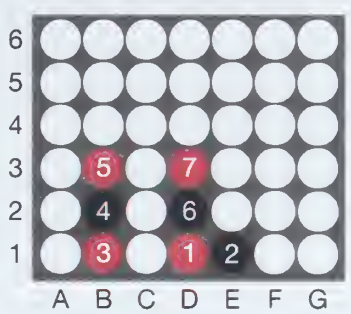
An undercutting threat usually renders the threat above it almost worthless. The undercutting threat need not be a major threat; indeed, interesting examples usually involve two minor threats along a diagonal. Sometimes one disk (three minor-threat fulfillments needed for four-in-row) is enough for a successful undercut.

The position in Diagram 39 will lead to some undercutting very early in the game. Where should Black play? (This can be phrased as a quick victory problem—Black to play, and mate in eight—where the moves for the quick victory are also the only moves that win at all!)

Black 8 e2 may appeal since a major second-row threat at c2 would undercut any future Red threat at C3 or C5. But Diagram

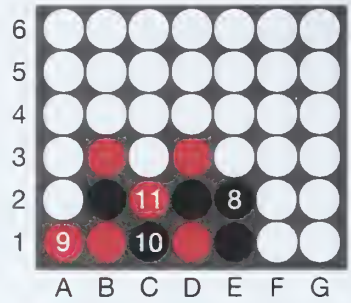
DIAGRAM 39

Black to play, with only one move to win



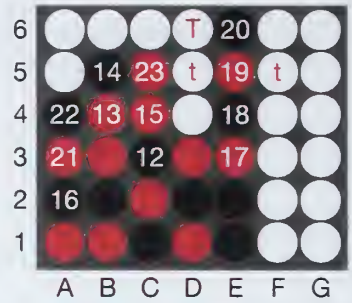
Moves: D1! e1 B1 b2 B3?? d2! D3 (8) __

DIAGRAM 40
Black to play (will lose)



Moves: D1! e1 B1 b2 B3?? d2! D3 (8) e2?? A1 c1 C2! (12) __

DIAGRAM 41
Black to play (will lose)

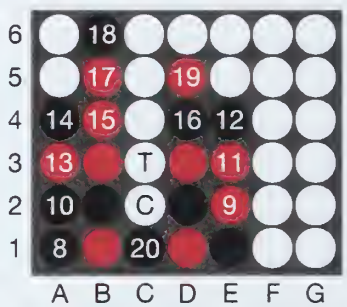


Moves: D1! e1 B1 b2 B3?? d2! D3 e2?? A1 c1 C2! (12) c3 B4! b5 C4 a2 E3 e4 E5 e6 A3 a4 C5! (24) __

40 shows that this doesn't work; Red can undercut the c2 threat with a check at C1. (Diagram 41 shows a possible victorious continuation for Red; obviously, Red couldn't achieve this without getting Black to open up column C.)

The winning play for Diagram 39 is shown in Diagram 42. Black's 8 a1 prevents Red's first-row check and leaves Black with *two* ways to build a major threat at c2. Red cannot prevent both ways, so Black gets a second-row undercutting threat and, as it turns out, victory early in the game. Black's victory is very straightforward but is a "mate in eight" instead of "mate in five" because Black had to spend three disks (12 e4, 14 a4, 18 b6) foiling Red's simple counterplays. (Red can't be allowed to build a major threat of his or her own at C2; it would undercut Black's win at c3.)

DIAGRAM 42
Red to play (will lose)



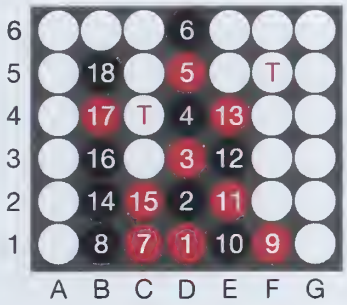
Moves: D1! e1 B1 b2 B3?? d2! D3 (8) a1! E2 a2! E3 e4! A3 a4! B4 d4! B5 b6! D5 c1 (21) __

Instead of a major threat, undercutting threats will often comprise a pair of minor threats along a diagonal. Moreover, Black may have such a diagonal (involving, say, f4) that undercuts a Red threat at F5, but with Red developing a diagonal threat (including F3) that in turn undercuts Black's diagonal. Such a *diagonal undercutting diagonal undercut* is a very common theme in CONNECT 4 games.

Diagram 43 shows a game that has been very well played on both sides. (Note Black's push-up play b3 B4 b5, which eliminates any possible left-side win for Red.) It's Red's turn, but Red has already established a winning major odd-row threat at F5. Can you figure out what counterplay Black intends and how to defend against it? We have not discussed this kind of situation yet, but try anyway.

DIAGRAM 43

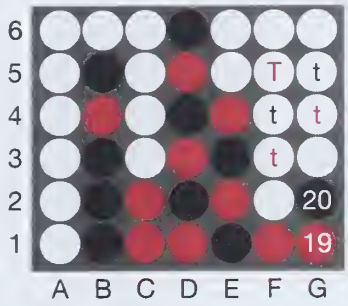
Red to play, with only one move to win



Moves: D1! d2 D3! d4 D5! d6 C1! b1 F1! e1 E2! e3 E4! b2 C2! b3 B4! b5 (19) __

DIAGRAM 44

Red to play and win



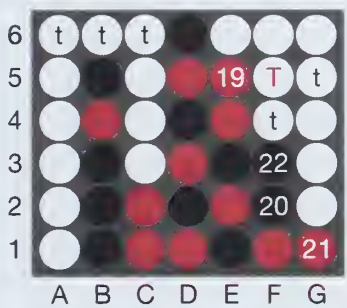
Moves: D1! d2 D3! d4 D5! d6 C1! b1 F1! e1 E2! e3 E4! b2 C2! b3 B4! b5 (19) G1! g2 (21) __

Red isn't worried about Black playing b6 to build a major threat at c5; Red has that undercut already with a threat at C4. Red's key move is 19 G1. In Diagram 44, Red has a winning major odd-row threat at F5. Black undercuts it with a diagonal threat, f4-g5, but to no avail since Red has the lower undercut at F3-G4.

Red's only winning move in Diagram 43 is 19 G1. If Red plays elsewhere, Black gets to capture f3, as shown in Diagram 45; Black's undercut at f4-g5 now works, and Red will have to remember to block at B6 just to salvage a draw. (Red can delay the B6 play somewhat; in fact, Red should next try 23 G2 to see whether Black is foolish enough to fall for the push-up play 23 G2 b6? G3! g4 G5!, with the F5 threat now a winner.)

DIAGRAM 45

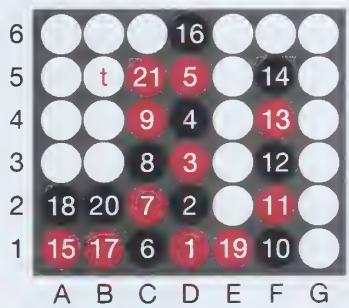
Red to play and draw



Moves: D1! d2 D3! d4 D5! d6 C1! b1 F1! e1 E2! e3 E4! b2 C2! b3 B4! b5 (19) E5? f2 G1! f3 (23) __

DIAGRAM 46

Black to play, with only one move to draw



Moves: D1! d2 D3! d4 D5! c1 C2 c3 C4! f1 F2 f3 F4 f5 A1 d6 B1! a2 E1! b2 C5? (22) __

Diagram 46 shows a very common situation involving an undercut. One way or another, Red is destined to get a major odd-row threat at B5 or A5. Where should Black play to circumvent this?

Black has the undercutting diagonal threats a5-b4 to neutralize Red's fifth-row threat. But Red in turn undercuts that play with diagonal threats A4-B3.

DIAGRAM 47

Red to play and draw



Moves: D1! d2 D3! d4 D5! c1 C2 c3 C4! f1 F2 f3 F4 f5 A1 d6 B1! a2 E1! b2 C5? (22) b3! F6 g1 E2 g2 G3 g4 G5 g6 C6 a3! E3 e4! E5! e6! (37) __

Black neutralizes that play by grabbing b3 immediately. Play might continue as shown in Diagram 47, which leads to the classic undercut endgame and therefore a draw. (Two columns have odd-row threats; Red likes odd numbers to win, but two isn't an odd number.) Red could have played 23 B4 b5 F6 instead of just 23 F6, but F6 is essential for Red just to salvage a draw; if Black gets an even-row threat anywhere, Black wins.

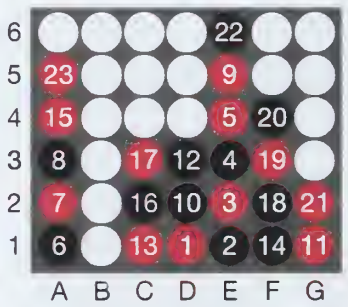
A Black major third-row threat normally doesn't neutralize a Red major fifth-row threat in the same column, but it will still be effective in some cases (see Diagram 210 in Chapter 12).

Where must Black play in Diagram 48? Black is almost in zugzwang, and a6 is the only safe cell. Play continues, and in Diagram 49 Red's major odd-row threat clearly is undercut.

In most of these undercutting examples, Black has two disks along a diagonal, yielding two minor threats (one odd-row threat, one even-row threat) that undercut a Red major odd-row threat. Following, I give a final example of undercutting, in which Black has only one disk along his or her undercutting diagonal, so there are *three* minor threats that have to be fulfilled for the four-in-row, yet Black's undercut does work. The author is indebted to Alan Pruce, who provided this interesting and instructive game.

DIAGRAM 48

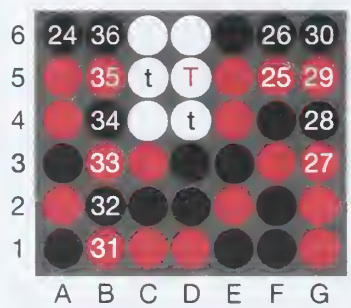
Black to play, with only one move to draw



Moves: D1! e1 E2? e3! E4 a1! A2 a3 E5 d2 G1 d3! C1! f1 A4 c2 C3! f2! F3 f4! G2 e6 A5 (24) __

DIAGRAM 49

Red to play and draw



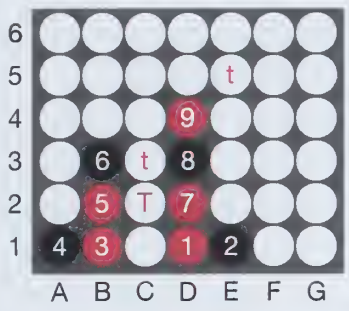
Moves: D1! e1 E2? e3! E4 a1! A2 a3 E5 d2 G1 d3! C1! f1 A4 c2 C3! f2! F3 f4! G2 e6 A5 (24) a6! F5 f6! G3 g4! G5! g6! B1 b2! B3 b4 B5 b6! (37) __

Diagram 50 is a well-played (if perhaps slightly unusual) opening in which Red may play 11 A2 or E2 next and form the indicated triple threat.

If Black plays 10 a2, Red is likely to answer with E2, and vice versa. If Black plays

DIAGRAM 50

Black to play (will lose)



Moves: D1! e1 B1 a1 B2! b3 D2 d3 D4 (10) __

10 d5 Red can win with either 11 A2 or E2 (or with 11 B4). But if Black plays 10 b4, even an expert Red may answer incorrectly, as shown in Diagram 51.

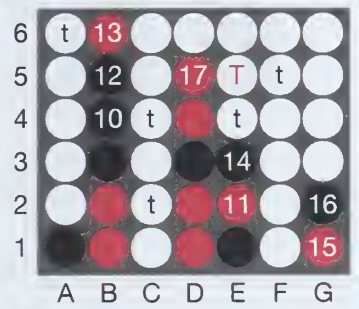
Red still has a triple threat, which we've indicated at E5 (Red can march up column C, capture C3, and promote E5 to a major threat whenever he or she wishes); however, we've focused attention on Black's minor threats. Black's threats along the diagonal c2-f5 successfully undercut Red's E5 threat, *even though Black as yet has only one of the four disks needed for the four-in-row!* (Black's even-row threats along the a6-d3 diagonal are relevant because they give Black a win instead of just a draw.)

Although Black's major threat at c3 is irrelevant in this game, Black's possession of the cell e3 is *essential*; if Red had this cell, Red's diagonal C1-F4 would undercut Black's diagonal c2-f5.

Since Black's odd-row threat at f5 is key to his or her undercut, Black must be alert for any Red attempt to undercut it with an even-row threat in column F. This is why Black played 16 g2. Do you see the more subtle Red threat possibility? Where should Black play next in Diagram 51?

DIAGRAM 51

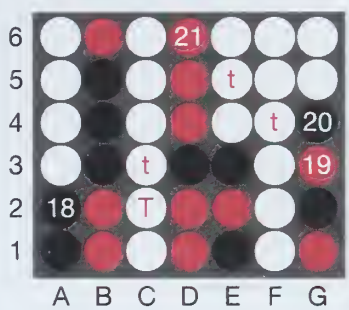
Black to play, with only one move to win



Moves: D1! e1 B1 a1 B2! b3 D2 d3 D4 (10) b4 E2?? b5! B6 e3! G1 g2! D5 (18) __

DIAGRAM 52

Black to play (will lose)



Moves: D1! e1 B1 a1 B2! b3 D2 d3 D4 b4 E2?? b5! B6 e3! G1 g2! D5 (18) a2?? G3 g4 D6! (22) __

Black will win in this diagram if he or she is careful, but the reader may still wonder what was special about Black's 10 b4. The answer is that when Black plays well (and this Black is playing very well), Red will often need to develop chances on both sides of the board, but with b3-b4-b5 all in Black's hands (and Red wasting a turn to answer b5 at B6), Red has no chance for any winning threat on the left side.

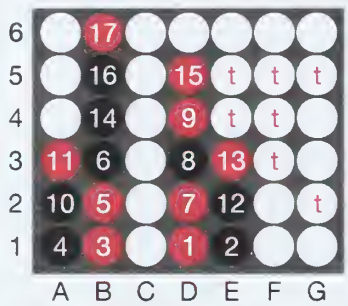
The correct move for Black in Diagram 51 is 18 d6 because if Red captures that cell,

Red will have threats along the D6-G3 diagonal, as shown in Diagram 52. Red's minor threat pair E5-F4 cannot win by itself of course (F4 is an even-row cell), but it does serve to undercut Black's minor threat at f5. Red now wins easily, for example, by playing up column F and capturing F5 if Black takes f4.

By the way, in Diagram 50 Red need not actually form a triple threat immediately. If play goes as shown in Diagram 53, Red gets a winning plethora of minor threats on the right side similar to the *sextuple threat* defined in Chapter 8.

DIAGRAM 53

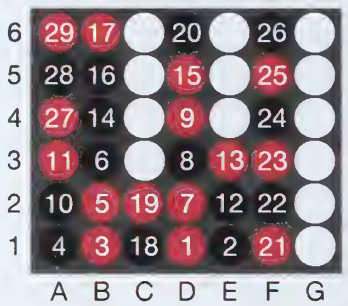
Black to play (will lose)



Moves: D1! e1 B1 a1 B2! b3 D2 d3 D4
a2 A3 e2 E3! b4 D5! b5 B6! (18) __

DIAGRAM 54

Black to play (will lose)



Moves: D1! e1 B1 a1 B2! b3 D2 d3 D4
a2 A3 e2 E3! b4 D5! b5 B6! c1 C2 d6
F1 f2 F3 f4 F5! f6 A4! a5 A6! (30) __

Diagram 54 shows a possible continuation from Diagram 53. Black is now in zugzwang and will soon be forced to let Red play E5 to form a major odd-row threat at G5. (Red's existing C5 threat is undercut by Black's c4 threat.)

Chapter 6

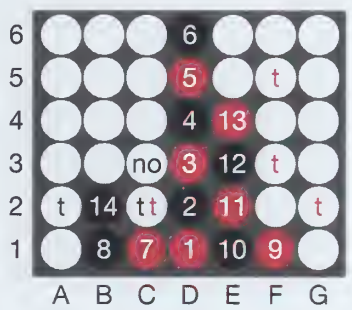
Counterthreats

When Red acquires a major odd-row threat on one side of the board that Black is unable to undercut, Black must try to build his or her own threat on the other side of the board. If Black builds no such threats, or only even-row threats, Black loses. If Black builds one or two odd-row threats in columns separate from Red's threat, Black draws. If Black builds two odd-row threats, including one in the same column as Red's threat, Black wins. Black also wins with odd-row threats in any three columns, but this is rarely achieved. The most common way for Black to win when Red has a major odd-row threat is for Black to construct an odd-row threat *and* an even-row threat.

A Black odd-row threat that neutralizes a Red odd-row threat is called a *counterthreat*. Black tries to form a counterthreat in many of our example games. For the example in this chapter, let's look at an exciting game where Black's even- and odd-row threats are adjacent, and it might appear he or she cannot get both and will have to choose one or the other. But Black plays carefully and gets the cake (his or her odd-row threat) and eats it too (wins with an even-row threat). This game teaches an important lesson: Sometimes a player needs to delay the completion of a threat.

DIAGRAM 55

Red to play, with only one move to win



Moves: D1! d2 D3! d4 D5! d6 C1! b1 F1! e1 E2! e3 E4! b2 (15) __

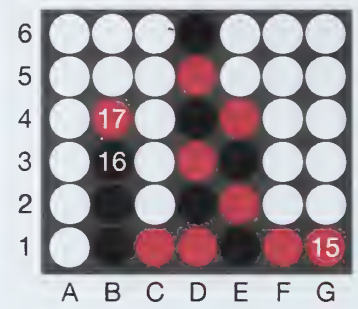
Our illustrative game arises in the standard opening shown in Diagram 55. The board abounds with relevant potential threats, only a few of which are marked. Where should Red play next? If Red fails to find his or her key play, where should Black play? Try to answer both questions before reading ahead.

What may make this problem seem hard is that Red and Black each have multiple needs that must be prioritized. Red is likely to establish F5 as a major threat, but Black's d2-g5 diagonal may undercut that threat. Red's d1-g4 diagonal may in turn undercut *that*, but Black hopes to play f2-G1-f3 and destroy any Red threat at F3. For this reason, Red might be tempted to play 15 G1, with the assumption that Black will answer 16 g2 (since if Red plays

there he or she gets a major odd-row threat at F3). Red would then continue play, feeling that his or her situation has improved on the right side of the board.

But in Diagram 55, 15 G1 would be a mistake. Playing Black in Diagram 56, can you make Red pay for this blunder? Be careful; there is only one winning move.

DIAGRAM 56
Black to play, with only one move to win



Moves: D1! d2 D3! d4 D5! d6 C1! b1 F1! e1 E2! e3 E4! b2 (15) G1?? b3 B4 (18) __

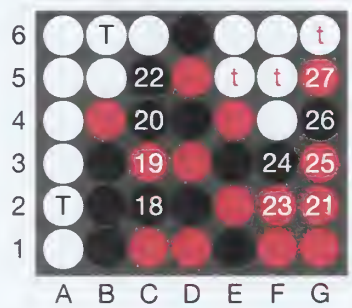
The four most common choices for Black's next play are as follows:

- Play 18 c2 to achieve an even-row threat at a2, and prevent Red from playing C2 to build an F5 threat.
- Play 18 g2 to prevent Red from playing there to build an F3 threat.
- Play 18 e5 to prevent Red from getting this key cell and to build an odd-row threat at c3.
- Play 18 a1, building an odd-row threat at c3 and gaining a turn since Red must answer at A2.

I will consider these moves one by one. If Black plays c2, the continuation might be as shown in Diagram 57.

Black has two even-row threats on the left side of the board, but they're useless. Whether Black plays now at E5 or G6, Red will take the other cell and build a winning major threat at F5. Black needs an odd-row threat to neutralize this, but Red destroyed Black's potential odd-row threat with C3 as soon as Black played c2.

DIAGRAM 57
Black to play (will lose)



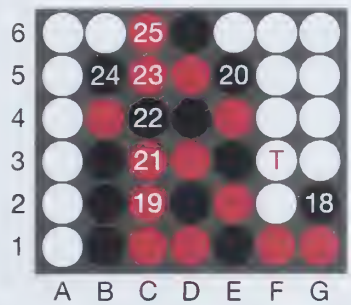
Moves: D1! d2 D3! d4 D5! d6 C1! b1 F1! e1 E2! e3 E4! b2 G1?? b3 B4 (18) c2?? C3! c4 G2 c5 F2! f3 G3! g4 G5! (28) __

Returning to the problem in Diagram 56, Black might stop the threat at F3 by answering G1 with g2. But now Red will wake up and play C2 as he or she should have in the first place! The continuation might be as shown in Diagram 58.

Although Black's g2 play makes it harder for Red to establish a threat at F3, it has no value in countering Red's strong left-side attack. While it is usually right to prevent Red from establishing an easy major odd-row threat, this game abounds with other threats, and Black needs to look for a move that has more than one purpose. (Black's g2 move doesn't even completely

DIAGRAM 58

Black to play (will lose)



Moves: D1! d2 D3! d4 D5! d6 C1! b1 F1! e1 E2! e3 E4! b2 G1?? b3 B4 (18) g2?? C2! e5 C3! c4 C5! b5 C6 (26) __

destroy Red's F3 threat, as shown in Diagram 58.)

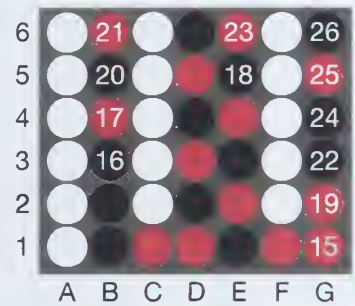
Black's winning move in Diagram 56 is 18 e5. Black lets Red get a major odd-row threat at F3, and then he or she defeats Red with an odd-row counterthreat at c3 and even-row minor threats at c2 and a2. Diagram 59 shows the result. (I've chosen some slightly weak moves for Black just to demonstrate how easy Black's victory is once he or she plays 18 e5.) Red is in zugzwang in Diagram 59 and will play up column F, after which Black plays c2 and wins the ending at a2.

A key tactic to note here is that although Black's victory depends on his or her second-row threats at a2-b2-c2-d2, Black must not play c2 until the ending. Black also needs a threat at c3; playing c2 would destroy that.

Finally, 18 a1 is Black's second-best move in Diagram 56; it leads to a draw, while all other moves but e5 lead to a loss. But Black's second-row threat is destroyed after a1 A2, and why settle for a draw when you can get a win?

DIAGRAM 59

Red to play (will lose)

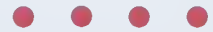


Moves: D1! d2 D3! d4 D5! d6 C1! b1 F1! e1 E2! e3 E4! b2 (15) G1?? b3 B4 e5! G2 b5 B6 g3 E6 g4! G5 g6! (27) __

Return to Diagram 55 so you can see that C2 is the key point; if Red plays 15 C2, he or she gets a major threat at F5 and eventual victory. If Red plays elsewhere, Black will play e5 or a1, after which Red will no longer be able to take C2. (A variation of this opening in which both players play well is shown in Diagram 43.) Red might be reluctant to play C2 because of Black's presence along the a1-e5 diagonal, but Red needs to overcome that fear! Black can't play c3 (unless it wins at once), and Red should be happy to see a continuation like C3 c4 C5, with Red getting fifth-row threats.

Chapter 7

Minor Threats



So far I have dealt mostly with major threats (cells whose capture would complete a four-in-row), but minor threats (cells whose capture leads to a major threat) are also important. And of course, every major threat starts life as a minor threat.

I can't provide detailed transformation rules for minor threats as I did for major threats; there are too many cases, and sometimes six or more minor threats interact to create a win. Instead I explore minor threats by looking at examples. There are a few basic patterns you will see over and over.

One can always hope that a minor threat will become a major threat, perhaps due to one's opponent's error, but I usually do not speak of minor threats unless they already have strategic significance. I divide such strategic threats into six categories:

- Simple Black even-row threat.
- Mixed diagonal undercutting threat. A mixed threat pair (even row and odd row) that has no offensive value may nonetheless undercut an odd-row threat above it.
- Triple threat. Three (or four) threats sometimes work together to have the same strategic value as one major odd-row threat. Such configurations are so common and important that I give them the name *triple threat.*
- Winning quadruple threat. This is a specific but common configuration of four threats that win.
- Sextuple threat. This is a common configuration of six threats that are equivalent to a triple threat.
- Plethora of threats. Sometimes five or more threats work together to create or counter a winning threat. Such situations may arise when one player has a preponderance of disks in the upper center of the board.

I have already discussed Black minor even-row threats. I mention them here just for completeness. These threats are easy to understand, but they are important since they lead directly to Black victory whenever Red cannot form a winning

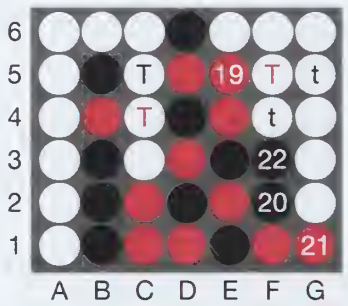
odd-row threat. I will now clarify the other five categories of strategic minor threat with examples.

Mixed Diagonal Undercutting Threat

In Diagram 60, Black gets a draw despite the Red major threat at F5, because Red's threat is undercut with Black's mixed diagonal threats f4-g5. Black is happy to play up column F and let Red play F4 whenever he or she wants—Black isn't trying to win with this diagonal, just kill Red's threat at F5—so it is Black who has zugzwang control (columns F and G comprise *two* odd-threated diagonals). This position arises from a continuation of Diagram 43, but with Red making a mistake; Red should have maintained his or her own mixed diagonal threats at F3-G4 to undercut Black's diagonal.

On the opposite side of the board in Diagram 60, Red is undercutting Black's c5 threat with a major threat at C4, but the topic here is undercutting with *minor* threats. In Diagram 60, by the way, Red should next try 23 G2 (hoping Black overlooks the winning push-up play G3 g4 G5) and remember to play B6 so that Black can't form a horizontal sixth-row threat. (No need to worry about Black playing e6; Red's fifth-row threat isn't good enough to win but is good enough to undercut any Black threat at f6.)

DIAGRAM 60
Red to play and draw

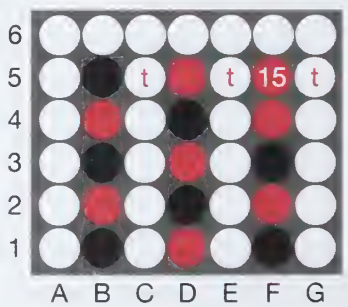


Moves: D1! d2 D3! d4 D5! d6 C1! b1 F1! e1 E2! e3 E4! b2 C2! b3 B4! b5 (19) E5? f2 G1! f3 (23) __

Triple Threat

In Diagram 61, Red has formed minor odd-row threats in three separate columns. Although Red has only minor threats, Red has zugzwang control (due to the presence of an odd number of odd-threated columns) and will win. Black will eventually be in zugzwang, but instead of then being forced to give Red an immediate four-in-row, Black will be forced to give Red a winning major threat. (Later Black will be in zugzwang again.) The threat configuration in Diagram 61 is called a *triple minor odd-row threat.*

DIAGRAM 61
Black to play (will lose)

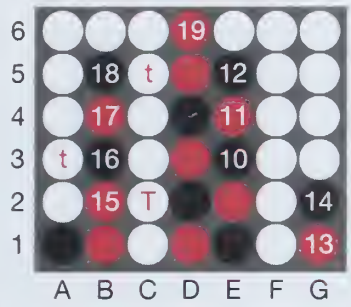


Moves: D1! d2 D3! d4 D5! b1 B2! b3 B4! f1 F2! f3 F4! b5 (15) F5! (16) __

In Diagram 62, Red has three threats on the left side, and again the three threats

DIAGRAM 62

- Black to play (will lose)



Moves: D1! d2 D3! d4 D5! e1 B1 a1 E2 (10) e3 E4! e5 G1 g2 B2 b3 B4 b5 D6 (20) __

will lead to Red victory. This time only two of Red's threats are odd-row threats, so this three-threat configuration is called a *triple mixed threat.*

In Diagram 62, there are only two odd-threated columns, but it is still Red who has zugzwang control. This is because of Red's even-row threat; Black can't play c1 so will have to play a2 eventually. The simplest way to remember this is to recall transformation rule T3; column C, with threats on both the second and fifth rows, should be counted as *two* odd-threated columns.

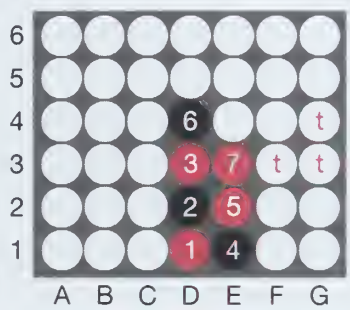
Instead of the second-row/fifth-row configuration, the even-row threat in a triple mixed threat can simply be in a row *adjacent* to one of the odd-row threats.

Diagram 63 shows a variation of the triple mixed threat. It is an effective triple threat because even though the even-row threat (G4) is just a minor threat, the minor threat (F3) that would convert G4 to a major threat is itself one of the very cells that form an odd-row horizontal threat. The configuration shown in Diagram 63 is called the *J shape* because the four red disks form a shape like the letter *J*.

When play goes up a column, the players take alternating cells (if there are no other tactical issues), so sooner or later Red must get either G3 or G4 in Diagram 63. In either case, F3 becomes a major threat, so it is easy to see that this triple threat is equivalent to a major odd-row threat.

DIAGRAM 63

Black to play (will lose)



Moves: D1! d2 D3! e1 E2 d4 E3 (8) __

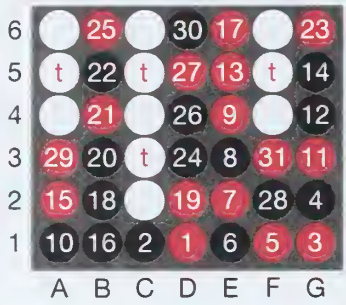
Black played poorly in this diagram. As a general rule, Black shouldn't let Red build horizontal third-row threats so easily, but there are exceptions.

Both the triple minor odd-row threat and the triple mixed threat are equivalent to single major odd-row threats. I use the term *triple threat* as shorthand to refer to either of these configurations.

Diagram 64 shows a triple threat comprised by *four* minor odd-row threats. It's still called a triple threat because there are three odd-threated columns, and it functions exactly as an ordinary triple minor odd-row threat does. (When Red has two odd-row threats in the same column, as shown by C3 and C5 here, the threats are linked and usually behave like a single threat; if Red occupies C3, Red

DIAGRAM 64

Black to play (will lose)



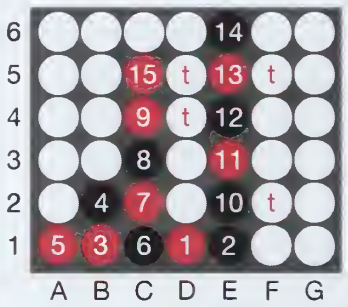
Moves: D1! c1 G1 g2 F1 e1 E2 e3 E4 a1 G3 g4 E5 g5 A2 b1 E6 b2 D2 b3 B4! b5 G6 d3 B6 d4 D5 f2 A3 d6 F3! (32) __

will usually end up occupying C5 as well. I am grateful to Johan "Eliten" Nordlund, CONNECT 4 champion of Sweden, who provided Diagram 64.)

In Diagram 65, Red has two minor even-row threats, each in one of Red's odd-threated columns, and has a winning endgame. In the eventual zugzwang, Red will get to play either D5 (forming a major odd-row threat) or F2 (forming an ordinary triple threat). Thus, the configuration of threats in Diagram 65 works just the same as a triple threat, and I call it a triple threat even though there are actually four threat cells. (I do *not* call it *quadruple threat*—that term is reserved for the very special configuration I examine in the next section.)

DIAGRAM 65

Black to play (will lose)



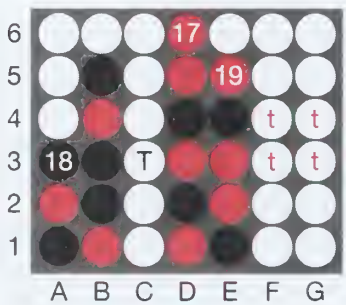
Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 E3?? e4 E5 e6?? C5 (16) __

Winning Quadruple Threats

Diagram 66 (which is repeated as Diagram 110 in Chapter 8) shows a winning position for Red. Either F3-G3-G4 or F4-F3-G3 would constitute a triple threat, but a triple threat isn't good enough; Black has built a counterthreat at c3. The *four* threats working together give victory, however. When Red eventually captures F3 or G3 due to zugzwang, Red doesn't get just a major odd-row threat; Red gets an adjacent major threat pair.

DIAGRAM 66

Black to play (will lose)



Moves: D1! d2 D3! d4 D5! e1 B1 a1 E2 b2 E3 e4 A2! b3 B4! b5 (17) D6! a3 E5! (20) __

In Diagram 66, if Red gets F3, he or she also establishes a C3 threat. For this reason, Red wins even if G4 (but not both G4 and C3) is a "blue cell." (To handicap Red, sometimes a specific cell is designated as a *blue cell*, which Red cannot use to win; see Chapter 11.)

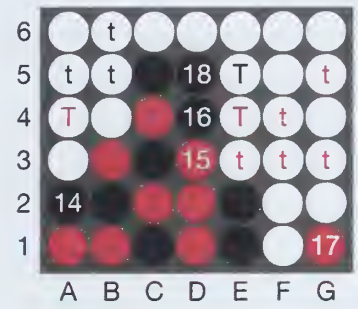
Black could get a draw against this threat configuration if Black had *two* major odd-row threats in columns separate from the quadruple threat. But that would be

exceedingly rare, so I will simply call the configuration a *winning quadruple threat.*

Sextuple Threat

In Diagram 67 (which is based on Diagram 74 in Chapter 8), Red has six threats on the right side, which are equivalent to a triple threat. When column F is played, Red will get F3 or F4; in the former case, E4-E3-G3 will form a triple threat; in the latter case, E4-E3-G5 will form a triple threat. This configuration is common enough to deserve its own name: *sextuple threat.*

DIAGRAM 67
Red to play and win



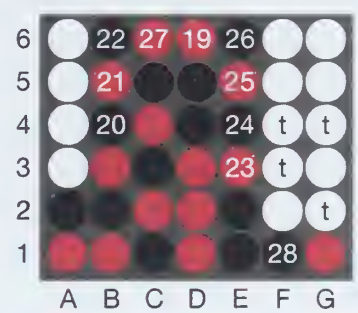
Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 B3? c5 D2 (14) a2? D3 d4 G1 d5 (19) __

Although Red's win will depend on eventually playing up column F, Red must be patient and let Black play f1 (as he or she will eventually, in zugzwang, when e3 or g2 would be more disastrous). (If play continues 19 F1 f2, Black will be able to create a double check with his or her disk 24—do you see how?) Red mustn't play E3 in Diagram 67 either; Black then gets a key minor threat at f3. Finally, Red mustn't play G2; Red needs to maintain *both* his or her odd-row threats in column G.

In addition to the six right-side threats that form the sextuple threat, Red's A4 threat is also essential in Diagram 67; it undercuts a possible Black counterthreat at a5. (Without this threat, Red would still have zugzwang control and be able to block at B5; but Black would then take b6, which upsets Red's plans on the right side). Thus, Red needs two major threats and five minor threats to guarantee victory here!

In addition to the Black left-side minor threats (needed only to explain why Red's A4 threat is essential), I have also marked Black's major threat e5, even though it is undercut, because if Red is forced to block there, the wasted turn could spell defeat for him or her. Black has several other relevant threats in Diagram 67 (though I didn't clutter it up by trying to show them all) that could come into play if Red isn't careful. In Diagram 68, I show how these Black threats might come alive if Red errs in the continuation with 21

DIAGRAM 68
Red to play and draw

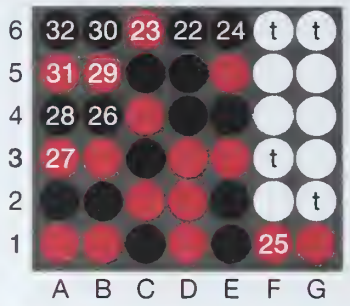


Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 B3? c5 D2 a2? D3 d4 G1 d5 (19) D6 b4 B5? b6 E3! e4! E5! e6 C6 f1! (29) __

B5. Red must settle for a draw in Diagram 68, since 29 F2? f3 G2 f4 would win for Black.

I mentioned that Red shouldn't play 19 E3 in Diagram 67. Diagram 69 shows the result if Black answers Red correctly. Red has finished in zugzwang; if Red plays 33 F2, Black f3 is check, so Black gets f4. The trick in the problem is the temptation to start with 22 c6 and make an immediate Black major threat at f3. That's a logical way to defeat Red's right-side threats, but Black would get only a draw, since Red answers with 23 D6. Black needs to go for broke with 22 d6, the only way to guarantee a horizontal sixth-row win.

DIAGRAM 69
Red to play (will lose)



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 B3? c5 D2 a2? D3 d4 G1 d5 E3?? e4! E5! (22) d6! C6 e6! F1 b4! A3 a4! B5 b6! A5 a6! (33) __

In the moves of Diagram 69, after 19 E3??, Black makes seven consecutive moves that are *uniquely* forced for Black's guaranteed win. This actually becomes *twelve* consecutive uniquely forced moves if the game then continues 33 G2 g3! G4 f2! G5 g6! F3 f4! F5 f6!. It isn't only Red who must play with utter precision in CONNECT 4!

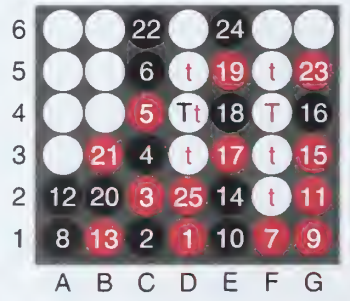
There were blunders in the moves leading up to the sextuple threat in Diagram 67, but Diagram 284 in Chapter 13 shows a sextuple threat that arises in a well-played opening.

Plethora of Threats

The winning quadruple threat and sextuple threat are very specific threat combinations, each of which has a very clear meaning (though the play continuation can get complicated). But there are several other ways that five or more threats can work together in a complicated fashion, and I can't cover them all. You'll just have to feel your way along in such positions.

Diagram 70 (which is an early position reached in the game shown in Diagram 177 in Chapter 10) shows a position where Red's win requires no less than seven threats (six of them minor) in columns D and F. In a handicap game, Red wouldn't win if *any* of these cells were nominated as the "blue cell"

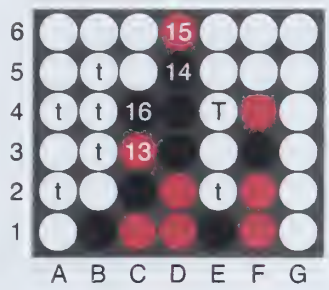
DIAGRAM 70
Black to play (will lose)



Moves: D1! c1 C2? c3! C4 c5? F1? a1? G1 e1 G2 a2 B1? e2! G3 g4! E3 e4! E5 b2 B3! c6 G5 e6? D2 (26) __

DIAGRAM 71

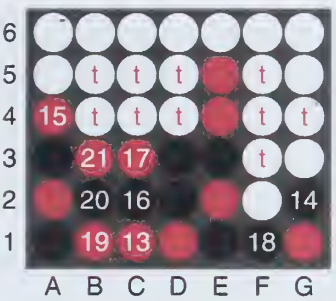
Red to play (will lose)



Moves: D1! b1 F1 e1 F2! f3 D2! d3 C1 d4 F4! c2 (13) C3?? d5! D6 c4! (17) __

DIAGRAM 72

Black to play (will lose)



Moves: D1! e1 E2? e3! E4 a1! A2 a3 E5 d2 G1 d3! (13) C1! g2? A4! c2 C3! f1 B1 b2 B3 (22) __

(see Chapter 11). Red's threat D5 is undercut by Black's major threat but still has good value; Black may not be able to answer D4 at d5 because of the check at F2.

In the game leading to Diagram 379 (see Chapter 13), the position in Diagram 71 is reached, with Black victory depending on seven distinct threats (six of them minor.)

Diagram 260 (repeated here as Diagram 72; see Chapter 12) shows a position where Red needs no less than ten minor threats (or potential minor threats) to win!

There are other examples of "plethora of threats" in Chapter 13. Diagram 390 shows a Red win based on nine minor threats. Diagram 393 shows Black winning with five minor threats despite a Red major odd-row threat.

Chapter 8

Review of Strategy and Tactics

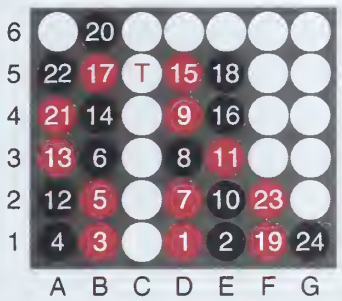
Earlier chapters covered the basic ideas of winning play. In this chapter I give further examples illustrating those ideas.

Red Must Force an Odd-Row Threat

Many openings feature an unrelenting quest by Red to establish an odd-row threat. Diagram 73 is an example of this; although some of Red's moves may seem weak, Red never blunders and does finally achieve the odd-row threat.

Red plays 5 B2, hoping to get B3 and pursue a third-row horizontal threat. The move 7 D2 aims at a future B4 with odd-row threats along the A5-D2 diagonal; similarly, 9 D4 aims at the B2-E5 diagonal and also threatens to play D5. The move 11 A2 might have led to an easier Red victory, since it would give Red a triple threat (C2-C3-E5) immediately. The move 13 A3 is needed to prevent a quick Black win (13 F1?? a3 A4 d5 with a Black adjacent threat pair, c3-c4). The moves 15 D5 and 17 B5 aim at a horizontal fifth-row threat; 19 F1 prevents an undercutting Black threat at c4; 21 A4 aims at A5 but just delays the key F2, with which finally Red establishes a C5 threat.

DIAGRAM 73
Red to play and win



Moves: D1! e1 B1 a1 B2! b3 D2 d3 D4 e2 E3 a2 A3! b4 D5! e4 B5 e5 F1! b6 A4 a5 F2! g1 (25) __

I have stated a purpose for each Red move in this opening, and with Red's intentions clarified, most of Black's moves have a clear purpose as well. Space constraints prohibit me from giving the same detail on other example games; the reader should work to understand them by him- or herself. (The question marks and exclamation points should be of much help, since all blunders in the example games are identified, along with special winning moves.)

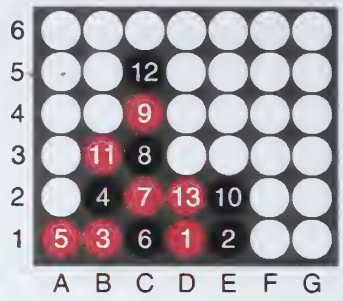
Diagram 74 shows a game that illustrates the great importance of the fifth row. Where should Black play?

I'm afraid many people will try the forcing move 14 e3, but it is Black who ends up being forced, as shown in Diagram 75. Here Red's disk at D5 gives Red

an easy win. Red can play C6 next with an obvious major odd-row threat at F3, or may prefer to tease Black and just make weak-looking moves, since there is no way for Black to escape. (If Red does let Black get c6, Red must *not* ever play G1. Red has a plethora of right-side threats that are equivalent to a winning triple threat, but, without C6, the threats F3-G2 are essential.)

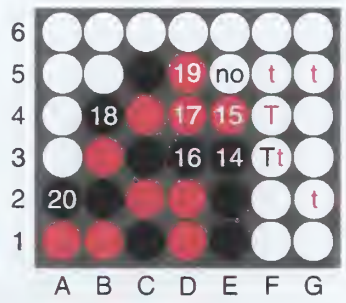
Black should have played 14 d3 in Diagram 74. Diagram 76 shows a correct continuation. Where should Red play next?

DIAGRAM 74
Black to play, with only one move to draw



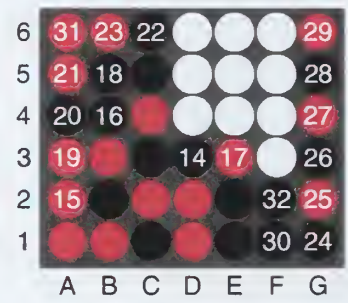
Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 B3? c5 D2 (14) __

DIAGRAM 75
Red to play and win



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 B3? c5 D2 (14) e3? E4! d3 D4! b4 D5! a2 (21) __

DIAGRAM 76
Red to play, with only one move to draw

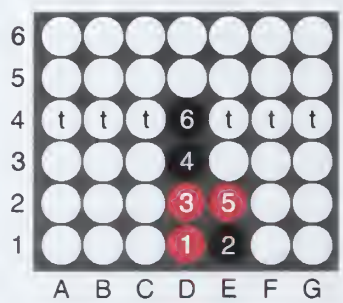


Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 B3? c5 D2 (14) d3! A2 b4 E3 b5 A3 a4 A5! c6 B6 g1 G2 g3 G4 g5 G6 f1 A6 f2! (33) __

Red cannot play E4 (Black's e5 creates an adjacent threat pair at d4-d5), but Red needs to avoid D4 as well, since Black wins on the sixth row after 34 d5! F3 d6!. Black also avoids d4 and e4 for now, so the game will continue 33 F3 f4 and result in a draw.

In Diagram 74, 14 a2 may be a tempting play since it stops Red from getting an easy D5 threat. But Red will still win easily with a sextuple threat (see Diagram 67).

DIAGRAM 77
Red to play (will lose)



Moves: D1! e1 D2? d3! E2? d4 (7) __

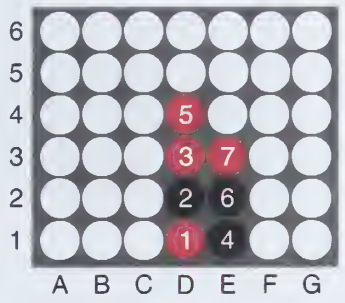
Black Even-Row Wins

I give a few examples here showing how easy it is for Black to win if Red neglects to block Black's horizontal even-row threats. Playing Black, I think you will be able to achieve

the position in Diagram 77 quite often if your opponent hasn't read this book! Red dominates the second row, but even-row threats are useless for Red. As shown with threat symbols (*t*), Black will eventually win on the (even) fourth row, even though Black has only one disk there so far.

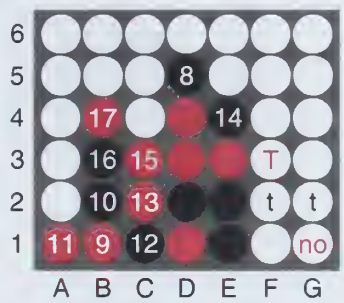
In Diagram 78, Black will win on the second row if he or she plays correctly. Red has established an odd-row (third-row) threat, but it's almost useless because it's undercut by Black's second-row threat. Of Black's seven possible plays in Diagram 78, one wins, and the other six lose. Where should Black play next in Diagram 78?

DIAGRAM 78
Black to play, with only one move to win



Moves: D1! d2 D3! e1 D4 e2 E3?? (8) __

DIAGRAM 79
Black to play (will lose)



Moves: D1! d2 D3! e1 D4 e2 E3?? (8) d5?? B1 b2 A1! c1 C2! e4 C3! b3 B4 (18) __

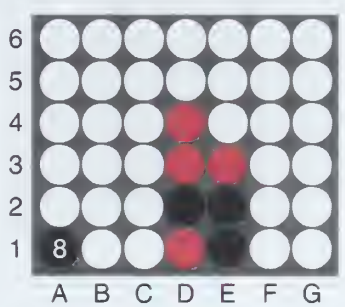
If you're puzzled, examine the continuation in Diagram 79 and then return to the problem.

After Red 9 B1, Black 10 b2 preserves Black's second-row threat but only for a moment; Red's 11 A1 is check, so Red gets C2 and kills Black's threat. C2 is also check, so Red gets C3 and forms an odd-row threat at F3. I've shown a Black minor threat at f2, but it does *not* undercut Red's winning threat at F3 because Red will *not* be so foolish as to play G1. Study this, then return to Diagram 78 and find Black's only winning move.

The solution is shown in Diagram 80. Did you guess right? Black's 8 a1 kills Red's first-row threat and so preserves Black's second-row threat. (Red can take A2 if Red wants, but Black's e2 means A2 would be worthless for Red; Column A will be irrelevant in this game.)

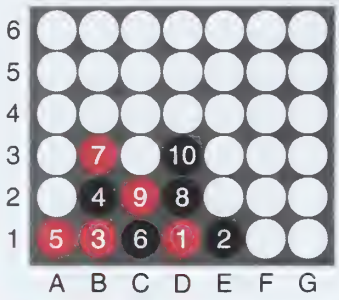
In Diagram 81 on the next page, Red had overlooked an impending second-row threat by Black. Red was able to block Black's second-row threat, of course, but Black then captured d3, securing a win.

DIAGRAM 80
Red to play (will lose)



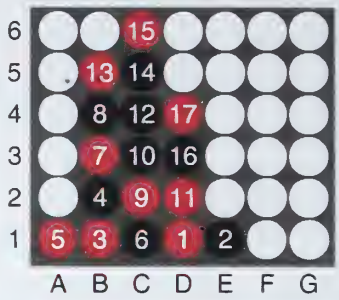
Moves: D1! d2 D3! e1 D4 e2 E3?? (8) a1! (9) __

DIAGRAM 81
Red to play (will lose)



Moves: D1! e1 B1 b2 A1! c1 B3?? d2! C2 d3! (11) __

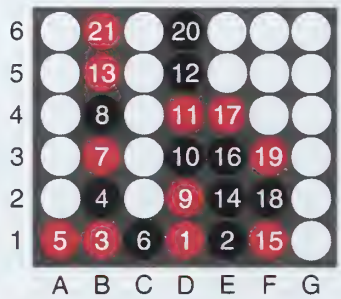
DIAGRAM 82
Black to play and draw



Moves: D1! e1 B1 b2 A1! c1 B3?? b4? C2! c3! D2! c4 B5! c5 C6! d3 D4! (18) __

Diagram 82 shows a game where Black blunders back with 8 b4 and Red plays with utmost precision to get a draw. Red must play 9 C2. Other moves lose, as shown, for example, in Diagram 83. A key difference between the moves 9 C2 and 9 D2 in these diagrams is that C2 loses value if delayed.

DIAGRAM 83
Black to play and win

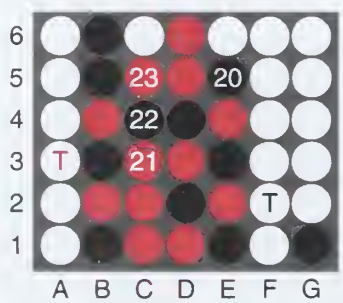


Moves: D1! e1 B1 b2 A1! c1 B3?? b4? D2? d3! D4 d5 B5 e2 F1 e3! E4 f2 F3 d6! B6 (22) __

Push-Up Play

In Diagram 84, Red will win the ending with a major odd-row threat at A3. (Black's even-row threat on the opposite side of the board is irrelevant.) Red obtained this threat by playing C1-C2-C3, which let Red take C5 in a so-called push-up play. Black's play 20 e5 was a mistake. Black would have won if he or she had played 20 c3 instead and prevented the push-up play.

DIAGRAM 84
Black to play (will lose)



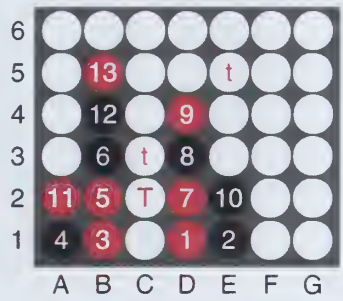
Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 D6 g1 C1?? b6! C2 (20) e5?? C3! c4 C5! (24) __

In Diagram 85, Red has just blundered, and Black can destroy Red's winning threat with a simple push-up play beginning 14 e3. (Black could have played the push-up play 12 e3 E4 e5 instead of 12 b4, but then Red would play 13 B4 and get a different triple threat.)

In Diagram 86, Red doesn't yet have an even-row threat and has just erred with the weak move 13 F1. Where should Black play? (Don't jump to conclusions!)

DIAGRAM 85

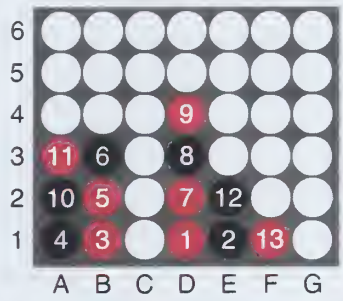
Black to play, with only one move to draw



Moves: D1! e1 B1 a1 B2! b3 D2
d3 D4 e2 A2 b4 B5? (14) __

DIAGRAM 86

Black to play and win

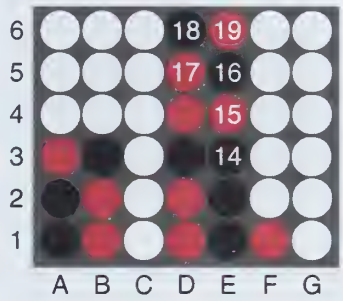


Moves: D1! e1 B1 a1 B2! b3 D2
d3 D4 a2 A3 e2 F1?? (14) __

Black may try the push-up play e3 E4 e5 to kill Red's odd-row threat in column C, as shown in Diagram 87. Black isn't worried about Red threats in other columns because Black has a major odd-row threat. However, Black has no place to build an even-row threat and will have to settle for a draw.

DIAGRAM 87

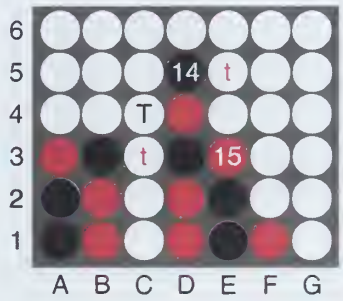
Black to play and draw



Moves: D1! e1 B1 a1 B2! b3 D2 d3 D4 a2 A3
e2 F1?? (14) e3? E4! e5 D5! d6 E6! (20) __

DIAGRAM 88

Black to play, with only one move to win



Moves: D1! e1 B1 a1 B2! b3 D2 d3 D4
a2 A3 e2 F1?? (14) d5 E3 (16) __

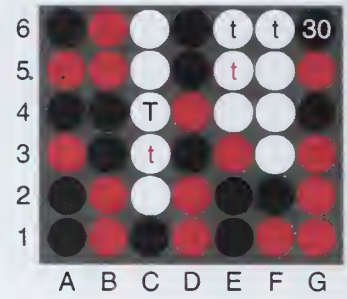
Black does better to play 14 d5, as shown in Diagram 88, and secure an even-row threat. Red answers 15 E3 to stop the push-up play, of course. Where should Black play next?

Black must play 16 b4 at once in Diagram 88 because, if Red gets to play there, Red will threaten to make C4 a major even-row threat (and Black defenses against that at a4 or e4 lead to Red A5 or E5 and a major odd-row threat).

After Black's correct 16 b4 move, the game may develop into the position shown in Diagram 89, where Black has just played g6 and will eventually win the endgame on the sixth row. Some players might be tempted to play c2 (followed by

c3) instead of g6, but this serves no purpose. I have marked two Red threats in Diagram 89 just to emphasize that the Red's threat at C3 is *not* part of a triple threat.

DIAGRAM 89
Red to play (will lose)

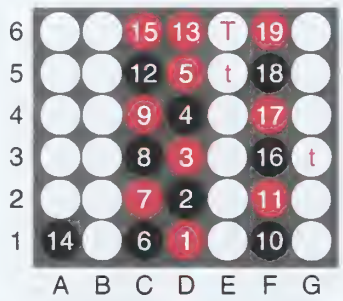


Moves: D1! e1 B1 a1 B2! b3 D2 d3 D4 a2 A3 e2 F1?? d5 E3 b4! B5 d6 G1 f2! B6 c1 G2 a4 A5 a6 G3 g4! G5 (30) g6! (31) __

Triple Threats

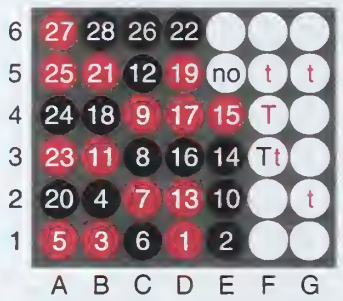
If Black prevents Red from building a major odd-row threat in the opening, Red will need a triple threat to win, so the establishment of such a threat configuration is a very common theme. Diagram 90 is an example; Red's 13 D6 may seem weak, but this strong point ends up key to Red's winning threats. (The move 13 A1 or F3 would also win, but the follow-up moves are not so easy to see.)

DIAGRAM 90
Black to play (will lose)



Moves: D1! d2 D3! d4 D5! c1 C2 c3 C4! f1 F2 c5 D6 a1 C6! f3 F4! f5 F6 (20) __

DIAGRAM 91
Red to play, with only one move to win



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 B3? c5 D2 e3? E4! d3 D4! b4 D5! a2 B5 d6 A3! a4 A5 c6 A6 b6 (29) __

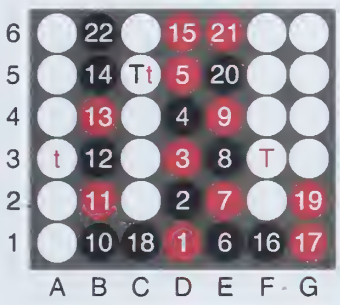
Diagram 91 (derived from Diagram 75) illustrates a rather complicated configuration of threats. Red will win; try to understand why. A few poor moves were made in this game: Red's 11 B3 leads to a draw with perfect play (Red should have played 11 C5); Black's 14 e3 loses (d3 was needed); Red's plays 21, 25, and 27 were weak, but Red was just taunting Black, since Red still has a won game.

Red's threats at G2, F3, and F4 are not by themselves enough to win; for a winning triple threat, at least two of the threats must be odd-row threats. Red's threats at F5, G5, and F4 *would* be a winning triple threat, since two of the threats are odd, but Black's f3 threat undercuts this. *Working together*, however, the five threats at F5, G5, F4, F3, and G2 do give Red victory here. Column F will be

played first, and Red will get F3. A way to understand that Red's configuration wins is to treat F3-G2-G5 as an ordinary triple threat, noticing that whenever Red gets F3, Red gets F5 as well.

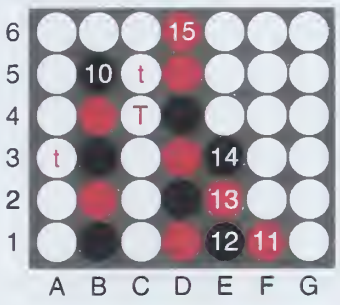
Since three minor odd-row threats in three separate columns constitute a win for Red, it's also a win if one of the threats is a major threat! Even though in Diagram 92 Red and Black have one major odd-row threat apiece, it is *not* a draw. Rather, it is a very easy win for Red because of Red's triple threat.

DIAGRAM 92
Red to play and win



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 D6 f1 G1 c1 G2! e5 E6 b6 (23) __

DIAGRAM 93
Black to play (will lose)



Moves: D1! d2 D3! d4 D5! b1 B2! b3 B4! (10) b5 F1! e1 E2! e3 D6 (16) __

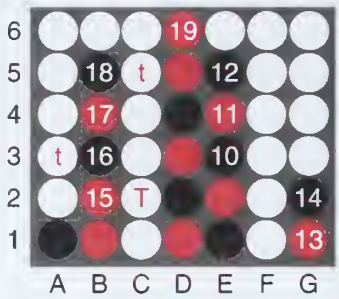
In Diagram 93, Black plays 10 b5 to avoid a triple odd-row threat similar to the one shown in Diagram 61. Red responds by building a triple mixed threat, as shown.

Because the threat at C4 is a major threat adjacent to a minor threat, the endgame win in Diagram 93 is rather trivial; Red can play up column C, capture C5, and establish a major threat at A3. There is no way for Black to build a significant counterthreat. Red's play has been quite precise (though Red can substitute 15 G1 for D6). If Black plays to prevent the Red triple threat in Diagram 93 by playing 14 d6 instead of e3, Red plays 15 E3 and gets a different triple threat.

In the preceding example, Red's even-row threat C4 was adjacent to Red's odd-row threat C5. Alternatively, as shown in Diagram 94 on the next page, a threat at C2 can be combined with a threat at C5. In other words, the tactical value of an adjacent threat pair is not essential in these threat configurations; the column C even-row threat leads to zugzwang control, and Black will eventually be forced to play up column A and allow Red to occupy his or her minor odd-row threat at A3. Once Red achieves that, Red cheerfully relinquishes the column C even-row threat. (The one even-odd threat combination of no value to Red is third row and sixth row.)

DIAGRAM 94

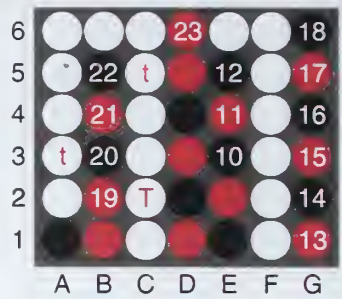
Black to play (will lose)



Moves: D1! d2 D3! d4 D5! e1 B1 a1 E2 (10) e3 E4! e5 G1 g2 B2 b3 B4 b5 D6 (20) __

DIAGRAM 95

Black to play (will lose)

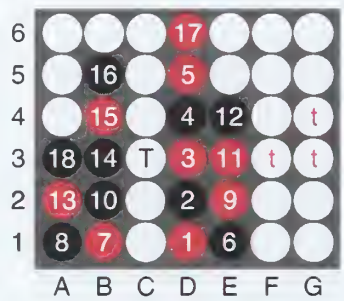


Moves: D1! d2 D3! d4 D5! e1 B1 a1 E2 (10) e3 E4! e5 G1 g2 G3 g4 G5 g6 B2! b3 B4! b5 D6! (24) __

Diagram 95 is essentially the same as Diagram 94 except that Red has interposed two moves (15 G3, 17 G5), which accomplish little but do more or less force Black's replies (16 g4, 18 g6). Once these temporizing moves are played out, the key moves at B2, B4, and D6 are only-way-to-win moves, as shown with exclamation points.

DIAGRAM 96

Red to play, with only one move to win



Moves: D1! d2 D3! d4 D5! e1 B1 a1 E2 b2 E3 e4 A2! b3 B4! b5 D6! a3 (19) __

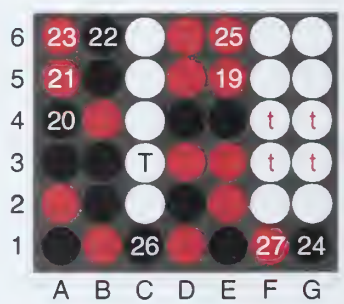
Where should Red play in Diagram 96? (This position is also discussed at Diagram 110.)

Both players played well in Diagram 96; the only deviations permitted to Red would be 11 A2, 9 B2, or 7 E2. Red already has a triple threat, but Black has a counterthreat at c3. Red's winning move now is 19 E5, which builds an invincible quadruple threat!

After the key 19 E5, players will make temporizing moves until Diagram 97 (or something similar) is reached. Black is in zugzwang and must either relinquish the threat at c3, play g2 G3, with Red getting the winning adjacent threat pair F3-F4, or play f2 F3, with Red getting the winning adjacent threat pair G3-G4. (The threat at G4 isn't strictly necessary, since Red also gets a major threat at C3 after f2 F3. In other words, Red needs to use *either* the threat at C3 or the one at G4; having both is an unneeded luxury.)

DIAGRAM 97

Black to play (will lose)

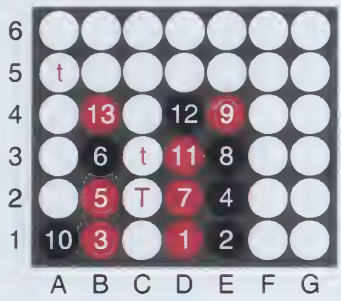


Moves: D1! d2 D3! d4 D5! e1 B1 a1 E2 b2 E3 e4 A2! b3 B4! b5 D6! a3 (19) E5! a4 A5 b6 A6 g1 E6 c1 F1! (28) __

Diagram 98 shows another triple threat. Black has little chance here for a relevant

DIAGRAM 98

Black to play (will lose)



Moves: D1! e1 B1 e2 B2 b3 D2 e3 E4! a1 D3 d4 B4 (14) __

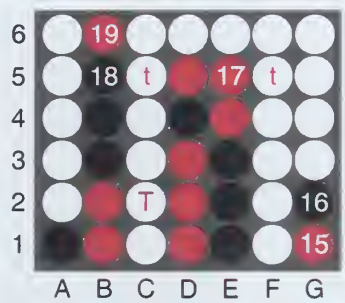
counterthreat. (Diagram 38 is derived from this position.) Black can play g1 and claim f2 as a threat, but a right-side even-row threat will have no effect on Red's left-side win. (Black would like a major odd-row threat on the right side as a counterthreat to negate Red's threat, but there is no real chance for a Black odd-row threat in Diagram 98.)

Red's move 13 B4 seems clear-cut; it establishes the minor odd-row threats along a diagonal. But it is interesting to note that Red had other winning moves. One possibility is to substitute 13 D5. Black will then probably take b4 to prevent the easy Red win seen in Diagram 98. Where should Red then play his or her next two disks?

Red should play as shown in Diagram 99. Now a Black threat at f2 *would* be relevant, so Red kills that with G1. G1 is a forcing move; Black must play g2 to prevent an easy Red major odd-row threat at F3. Black ends up with b5, but Red gets the winning triple threat C2-C5-F5.

DIAGRAM 99

Black to play (will lose)

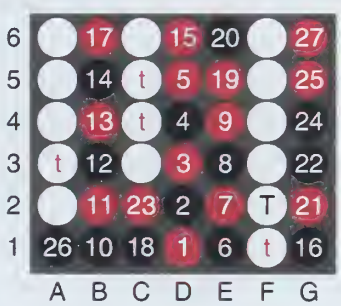


Moves: D1! e1 B1 e2 B2 b3 D2 e3 E4! a1 D3 d4 D5 b4 (15) G1! g2 E5! b5 B6! (20) __

Red can change the order of his or her moves slightly in Diagram 99, for example, playing 13 G1 and delaying D5. Red can also play 13 B4 and win on the left side with the triple threat A5-C3-C2.

DIAGRAM 100

Black to play, with only one move to draw



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 D6 g1 B6! c1 E5 e6 G2? g3! C2 g4 G5 a1 G6 (28) __

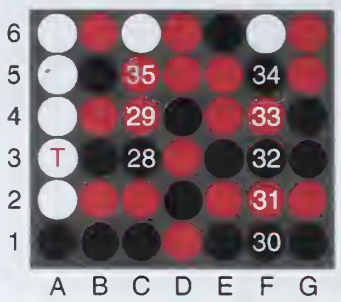
Diagram 100 shows a position that can arise in one of the standard openings if Red makes a subtle blunder. I've marked A3-C5-C4 as though it were a (winning) triple threat for Red, even though Red cannot play F1 to convert C4 to a true threat. Be aware of these subtle threat configurations that can play a role similar to a more ordinary triple threat. Where must Black play next in Diagram 100?

Black is almost in zugzwang but can salvage a draw by playing F1 to kill Red's

threat at C4. Otherwise, the play will be as shown in Diagram 101, with Red getting full value for the C4-F1 threat.

Next I take a look at a triple minor threat configuration so common that it warrants its own section.

DIAGRAM 101
Black to play (will lose)

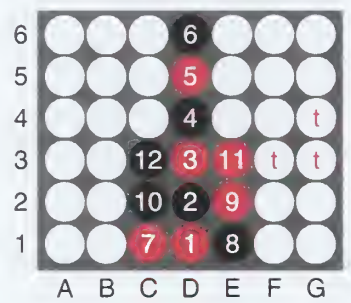


Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 D6 g1 B6! c1 E5 e6 G2? g3! C2 g4 G5 a1 G6 (28) c3? C4! f1 F2! f3 F4! f5 C5! (36) __

J Shape

In Diagram 102, Red has established three minor threats on the right side that will lead to Red's victory. In this case, it is easy to see how Red can convert the triple minor threat to a major odd-row threat; Red marches up column G and is sure to capture either G3 or G4. In either case, F3 will become a major odd-row threat. Red wins.

DIAGRAM 102
Red to play and win

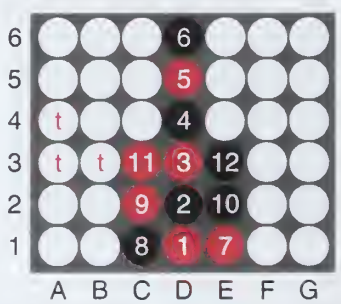


Moves: D1! d2 D3! d4 D5! d6 C1! e1 E2 c2 E3 c3 (13) __

The four red disks at D3, E3, E2, and D1 (disks 3, 11, 9, and 1 in Diagram 102) look a bit like a capital letter *J*, so I call this common configuration the *J shape*. (Of course, sometimes a mirror image of this shape will arise and have the same effect. Still other configurations of Red disks will give rise to a similar group of Red minor threats.)

Against an incautious opponent, Red will often be able to establish a J shape early in the game and win easily. However a J shape does not guarantee victory, and there are ways Black can counterattack or foil Red's J shape. It's Red's turn to play in Diagram 102, and six of Red's moves win, but one loses. See whether you can figure out what Red's *losing* move is in Diagram 102. In the next section I give the answer to this, along with more discussion of the J shape.

DIAGRAM 103
Red to play and win



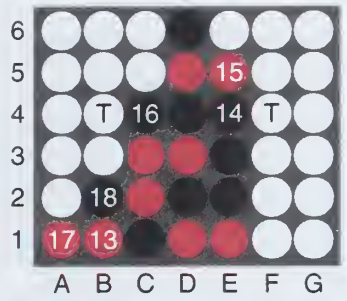
Moves: D1! d2 D3! d4 D5! d6 E1! c1 C2 e2 C3 e3 (13) __

J-Shape Races

Diagram 103 is Diagram 102 presented in its mirror-image form so you get used to seeing the mirror J. Have you figured out what Red's one losing move is?

Red must not play 13 B1 in Diagram 103 (or 13 F1 in Diagram 102). This would lead to the position shown in Diagram 104.

DIAGRAM 104
Red to play (will lose)

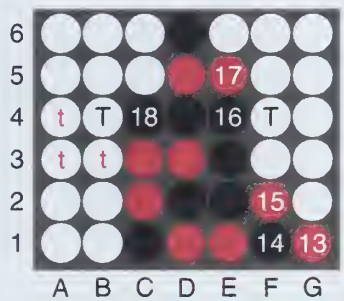


Moves: D1! d2 D3! d4 D5! d6 E1! c1 C2 e2 C3 e3 (13) B1?? e4! E5 c4! A1 b2 (19) __

Black will soon play b3 to kill Red's left-side J shape. Black's major even-row threat at f4 will then give Black an eventual victory. Red cannot play B3, of course, because of Black's major threat at b4. I have discussed undercutting threats, where Red's fifth-row threat is neutralized by a Black fourth-row threat underneath it, but here we have an *overcutting* threat, where Red's minor threat on the third row is neutralized by Black's major threat above it.

The reader may well ask why 13 B1 was Red's *only* losing move. It was a forcing move that provoked Black to play e4 ... c4, but couldn't Black play that way anyway? Let's see. Examine Diagram 105, but now look for Red's only *winning* move. (Remember the specific reason why Red's triple minor threat can be converted to a major odd-row threat.)

DIAGRAM 105
Red to play, with only one move to win

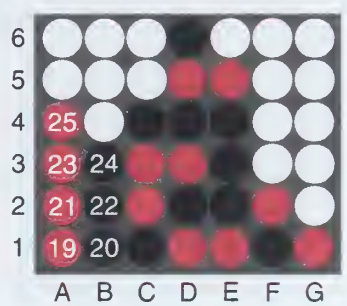


Moves: D1! d2 D3! d4 D5! d6 E1! c1 C2 e2 C3 e3 (13) G1 f1 F2 e4 E5! c4 (19) __

I call the configuration in Diagram 105 an *overcut J shape.* Red has achieved a triple minor threat, but Black has major fourth-row threats in compensation. The best strategy for Red, of course, is to prevent Black from achieving the disk trio c4-d4-e4, but often this cannot be prevented. In Diagram 105, Red must start with A1 and continue to play up column A. Major threats cannot be overcut, so Red needs to hurry and convert Red's B3 minor threat to a major threat.

Diagram 106 shows clearly that the players are engaged in a race. Red wants to play up column A quickly and convert the B3 threat to a major threat. Black wants to play up column B and capture b3 (or b4) before Red achieves Red's column A goal. Looking at Diagram 106, we see that Black has, in a

DIAGRAM 106
Red has won



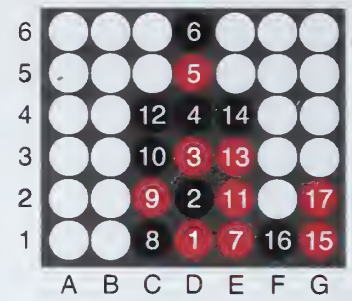
Moves: D1! d2 D3! d4 D5! d6 E1! c1 C2 e2 C3 e3 G1 f1 F2 e4 E5! c4 (19) A1! b1 A2! b2 A3! b3 A4!

sense, won this race but lost the war; Red gets victory at A1-A2-A3-A4.

Let's look at some more openings where Red achieves the J shape.

In Diagram 107, Black plays 10 c3 to avoid giving Red an easy J shape, but then has to let Red build a J shape on the right side (12 e3 loses immediately to 13 F1). Red gets a J shape with 13 E3; Black gets an overcutting threat with 14 e4; Red plays 15 G1 and 17 G2 and wins the race between column G and column F.

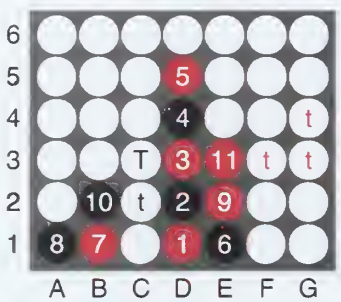
DIAGRAM 107
Black to play (will lose)



Moves: D1! d2 D3! d4 D5! d6 E1! c1 C2 c3 E2 c4 E3 e4 G1! f1 G2! (18) __

One lesson to take from Diagram 107 is that Black's 8 c1 was a weak move; it led directly to Red's winning J shape. Black would do better to play 8 f1, in accordance with a general principle that b1 and f1 are often better early plays than the more central c1 and e1.

DIAGRAM 108
Black to play (will lose)

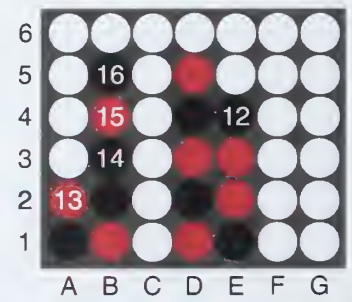


Moves: D1! d2 D3! d4 D5! e1 B1 a1 E2 b2 E3 (12) __

When Red achieves a J shape and Black is unable to get the overcutting c4-d4-e4, Black should strive to counter Red's threat with a major odd-row threat on the opposite side of the board. For example, consider Diagram 108.

As shown, Black has a left-side major odd-row threat, and Red has a right-side triple threat equivalent to a major odd-row threat. If Red cannot establish a second significant threat, Red will draw or even lose. Diagram 108 has immediate tactical possibilities, since Black threatens a2, which would give Black the adjacent threat pair c2-c3, while Red threatens E4, which would give Red an undercutting major threat at C2. This makes the players' next moves pretty clear-cut.

DIAGRAM 109
Red to play, with only one move to win



Moves: D1! d2 D3! d4 D5! e1 B1 a1 E2 b2 E3 (12) e4 A2! b3 B4! b5 (17) __

The moves shown in Diagram 109 are all absolutely essential: 12 e4 and 13 A2 prevent threats at C2; 14 b3 prevents a Red major threat at C3; 15 B4 prevents a Black adjacent

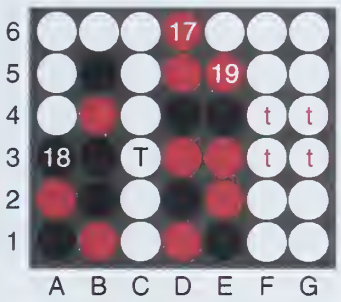
threat pair c3-c4; 16 b5 dampens Red's hopes for threats at C4-C5. (Although "essential," Black's moves don't qualify for an exclamation point by definition; against perfect Red play, Black will lose.) In Diagram 109, Red's next move (disk 17) must also be placed with precision. Can you see where? (After making this best play, assume Black's obvious reply and find the proper disk 19 for Red as well.)

With each player having a major odd-row threat or the equivalent, the game will be drawn if neither player establishes another significant threat. Red's play 17 D6 in Diagram 110 makes it impossible for Black to establish a sixth-row threat, but more important, Red threatens A3, which would establish C5 as a second major odd-row threat for Red. Black's counter at 18 a3 is mandatory, of course; Red's follow-on at 19 E5 is also the only winning move. Red ends up with a winning quadruple threat essentially identical to the one in Diagram 97.

Diagram 111 shows a possible continuation from Diagram 110.

DIAGRAM 110

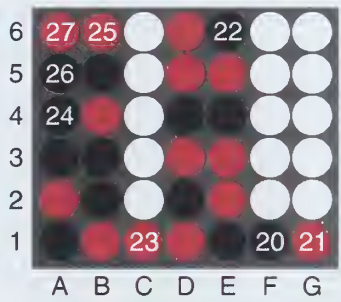
Black to play (will lose)



Moves: D1! d2 D3! d4 D5! e1 B1 a1 E2 b2 E3 e4 A2! b3 B4! b5 (17) D6! a3 E5! (20) __

DIAGRAM 111

Black to play (will lose)



Moves: D1! d2 D3! d4 D5! e1 B1 a1 E2 b2 E3 e4 A2! b3 B4! b5 D6! a3 E5! (20) f1 G1 e6 C1 a4 B6 a5 A6! (28) __

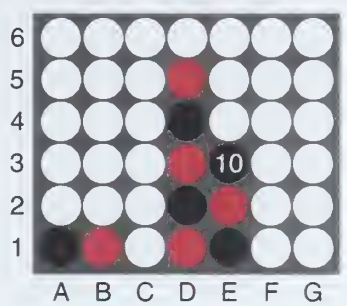
Before looking at more variants of J-shape openings, let's return to Diagram 108 and see what might have happened if Black had prevented Red from getting the J shape by playing 10 e3, leading to the position in Diagram 112.

Red should play 11 E4 at once to establish a major threat at C2. Although Red may not get a major odd-row threat in the opening, with the threat at C2 already established, it will be rather easy for Red to achieve a winning triple threat combination. A likely result is shown in Diagram 94.

Diagram 113 (related to Diagram 218; see Chapter 12) shows another opening where

DIAGRAM 112

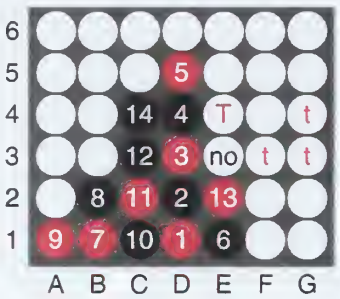
Red to play, with only one move to win



Moves: D1! d2 D3! d4 D5! e1 B1 a1 E2 (10) e3 (11) __

both players play well. Black can't play e3, of course, so Red can (more or less) treat E3 as if he or she already occupied it.

DIAGRAM 113
Red to play and win

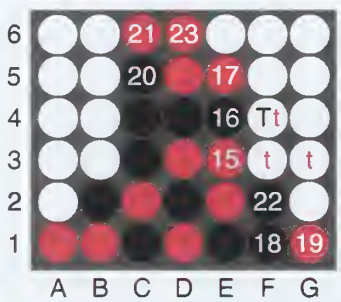


Moves: D1! d2 D3! d4 D5! e1 B1 b2 A1 c1 C2 c3 E2 c4 (15) __

In Diagram 113, although Red hasn't yet formed a J shape, his or her threat at E4 guarantees that Black will not be able to prevent him from doing so; thus I call this configuration a *future J. shape.* With this shape, Red can play E3 early on to complete the actual J, or Red can delay—either way, E3 belongs to Red. It is most logical for Red to delay the play, so that Red need not worry about Black playing e4 and getting an overcutting threat at f4. However it may be slightly easier to understand the position once Red actually places a red disk on E3, so Red will be forgiven for making that play right away, *provided it doesn't sacrifice Red's victory.* Look at Diagram 113. Can Red play E3 at once and still win? If you can figure this out without reading ahead or experimenting with your CONNECT 4 set, you will have demonstrated excellent visualization.

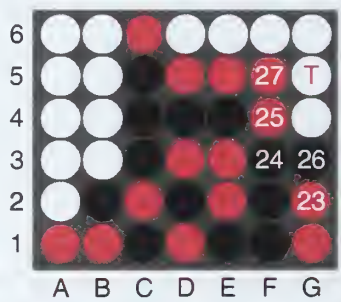
As shown in Diagram 114, when Red plays 15 E3, Red allows Black to construct the overcutting threat f4, apparently loses the resultant race in columns F and G, yet still wins the game, with a *different* triple minor threat. (If Black next

DIAGRAM 114
Black to play (will lose)



Moves: D1! d2 D3! d4 D5! e1 B1 b2 A1 c1 C2 c3 E2 c4 (15) E3 e4 E5! f1 G1 c5 C6! f2 D6! (24) __

DIAGRAM 115
Black to play (will lose)



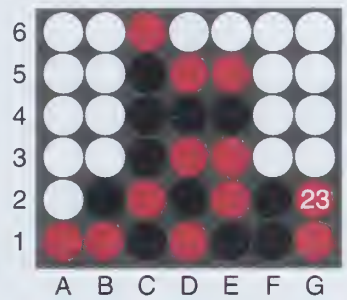
Moves: D1! d2 D3! d4 D5! e1 B1 b2 A1 c1 C2 c3 E2 c4 E3 e4 E5! f1 G1 c5 C6! f2 (23) G2?? f3?? F4! g3 F5! (28) __

plays g2, Red's G3 gives Red an adjacent threat pair at F3-F4; if Black instead plays f3, Red's F4 gives Red a winning major odd-row threat at G3.)

Instead of 23 D6 in Diagram 114, Red might be tempted to play as shown in Diagram 115, where Red has a winning major odd-row threat; however, Black's 24 f3 was an error. How should Black have responded in Diagram 116?

DIAGRAM 116

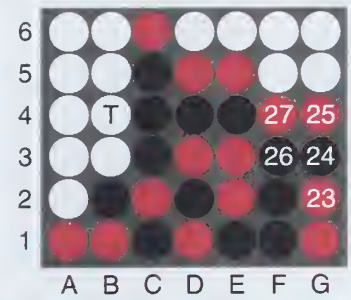
Black to play, with only one move to win



Moves: D1! d2 D3! d4 D5! e1 B1 b2 A1 c1 C2 c3 E2 c4 E3 e4 E5! f1 G1 c5 C6! f2 (23) G2?? (24) __

DIAGRAM 117

Black to play and win



Moves: D1! d2 D3! d4 D5! e1 B1 b2 A1 c1 C2 c3 E2 c4 E3 e4 E5! f1 G1 c5 C6! f2 (23) G2?? g3! G4 f3! F4 (28) __

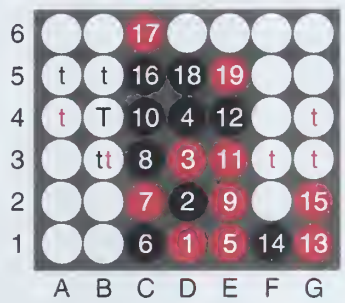
Black must play as shown in Diagram 117, with an endgame win at b4.

As implied earlier, Red could have avoided complications like those in Diagram 117 by simply delaying the play 15 E3 in Diagram 113, playing up column G instead.

Diagram 118 presents some interesting problems. As usual, try to solve these problems before reading on. (1) Red played well except for one blunder. What was the blunder? Where should Red have played instead? (2) Red still has a strong tactical threat available. What is it? (3) Where should Black play next? (There are several possibilities.)

DIAGRAM 118

Black to play and win



Moves: D1! d2 D3! d4 E1?? c1 C2 c3! E2 c4! E3 e4! G1 f1! G2 c5 C6 d5 E5 (20) __

Diagram 118 shows a position where Red has achieved a J shape, won the race in column G to establish a triple threat on the right-hand side, and yet still loses to Black's powerful threats on the left side. Assuming that Red's right-side configuration is equivalent to a major odd-row threat, on the left side Black will need an odd-row threat (to cancel Red's odd-row threat) and an even-row threat (to get a win instead of just a draw). Black has an even-row threat (at b4) already (unless Red establishes a B3 threat to undercut it); the question is: Will Black get an odd-row threat on the left side?

I've indicated several Black and Red threats in Diagram 118 to show how the analysis proceeds. The triple threat at a5-b5-b4 would provide Black with the equivalent of a major odd-row threat, but Red's diagonal threats at A4-B3 would normally undercut Black's a5 threat and thus thwart Black. (When the A column is played, Black will want a4 so Red doesn't get the B3 major threat, but Red then takes A5 to kill Black's odd-row threat.) However, Black can undercut Red's

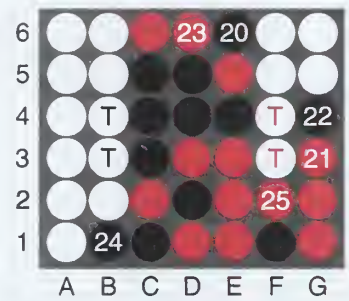
diagonal by playing a1, and if Red plays A1, Black plays a2 to build a major odd-row threat at b3. (I've tried to depict this last possibility by showing b3 as Black's threat cell.)

In other words, Black might wish to play a1 and construct a threat at b2, *not* to build an even-row threat in column B (Black already has one at b4), but to neutralize any possible Red threat at B3.

Did you guess where Red's blunder was in Diagram 118? The cell D5 is one of the most powerful on the board; here it leads to Black's fifth-row counterthreat and therefore victory. It was essential for Red to take D5 at the first possible chance, answering 4 d4 with 5 D5. (As the game played out, Black had to play almost exactly as he or she did to win and never got a chance to take d5 until disk 18. However, once Red passed up the first chance, Red never got an effective opportunity to play D5 either.)

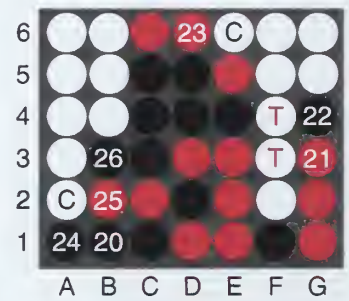
I asked you to guess Black's next move in Diagram 118. Black has several winning moves (though not 20 f2 G3, with Red getting double check). Did you guess 20 e6 to build an adjacent threat pair at b3-b4? Sorry, that doesn't work. In Diagram 119, you will see that Red would then get an adjacent threat pair and win the resultant race.

DIAGRAM 119
Black to play (will lose)



Moves: D1! d2 D3! d4 E1?? c1 C2 c3! E2 c4! E3 e4! G1 f1! G2 c5 C6 d5 E5 (20) e6?? G3! g4 D6! b1 F2! (26) __

DIAGRAM 120
Red to play (will lose)



Moves: D1! d2 D3! d4 E1?? c1 C2 c3! E2 c4! E3 e4! G1 f1! G2 c5 C6 d5 E5 (20) b1 G3 g4! D6 a1 B2 b3 (27) __

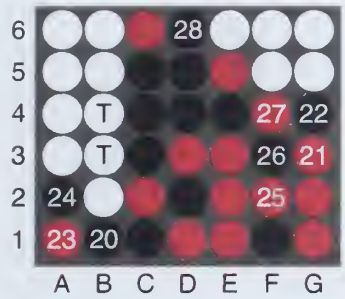
Diagram 119 shows Red's tactical possibility alluded to previously. Black can prevent this with 20 g3 or 20 d6, but 20 a1 or 20 b1 also works for Black since now Black wins this race, as shown in Diagram 120.

In fact, Black has a forced mate in seven in Diagram 118. As shown in Diagram 121, the reason it's mate in seven instead of mate in three is that Red has tactical threats that can delay Black's victory.

Diagram 122 shows a different variation on this opening. Where must Black now play to maintain victory?

DIAGRAM 121

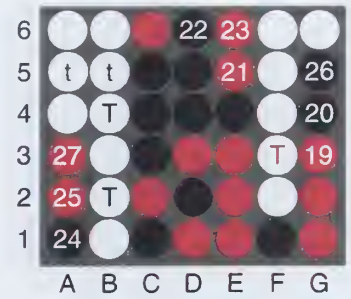
Red to play (will lose)



Moves: D1! d2 D3! d4 E1?? c1 C2 c3! E2 c4! E3 e4! G1 f1! G2 c5 C6 d5 E5 (20) b1 G3 g4! A1 a2 F2 f3! F4 d6! (29) __

DIAGRAM 122

Black to play, with only one move to win

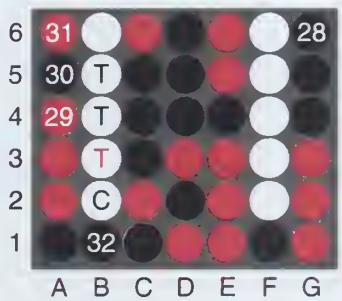


Moves: D1! d2 D3! d4 E1?? c1 C2 c3! E2 c4! E3 e4! G1 f1! G2 c5 C6 d5 (19) G3 g4! E5 d6 E6 a1 A2 g5 A3 (28) __

If you played 28 a4 to block Red from playing there and making a major odd-row threat at B3, please give yourself a big ... *thumbs-down*! Red will take A5; Black ends up with no odd-row counterthreat; Red wins with an F3 threat. Playing 28 b1 B2 b3 B4 b5 doesn't work either; it leads quickly to the establishment of a Black major odd-row threat, but Black's even-row threat evaporates, and Black needs *both* the odd- and the even-row threat to win. Black's only winning move is to play 28 g6 to temporize; the game will continue as shown in Diagram 123; Red's threat at B3 is neutralized by Black's threat at b2. Black wins with the adjacent threat pair b4-b5.

DIAGRAM 123

Red to play (will lose)



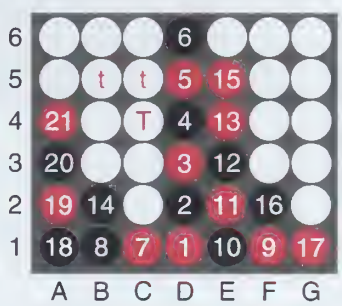
Moves: D1! d2 D3! d4 E1?? c1 C2 c3! E2 c4! E3 e4! G1 f1! G2 c5 C6 d5 G3 g4! E5 d6 E6 a1 A2 g5 A3 (28) g6! A4 a5! A6 b1! (33) __

The Push-Up Play

There is an important tactical maneuver that you need to be alert for, both using it when you can and blocking it when your opponent is about to use it. I introduce this topic with Diagram 124. Red has built a triple threat in the upper left and will win unless Black is careful. Where should Black play in Diagram 124?

DIAGRAM 124

Black to play, with only one move to draw

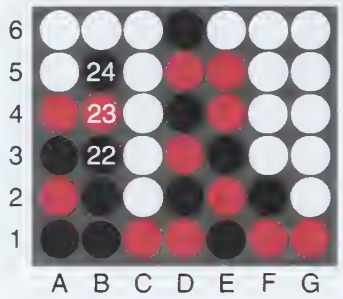


Moves: D1! d2 D3! d4 D5! d6 C1! b1 F1! e1 E2! e3 E4! b2 E5? f2 G1! a1 A2 a3 A4 (22) __

Black should play 22 b3, as shown in Diagram 125, and, because this threatens an immediate vertical win, Black can follow

DIAGRAM 125

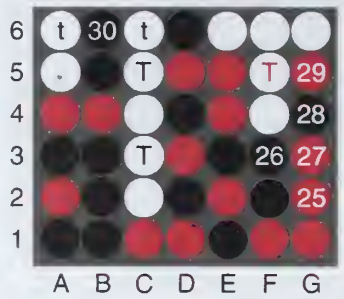
Red to play and draw



Moves: D1! d2 D3! d4 D5! d6 C1! b1 F1! e1 E2! e3 E4! b2 E5? f2 G1! a1 A2 a3 A4 (22) b3! B4! b5! (25) __

DIAGRAM 126

Red to play (will lose)



Moves: D1! d2 D3! d4 D5! d6 C1! b1 F1! e1 E2! e3 E4! b2 E5? f2 G1! a1 A2 a3 A4 b3! B4! b5! (25) G2 f3! G3 g4! G5? b6! (31) __

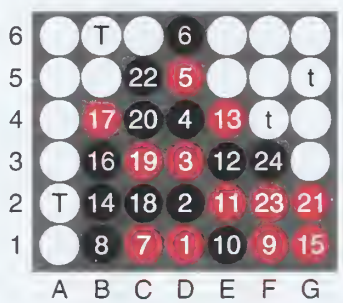
with 24 b5 to destroy Red's left-side threat. Starting with disks at b1 and b2, Black is able to push up column B all the way to b5—that's why I call this the *push-up play.*

In Diagram 125, it is now Red who must be careful just to get a draw. If Red plays as shown in Diagram 126 to construct a right-side major odd-row threat, Black plays b6 and forms a winning sixth-row threat. Red should have played 29 F4, forcing Black to reply with 30 f5, and then taken B6 before Black got a chance to play there.

Diagram 127 is an interesting position. Black has achieved major even-row threats on the left side and an undercutting diagonal threat on the right side. Where should Red play next?

DIAGRAM 127

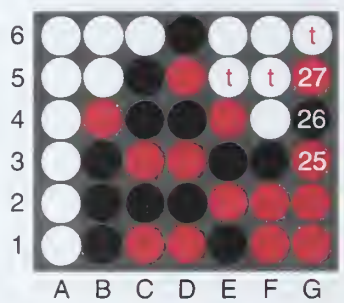
Red to play, with only one move to win



Moves: D1! d2 D3! d4 D5! d6 C1! b1 F1! e1 E2! e3 E4! b2 G1?? b3 B4 c2?? C3 c4 G2 c5 F2! f3 (25) __

DIAGRAM 128

Black to play (will lose)



Moves: D1! d2 D3! d4 D5! d6 C1! b1 F1! e1 E2! e3 E4! b2 G1?? b3 B4 c2?? C3 c4 G2 c5 F2! f3 (25) G3! g4 G5! (28) __

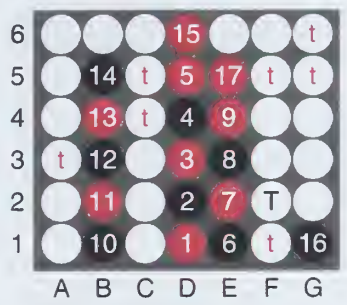
Red makes the push-up play G3 g4 G5, as shown in Diagram 128. Red will take either E5 or G6 next and build a winning major odd-row threat.

Overcutting Threats

A major threat can neutralize the opponent's minor threat underneath it. Whereas an undercutting threat can neutralize both major and minor threats, an *overcutting* threat lies *above* the opponent's threat and will have little value unless the opponent's threat is just a minor threat. (I write "little value" rather than "no value" because a blocking disk must be spent when the column is played, and sometimes the waste of a move is critical.)

DIAGRAM 129

Black to play, with only one move to win



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 D6 g1 E5?? (18) __

Here's a rather difficult problem in which a key theme is an *overcutting* threat. Where should Black play in Diagram 129?

Diagram 129, which is likely to arise after a very common opening, is a complicated position, and it may seem far too early to predict the outcome just by examining the strategic threats. Nevertheless, we can understand how Red intends to win using the threats I've indicated in the diagram. Before discussing those Red threats, we note that Black has two ways to build a major threat in Diagram 129: The move 18 c1 would build a threat at f4, and 18 b6 would build a threat at c5. But just one of those moves wins; the other loses.

If C4 were a Red major threat, Red would have a routine win with a triple threat A3-C5-C4. But of course Red can't play F1 to establish C4 as a major threat; Red's minor threat at F1 is *overcut* by Black's major threat at f2.

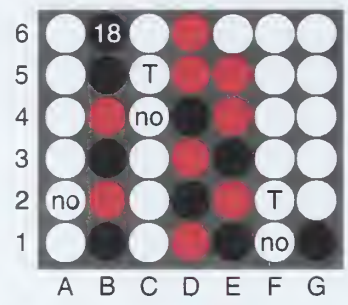
But Red's C4 is still a valid threat. Red will play up column C eventually and take C4. (Black can't take c4 because Red C5 turns A3 into a *major* odd-row threat.) As soon as Red takes C4, Black must answer at f1, and Black's f2 threat then disappears. After this, Red's *right-side* triple threat F5-G5-G6 becomes active and gives Red victory.

But Black can foil this plan! Return to Diagram 129, and figure out Black's winning move. Diagram 130 shows the result of Black's play 18 b6; Diagram 131 shows the result of Black's play 18 c1.

Diagram 130 shows the solution to Diagram 129. Black's 18 b6 constructs a threat at c5 and thus overcuts Red's threat to play c4. As long has Black never plays in any of the cells marked "no" (and plays above Red when Red plays in such a cell) Black will win easily with the major even-row threat at f2. (The same key move also occurs in Diagram 84.)

DIAGRAM 130

Red to play (will lose)

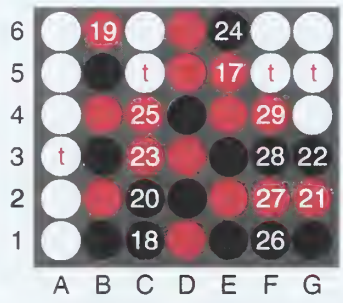


Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 D6 g1 E5?? (18) b6! (19) __

Diagram 131 shows why 18 c1 (or any play other than 18 b6) loses for Black. Red will play C4, forced moves will follow in column F, but after Red 29 F4, Black has no further forcing moves. Black will play f5; Red will reply C5 and win eventually at A3. In Diagram 131, Red took G2 to build an F3 threat, but Red doesn't really need it; if Black takes g2, Red gets G3 and wins another way with an F4 threat. If Black plays 20 e6 instead of 20 c2, Red's 21 G2 would be a subtle blunder (shown in Diagram 100); instead, Red should play as shown in Diagram 132.

DIAGRAM 131

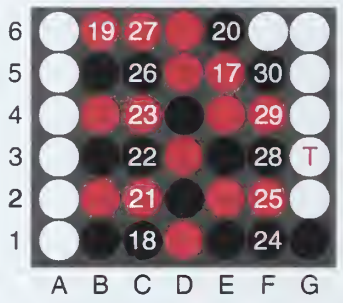
Black to play (will lose)



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 D6 g1 (17) E5?? c1?? B6! c2 G2 g3 C3 e6 C4! f1 F2! f3 F4! (30) __

DIAGRAM 132

Red to play and win



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 D6 g1 (17) E5?? c1?? B6! e6 C2 c3 C4 f1 F2! c5 C6 f3 F4! f5 (31) __

(Diagrams 131 and 132 contain the move sequence 17 E5 c1 B6, where both E5 and c1 are blunders. The blunders cancel each other, however, and lead to the same positions achieved in "perfect" openings with the moves 17 B6 c1 E5.)

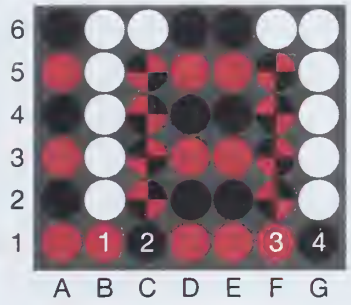
"I Play Where You Play"

Surprisingly often, the best play for a one player will be to play above the opposing player's just-played disk; the Candlesticks Opening, in Diagram 236 (see Chapter 12), is one example.

When you can identify a situation where this imitative play is good enough, it makes successful play *very* easy. There are some proofs about openings based on imitative play. For example, Black never loses against opening Red 1 B1. In Diagram 133, the bicolored disks may be either red or black, but they do alternate in color up the columns. That is, Black takes either c2 or c3, and so on. (Similarly, Black waits for Red to take F1 or G1, then takes the other.) You will see that if

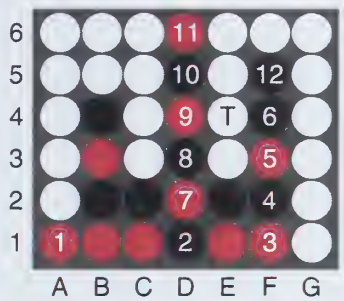
Red gets four-in-row on, say, the third row, Black must have achieved four-in-row earlier on the second row. Victor Allis was the first to prove, in this simple way, that Black can be assured of at least a draw after Red 1 B1 or Red 1 C1.

DIAGRAM 133
Red to play, does not win



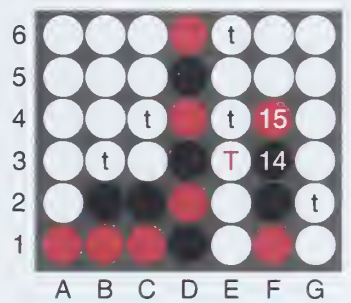
Similarly, Black can be assured of victory if Red starts with 1 A1. Black does this by answering A1 at d1; playing in column D (and then "switching off the autopilot") as soon as Red plays at A4, B5, C3, E3, F5, or G1; making a winning move (in column F or B) after D6; and otherwise playing immediately above Red's last disk. Red's only counter to this Black strategy is to play up column D to prevent Black from forming any horizontal even-row threat. To prove that Black always wins, therefore, I must show that Black always has a winning move in response to Red D6.

DIAGRAM 134
Red to play (will lose)



If Black follows the strategy just stated, when Red eventually plays D6 there will be zero, two, or four disks in column F. If Black can play f1 or f5, that play will win; Diagram 134 shows an example; there's no place left for Red to build a winning odd-row threat. Look at Diagram 134, and you will see plainly that if Red had omitted D2, D4, or D6, Black could have taken that cell and gotten a horizontal even-row win. The author was the first to prove (without computer assistance) that Black can be assured of a win after Red 1 A1.

DIAGRAM 135
Black to play, with only one move to draw



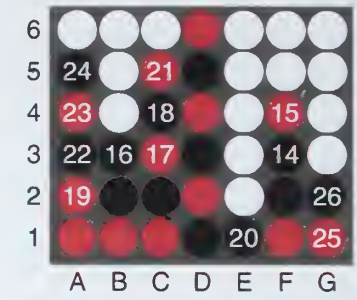
Moves: A1?? d1 C1 c2 D2 d3 F1 f2 D4 d5 B1 b2 D6 (14) f3? F4! (16) __

There is one complication to this proof. If there are exactly two disks in column F when Red plays D6, and Red has already captured C1, then f3 may not win for Black, since Red gets a major odd-row threat, as shown in Diagram 135. Instead, in such cases, Black should answer D6 by playing in column B. (As depicted by the threat b3-c4-e6, Black would have won in Diagram 134 if Black had played 14 b3 instead of f3.)

The continuation from Diagram 135 is interesting. Red has a major odd-row threat, so Black wants to build either undercutting diagonal threats along the b5-e2 diagonal or an odd-row counterthreat. Red won't be able to prevent this, as it turns out, so will want to salvage a draw by preventing Black from getting an even-row threat. Diagram 136 shows the continuation; note that ten consecutive plays (five by Red, five by Black) are uniquely forced to achieve the drawn result.

DIAGRAM 136

Red to play, with only one move to draw



Moves: A1?? d1 C1 c2 D2 d3 F1 f2 D4 d5 B1 b2 D6 (14) f3? F4! b3! C3! c4! A2! e1! C5! a3! A4! a5! G1 g2 (27) __

Chapter 9

Opening Tables

Of course, there are far too many possible CONNECT 4 openings to show the outcome of each one here, but the following tables may be useful. Table 1 shows the result of every possible three-move opening. (Only one form of each mirror-image pair is shown.) A red *R* means that First Player can force victory after the indicated three-move opening; a black *b* means that Second Player can force victory; an equal sign (=) means the game is drawn with best play.

Table 1 (Three-Move Openings)

Moves 1, 2	Move 3						
	A	B	C	D	E	F	G
A1 a2	b	=	=	R	b	R	b
A1 b1	b	b	b	b	b	b	b
A1 c1	b	b	b	b	b	b	b
A1 d1	b	b	b	b	b	b	b
A1 e1	b	b	b	R	b	b	b
A1 f1	b	b	b	b	b	b	b
A1 g1	b	b	b	R	b	b	b
B1 a1	=	R	=	b	R	b	b
B1 b2	b	b	b	b	=	b	b
B1 c1	b	b	b	b	b	b	b
B1 d1	b	=	b	=	b	b	b
B1 e1	b	b	b	R	b	b	b
B1 f1	b	R	=	R	R	=	b
B1 g1	b	R	=	R	=	b	b
C1 a1	=	=	=	R	=	=	b
C1 b1	b	=	b	b	=	R	b
C1 c2	b	=	b	=	b	b	b
C1 d1	b	b	=	=	b	b	b
C1 e1	b	b	=	b	=	b	b
C1 f1	b	=	=	=	=	=	b
C1 g1	b	=	R	R	=	R	b
D1 a1	=	b	R	R	R	R	R
D1 b1	b	R	b	=	=	R	b
D1 c1	b	b	=	=	b	R	R
D1 d2	b	b	b	R	b	b	b

Tables 2 and 3 are similar to Table 1 but show the results of some five- and seven-move openings. To save space, most poorly played openings are omitted.

Table 2 (Five-Move Openings)

Moves 1, 2, 3, 4	Move 5						
	A	B	C	D	E	F	G
D1 a1 G1 a2	R	=	R	=	R	R	R
D1 a1 G1 b1	R	R	b	R	R	R	R
D1 a1 G1 c1	R	b	R	R	R	R	R
D1 a1 G1 d2	b	b	b	R	=	b	b
D1 a1 G1 e1	b	b	b	b	b	b	R
D1 a1 G1 f1	b	b	=	R	=	R	=
D1 a1 G1 g2	=	R	R	R	R	R	R
D1 b1 B2 a1	R	R	R	=	R	R	R
D1 b1 B2 b3	b	R	b	R	R	R	b
D1 b1 B2 c1	R	R	=	b	R	R	R
D1 b1 B2 d2	b	R	b	R	=	=	b
D1 b1 B2 e1	R	R	b	R	R	=	=
D1 b1 B2 f1	R	R	b	R	=	R	R
D1 b1 B2 g1	R	R	R	R	R	R	R
D1 b1 F1 e1	b	=	b	=	b	R	b
D1 b1 F1 g1	b	R	R	R	R	R	b
D1 c1 F1 a1	R	b	R	R	R	R	R
D1 c1 F1 b1	=	R	R	R	R	R	R
D1 c1 F1 c2	b	b	R	b	b	R	R
D1 c1 F1 d2	b	b	b	R	b	b	R
D1 c1 F1 e1	=	b	R	R	R	R	=
D1 c1 F1 f2	b	b	=	b	b	b	R
D1 c1 F1 g1	b	b	=	=	b	R	b
D1 c1 G1 a1	R	b	R	R	R	R	R
D1 c1 G1 b1	=	R	R	=	R	R	=
D1 c1 G1 c2	b	b	R	=	b	R	b
D1 c1 G1 d2	b	b	b	R	b	R	b
D1 c1 G1 e1	b	=	=	=	R	=	b
D1 c1 G1 f1	b	b	b	b	b	R	b
D1 c1 G1 g2	R	=	R	R	R	R	R
D1 d2 D3 a1	R	b	R	R	R	R	R
D1 d2 D3 b1	b	R	b	R	R	R	b
D1 d2 D3 c1	b	b	R	R	b	R	R
D1 d2 D3 d4	b	b	b	R	b	b	b

Table 3 (Seven-Move Openings)

Moves 1, 2, 3, 4, 5, 6	Move 7						
	A	B	C	D	E	F	G
D1 b1 B2 b3 F1 e1	b	=	b	R	b	b	b
D1 b1 B2 b3 D2 d3	R	R	b	R	R	R	R
D1 b1 B2 d2 B3 b4	b	b	b	R	R	=	b
D1 b1 B2 d2 D3 b3	b	R	b	R	R	R	b
D1 b1 B2 d2 D3 d4	b	R	b	R	=	R	b
D1 b1 B2 d2 D3 f1	b	R	=	R	=	R	=
D1 b1 F1 e1 F2 d2	b	b	b	R	b	R	b
D1 b1 F1 e1 F2 f3	b	b	b	R	b	b	b
D1 c1 F1 c2 C3 c4	b	b	b	R	=	R	R
D1 c1 F1 c2 C3 d2	b	b	b	R	b	R	b
D1 c1 F1 c2 C3 f2	b	b	b	=	R	b	R
D1 c1 F1 c2 G1 e1	b	b	b	b	b	R	b
D1 c1 F1 d2 D3 a1	R	b	R	R	R	R	R
D1 c1 F1 d2 D3 b1	R	R	R	R	R	R	R
D1 c1 F1 d2 D3 c2	b	b	R	R	b	R	R
D1 c1 F1 d2 D3 d4	b	b	R	R	b	b	R
D1 c1 F1 d2 D3 e1	R	b	R	R	R	R	R
D1 c1 F1 d2 D3 f2	b	b	R	R	b	b	R
D1 c1 F1 d2 D3 g1	b	b	R	R	b	R	b
D1 c1 F1 d2 G1 e1	b	b	b	R	b	b	b
D1 c1 F1 f2 G1 e1	b	b	R	b	R	b	b
D1 c1 F1 g1 F2 d2	b	b	b	R	b	R	b
D1 c1 F1 g1 F2 f3	b	b	b	R	b	R	b
D1 c1 G1 c2 C3 c4	b	b	b	R	b	R	b
D1 c1 G1 c2 C3 d2	b	b	b	R	b	b	b
D1 c1 G1 c2 C3 f1	b	b	=	R	b	=	b
D1 c1 G1 c2 F1 e1	b	b	b	b	b	R	b
D1 c1 G1 d2 D3 c2	b	b	R	R	b	R	b
D1 c1 G1 d2 D3 d4	b	b	b	b	b	R	=
D1 c1 G1 d2 F1 e1	b	b	b	R	b	b	b
D1 c1 G1 e1 E2 c2	b	R	R	b	R	b	b
D1 c1 G1 e1 E2 d2	b	b	b	R	b	b	b
D1 c1 G1 e1 E2 e3	b	b	=	=	R	b	b
D1 c1 G1 f1 F2 c2	b	b	=	R	b	b	=
D1 c1 G1 f1 F2 d2	b	b	b	R	b	b	b
D1 c1 G1 f1 F2 f3	b	b	b	R	b	b	b

D1 d2 D3 c1 C2 c3	b	b	R	R	b	b	b
D1 d2 D3 c1 C2 d4	b	b	R	R	b	R	b
D1 d2 D3 c1 D4 c2	b	b	b	R	b	R	R
D1 d2 D3 c1 F1 a1	R	b	R	R	R	R	R
D1 d2 D3 c1 F1 b1	R	R	R	R	R	R	R
D1 d2 D3 c1 F1 c2	b	b	R	R	b	R	R
D1 d2 D3 c1 F1 d4	b	b	R	R	b	b	R
D1 d2 D3 c1 F1 e1	R	b	R	R	R	R	R
D1 d2 D3 c1 F1 f2	b	b	R	R	b	b	R
D1 d2 D3 c1 F1 g1	b	b	R	R	b	R	b
D1 d2 D3 d4 D5 a1	=	b	R	b	R	=	=
D1 d2 D3 d4 D5 b1	b	R	b	b	=	=	b
D1 d2 D3 d4 D5 c1	b	b	R	b	b	R	b
D1 d2 D3 d4 D5 d6	b	=	R	.	R	=	b

Chapter 10

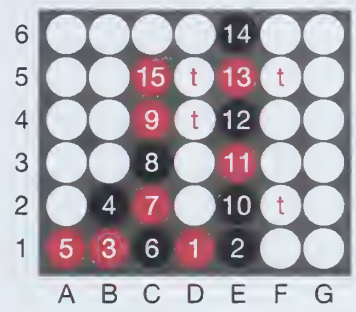
Sample Games

In this chapter I review a few example games in detail. For some games, I discuss several variations; keep in mind that the section title (for example, "Triple Threat Versus Triple Threat,") might not apply to all of the variations. (Many of these games are based on standard openings, reviewed in later chapters, such as the 5-4 Opening and the 3-4 Opening.)

Triple Threat Versus Triple Threat

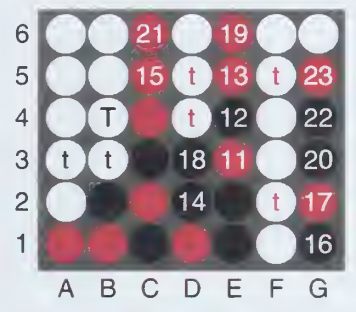
I begin by reviewing games related to the triple threat in Diagram 137 (also shown as Diagram 65 in Chapter 7).

DIAGRAM 137
Black to play (will lose)



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 E3?? e4 E5 e6?? C5 (16) __

DIAGRAM 138
Black to play, with only one move to win



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 (11) E3?? e4 E5 d2! C5 g1 G2 d3! E6 g3 C6 g4! G5 (24) __

Instead of the blunder 14 e6, in Diagram 138 Black plays 14 d2 and should win. (What is Black's next correct move? The solution is shown in Diagram 358 in Chapter 13.)

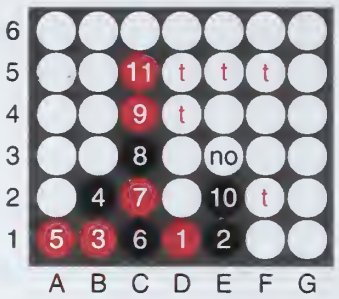
Red's move 11 D2 earlier in this opening would also be a blunder and would allow Black to announce mate in seven with a series of forcing moves! Find it. (The solution is shown in Diagram 354 in Chapter 13.)

I've shown Diagram 137 to help you understand Diagram 139, where an expert might understand that Red has the same winning endgame as shown in Diagram 137.

In Diagram 139, I've indicated the same winning threats as shown in Diagram 137, even though in each case Red so far has only one of the four necessary

disks. These six Red threats become equivalent to a triple odd-row threat, so the effect is similar to the sextuple threat I defined earlier, but this game play is actually somewhat more intricate. Red's diagonal threats at D3 and E4 are also essential (perhaps I should call this an *octuple threat*!). Because of D3, Black cannot allow Red to capture E4; this means Black will have to let Red capture both E3 and E5.

DIAGRAM 139
Black to play (will lose)



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 C5! (12) __

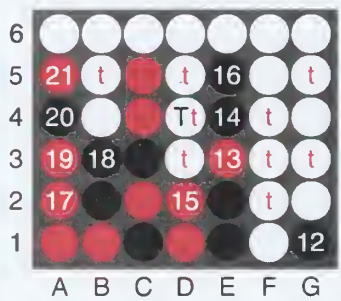
The moves in Diagram 139 were played in topnotch expert fashion. (Red can substitute 7 E2 for C2.) It is quite common for a fifth-row play to be best, as Red's 11 C5 is here, because it gives up only a comparatively valueless sixth-row point (D2 would give up d3, E3 would give up e4, and so on).

By trying to show the winning threat configuration that Red intends to achieve, I may have misled the reader into thinking that Diagram 139 is a cut-and-dried position with no interesting play left. This is far from the case. Black may get a counterthreat on the left side, and there's another Black threat that might cause Red to waste a key turn. Suppose Black plays 12 g1 in Diagram 139; can you figure out where Red must then play?

Red must answer 12 g1 with 13 G2. Otherwise, Diagram 140 or 141 might result. In Diagram 140, Red has a plethora of threats that, though not good enough to win, salvage a draw in the face of Black's major threat at d4. In Diagram 141, Red makes a second blunder (15 E5), after which Black 16 d2 threatens to play d3 and a2, but Red can block only one of those.

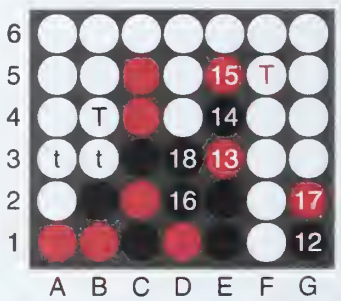
Red played 17 G2 in Diagram 141 because if Black got that cell, Black's major threat at f2 would negate Red's right-side threats. But this gave Black the chance

DIAGRAM 140
Black to play and draw



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 C5! (12) g1 E3? e4! D2! e5 A2 b3 A3! a4! A5 (22) __

DIAGRAM 141
Red to play (will lose)



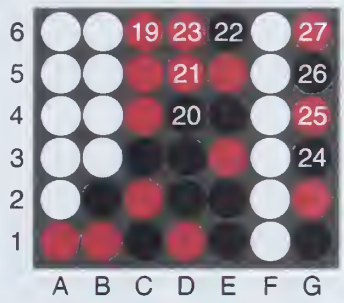
Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 C5! (12) g1 E3? e4! E5? d2! G2 d3! (19) __

to play 18 d3 and establish a winning threat configuration on the left side. (Black's odd-row threat at a3 neutralizes Red's future major odd-row threat at F5, so Black wins the endgame at b4.)

From Diagram 141, the game might continue as shown in Diagram 142 or 143. In each case, where must Black play next?

DIAGRAM 142

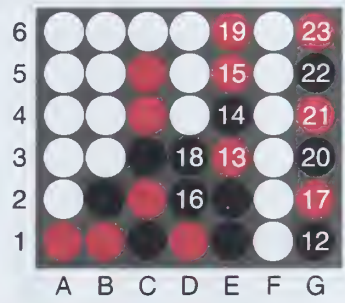
Black to play, with only one move to win



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 C5! g1 E3? e4! E5? d2! G2 d3! (19) C6 d4 D5 e6 D6 g3 G4 g5 G6 (28) __

DIAGRAM 143

Black to play, with only one move to win



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 C5! (12) g1 E3? e4! E5? d2! G2 d3! E6 g3 G4 g5 G6 (24) __

I'm sure the reader knew that Black had to play 28 f1 in Diagram 142, since alternatives sacrifice Black's threats. But since column F has an even number of empty cells in Diagram 142, it might seem that play there just postpones the inevitable; Black will be forced to play a2 anyway, after column F fills up. But the play in column F *is* essential because it is Red who suffers zugzwang and who must let Black capture Red's threat cell F5. Once Black accomplishes this, Black no longer needs an odd-row threat and is happy to play a2.

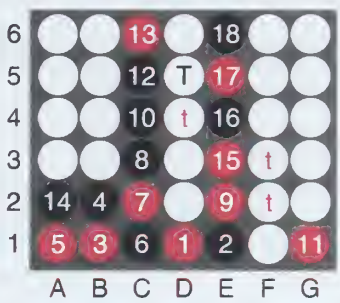
In Diagram 143, Black 24 c6 is the winning move that places Red in zugzwang, forced to give up part of a triple threat or to let Black win quickly. Perhaps one might analyze this position as having no less than *six* odd-threated columns; Black's left-side triple threat and Red's right-side triple threat each count as three odd-threated columns.

Diagram 144 shows another instructive variation, where Red takes 9 E2 instead of C4. The game can then develop in various ways; Diagram 144 is just one example.

Again Red's intended threats are shown with the letter *t*. The threat at F3 isn't developed yet, but Black can't stop it. Similarly, a Black major threat at d5 is shown;

DIAGRAM 144

Red to play and win



Moves: D1! e1 B1 b2 A1! c1 C2 c3 E2 c4 G1 c5 C6! a2 E3 e4 E5! e6 (19) __

Black can establish that whenever Black chooses to play b3. (Red doesn't want to take B3 and see B3 b4 D2 d3.) The Red configuration may not seem to fulfill victory requirements, given Black's counterthreat, but Red will end up getting a winning adjacent threat pair in the endgame. Red's 11 G1 (for which C5 could be substituted successfully) may seem odd, but experts understand the special value of the corner diagonals, such as G1-F2-C3-D4.

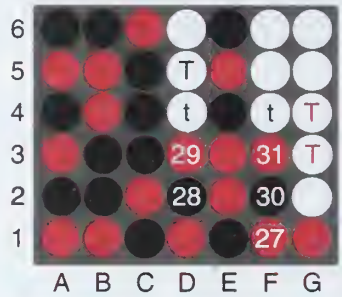
The game might continue as shown in Diagram 145. Black is now in zugzwang. If Black plays 34 d4 D5, Red wins with a major odd-row threat at F3. If Black plays 34 f1 F2, Red wins immediately with double check. (Red 19 A2 also succeeds in Diagram 144, with a possible continuation shown in Diagram 297; see Chapter 13.)

DIAGRAM 145
Black to play (will lose)



Moves: D1! e1 B1 b2 A1! c1 C2 c3 E2 c4 G1 c5 C6! a2 E3 e4 E5! e6 (19) A3 a4 A5 b3 B4! a6 B5 b6 G2! d2 G3 g4 D3! g5 G6 (34) __

DIAGRAM 146
Black to play and win

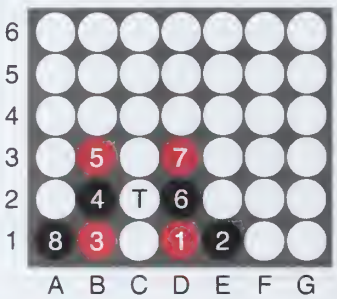


Moves: D1! e1 B1 b2 A1! c1 C2 c3 E2 c4 G1 c5 C6! a2 E3 e4 E5! e6 A3 a4 A5 b3 B4! a6 B5 b6 (27) F1? d2! D3 f2! F3? (32) __

Diagram 146 shows a continuation in which Red blunders with 27 F1, and then blunders again with 31 F3. Red gets an adjacent threat pair, G3-G4, but it's Black's move, and 32 d4 (or f4) wins quickly.

Red's early play 5 A1 was essential in the games we've just considered. If Red omits this play, the game might go as shown in Diagram 147. There, Black will play a2 or e2 (whichever Red doesn't take) at Black's next turn and get a major threat at C2; this will effectively lock up the game. Black's moves at 6 d2 and 8 a1 were key; the moves 6 b4 (and 8 b4) would draw; other Black plays would lose.

DIAGRAM 147
Red to play (will lose)



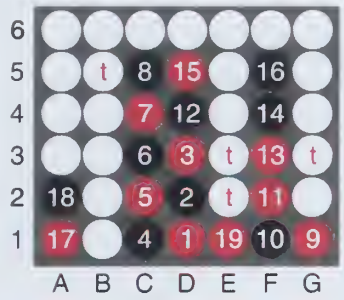
Moves: D1! e1 B1 b2 B3?? d2! D3 a1! (9) __

Missed Opportunities

Diagram 148 shows an example of a triple odd-row threat where a fourth threat is actually needed: E2, where neither Red nor Black can afford to play. Red needn't worry about Black getting a5 to undercut the B5 threat

DIAGRAM 148

Black to play (will lose)



Moves: D1! d2 D3! c1 C2 c3 C4 c5 G1? f1! F2 d4! F3? f4 D5 f5?? A1! a2 E1! (20) __

with Black threat b4—if Black gets a5, that means that Red got A4 and a major threat at B3. In other words, Red's partial diagonal C4-D3 is undercut by Black's partial diagonal c3-d2, but that in turn is undercut by the Red partial diagonal at C2-D1. Finding the bottommost such diagonal threat is an essential part of most endgame analyses.

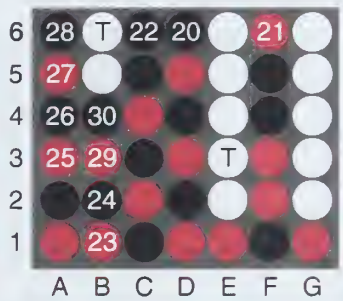
There are three errors in the game shown in Diagram 148. Red should have played 9 E1 to win. Black should often hesitate before playing d4, but of course 12 d4 was mandatory here to prevent a Red major odd-row threat. Red should have responded with 13 D5 (Red's only way to draw). Later, Black 16 a1 was the correct play to win; all other moves lose, including Black's actual play 16 f5

The position still has interesting play. If Black plays 20 c6 in Diagram 148, Red must respond at D6, and vice versa. Otherwise, the position in Diagram 149 may develop, where Red's minor threat B5 is overcut by Black's winning major threat at b6.

It would be inconsistent for Black to substitute 8 d4 as shown in Diagram 150. Red can then play 9 D5 and reach the same position as the 5-4 opening (although the moves would be in a different order), but will prefer to play 9 F1, building a threat at F2 and winning easily, as shown in Diagram 150, for example.

DIAGRAM 149

Red to play (will lose)

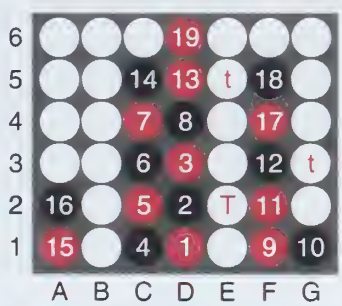


Moves: D1! d2 D3! c1 C2 c3 C4 c5 G1? f1! F2 d4! F3? f4 D5 f5?? A1! a2 E1! (20) d6 F6?? c6! B1 b2! A3 a4! A5 a6! B3 b4! (31) __

Diagram 151 shows another variation of the 3-4 opening.

DIAGRAM 150

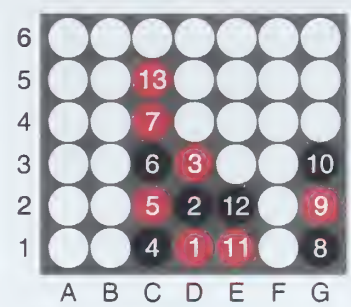
Black to play (will lose)



Moves: D1! d2 D3! c1 C2 c3 C4 d4 F1 g1 F2 f3 D5! c5 A1 a2 F4 f5 D6 (20) __

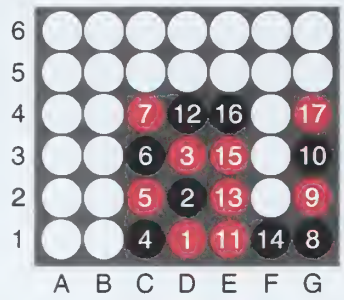
DIAGRAM 151

Black to play (will lose)



Moves: D1! d2 D3! c1 C2 c3 C4 g1 G2! g3 E1! e2 C5! (14) __

DIAGRAM 152
Black to play (will lose)

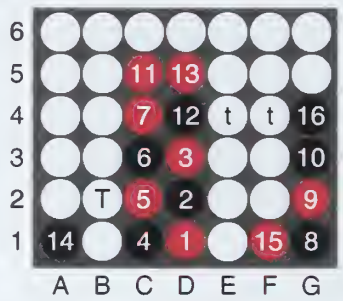


Moves: D1! d2 D3! c1 C2 c3 C4 g1 G2! g3 E1! d4 E2! f1 E3! e4 G4 (18) __

As indicated in the move list with exclamation points, several of Red's moves are forced. The play 13 C5 may not be obvious—a way to understand this move is that it gives up nothing (c6 is not an important cell, while 13 D4 or E3 would give Black a prime location [d5 or e4] for Black's disk 14). Diagram 152 shows another variation where Red plays well.

The play 11 E1 (threatening E2 with an eventual win at B5) was a key move. If Red omits it, Black will have a chance to show his or her skill, as shown in Diagram 153, where Black's even-row threats will lead to victory.

DIAGRAM 153
Red to play (will lose)



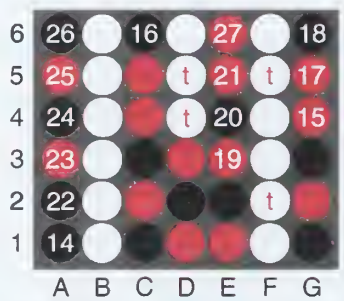
Moves: D1! d2 D3! c1 C2 c3 C4 g1 G2! g3 C5?? d4! D5 a1! F1 g4! (17) __

D4 Is Often Poison

Return to Diagram 151; what should Black's plan be? (Of course, Black will lose no matter what his or her plan if Red continues to play perfectly, but if Red is sure to be *perfect*, Black would have given up in response to 1 D1! Black does have a plan likely to succeed against all but the cleverest Red.)

Black can't very well take a good cell such as d4 or e3, since Red will get the even better cell above it (D5 or E4). Since Black is forced to play on the periphery, he or she should make the most of it. If Black can grab a1 and g4, followed by d4, Black will have both sides of the board locked up with even-row threats. Of course, Red will not permit this. In fact, Red must catch on to this plan at once (to avoid being forced to take D4 later) and play A1 (or B1) in response to g4 and specifically G4 in response to a1.

DIAGRAM 154
Black to play (will lose)



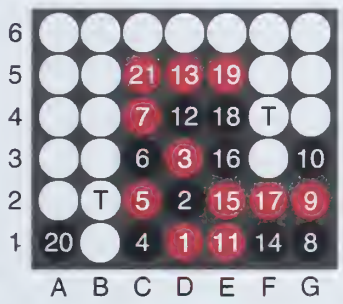
Moves: D1! d2 D3! c1 C2 c3 C4 g1 G2! g3 E1! e2 C5! (14) a1 G4! c6 G5 g6 E3! e4 E5! a2 A3 a4 A5 a6 E6! (28) __

Diagram 154 shows a game where Red figures out Black's plan in time and wins with clever play. Black is in zugzwang and will next see his or her forlorn threats at b2-d4 disappear.

I call this instructive game "D4 Is Poison" because Red cannot afford to play D4 once Black gets g1, and (with an exception) if Black ever plays there, Red answers with D5, resulting in an easy win. (The exception is shown in Diagram 152; if Red tries 13 D5 there, Diagram 155 might result, with Black's major even-row threats in control.)

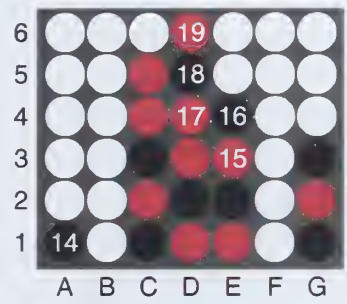
In Diagram 156, neither player has any threats. Red has salvaged a draw, but that's all.

DIAGRAM 155
Black to play and win



Moves: D1! d2 D3! c1 C2 c3 C4 g1 G2! g3 E1! d4 D5? f1! E2? e3 F2 e4! E5 a1! C5 (22) __

DIAGRAM 156
Black to play and draw



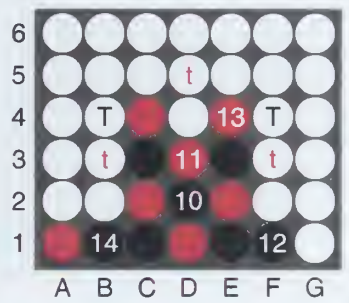
Moves: D1! d2 D3! c1 C2 c3 C4 g1 G2! g3 E1! e2 C5! (14) a1 E3? e4! D4! d5! D6 (20) __

The Computer's Game

The game shown in Diagram 157 seems to start much like that in Diagram 269 (Chapter 13). Red substitutes 5 E2 for the 5 C2 move in that game, but this turns out not to give Red victory. By the way, if Black substitutes 10 f1 in Diagram 157, Red has a mate in four—do you see it?

Diagram 157 is very difficult to analyze, but it strongly favors Black, who has the upper part of the board locked up with major even-row threats at b4 and f4. Either of these threats is enough for Black victory if Red cannot counterattack. Of Red's seven possible moves in Diagram 157, six lose and only one leads to a draw with perfect play. If you can guess Red's perfect (drawing) move in this *extremely* difficult diagram, give yourself a *big* pat on the back. (I call this "The Computer's Game" because I once thought only a computer would be able to guess Red's best move, but these days there are some *very* smart and practiced players who may not find this so difficult!)

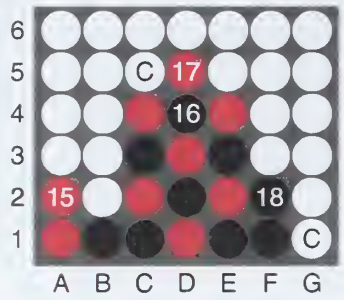
DIAGRAM 157
Red to play, with only one move to draw



Moves: D1! e1 A1 c1 E2? e3 C2 c3 C4 (10) d2! D3! f1! E4! b1! (15) __

DIAGRAM 158

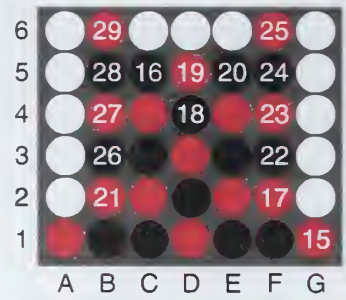
Red to play (will lose)



Moves: D1! e1 A1 c1 E2? e3 C2 c3 C4 d2! D3! f1! E4! b1! (15) A2? d4 D5 f2 (19) __

DIAGRAM 159

Black to play and draw



Moves: D1! e1 A1 c1 E2? e3 C2 c3 C4 d2! D3! f1! E4! b1! (15) G1! c5 F2! d4 D5! e5 B2! f3 F4! f5! F6! b3 B4! b5! B6! (30) __

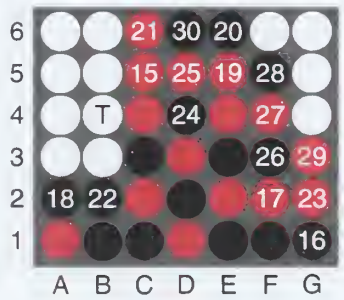
To achieve a draw in the face of Black's major even-row threats, it is essential that Red threaten a winning counterattack. Red's major threats at B5 and F5 are worthless since they're undercut. Some possible future minor odd-row diagonal threats for Red are marked in Diagram 157, but it will take a lot of work for Red to develop these. In particular, D5 is essential, but because of a variation, shown in Diagram 158, Red may be unable to afford to take D5 even when Black plays d4.

To prevent Black from getting D5 or the double check shown in Diagram 158, Red must play on the C5-F2-G1 diagonal, but not at F2, which lets Black win easily. The choices for Red in Diagram 157 are therefore 15 C5 or 15 G1. Before considering either move, note that (if Black is a perfect player) Red can't hope for better than a draw, since if Black is worried about the counterattack, Black can force a draw with ease, as shown in Diagram 159, for example.

Now, let's suppose that Red plays 15 C5 in Diagram 157. Black, if he or she is clever, can now win with the plays shown in Diagram 160. Black's move 18 a2 was

DIAGRAM 160

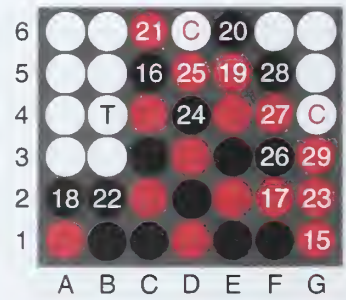
Red to play (will lose)



Moves: D1! e1 A1 c1 E2? e3 C2 c3 C4 d2! D3! f1! E4! b1! (15) C5? g1 F2 a2! E5 e6! C6 b2! G2 d4! D5 f3! F4 f5! G3 d6! (31) __

DIAGRAM 161

Black to play (will lose)



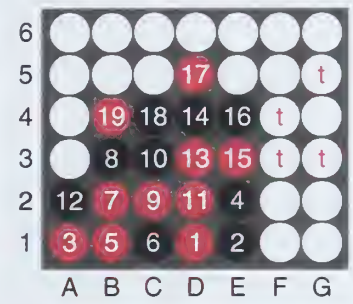
Moves: D1! e1 A1 c1 E2? e3 C2 c3 C4 d2! D3! f1! E4! b1! (15) G1! c5 F2! a2? E5! e6 C6 b2 G2 d4 D5 f3 F4! f5 G3! (30) __

especially smart and certainly far from obvious. The move 24 d4 was also essential to avoid variations such as 24 a3 A4 b3 B4 d4 B5.

In Diagram 160, Red made some forcing moves, but this attack came to nothing. If Red substitutes 15 G1 for C5 and the game continues as shown in Diagram 160, Diagram 161 develops.

In Diagram 161, Red wins with double check! The disk at G1 turned out to have special significance, namely, the possible four-in-row G1-G2-G3-G4. Be on the lookout for such vertical pushes in columns A and G. In Diagram 160, 18 a2 was a clever move and the only way for Black to win. In Diagram 161, however, 18 a2 was a mistake—Black should instead settle for a draw.

DIAGRAM 162
Black to play (will lose)



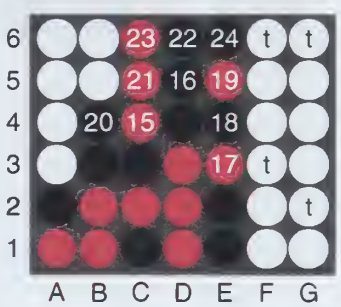
Moves: D1! e1 A1 e2 B1! c1 B2! b3 C2! c3 D2! a2 D3! d4 E3 e4 D5! c4 B4! (20) __

My Favorite Game

Diagram 162 is a lovely game worth memorizing. With plays 8 and 10, Black prevents Red from playing on the third row; Red would win easily if Black didn't defend this way. Red gets two disks on the third row, and because of the adjacent minor threat pair F3-F4, those two disks are enough for a winning minor odd-row threat combination. (The threat pair F3-F4 means that either G3 or G5 will become a major threat.) The play on the third row is both instructive and surprising; that's why I call these moves "My Favorite Game."

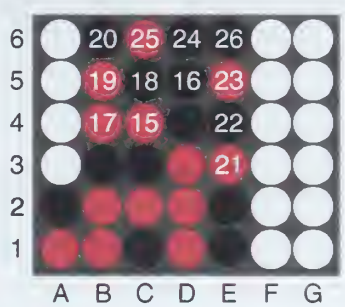
I consider Diagram 162 to be "lovely" because, after 14 d4, Red has only two disks on the entire board that can possibly contribute to a four-in-row, while Black seems to have powerful possibilities in the upper left. Yet Red has the strategic victory, with a sextuple threat. (The sextuple threat included four threats still seen in the

DIAGRAM 163
Red to play (will lose)



Moves: D1! e1 A1 e2 B1! c1 B2! b3 C2! c3 D2! a2 D3! d4 (15) C4 d5 E3?? e4! E5 b4 C5 d6! C6 e6 (25) __

DIAGRAM 164
Red to play (will lose)



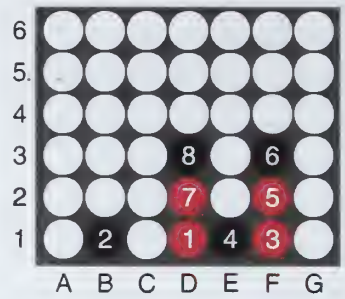
Moves: D1! e1 A1 e2 B1! c1 B2! b3 C2! c3 D2! a2 D3! d4 (15) C4 d5 B4?? c5! B5 b6! E3 e4! E5 d6! C6 e6! (27) __

diagram, as well as E3 and E4, which are now occupied.) Red's play 15 E3 was not essential but was the simplest way to settle the game. If Red neglects to make this simplifying move now, it will be quite easy for Red to go wrong later, as shown in Diagram 163.

Red's mistake in Diagram 163 was to play E3 and let Black get both d5 and e4. Another way for Red to err is shown in Diagram 164.

In Diagram 164, 17 B4 was a mistake because it eventually allowed Black to force Red to play E3 when Red didn't want to. All in all, Red should just play 15 E3 (or 15 D5) and avoid these unnecessary complications.

DIAGRAM 165
Red to play and win



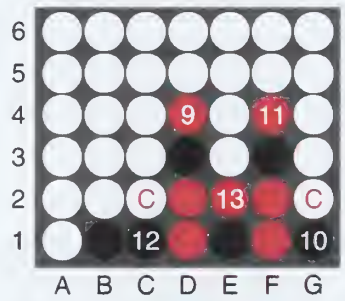
Moves: D1! b1 F1 e1 F2! f3 D2! d3 (9) __

Out of Nowhere

The game in Diagram 165 seems to contradict our basic principles. Red has allowed Black to get two disks on the third row, while Red hasn't a single disk there. Yet Red has played excellently and should win this game easily. In addition to future prospects, Red threatens to play on the C1-F4 or G1-D4 diagonals and establish E3 as a threat.

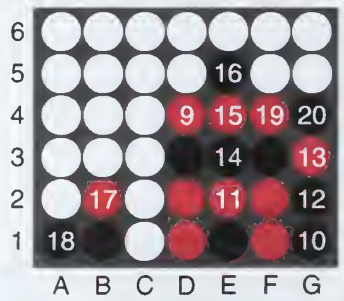
In fact, if Black refuses to allow Red to establish the E3 threat, the game ends quickly, as shown in Diagram 166 ...

DIAGRAM 166
Black to play (will lose)



Moves: D1! b1 F1 e1 F2! f3 D2!
d3 (9) D4 g1 F4 c1 E2 (14) __

DIAGRAM 167
Red to play, with only one move to win



Moves: D1! b1 F1 e1 F2! f3 D2! d3 (9) D4
g1 E2 g2 G3! e3 E4! e5 B2 a1 F4 g4 (21) __

... with Red winning easily with two immediate threats.

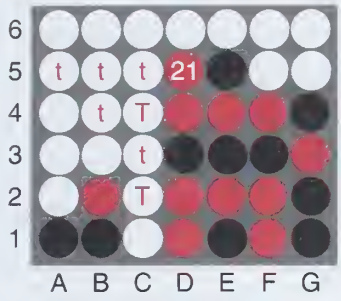
Surely Red should take the easy win just demonstrated, but it may be instructive to look at a slower way for Red to win. In Diagram 167, Red has built only even-row threats (and no matter how many of those Red gets, he or she can't win). Can Red eventually build the odd-row threat Red needs? Where should Red play next

in Diagram 167? With good visualization, you may be able to find Red's winning threat after Red plays one more disk.

The solution is shown in Diagram 168.

Did you get this one right? Red will obviously get C3 and C5 when column C is played and will certainly get either B4 or B5. Therefore, A5 will become Red's major odd-row threat no matter what happens! Almost all of Red's disks are on even-row cells, yet Red obtains a fifth-row horizontal win from out of nowhere.

DIAGRAM 168
Black to play (will lose)



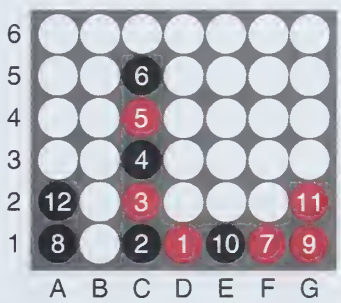
Moves: D1! b1 F1 e1 F2! f3 D2! d3 D4 g1 E2 g2 G3! e3 E4! e5 B2 a1 F4 g4 (21) D5! (22) __

Threats Everywhere

Starting from Diagram 169, Red has two moves that lead to mate in seven, four other winning moves, and one bad move that lets Black salvage a draw. See whether you can guess the moves that lead to an early four-in-row, and guess the weakest move.

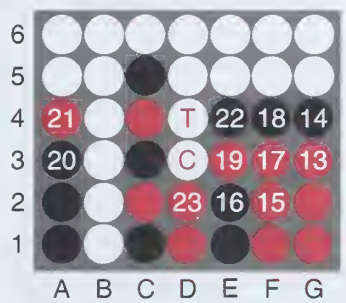
The quick win begins with F2 or G3. One variation is shown in Diagram 170.

DIAGRAM 169
Red to play and win



Moves: D1! c1 C2? c3! C4 c5? F1? a1? G1 e1 G2 a2 (13) __

DIAGRAM 170
Black to play (will lose)



Moves: D1! c1 C2? c3! C4 c5? F1? a1? G1 e1 G2 a2 (13) G3 g4 F2 e2 F3 f4 E3! a3 A4! e4 D2 (24) __

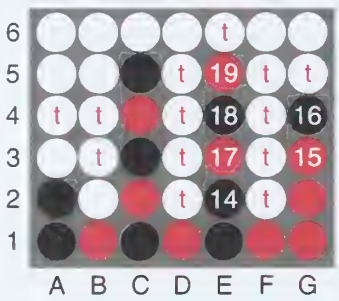
The bad move from Diagram 169 is 13 B1, after which Black will have to play very well to claim a draw, and Red will have to play very well to take advantage of any Black weakness. If you can guess most of the follow-up moves, you are an expert!

Diagram 171 shows the continuation of play after 13 B1. Black's 14 e2 and 18 e4 prevent Red from getting a major threat at D3. Red ends up with no major

threats at all, but, as indicated in Diagram 171, Red has a plethora of minor threats that might develop into victory. Black has two moves that can salvage a draw. Take full credit if you find either one.

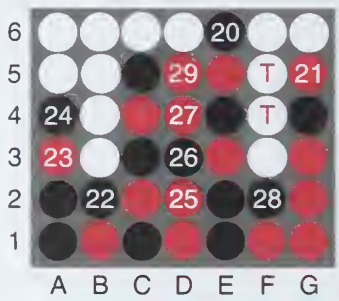
Before showing Black's successful moves, let's look at Black's third-best move. This result is shown in Diagram 172. Observe how Red uses his or her threats in the lower right side as a springboard to build a winning threat in the upper right side. (Do you see what happens if Red omits 23 A3?)

DIAGRAM 171
Black to play and draw



Moves: D1! c1 C2? c3! C4 c5? F1? a1? G1 e1 G2 a2 B1? (14) e2! G3 g4! E3 e4! E5 (20) __

DIAGRAM 172
Black to play (will lose)

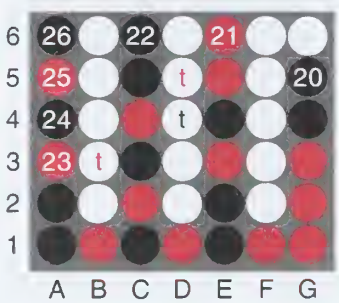


Moves: D1! c1 C2? c3! C4 c5? F1? a1? G1 e1 G2 a2 B1? e2! G3 g4! E3 e4! E5 (20) e6? G5 b2 A3! a4 D2! d3 D4! f2 D5! (30) __

Return to Diagram 171; Black must start with 20 g5 or 20 b2. In Diagram 173, I look at 20 g5.

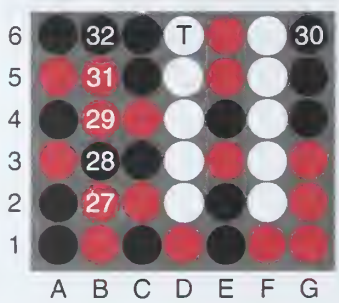
This position will be drawn with perfect play, but I've marked two Red minor odd-row threats that are relevant though undercut by Black's eventual threat at d4. If Red grabs B2, thinking to prevent Black from playing there and getting a threat at d4, Black ends up with an easy win, shown in Diagram 174.

DIAGRAM 173
Red to play and draw



Moves: D1! c1 C2? c3! C4 c5? F1? a1? G1 e1 G2 a2 B1? e2! G3 g4! E3 e4! E5 (20) g5 E6 c6 A3 a4! A5 a6 (27) __

DIAGRAM 174
Red to play (will lose)



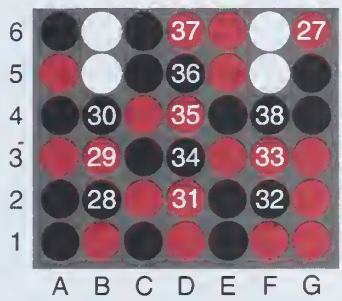
Moves: D1! c1 C2? c3! C4 c5? F1? a1? G1 e1 G2 a2 B1? e2! G3 g4! E3 e4! E5 g5 E6 c6 A3 a4! A5 a6 (27) B2? b3! B4 g6! B5 b6! (33) __

Now Red has no useful threat; Black will win in the sixth row. Why did this happen? Red's diagonal threats B3-D5, though too feeble to win, are the undercutting threat Red needs to prevent Black from winning. Red wants to force Black to block at d5 eventually, then take D6 and kill Black's sixth-row threat. This gameplay is shown in Diagram 175.

In Diagram 171, Black may play 20 b2. If the game then continues as shown in Diagram 176, where should Black play next? As in the variations we've just seen, Red has plenty of minor threats in columns D and F; Black is not out of the woods yet.

DIAGRAM 175

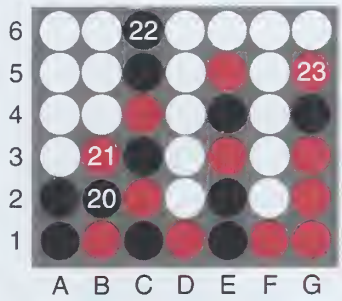
Red to play and draw



Moves: D1! c1 C2? c3! C4 c5? F1? a1? G1 e1 G2 a2 B1? e2! G3 g4! E3 e4! E5 g5 E6 c6 A3 a4! A5 a6 (27) G6 b2! B3! b4 D2! f2 F3! d3! D4! d5! D6 f4! (39) __

DIAGRAM 176

Black to play, with only one move to draw



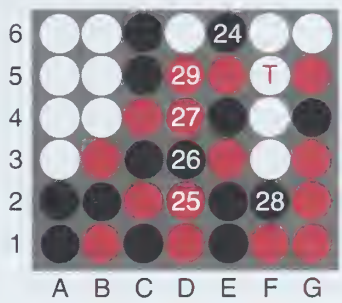
Moves: D1! c1 C2? c3! C4 c5? F1? a1? G1 e1 G2 a2 B1? e2! G3 g4! E3 e4! E5 (20) b2 B3! c6 G5 (24) __

First let's look at an unsuccessful effort. In Diagram 177 Red drives up column D and establishes a winning threat at F5.

This win by Red relied on a plethora of threats in columns D and F, including the diagonal threat D2-F4. That's the key to Black's defense; it may seem like the least of Black's worries, but it's the only threat Black can kill. Black must play 24 d2 in Diagram 176 to get a draw, as shown in Diagram 178 on the next page.

DIAGRAM 177

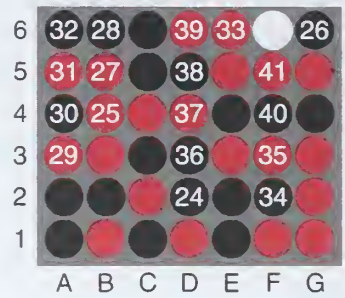
Black to play (will lose)



Moves: D1! c1 C2? c3! C4 c5? F1? a1? G1 e1 G2 a2 B1? e2! G3 g4! E3 e4! E5 b2 B3! c6 G5 (24) e6? D2 d3 D4! f2 D5 (30) __

DIAGRAM 178

Black to play and draw



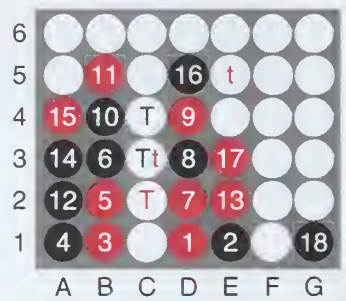
Moves: D1! c1 C2? c3! C4 c5? F1? a1? G1 e1 G2 a2 B1? e2! G3 g4! E3 e4! E5 b2 B3! c6 G5 (24) d2! B4! g6 B5 b6! A3 a4! A5 a6 E6! f2! F3! d3! D4! d5! D6! f4 F5 (42) __

Threat in Reserve

The opening in Diagram 179 was rather well played by both sides. Red has a winning triple threat but must be wary of Black's adjacent threat pair, even though it is undercut. For example, 19 B6 (building a C5 threat) would be the easy way for Red to win now, if not for Black's c4 threat.

DIAGRAM 179

Red to play, with only one move to win

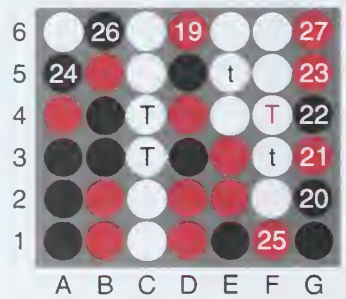


Moves: D1! e1 B1 a1 B2! b3 D2 d3 D4 b4 B5 a2 E2 a3 A4! d5 E3 g1 (19) __

Red should play 19 G2 in Diagram 179 to prevent Black threats on the d5-g2 diagonal. But we'll suppose Red overlooks this need and wastes his or her next turn as shown in Diagram 180. After this, Black's g4 was mandatory to prevent a Red major threat at F3, but otherwise the players just fill in neutral cells, waiting for zugzwang. Neither Red nor Black can afford to play E4; if Black captures e5, he or she kills Red's triple threat, while a Red disk at E5 gives quick victory with the adjacent threats C2-C3. I've marked several threats in Diagram 180, but, as we will see, the winning Black threat is a different one altogether, still almost invisible in this diagram.

DIAGRAM 180

Black to play, with only one move to win

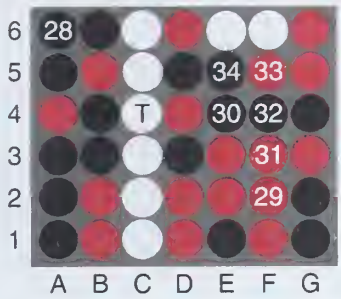


Moves: D1! e1 B1 a1 B2! b3 D2 d3 D4 b4 B5 a2 E2 a3 A4 d5 E3 g1 (19) D6?? g2! G3 g4! G5 a5 F1 b6 G6 (28) __

Black's winning move in Diagram 180 is of course to take the last neutral cell with 28 a6; the continuation is shown in Diagram 181.

I call this game "Threat in Reserve" because two different threats involving e4 are the keys to Black's victory (specifically, the threats Red blocked with E1 and E5,) yet Black loses quickly if he or she places a disk on e4 prematurely.

DIAGRAM 181
Red to play (will lose)



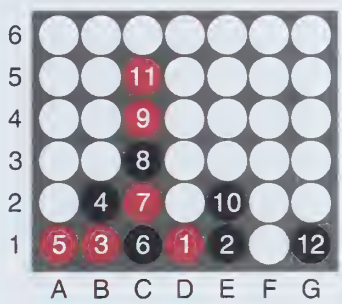
Moves: D1! e1 B1 a1 B2! b3 D2 d3 D4 b4 B5 a2 E2 a3 A4 d5 E3 g1 D6?? g2! G3 g4! G5 a5 F1 b6 G6 (28) a6! F2 e4! F3 f4! F5 e5! (35) __

Don't Play in Column D; Win There!

The cells D3 and D5 are usually the best ones on the board. Earlier we saw that "D4 Is Often Poison," but at least D4 occupies an important long diagonal. The cell D2 is even more often a bad cell to play than D4. Thus, in Diagram 182, Red wants to eliminate Black's second-row horizontal threats immediately but mustn't do so by playing D2.

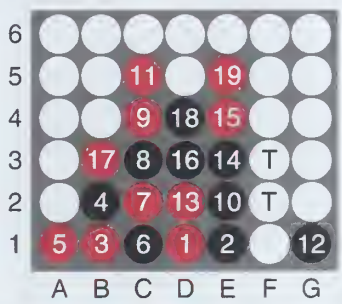
As shown in Diagram 183, Black can win quickly if Red plays 13 D2 in Diagram 182.

DIAGRAM 182
Red to play, with only one move to win



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 C5! g1 (13) __

DIAGRAM 183
Black to play and win



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 C5! g1 D2?? e3! E4 d3! B3 d4! E5 (20) __

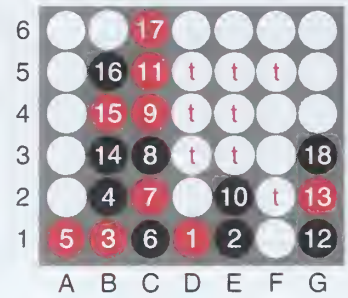
Red should have played 13 G2. This might lead to the position shown in Diagram 184, although Red's 17 C6 was a weak move. I've depicted Red's key future threats: the fifth row and two of the central diagonals. It appears that Red can play E3, followed by either E4 or E5, in either case getting a (sort of) triple threat, but Red's job isn't quite that easy.

If Red hurries to get a disk on the fifth row, as shown in Diagram 185, Black gets an undercutting threat on the fourth row.

Instead, Red must reply to 18 g3 with 19 G4 to block Black's fourth-row threat. Where should Red play next if Black answers 19 G4 with 20 a2? Red must not allow Black to play a3 and establish a major threat at d3, as shown in Diagram

DIAGRAM 184

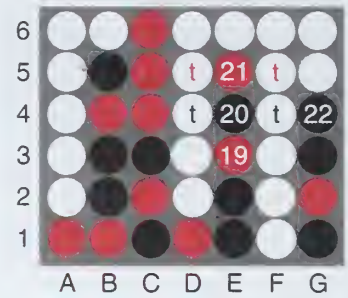
Red to play, with only one move to win



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 C5! g1 G2! b3 B4! b5 C6 g3 (19) __

DIAGRAM 185

Red to play (will lose)



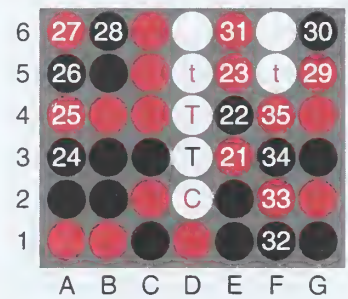
Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 C5! g1 G2! b3 B4! b5 C6 g3 (19) E3?? e4 E5 g4! (23) __

186, where Red's threats no longer work. Instead, Red must block with 21 A3 immediately. Diagram 187 shows one possible continuation.

In Diagram 187, for the first time in the entire game, Red can play safely at D2.

DIAGRAM 186

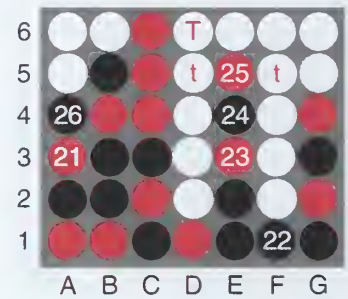
Black to play, with only one move to draw



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 C5! g1 G2! b3 B4! b5 C6 g3 G4! a2 (21) E3? e4! E5! a3! A4 a5 A6 b6 G5 g6 E6 f1! F2! f3! F4! (36) __

DIAGRAM 187

Red to play and win



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 C5! g1 G2! b3 B4! b5 C6 g3 G4! a2 (21) A3! f1 E3 e4 E5! a4 (27) __

Chapter 11

Rule Variations

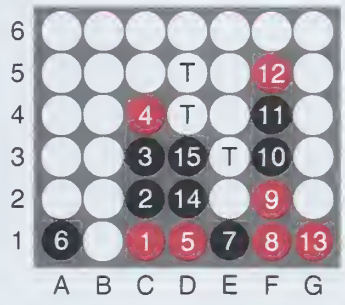
You may start to lose interest in CONNECT 4 playing by the standard rules if the same positions keep recurring. Fortunately, there are other games that can be played with the standard CONNECT 4 game set. A few of these alternate games are described here.

Two Disks at a Time

The rules of this variant are the same as those for regular CONNECT 4, *except* that a player plays *two* disks at each turn instead of one. The first player to get four-in-row wins. (To avoid giving Red a big advantage, only one disk is played at the very first turn.)

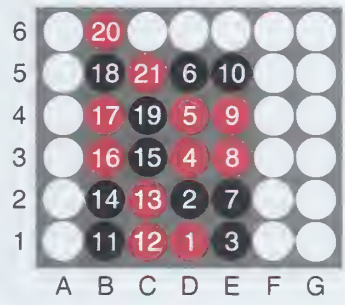
This game can always be won by Black, but it isn't easy, and Red will often win in practice. Diagram 188 shows a game where Black plays very precisely to win; two threats (d4-d5 and e2-e3) cannot both be stopped. (The two-disk games shown in this section are *not* annotated with exclamation points and question marks.)

DIAGRAM 188
Red to play



Moves: C1 c2 c3 C4 D1 a1 e1 F1
F2 f3 f4 F5 G1 d2 d3 (16) __

DIAGRAM 189
Red has won



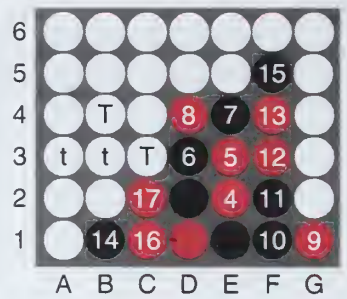
Moves: D1 d2 e1 D3 D4 d5 e2 E3 E4 e5
b1 C1 C2 b2 c3 B3 B4 b5 c4 B6 C5

To show that victory is by no means automatic for Black, consider Diagram 189, where Black's only mistake was to start with d2 e1 instead of d2 d3.

Diagram 190 shows a variation where Black errs as mentioned previously but wins anyway, as Red fails to follow through properly.

In the author's opinion, this game isn't as much fun as regular CONNECT 4; four-in-row is rather easy to get, and the game ends too quickly.

DIAGRAM 190
Black to play



Moves: D1 d2 e1 (4) E2 E3 d3 e4 D4
G1 f1 f2 F3 F4 b1 f5 C1 C2 (18) __

Five in a Row

This game is the same as regular CONNECT 4, except that it takes five in a row to win. In the author's opinion, this game isn't much fun; it's too hard to win, so draws become common.

Two Disks at a Time, Five in a Row

By combining the two rule variations above, a very good game arises. Playing two disks at a time makes the game fast-moving and often exciting. (Remember that only a single piece is played at the *very first* move.) When both players play carefully, the game is likely to end in a draw, but, as far as I know, there is as yet no proof or computer solution for this variation.

Continue Play Until the Board Is Filled

In this variation, the game doesn't end when one player gets four-in-row. Instead, play continues until the board is filled. Score 1 point for four-in-row, 2 points for five-in-row, 3 points for six-in-row, and 4 points for seven-in-row; whoever has the most points wins. (Most players find this variation to be less exciting than ordinary CONNECT 4.)

Two-by-Two Square Also Wins

A two-by-two square of same-colored disks wins, along with, as usual, any four-in-row. (Or, to negate first-move advantage, players can let the two-by-two square win only for Black.) As far as I know, this variant is unsolved.

Red Handicap

Since First Player (Red) has the advantage, players can try to make the game fair by imposing a handicap on Red. There are various ways to do this; many of them have the added advantage that the standard openings are no longer effective, so players will avoid the always-the-same-opening rut.

Here are six possible Red handicaps. For the first three Red can still force victory with perfect play. For the last three the game will be drawn with perfect play.

- Any four-in-row involving A6 does not count as a win for Red (but would count for Black). This is called *Blue A6* because, if you have a second set with disks of a different color, say, blue, then you'll have Red use a blue disk when he plays at A6—a four-in-row using the blue disk won't even look like a four-in-row. Red can win despite this handicap if he plays perfectly. In this variant, A6 is called a *blue cell* for Red.
- A horizontal sixth-row four-in-row does not count as a win for Red (but would count for Black). Diagram 192 shows a familiar opening that no longer wins for Red if horizontal sixth-row four-in-rows don't win.
- Both A6 and G6 are blue cells.
- Any four-in-row using A2 does not count as a win for Red (but would count for Black). (With this handicap, Black can draw after 1 D1 e1.)
- Certain initial moves are required or forbidden (see the next section, "Opening Lottery").
- D6 is a blue cell. (Black draws with DdDdDcFg or DdDdDcCc [the columns in the order they were played].)

Since Red just barely can force victory when Black plays perfectly, it may be surprising that Red can still win when A6 and G6 are blue, or when all Red's sixth-row horizontal wins are excluded. In fact, I've verified with computer assistance that Red can win with as many as *nine* cells treated as blue simultaneously! (Red can force victory only when the nine blue cells are A6, B6, C6, E6, F6, G6, B4, F4, and either A5 or G5.)

Opening Lottery

You and your partner may play the same openings over and over again, perhaps DbB or DcG. One way to regain interest quickly is to switch openings. Do as they do in checkers tournaments: Use dice, coins, or lottery to choose a forced opening. In Chapter 9 you can find which openings lead, theoretically, to a draw or to Red or Black victory, or you can fashion a set of six three-disk openings from the following lists.

Choose among these six *drawing* openings by throwing a die.

1. B??b?E!
2. B??d?D
3. C?eC
4. C?fF
5. D!cC?
6. D!bE?

Alternatively, choose among these six openings that *win for Red*.

1. A??a??F
2. B??f??E
3. C?b?F!
4. C?g?C
5. C?g?F
6. D!aG

Alternatively, choose among these six openings that *win for Black*.

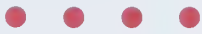
1. B??d?G?
2. B??e??E??
3. C?a?G??
4. D!cA??
5. D!dC??
6. D!gF??

PopOut

In Chapter 14 I cover variations that can be played on Hasbro's new CONNECT 4 set with the PopOut bar.

Chapter 12

Central-Column Openings



In this chapter and the next, I review some well-played openings of CONNECT 4 games. In most of the example games, Red ends up winning, but that doesn't mean opening study is of no interest to Black. As we will see, Red will need to play *very* precisely to prevail; Black must train him- or herself to watch for a misplay by Red, however slight, and then take advantage.

Openings can be divided broadly into two categories: *central-column openings*, in which players first play up column D, and *first-row openings*, in which three or more of the early disks are played in the first row.

In central-column openings, many issues are settled immediately; after the initial moves D1 d2 D3 d4, only Red has a chance for third-row horizontal threats and only Black can create threats on the important diagonals that pass through d4. In first-row openings, in contrast, players jockey for long-term position before these matters are resolved. For this reason, first-row openings lead to greater variety and are more difficult to master than central-column openings.

5-4 Opening

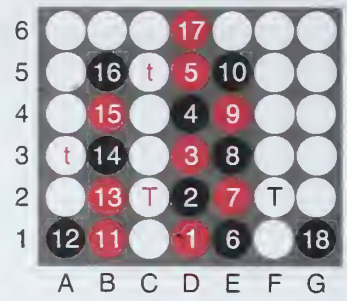
The *5-4 opening* gets its name because it starts with *five* disks played in column D (D1 d2 D3 d4 D5), followed by *four* disks played in a neighboring column (e1 E2 e3 E4). By mirror-image symmetry, the same analysis applies if column C is played first instead of column E.

In the 5-4 opening, Red's plays at D1, D3, and D5 are all quite clear-cut; odd-row central cells are the best cells on the board. When play shifts to column E, players continue to play right above the opposing player's previous disk ("I play where you play"); the plays E2 e3 E4 are logical because they block horizontal threats.

It would be a mistake for Black to play e5 on top of Red's E4. Diagram 191 shows this *5-5 opening*, in which Red gets a routine winning triple threat on the left side. (Black locks up the right side with a major threat at f2, but to no avail. Red's plays in Diagram 191 are basically the only way to win; the reason they are not shown with exclamation marks is that Red could have interposed temporizing moves in column G.)

The 5-5 opening is no good for Black because 11 B1 gives Red a major threat at C2; moreover, Black must waste disk 12 at a1 since if Red plays there, Red wins at once with the adjacent threats C1-C2. Therefore, Black should take b1 instead of playing 10 e5. A typical continuation is shown in Diagram 192, with play then going up column B. Just as E4 had to be answered at b1, so B4 has to be answered at b5; but the plays B2 b3 B4 are less clear-cut, and a play at the useful cell E5 is likely to be substituted at one of those turns.

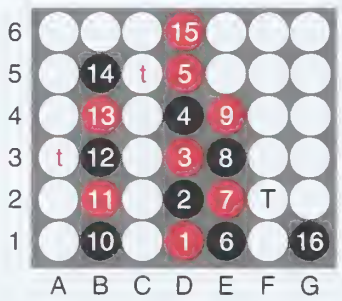
DIAGRAM 191
Red to play and win



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! e5 B1 a1 B2 b3 B4 b5 D6 g1 (19) __

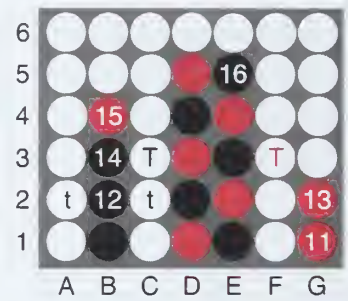
Before continuing the review of this important opening, I will point out that Red must not play as shown in Diagram 193, thinking only to make a major odd-row threat at F3, but ignoring Black's counterplay.

DIAGRAM 192
Red to play, with only one move to win



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 D6 g1 (17) __

DIAGRAM 193
Red to play (will lose)



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 (11) G1? b2! G2? b3 B4 e5! (17) __

Once Black takes 14 b5 in Diagram 192 to block the fifth-row horizontal threat, E5 loses much of its value; G1 and D6 become the important cells, and Red will play disk 15 at one of them. If Red plays 15 G1, Black should answer with 16 d6, and vice versa. (These are the two most valuable cells, so it is logical that each player gets one of them.) Diagram 192 shows the latter possibility. Red already has two minor odd-row threats but needs a third threat to win. Black's g1 has prevented Red F1 (which would build the third threat at C4) and neutralized Red G2 (a Red threat at F3 is undercut). In other words, it overcuts Red's F1 threat and undercuts Red's F3 threat. Black's g1 accomplishes a third important purpose as well: It builds a major even-row threat at f2.

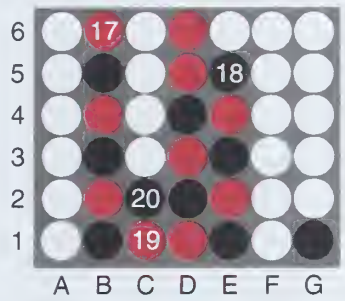
Although Black's g1 was an excellent play, Red can still win. (Red always wins if he or she makes no mistake, and Red has made no mistake yet.) Where should

Red play next in Diagram 192? This is a challenging problem, so try to figure it out before reading the next paragraph. (Or read just the hint if you wish.) If this opening is new to you, perhaps you should set the position up on your own board and play it out a few times with your partner until you understand how the game will develop.

Hint: Black wants to build an important overcutting threat. This same problem was considered from Black's viewpoint in Diagrams 129 and 130, and Black's winning move there is Red's winning move in Diagram 192. That's a simple enough principle: If one player needs to occupy a certain cell, the other player should capture it first!

DIAGRAM 194

Red to play, with only one move to win

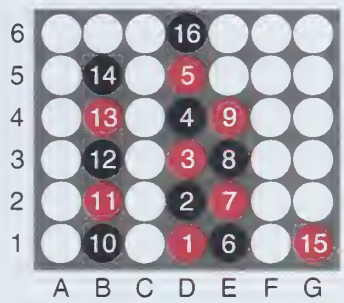


Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 D6 g1 (17) B6! e5 C1 c2 (21) __

When Red plays correctly, the game is likely to continue exactly as shown earlier, in Diagram 131 (although the moves are in a slightly different order). This analysis shows that, although Red can win in Diagram 192, the win is *far* from clear-cut. Red has a slight advantage on each side of the board and must make all his or her threats *work together* to win. Black's threat at f2 appears to neutralize the right side, but Red can wake up his or her right-side threats by playing C4 eventually.

DIAGRAM 195

Red to play, with only one move to win



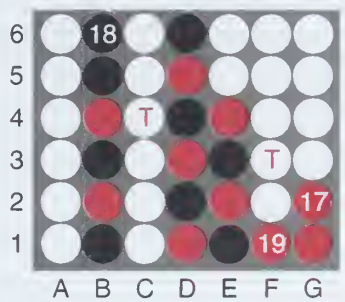
Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 G1 d6 (17) __

In Diagrams 130 and 131, Red had already played E5 and built a useful right-side threat combination, but what if Black plays as shown in Diagram 194? Where must Red then play?

Red should play 21 E6 and gets a routine triple threat at A3-C5-C6. In Chapter 11 I suggested handicapping Red by disallowing sixth-row horizontal wins for Red. With that handicap, Diagram 192 results in a draw; Red would have needed to play 15 G1 to win. Let's look at that variation now. In Diagram 195, where must Red play next?

DIAGRAM 196

Black to play (will lose)

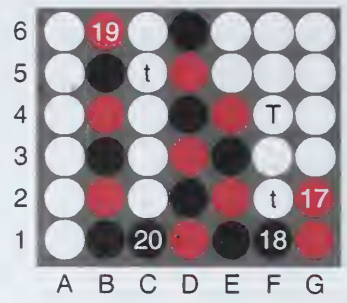


Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 G1 d6 (17) G2? b6? F1 (20) __

Many Red players will build a major odd-row threat immediately, perhaps expecting the play to continue as shown in Diagram 196 on the previous page.

The moves in Diagram 196 might appear logical: 17 G2 builds a major threat at F3; 18 b6 builds a counterthreat at c5; 19 F1 undercuts the counterthreat and renders it worthless. Red has a winning endgame in Diagram 196.

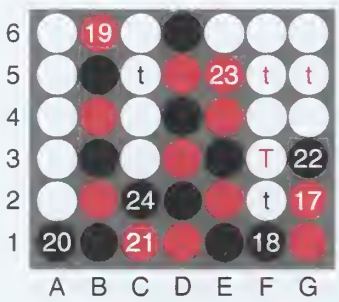
DIAGRAM 197
Red to play and draw



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 G1 d6 (17) G2? f1! B6 c1 (21) __

But both 17 G2 and 18 b6 were errors in Diagram 196. First examine Diagram 197 to see how Black *should* have played disk 18.

DIAGRAM 198
Red to play and win



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 G1 d6 (17) G2? f1! B6 a1? C1! g3 E5 c2 (25) __

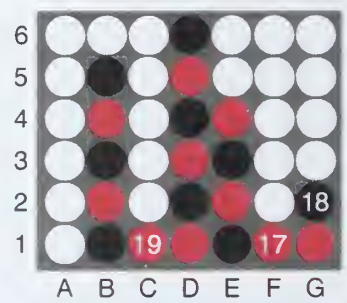
Red's play 19 B6 may have been intended to prevent Black from getting a counterthreat at c5, but it doesn't really work; the minor threat pair c5-f2 is enough to neutralize Red's F3 threat. With Red's F3 threat already neutralized, Black plays 10 c1 to build a major threat at f4; this neutralizes any future Red threat at F5. Diagram 197 is a drawn game.

Black 20 c1 isn't strictly necessary in Diagram 197, but Black must do something to prevent Red from building a threat at F5. (Red gets such a threat by playing either G5 or G6.) If Black wastes disk 20, Red prevails, as shown in Diagram 198. The strategic outcome from this diagram may not seem clear-cut, but the key point is that there are relevant odd-row threats in three columns. Three is an odd number, so zugzwang favors Red. In other words, Black will eventually be forced to play c4, f2, or g4; in each case, Red wins easily.

Now that we've seen how Black *should* respond to 17 G2, return to Diagram 195 and find Red's best move.

Diagram 199 shows the correct Red continuation. Rather than finalize his or her own odd-row threats, Red destroys Black's

DIAGRAM 199
Black to play (will lose)

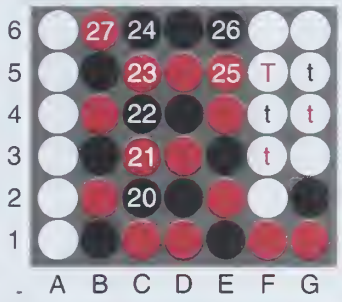


Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 G1 d6 (17) F1! g2 C1! (20) __

undercutting threats. Red expects to have no problem constructing a major threat at F3 or F5 later on. Where should Black play disk 20 in Diagram 199? Although all of Black's moves lose, there is one that puts Red to the test.

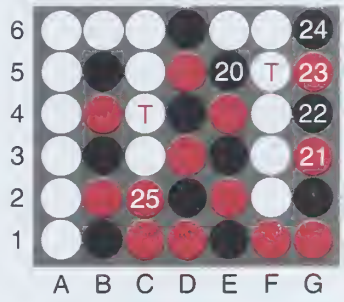
In Diagram 200, Red gets an easy win with a major odd-row threat at F5. Black has undercutting diagonal threats, but Red undercuts *these* with his or her own. In Diagram 201, Red again gets an easy win.

DIAGRAM 200
Black to play (will lose)



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 G1 d6 F1! g2 C1! (20) c2 C3 c4 C5! c6 E5 e6 B6! (28) __

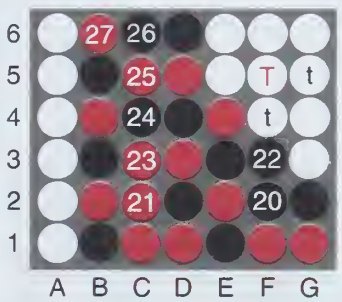
DIAGRAM 201
Black to play (will lose)



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 G1 d6 F1! g2 C1! (20) e5 G3 g4 G5! g6 C2! (26) __

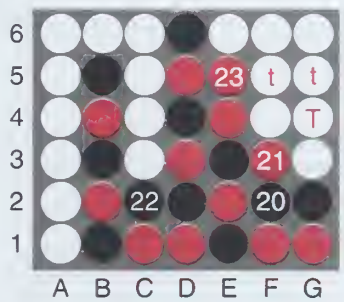
In Diagram 202, Black obtains the undercutting diagonal threats, and after blundering with 21 C2, Red has to play carefully just to get a draw. The play 20 f2 was Black's excellent move, and Red flunked the test. Where should Red have played disk 21?

DIAGRAM 202
Black to play and draw



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 G1 d6 F1! g2 C1! (20) f2 C2? f3 C3 c4! C5! c6 B6! (28) __

DIAGRAM 203
Black to play (will lose)



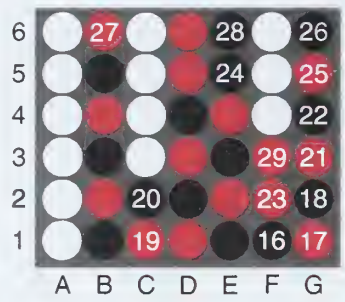
Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 G1 d6 F1! g2 C1! (20) f2 F3! c2 E5 (24) __

Red should have answered f2 with F3, as shown in Diagram 203. This is the way to undercut Black's threat at g5 with a Red threat at G4.

Look at Diagram 192 again; instead of 16 g1, Black might try 16 f1. This might lead to Diagram 204 or 205, worthy of study since Red must play skillfully.

DIAGRAM 204

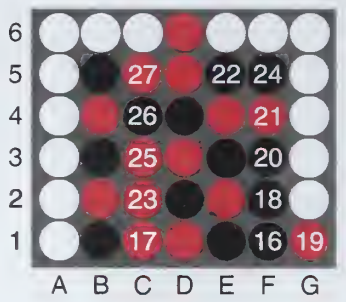
Black to play (will lose)



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 D6 (16) f1 G1 g2 C1! c2 G3 g4 F2 e5 G5 g6 B6! e6 F3! (30) __

DIAGRAM 205

Black to play (will lose)



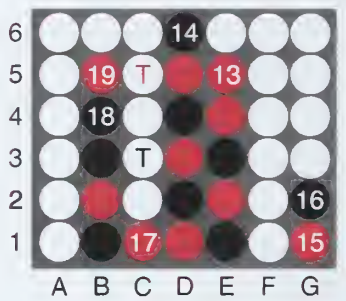
Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 B4 b5 D6 (16) f1 C1 f2 G1! f3 F4! e5 C2! f5 C3! c4 C5! (28) __

(Diagram 92 also arises from this opening, though there Black's play is weak.)

As mentioned earlier, in the 5-4 opening, E5 might be played at disk 11, 12, or 13. I discuss a few of those variations here. In Diagram 206, Red has an easy win, but Red's play had to be this precise. (Variations of this game are shown beginning at Diagram 212.)

DIAGRAM 206

Black to play (will lose)

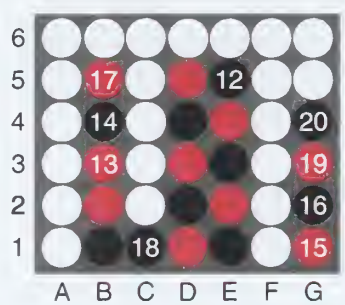


Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 (13) E5 d6 G1! g2 C1! b4 B5! (20) __

If Black plays 12 e5, the game can develop in various ways, with the continuation in Diagram 207 providing one of my favorite puzzles.

DIAGRAM 207

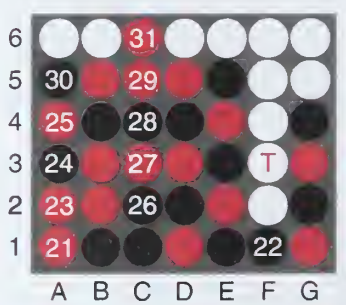
Red to play, with only one move to win



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 (12) e5 B3 b4 G1! g2 B5 c1 G3 g4 (21) __

DIAGRAM 208

Black to play (will lose)



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 e5 B3 b4 G1! g2 B5 c1 G3 g4 (21) A1! f1 A2! a3 A4! c2 C3! c4 C5! a5 C6! (32) __

The solution to Diagram 207 is shown in Diagram 208. Red plays an elegant sequence of forcing moves, which lead to Red's winning a major odd-row threat at F3. Less obvious, perhaps, is that Red must begin this attack immediately; otherwise, Black will get an early f2, as shown in Diagram 209, and be able to answer 33 C6 with 34 f3.

DIAGRAM 209

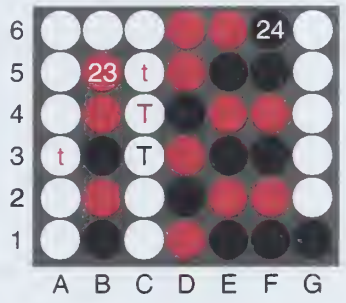
Black to play, with only one move to draw



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 e5 B3 b4 G1! g2 B5 c1 G3 g4 (21) D6? f1! A1 f2! A2 a3 A4! c2! C3! c4! C5! a5! C6! (34) __

DIAGRAM 210

Red to play and draw



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 e5 D6 g1 E6! b3 B4! f1 F2! f3 F4! f5 (23) B5? f6! (25) __

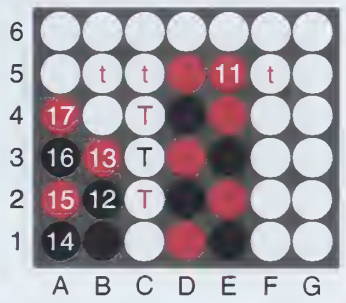
Another 12 e5 variation is shown in Diagram 210. Red's plays were very precise in this game, but Red should have blocked with 23 F6.

The last variation I consider in the 5-4 opening is Diagram 211, where Red played 11 E5 and follows with very precise play. Red's 15 A2 prevents Black from winning quickly with an adjacent threat pair; 16 a3 prevents Red from getting a major third-row threat. Red's 17 A4 forms a Red threat at C2 to undercut Black's c4 threat, which would otherwise successfully undercut Red's key threat at c5.

The opening shown in Diagram 206 may lead to Diagram 212. Where should Red play next?

DIAGRAM 211

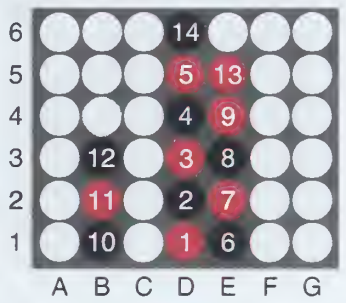
Black to play (will lose)



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 (11) E5 b2 B3! a1 A2! a3 A4! (18) __

DIAGRAM 212

Red to play, with only one move to win



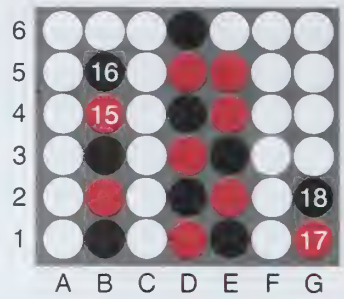
Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 E5 d6 (15) __

In the end, 15 C1 loses to 16 g1. Red's 15 B4 leads to Diagram 213 and at best a draw for Red. (Where must Red play next in that diagram to get a draw?) The play 15 G1 is Red's winning move, with Diagrams 214 and 206 showing possible continuations.

I asked how Red must play in Diagram 213 to get a draw. Red must start with 19 C1 (so that Black doesn't get an easy major threat at f4), and if the game continues as shown in Diagram 215, Red must continue with 23 B6 to kill Black's threats at c5 and c6. The final result might be something like Diagram 216.

DIAGRAM 213

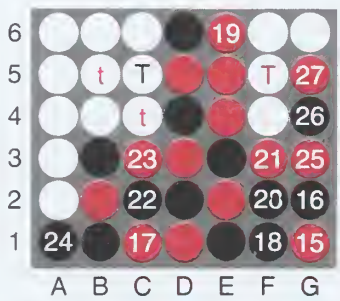
Red to play, with only one move to draw



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 E5 d6 (15) B4? b5! G1! g2 (19) __

DIAGRAM 214

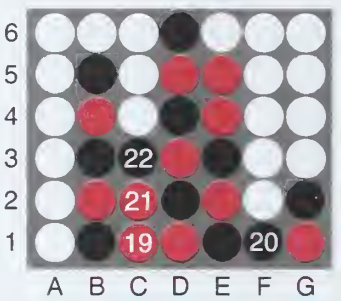
Black to play (will lose)



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 E5 d6 (15) G1! g2 C1! f1 E6 f2 F3! c2 C3! a1 G3 g4 G5! (28) __

DIAGRAM 215

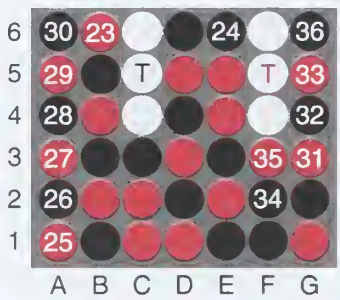
Red to play, with only one move to draw



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 E5 d6 B4? b5! G1! g2 (19) C1! f1! C2 c3 (23) __

DIAGRAM 216

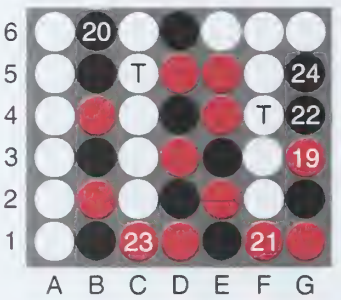
Red to play, with only one move to draw



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 E5 d6 B4? b5! G1! g2 C1! f1! C2 c3 (23) B6! e6 A1 a2 A3 a4 A5 a6 G3 g4 G5! f2! F3 g6! (37) __

DIAGRAM 217

Red to play (will lose)



Moves: D1! d2 D3! d4 D5! e1 E2 e3 E4! b1 B2 b3 E5 d6 B4? b5! G1! g2 (19) G3? b6 F1 g4! C1 g5! (25) __

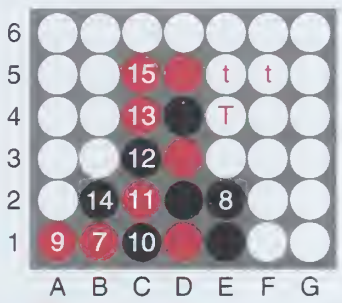
Red cannot deviate from the formula 19 C1, 23 B6. If Red plays B6 or F1 before C1, Black will take c1. If Red plays 19 G3 thinking that's a safe temporizing move, Black can win in various ways, as shown in Diagram 217, for example.

5-1 Opening

Red prevails in the 5-4 opening, but several difficult variations are possible, of which only a few are shown in the preceding section. Red may prefer to avoid this opening altogether by answering 6 e1 with 7 B1. This opening can be called the *5-1 opening*; it can also be called the *DdDdDeB opening* (listing the columns in the order they are played).

DIAGRAM 218

Black to play (will lose)



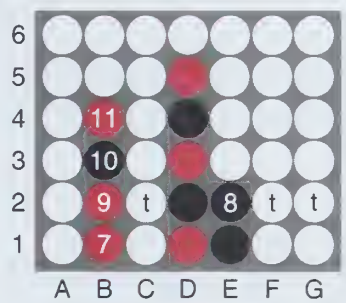
Moves: D1! d2 D3! d4 D5! e1 (7) B1 e2 A1! c1 C2! c3 C4! b2 C5! (16) __

In the 5-1 opening, Red will still have to play precisely if Black defends well, but there are fewer difficult variations, one of which is shown earlier, in Diagram 97. Diagram 218 shows another possibility, in which Red's moves 9, 11, 13, and 15 were all strictly necessary.

In Diagram 218, Black does not have an undercutting threat at e3-f4 because Red will play E3 e4 E5 whenever he or she wants. (Red had better play E3 as soon as Black plays f2 and, most simply, might as well play E3 the next chance he or she gets.)

DIAGRAM 219

Black to play, with only one move to win



Moves: D1! d2 D3! d4 D5! e1 (7) B1 e2 B2?? b3! B4 (12) __

Since Red *always* wins when both players play perfectly, opening study runs the risk of teaching good Red play but not good Black play! So let's suppose Red stumbles and plays 9 B2 instead of A1 in Diagram 218 and the game continues as shown in Diagram 219. Where should Black play next?

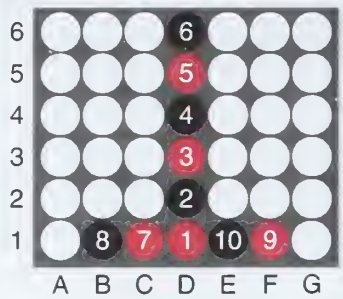
Black must play 12 b5 because Red would get a triple minor odd-row threat if Red played there. (After 12 d6 B5 e3 E4 e5, Red has the triple threat C2-C5-A5; if instead Black manages to get e4, Red E5 completes the major threat C5.) After 12 b5, Black will win eventually on the second row.

6-1 Opening

The 6-1 opening (also called the *column D opening*) begins with six disks in column D followed by one in column C, as shown in Diagram 220. Red plays 7 C1

DIAGRAM 220

Red to play, with only one move to win



Moves: D1! d2 D3! d4 D5! d6 C1! b1 F1! e1 (11) __

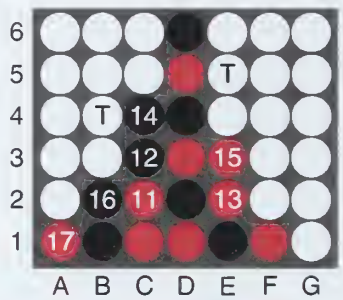
(or E1) rather than B1 because it leads to *two* forced Black plays in the first row (8 b1 and 10 e1), while after 7 B1 c1, Red has no forcing move.

However, after 7 C1, both players avoid playing at E1, preferring 8 b1 and 9 F1. This is in accordance with the general rule that B1 and F1 are better cells than E1 because the B1-E4 and F1-C4 diagonals are better than the E1-B4 diagonal. In Diagram 220, the logical Red moves seem to be 11 C2 (aiming at C3 and a J shape) or 11 E2 (aiming at E3 and a J shape). Which of these two moves is correct?

It may be obvious that 11 E2 is correct—it forms the major threat C4—but I review the flawed line anyway in Diagram 221. Red forms the J shape on the right side, but Black wins very easily. In fact, Black has the choice of two different ways to neutralize Red's J shape.

DIAGRAM 221

Black to play and win



Moves: D1! d2 D3! d4 D5! d6 C1! b1 F1! e1 (11) C2?? c3! E2 c4! E3 b2 A1 (18) __

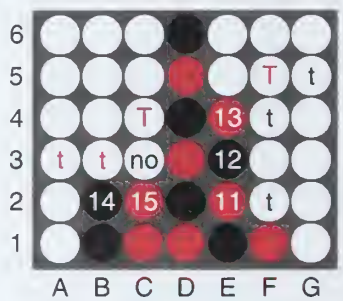
- Black can play 18 e4, sacrificing a major odd-row threat to overcut Red's J shape, responding to Red's G1 G2 with f2 f3.
- Black can play up column G eventually, allowing Red to convert his or her J shape into a major odd-row threat, which is neutralized because of Black's major odd-row threat at f5. Red's diagonal threats e4-f5 do not undercut Black's e5 threat because, with only two odd-threated columns, Black has zugzwang control.

In either case, Black's major even-row threat at b4 wins for Black.

Diagram 222 shows how Red should properly continue from Diagram 220; 11 E2 is correct because it forms a diagonal threat at C4. The diagram is rich with relevant

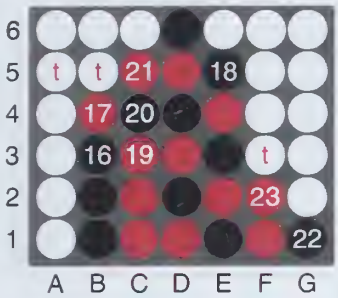
DIAGRAM 222

Black to play (will lose)



Moves: D1! d2 D3! d4 D5! d6 C1! b1 F1! e1 (11) E2! e3 E4! b2 C2! (16) __

DIAGRAM 223
Black to play (will lose)



Moves: D1! d2 D3! d4 D5! d6 C1! b1 F1! e1 E2! e3 E4! b2 C2! (16) b3 B4! e5 C3! c4 C5! g1 F2! (24) __

minor threats; I've shown so many that the diagram is cluttered, but I still haven't shown them all. Red has a major odd-row threat at F5, but after f2 G1 f3, Black's d2-g5 diagonal undercuts that threat. Red has possible third- and fifth-row horizontal wins on the left side, but (unless Red gets a chance to play B3) Black can kill these threats with the push-up play b3 B4 b5.

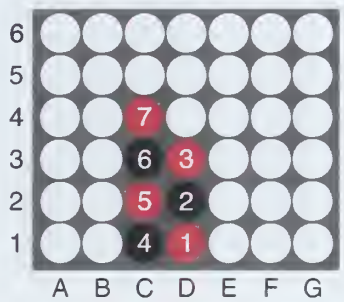
In addition to the strategic threats in Diagram 222, certain tactical ploys may develop. In Diagram 223, Red threatens both F3 (a triple check!) and B5; Black can block only one. Other continuations from Diagram 222 are shown in Diagrams 8 and 43.

3-4 Opening

In Diagram 224, players alternate play up the central columns just as in the 5-4 opening, except that Black doesn't play d4.

Although Red hasn't blundered yet, Red would probably have an easier win if he or she just played D4 as disk 5 or disk 7. Once Red delays D4, however, and plays as shown in the diagram, his or her game becomes difficult and future play at D4 is likely to be a blunder. Consequently, this opening is one that Black should try occasionally in order to probe Red's response but should probably avoid if Red plays an early D4

DIAGRAM 224
Black to play (will lose)



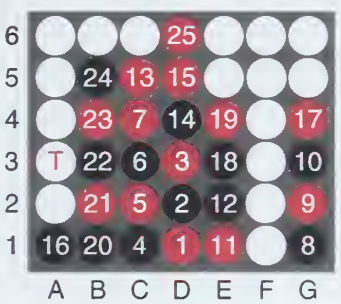
Moves: D1! d2 D3! c1 C2 c3 C4 (8) __

DIAGRAM 225
Black to play (will lose)



Moves: D1! d2 D3! c1 C2 c3 C4 g1 G2! g3 E1! e2 C5! a1 G4! c6 G5 g6 E3! e4 E5! a2 A3 a4 A5 a6 E6! b1 B2! (30) __

DIAGRAM 226
Black to play (will lose)



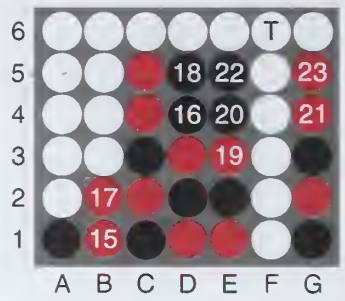
Moves: D1! d2 D3! c1 C2 c3 C4 g1 G2! g3 E1! e2 C5! d4 D5 a1 G4! e3 E4! b1 B2! b3 B4 b5 D6! (26) __

rather than playing as shown in Diagram 224.

Diagrams 225 and 226 show the precision required of Red in order to prevail in this opening. (Red's plays at E6 or E3 could have been made slightly earlier.) I can't show how Black achieves a draw in all the variations where Red blunders, but, of course, Black will almost always answer D4 with d5.

Red's 11 E1 weakens Black's c3-d2 diagonal and aims at E2.

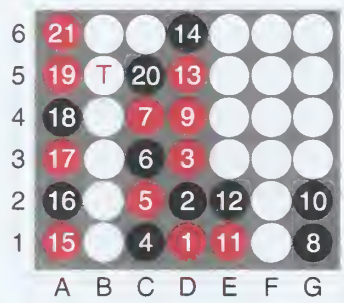
DIAGRAM 227
Black to play and win



Moves: D1! d2 D3! c1 C2 c3 C4 g1 G2! g3 E1! e2 C5! a1 (15) B1?? d4! B2 d5! E3 e4! G4 e5 G5 (24) __

Red needs 15 G4 (or 17 G4) to limit the threats Black creates with d4. If in Diagram 225 Red instead plays 15 D4 or E4, Black will answer with 16 d5 or e5 for a draw; other Red moves lose, as shown in Diagram 227, for example.

DIAGRAM 228
Black to play (will lose)



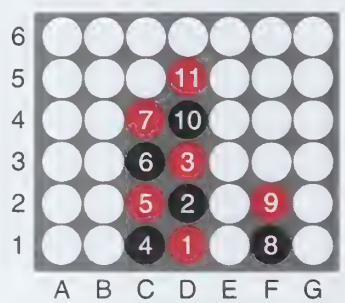
Moves: D1! d2 D3! c1 C2 c3 C4 g1 D4? g2? E1 e2 D5! d6 A1 a2 A3 a4 A5! c5 A6! (22) __

Red plays 9 G2 in these variations partly to prevent Black from playing there and locking out the right side with second-row horizontal threats, and partly to avoid giving up a key cell. If Red plays 9 C5 (or A1 or B1) instead of G2, Black should grab 10 g2 immediately, but 10 g2 would be a mistake in response to 9 D4. That would let Red win on the left side, as shown in Diagram 228. Instead, Black must defend with 10 c5 or d5, let Red get 11 G2, and get a draw, since Black's fifth-row disk is stronger than Red's D4.

An experienced Red will probably not allow the 3-4 opening to develop. Instead Red might play 5 F1 or 5 G1 or adopt the D3-D4 opening, which I consider next.

Instead of 8 g1, Black may try f1 in the 3-4 opening. Red answers at D4 or F2; in the latter case, a standard position from the 5-4 opening might be reached by transposition, as shown in Diagram 229. Red 11 F3, letting Black play f4 to make a major odd-row threat, would have been a mistake.

DIAGRAM 229
Black to play (will lose)



Moves: D1! d2 D3! c1 C2 c3 C4 f1 F2 d4 D5! (12) __

D3-D4 Opening

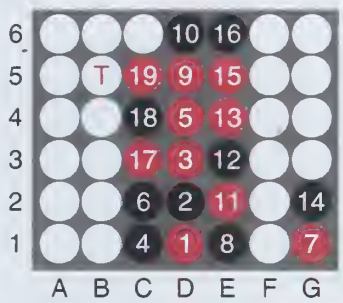
I use the term *D3-D4 opening* when Red gets both these key cells early in the game. This position often is very powerful for Red (although when Red already has C4 or E4, as shown in variations of the 3-4 opening, it's often better for Red to wait and try to get D5).

Diagram 230 indicates how easy it often is for Red to win when he or she gets D3-D4 early. Red starts by building an attack in the lower right side; by the time Black counters that, Red has overwhelming strength in the upper center of the board.

Diagram 231 shows another variant of the D3-D4 opening. Where should Red play next? I think even most experts will get this wrong.

DIAGRAM 230

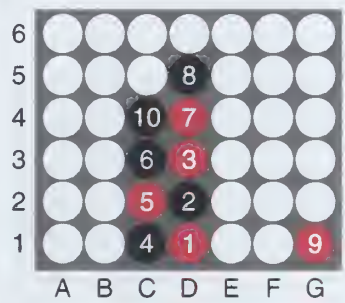
Black to play (will lose)



Moves: D1! d2 D3! c1 D4 c2 G1 e1 D5 d6 E2! e3 E4 g2 E5 e6 C3 c4 C5 (20) __

DIAGRAM 231

Red to play, with only one move to win

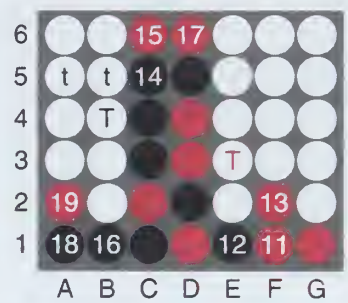


Moves: D1! d2 D3! c1 C2 c3 D4 d5 G1! c4 (11) __

Most Red players, I think, will play F1 F2, as shown in Diagram 232, to build a major odd-row threat quickly, but Black ends up with better threats in the upper left side and victory. This kind of position is called *Third-Row Gambit* because Black willingly lets Red get a major threat on the third row but seeks compensation on the fifth row. Black's play 18 a1 was a useless move, played just so I could pose another problem: Where must Black play next in Diagram 232 to preserve victory?

DIAGRAM 232

Black to play, with only one move to win



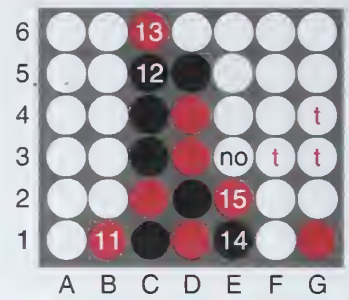
Moves: D1! d2 D3! c1 C2 c3 D4 d5 G1! c4 (11) F1?? e1! F2 c5! C6 b1! D6 a1 A2 (20) __

Red threatens to march up column A, getting either A4 (with major threat B3 then undercutting Black's threats) or A5 (eliminating Black's odd-row threat). Black's correct play is therefore the push-up play 20 b2 A3 b3 B4 b5.

Correct Red play following Diagram 231 is shown in Diagram 233. Red wins with the incomplete J shape, as depicted with threat marks.

Diagram 234 shows a variant of the D3-D4 opening in which Red has played *very* precisely, yet Black seems to have foiled Red's effort to get an odd-row threat. Red has a quick victory in this diagram, though; do you see it?

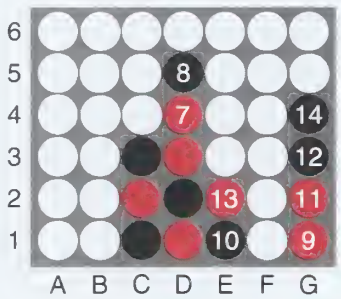
DIAGRAM 233
Black to play (will lose)



Moves: D1! d2 D3! c1 C2 c3 D4 d5 G1! c4 (11) B1! c5 C6! e1 E2! (16) __

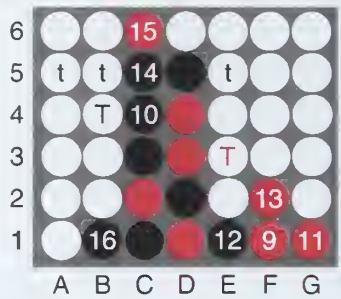
Red's move 9 G1 in this opening may be hard to find. The key idea is to recall the great offensive value of the D4-E3-F2-G1 diagonal. A losing alternative for Red is shown in Diagram 235.

DIAGRAM 234
Red to play and win



Moves: D1! d2 D3! c1 C2 c3 (7) D4 d5 G1! e1 G2! g3 E2! g4 (15) __

DIAGRAM 235
Red to play (will lose)



Moves: D1! d2 D3! c1 C2 c3 D4 d5 (9) F1?? c4! G1 e1! F2 c5! C6 b1! (17) __

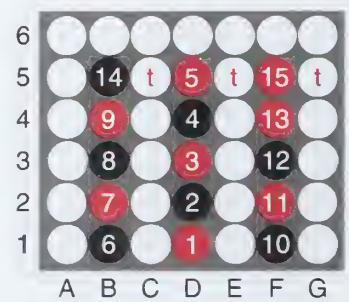
Candlesticks Opening

Diagram 236 is called a *Candlesticks Opening*, since separated columns are played up to look like candlesticks. Red wins easily with a triple threat. (This is the same as Diagram 61.)

In Diagram 237, Black tries 10 b5 instead of f1. Now Red will only have one forcing first-row play in any event, so 11 F1 is appropriate.

Diagrams 238 and 239 show some more variants of the Candlesticks Opening. Diagram 240 is an instructive variant, with one Red blunder.

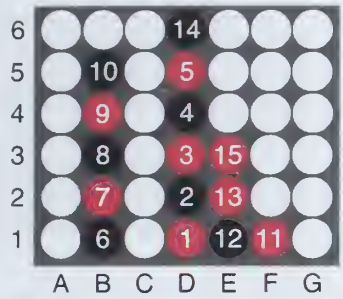
DIAGRAM 236
Black to play (will lose)



Moves: D1! d2 D3! d4 D5! b1 B2! b3 B4! f1 F2! f3 F4! b5 F5! (16) __

DIAGRAM 237

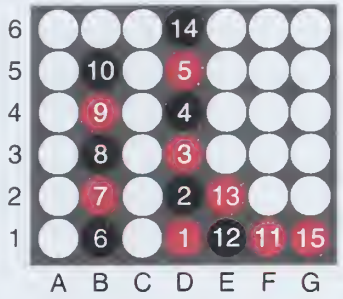
Black to play (will lose)



Moves: D1! d2 D3! d4 D5! b1 B2! b3 B4! b5 F1! e1 E2! d6 E3 (16) __

DIAGRAM 238

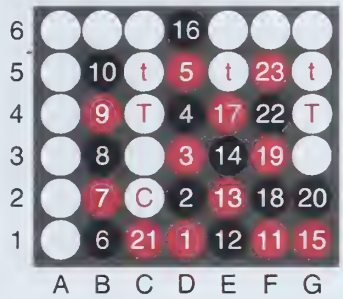
Black to play (will lose)



Moves: D1! d2 D3! d4 D5! b1 B2! b3 B4! b5 F1! e1 E2! d6 G1 (16) __

DIAGRAM 239

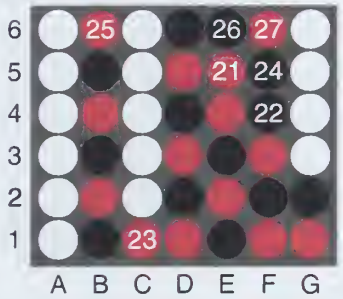
Black to play (will lose)



Moves: D1! d2 D3! d4 D5! b1 B2! b3 B4! b5 F1! e1 E2! e3 G1 d6 E4! f2 F3! g2 C1! f4 F5! (24) __

DIAGRAM 240

Black to play and draw

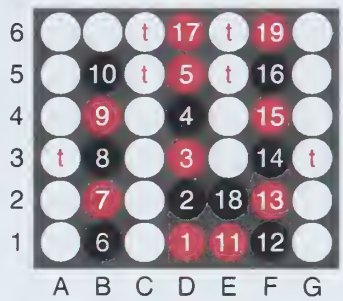


Moves: D1! d2 D3! d4 D5! b1 B2! b3 B4! b5 F1! e1 E2! e3 G1 d6 E4! f2 F3! g2 (21) E5? f4 C1! f5! B6! e6 F6! (28) __

Diagram 241 shows another variant. Note Red's unusual sextuple threat. Diagram 242 shows a typical continuation.

DIAGRAM 241

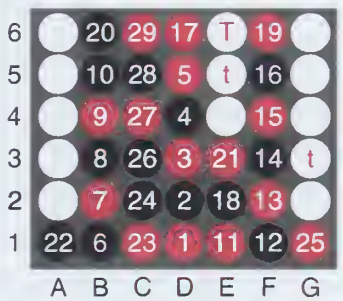
Black to play (will lose)



Moves: D1! d2 D3! d4 D5! b1 B2! b3 B4! b5 E1? f1! F2 f3 F4! f5! D6 e2? F6! (20) __

DIAGRAM 242

Black to play (will lose)

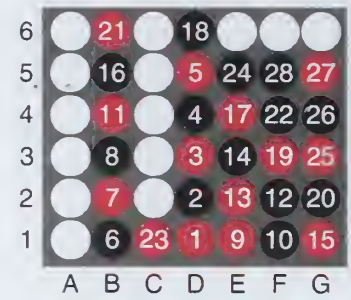


Moves: D1! d2 D3! d4 D5! b1 B2! b3 B4! b5 E1? f1! F2 f3 F4! f5! D6 e2? F6! b6 E3! a1 C1 c2 G1 c3 C4 c5 C6! (30) __

The Candlesticks Opening positions can be reached with the moves in a different order; for example, D1! b1 B2 b3 B4 d2 D3! d4 D5! Also, as shown in Diagram 243, some openings will be nearly like the Candlesticks Opening.

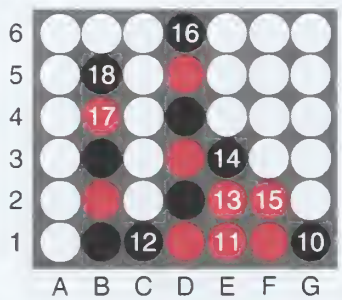
Diagram 244 shows another form of this opening. Diagram 245 shows the variant where Black fails to find the key move, 10 g1.

DIAGRAM 243
Red to play and draw



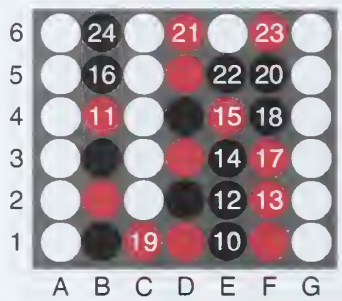
Moves: D1! d2 D3! d4 D5! b1 B2! b3 E1? f1! B4! f2 E2! e3! G1! b5! E4! d6 F3 g2! B6! f4 C1! e5 G3! g4! G5! f5 (29) __

DIAGRAM 244
Red to play (will lose)



Moves: D1! d2 D3! d4 D5! b1 B2! b3 F1?? (10) g1! E1 c1! E2 e3 F2 d6 B4 b5! (19) __

DIAGRAM 245
Red to play and draw

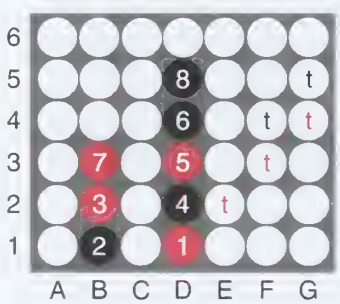


Moves: D1! d2 D3! d4 D5! b1 B2! b3 F1?? (10) e1? B4! e2! F2! e3 E4! b5 F3 f4! C1! f5 D6 e5 F6 b6! (25) __

Variations of the Candlesticks Opening

Openings can veer from one pattern to another. Consider Diagram 246. Black 2 b1 leads toward a first-row opening, but then 4 d2 and 6 d4 move back toward a Candlesticks Opening. Red plays 7 B3 to get minor third-row threats—this is an option Red doesn't have when the Candlesticks Opening develops with the normal move order—but now Black gets the key cell d5, so if Red isn't careful, he or she may regret this opportunity.

DIAGRAM 246
Red to play and win

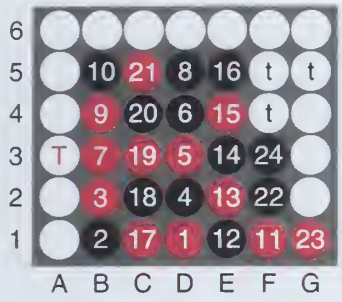


Moves: D1! b1 B2 d2 D3 d4 B3 d5 (9) __

Diagram 246 is a form of Third-Row Gambit, in that Red dominates the third row but Black dominates the fifth row. Rather than depicting Red's eventual winning threats, Diagram 246 shows the Red minor threats that are needed to undercut Black's

DIAGRAM 247

Red to play and draw



Moves: D1! b1 B2 d2 D3 d4 B3 d5 B4 b5 F1 e1 E2 e3 E4 e5 C1 c2 C3? c4! C5! f2 G1! f3! (25) __

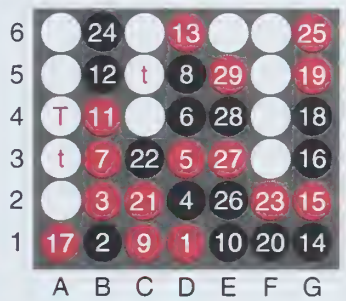
future threats in the upper right side. In Diagram 247, Black has obtained a fifth-row counterthreat and draws against Red's third-row threat. (Black doesn't win, but just to draw against an expert Red is a moral victory for Black.) Red should have played 19 G1 instead of C3; this is the only way for Red to preserve the important undercutting threats F3-G4.

There are *many* other patterns of play possible after Diagram 246. Problem 280 (on page 81) is taken from one possibility; if Red fails to solve that problem, Diagram 248 results. In that diagram, Black must grab c4 (for an easy win) before Red wakes up and plays A2.

Diagram 249 shows another opening similár to the Candlesticks Opening, where Red has made a series of temporizing moves and achieves a plethora of threats somewhat similar to a winning quadruple threat. In Diagram 249, Red can announce mate in four moves.

DIAGRAM 248

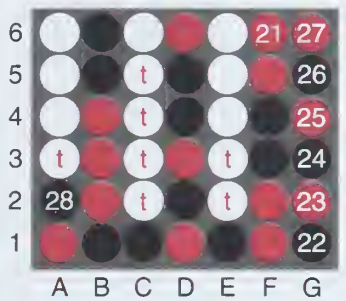
Black to play, with only one move to win



Moves: D1! b1 B2 d2 D3 d4 B3 d5 C1 e1 B4! b5 D6! g1 G2! g3 A1 g4 G5! f1 C2 c3 F2! b6 G6 e2 E3 e4 E5?? (30) __

DIAGRAM 249

Red to play



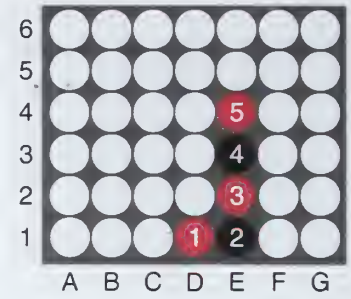
Moves: D1! b1 B2 d2 D3 d4 F1 e1 F2! f3 B3! d5 B4 b5 D6! f4 F5! c1 A1 b6 (21) F6! g1 G2! g3 G4! g5 G6! a2 (29) __

1-4 Opening

After 1 D1 e1, Red must play 3 A1 or 3 B1, but many players don't realize that. If Red plays 3 D2, Black will answer with 4 d3 and have little trouble winning, but the Red play 3 E2 presents more difficulty. Play will develop as shown in Diagram 250. This opening is neither a central-column opening (only one early disk is played in column D) nor a first-row opening (only two of the first five disks are played in the first row); I call this the *1-4 opening*. (This is the only "standard opening" I consider in which Red cannot force victory.) Black has only one move to

guarantee a draw in Diagram 250. Can you find it?

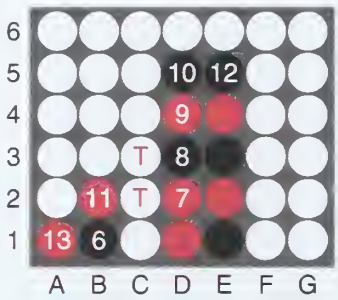
DIAGRAM 250
Black to play, with only one move to draw



Moves: D1! e1 E2? e3! E4 (6) __

Diagram 250 is of some historic interest. It may not pose much of a challenge to today's high-speed computers, but the computers of 1988 required *days* of computation to establish Black's best move here. And—if you're looking just for perfect Red play—all that computation is wasted, because Red has already blundered in Diagram 250! Both Victor Allis and I solved the game of CONNECT 4 in 1988, and each of us let our computers waste days looking at Diagram 250, because I assumed 3 E2 was the logical Red play. (The program Victor wrote even suggested 3 B1 as more likely to be the strong Red response, but Victor overrode it!)

DIAGRAM 251
Black to play (will lose)



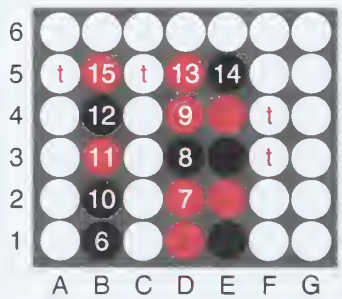
Moves: D1! e1 E2? e3! E4 (6) b1? D2 d3 D4! d5 B2! e5 A1 (14) __

Before I give the answer to the problem in Diagram 250, look at Diagram 251, where Red wins (along the A1-D4 diagonal).

Diagram 252 is a fun variation where Black gets b2 to break up the A1-D4 diagonal, but Red uses threats in the lower left as a springboard to get threats in the upper left. If Red doesn't end up occupying F3, he or she will get F2-F4 instead; in either case, Red forms a major even-row threat in column C to complete a winning triple threat.

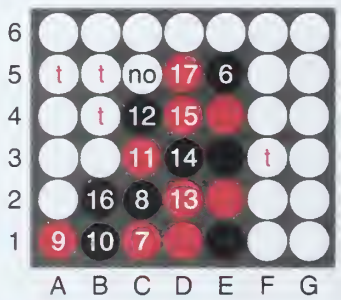
Diagram 253 shows yet another variation in the 1-4 opening where Red wins;

DIAGRAM 252
Black to play (will lose)



Moves: D1! e1 E2? e3! E4 (6) b1? D2 d3 D4! b2 B3! b4 D5! e5 B5 (16) __

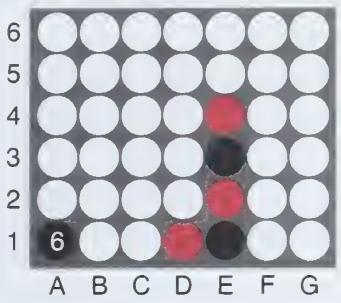
DIAGRAM 253
Black to play (will lose)



Moves: D1! e1 E2? e3! E4 (6) e5? C1! c2 A1 b1 C3! c4 D2 d3 D4! b2 D5! (18) __

again Red uses a threat involving the A1-D4 diagonal to get momentum for a win in the upper left. Red's winning triple threat may look quite incomplete, but note that if Black plays c5, C6 gives Red an easy win at F3.

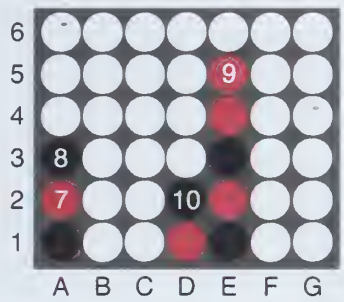
DIAGRAM 254
Red to play and draw



Moves: D1! e1 E2? e3! E4 (6) a1! (7) __

Have you guessed Black's correct play in Diagram 250 yet? Black must play as shown in Diagram 254. There's not a single Red disk on the A1-D4 diagonal yet, but there lie the threats Black must deal with! (Perhaps a more realistic way to understand why 6 a1 is the proper move is to note that A2 is a rather unimportant cell. If Black plays 6 b1 instead of a1, Red may get a strong disk at B2, as shown in Diagram 251, or cause Black to spend a turn taking b2, as shown in Diagram 252.)

DIAGRAM 255
Red to play and draw



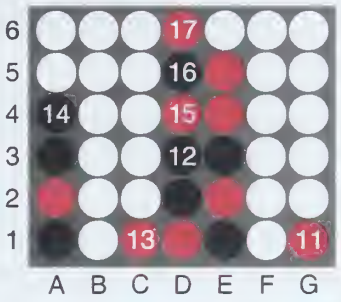
Moves: D1! e1 E2? e3! E4 a1! (7) A2 a3 E5 d2 (11) __

I just wrote that A2 is a rather unimportant cell, but do you know where Black should answer if Red plays 7 A2 in Diagram 254? If Black plays 8 d2, Red will play D3 and arrive at a winning position we've seen (though via a different move order) in the 3-4 opening. (See Diagrams 225 and 226.) Black must answer 7 A2 with 8 a3, as shown in Diagram 255. The play a3 seems even more unexpected than the play a1, but at least it means that Red has no chance for a horizontal third-row threat.

In Diagram 255, the game continues 9 E5 d2. These moves aren't absolute but are a likely expert continuation and lead to more interesting problems. For starters, five of Red's moves in Diagram 255 lose; what are the two plays that salvage a draw for Red?

If Red understands that he or she is trying just to salvage a draw, Red will play to block Black's even-row threats. Red must start with 11 C1 or 11 G1, each of which blocks a developing Black diagonal. In fact, Red is likely to get both C1 and G1, as shown in Diagram 256.

DIAGRAM 256
Black to play and draw

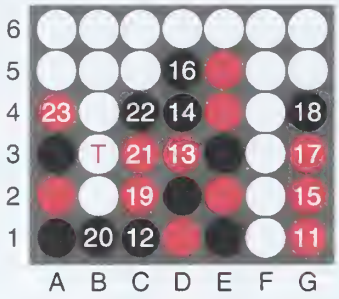


Moves: D1! e1 E2? e3! E4 a1! A2 a3 E5 d2 (11) G1 d3! C1! a4 D4! d5 D6! (18) __

It wouldn't do for Black to play 12 c1 to build an "easy" even-row threat after 11 G1. That would lead to Red victory, perhaps with moves like those shown in Diagram 257.

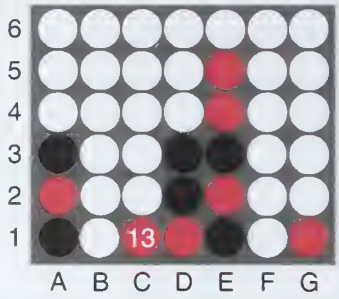
Diagram 258 is taken from Diagram 256 just before Black played 14 a4. Black has one other play to salvage a draw; I'll bet you won't guess where it is!

DIAGRAM 257
Black to play (will lose)



Moves: D1! e1 E2? e3! E4 a1! A2 a3 E5 d2 (11) G1 c1? D3 d4 G2 d5 G3 g4 C2! b1 C3! c4 A4! (24) __

DIAGRAM 258
Black to play and draw

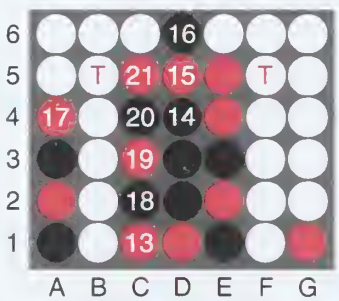


Moves: D1! e1 E2? e3! E4 a1! A2 a3 E5 d2 G1 d3! (13) C1! (14) __

Black 14 d4 doesn't work; it might lead to Diagram 259.

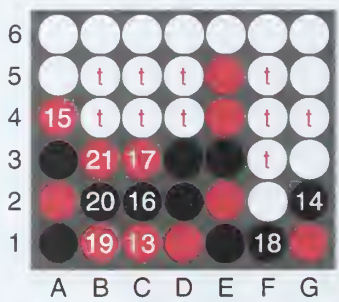
Black 14 g2 doesn't work; it might lead to Diagram 260. This diagram is cluttered with many threat indicators, but Red would *not* win if any of these ten cells were "blue" (not counting for four-in-row).

DIAGRAM 259
Black to play (will lose)



Moves: D1! e1 E2? e3! E4 a1! A2 a3 E5 d2 G1 d3! (13) C1! d4? D5! d6 A4! c2 C3! c4 C5! (22) __

DIAGRAM 260
Black to play (will lose)

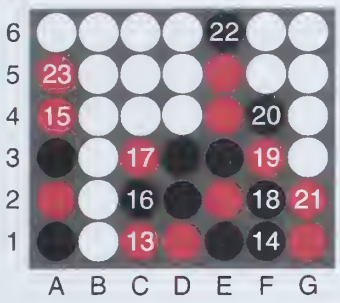


Moves: D1! e1 E2? e3! E4 a1! A2 a3 E5 d2 G1 d3! (13) C1! g2? A4! c2 C3! f1 B1 b2 B3 (22) __

Black 14 f1 is the only way to draw! A likely result after that play was shown (as a "Where must Black move?" problem) in Diagram 48, repeated here as Diagram 261. Note that after 18 f2, 20 f3 would be double check; this is the special significance of the otherwise weak-looking 14 f1.

DIAGRAM 261

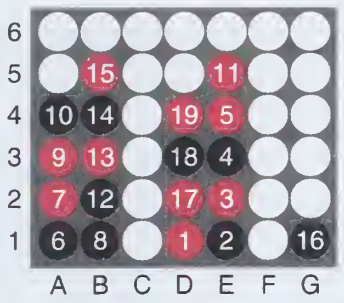
Black to play, with only one move to draw



Moves: D1! e1 E2? e3! E4 a1! A2 a3 E5 d2 G1 d3! (13) C1! f1 A4 c2 C3! f2! F3 f4! G2 e6 A5 (24) __

DIAGRAM 262

Black to play, with only one move to draw



Moves: D1! e1 E2? e3! E4 a1! A2 b1 A3 a4 E5 b2! B3 b4! B5 g1 D2 d3! D4! (20) __

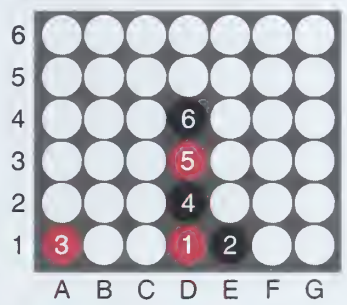
Black's 8 a3 is not mandatory in the opening we've just considered. The moves 8 b1 and 8 e5 are also good enough to draw, with Diagram 262 showing just one possible continuation.

Chapter 13

First-Row Openings

When Red has mastered the central-column opening, Black will usually start answering Red's D1 with 2 e1 or f1. (Black 2 g1 is too weak to consider.) Often more disks are soon played in the first row, giving these variants a generic name, *first-row opening* (defined arbitrarily as any game in which at least three of the first four disks are played in the first row). I refer, briefly, to the DcF, DcG, DbF, DbB, and DaG variants (others sometimes call these the "436," "437," "426," "422," and "417" openings). First-row openings admit more variety than central-column openings, and a line from one initial sequence may transpose into a line from a different sequence.

DIAGRAM 263
Red to play, with only one move to win

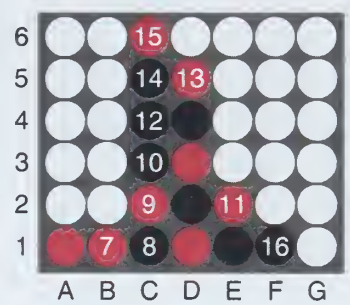


Moves: D1! e1 A1 d2 D3 d4 (7) __

Diagram 263 fulfills the definition of a first-row opening (the first three disks are played in the first row), but play then continues up column D. If Red were now to play 7 D5, the position would become a central-column opening position, specifically the 5-1 opening, but with Red responding to 6 e1 weakly with A1 instead of B1 or E2 as required. But the moves in Diagram 263, *in the order they were actually played*, were all strong moves; Red has a winning move, but it isn't 7 D5. Where is it?

Red forces on the first row with 7 B1 in Diagram 264 and builds a diagonal threat with 9 C2. Black blocks the left-side J shape with c3, Red plays 11 E2 to get a future J shape on the right side, but Black plays 16 f1, apparently winning the right-side race. Can Red still win?

DIAGRAM 264
Red to play, with only one move to win



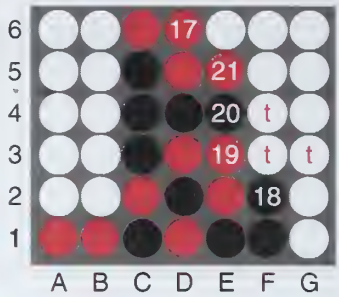
Moves: D1! e1 A1 d2 D3 d4 (7) B1! c1 C2! c3 E2 c4 D5! c5 C6! f1 (17) __

Black spent a valuable turn playing 16 f1, and, as shown in Diagram 265, Red then wastes no time grabbing D6, E3, and E5 to form a winning triple threat.

In these openings, Black 4 d2 is weak because Red will force with B1 and follow

with B2 or C2 to form a powerful diagonal. Black should prefer to prevent that with 4 c1. Diagram 266 shows a first-row opening where Red plays precisely after 4 c1; it ends up looking like a 1-4 or 3-4 opening, in which Red has built a major odd-row threat but seems to have wasted a disk at A1.

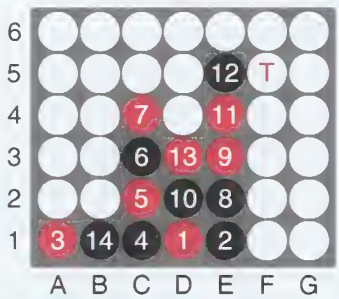
DIAGRAM 265
Black to play (will lose)



Moves: D1! e1 A1 d2 D3 d4 B1! c1 C2! c3 E2 c4 D5! c5 C6! f1 (17) D6! f2 E3! e4 E5! (22) __

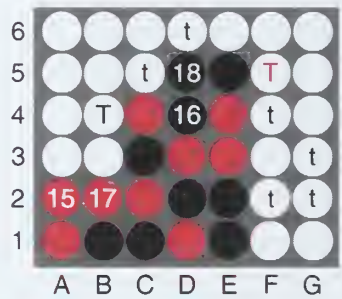
Red isn't home free in Diagram 266 and may make a subtle mistake on his or her very next move, as shown in Diagram 267. In that diagram, Red will next play C5 to prevent Black from getting multiple major odd-row threats, and Black will answer at d6. Black already has an even-row threat but

DIAGRAM 266
Red to play and win



Moves: D1! e1 A1 c1 C2! c3 C4! e2 E3! d2 E4 e5 D3 b1 (15) __

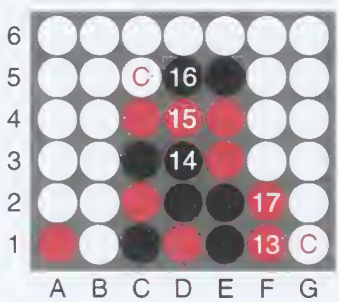
DIAGRAM 267
Red to play (will lose)



Moves: D1! e1 A1 c1 C2! c3 C4! e2 E3! d2 E4 e5 D3 b1 (15) A2?? d4! B2 d5! (19) __

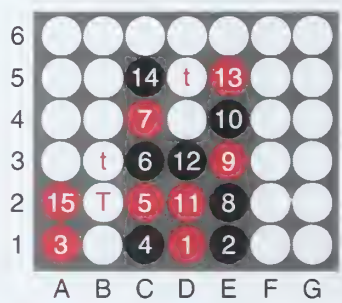
needs to build another one to undercut Red's major odd-row threat at F5. Sooner or later, Black will play up column G and capture either g2 or g3, forming the required major even-row threat in column F.

DIAGRAM 268
Black to play (will lose)



Moves: D1! e1 A1 c1 C2! c3 C4! e2 E3! d2 E4 e5 (13) F1 d3 D4! d5 F2! (18) __

DIAGRAM 269
Black to play (will lose)

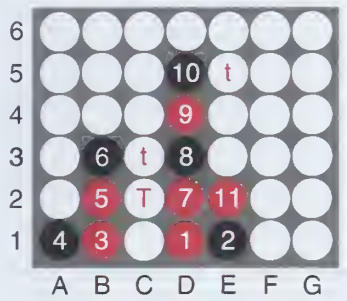


Moves: D1! e1 A1 c1 C2! c3 C4! e2 E3! e4 D2 d3 E5! c5 A2! (16) __

Instead of 13 D3, as shown in Diagram 266, Red has an interesting variant win, shown in Diagram 268.

Sometimes Red can win with an early D2, as shown in Diagram 269. Here, 15 A2 is a powerful move that reminds us how useful it often is to have occupied the cell A1 (or G1) in the opening. Note that Black's plays 8, 10, 12, and 14 each took the very cell Red was hoping to capture for a minor odd-row threat.

DIAGRAM 270
Black to play (will lose)



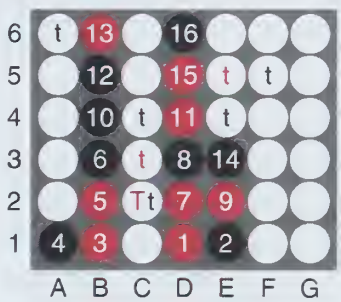
Moves: D1! e1 B1 a1 B2! b3 D2 d3 D4 d5 E2 (12) __

We've been looking at Red 3 A1, but 3 B1 is somewhat stronger. Diagram 270 gives an example; a variation is shown in Diagram 398.

Red could play 9 A2 instead of D4 in this game and get a different winning triple threat. (Red then answers 10 d4 with B4.) Red must *not* substitute the perhaps similar-seeming 9 E2. If he or she does, Black is likely to get e3, as shown in Diagram 271. By killing Red's C1-F4 diagonal, Black can develop an undercutting diagonal at c2-f5, as implied by the threats depicted in the diagram. This game, which I like to call the "Pruce Defense," has a particular elegance; Red has a triple threat, but Black's f5 minor threat in the undercutting diagonal c2-f5 provides the fourth odd-threated column to contain Red. Black doesn't just draw—Black wins with the a6-c4 even threats.

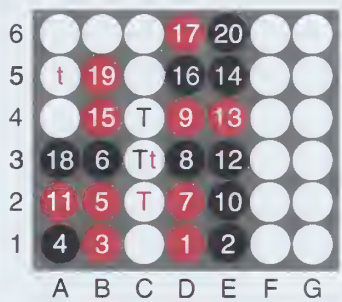
Diagram 272 shows another typical result in this opening. Black's threats in column C are irrelevant here; Red has zugzwang control and will be able to place a disk at A5 without any problem.

DIAGRAM 271
Red to play (will lose)



Moves: D1! e1 B1 a1 B2! b3 D2 d3 E2??
b4! D4 b5! B6 e3! D5 d6! (17) __

DIAGRAM 272
Red to play and win

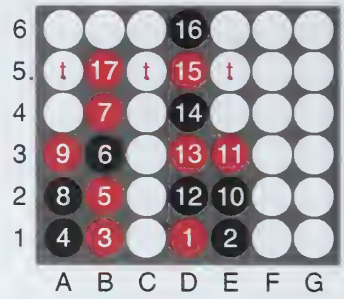


Moves: D1! e1 B1 a1 B2! b3 D2 d3 D4 e2
A2 e3 E4! e5 B4! d5 D6! a3 B5 e6 (21) __

Diagram 273 is a fine unusual-looking opening worth study. (It starts with the moves from Diagram 17.) As indicated with exclamation points, Red must play quite precisely to ensure victory. After this perfect play, Red has an obvious fifth-row win.

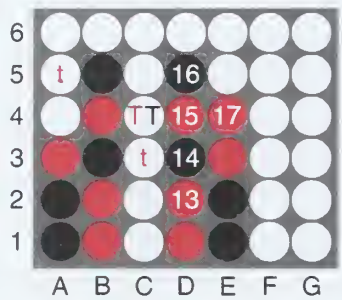
Since Red's win was fairly straightforward once Black played 12 d2 in Diagram 273, let's look at some alternatives to that move. If Black substitutes 12 b5, Red must answer with 13 D2. Red's D2 establishes minor odd-row threats on the A5-D2 diagonal; Red also has potential horizontal threats on both the third and fourth rows and along the B2-E5 diagonal; Red wins easily. Diagram 274 shows a continuation in which Red places a disk on E4 and wins with a routine triple threat; Black's major threat at c4 ends up being useless. (In other variations, Red might play a key disk at D3, D4, D5, or E5 instead of E4.)

DIAGRAM 273
Black to play (will lose)



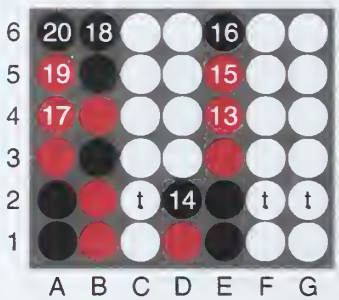
Moves: D1! e1 B1 a1 B2! b3 B4 a2 A3! e2 E3! d2 D3! d4 D5! d6 B5! (18) __

DIAGRAM 274
Black to play (will lose)



Moves: D1! e1 B1 a1 B2! b3 B4 a2 A3! e2 E3! b5 (13) D2! d3 D4 d5 E4 (18) __

DIAGRAM 275
Red to play (will lose)



Moves: D1! e1 B1 a1 B2! b3 B4 a2 A3! e2 E3! b5 (13) E4?? d2! E5 e6! A4 b6 A5 a6! (21) __

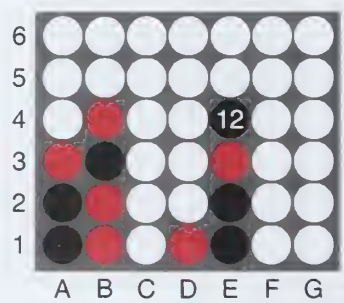
If Red neglects to answer 12 b5 with 13 D2, something like Diagram 275 might develop, where Red has no useful threats and is now in zugzwang.

Where should Red play in Diagram 276?

This is a trick problem! If Red plays 13 D2, which was the winning disk 13 in Diagram 275, Red ends up with nothing and Black wins, perhaps as shown in Diagram 277. In Diagram 276, Black has occupied the key cell e4, where Red got to play in Diagram 274.

Instead of 13 D2, Red should play E5, although Red can make a temporizing

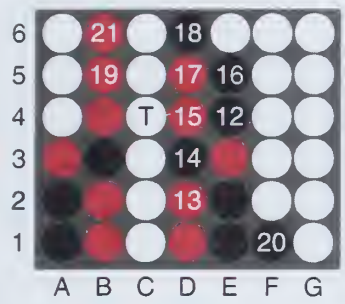
DIAGRAM 276
Red to play and win



Moves: D1! e1 B1 a1 B2! b3 B4 a2 A3! e2 E3! (12) e4 (13) __

DIAGRAM 277

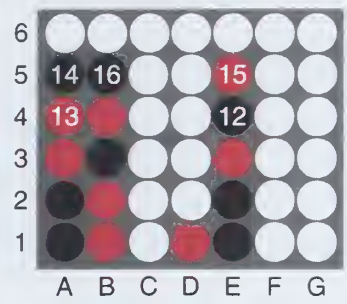
Black to play and win



Moves: D1! e1 B1 a1 B2! b3 B4 a2 A3! e2 E3! (12) e4 D2?? d3! D4 e5! D5 d6! B5 f1! B6 (22) __

DIAGRAM 278

Red to play, with only one move to win



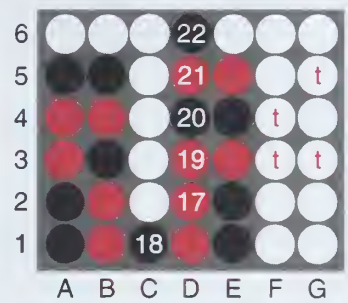
Moves: D1! e1 B1 a1 B2! b3 B4 a2 A3! e2 E3! (12) e4 A4 a5 E5! b5 (17) __

move at A4 first, as shown in Diagram 278. Where should Red play next in that diagram?

Now D2 is again the winning move, because if Black plays there, Black's second-row horizontal threats would lock up the game. After D2, Red D4 would claim a major threat at C3, so Black will probably let Red take D3 instead, as shown in Diagram 279.

DIAGRAM 279

Red to play and win

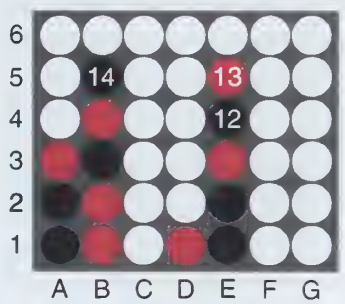


Moves: D1! e1 B1 a1 B2! b3 B4 a2 A3! e2 E3! e4 A4 a5 E5! b5 (17) D2! c1 D3 d4 D5! d6 (23) __

Very generally, central odd-row cells such as D3, E3, and D5 are so valuable that in most openings a player will drop a disk there as soon as he or she gets a chance. Plays such as D2, however, may leave the player in a quandary: To play or not play D2? The main reason *not* to play D2 is often that it lets the opposing player drop a key

DIAGRAM 280

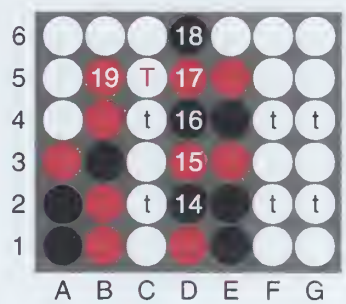
Red to play, with only one move to win



Moves: D1! e1 B1 a1 B2! b3 B4 a2 A3! e2 E3! (12) e4 E5 b5 (15) __

DIAGRAM 281

Black to play (will lose)



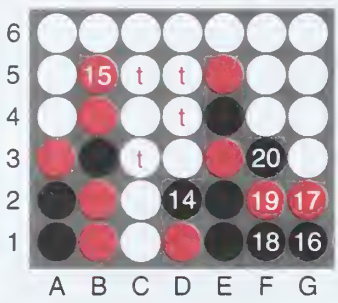
Moves: D1! e1 B1 a1 B2! b3 B4 a2 A3! e2 E3! e4 E5 (14) d2 D3 d4 D5! d6 B5! (20) __

disk at D3. Diagram 278 teaches a good lesson: Look to see whether the opposing player wants D3 or can even afford to play there. If not, D2 is likely to be a good move after all.

In Diagram 280 Red must also play 15 D2 next. (With a black disk at b5, Red cannot play A4 d2 A5 a6 B5, so the "temporizing" move A4 is no longer viable.)

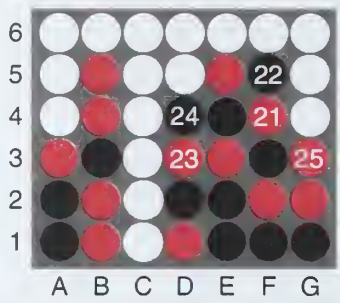
If Black plays 14 d2 instead of b5 in Diagram 280, the likely continuation is shown in Diagram 281. Black's even-row threats don't pose a problem for Red, because Red wins on the left side without needing any threats in column F.

DIAGRAM 282
Red to play, with only one move to win



Moves: D1! e1 B1 a1 B2! b3 B4 a2 A3! e2 E3! e4 E5 (14) d2 B5 g1 G2! f1 F2! f3 (21) __

DIAGRAM 283
Black to play (will lose)

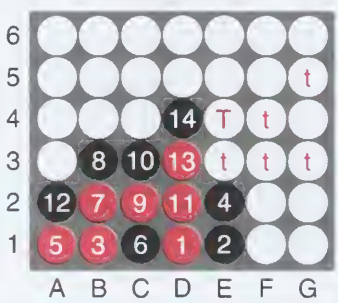


Moves: D1! e1 B1 a1 B2! b3 B4 a2 A3! e2 E3! e4 E5 d2 B5 g1 G2! f1 F2! f3 (21) F4! f5 D3! d4 G3 (26) __

Red can also succeed by substituting B5 for 15 D2 in Diagram 281, but this play will require more care. The position in Diagram 282 may result. Study this position; what is Red's winning threat combination? How will Black try to undercut it?

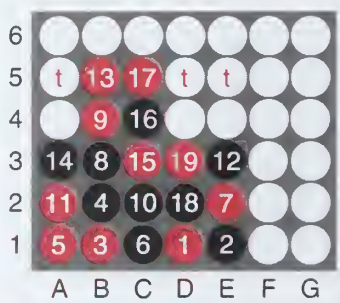
Black threatens to play f4 and undercut Red with a fourth-row horizontal threat. Red must play as shown in Diagram 283 to kill that threat, followed by D3 after 22 f5 (or vice versa) to kill Black's undercutting diagonal c2-f5. (D5 can be substituted for 25 G1, of course.)

DIAGRAM 284
Red to play and win



Moves: D1! e1 B1 e2 A1 c1 B2! b3 C2! c3 D2! a2 D3! d4 (15) __

DIAGRAM 285
Black to play (will lose)



Moves: D1! e1 B1 b2 A1! c1 E2 b3 B4! c2 A2! e3 B5! a3 C3 c4 C5! d2 D3! (20) __

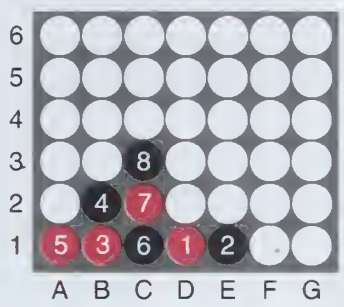
Diagram 284 shows one of my favorite openings. Red probes on the second row with B2 and C2, but Black answers at b3 and c3 to deny Red a presence on the third row. Finally Red plays D2 forcing Black to block at A2; then Red is able to seize D3. Red ends up getting only a single disk on the third row, but it's enough to win; Red has a sextuple threat.

Diagram 285 shows a well-played opening; Red will win, if he or she continues to be careful, because of Red's triple minor threat on the fifth row.

If Red substitutes 7 C2 in this opening, Black will probably take c3, as shown in Diagram 286.

DIAGRAM 286

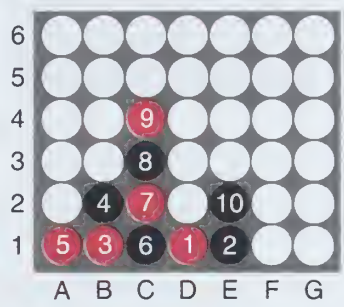
Red to play and win



Moves: D1! e1 B1 b2 A1! c1 C2 c3 (9) __

DIAGRAM 287

Red to play, with only one move to win



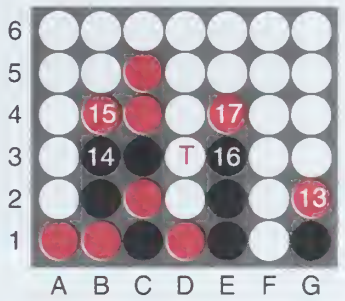
Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 (11) __

In Diagram 286, Red has two winning moves: 9 C4 and 9 E2. Since C4 and E2 are the two logical strong points, Black will probably answer 9 C4 with 10 e2, and answer 9 E2 with 10 c4. I consider each case in turn. Where should Red play next in Diagram 287?

The correct move is clearly C5. Black wants e4 (to stop Red's diagonal threat at D3), so Red will get E5 to go with C5 for a likely fifth-row win. Various possible continuations lead to Problem 283 (solution shown in Diagram 288), Problem 184 (solution shown in Diagram 290), Problem 282 (solution shown in Diagram 292), and Problem 180 (solution shown in Diagram 294).

DIAGRAM 288

Black to play (will lose)



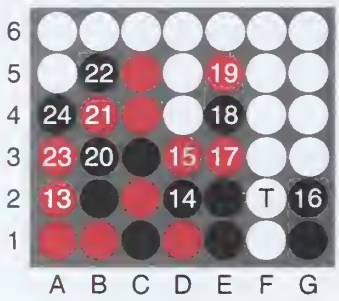
Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 C5! g1 (13) G2! b3 B4! e3 E4! (18) __

The solution to Problem 283 is as follows: Red must play 13 G2 and prevent Black from getting a second-row horizontal threat. Black is then helpless and will probably end up giving Red an easy win, as shown in Diagram

288. If Red omits 13 G2, Black will win, perhaps as shown in Diagram 289. If Red plays 15 G2 in that variation, Black can get a quick victory.

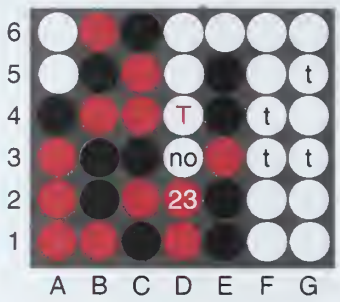
The solution to Problem 184 is as follows: As shown in Diagram 290, D2 is the key move. It prevents a possible Black second-row undercutting threat and establishes a winning (if unusual) minor threat configuration on the right side.

DIAGRAM 289
Red to play (will lose)



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 C5! g1 (13) A2?? d2! D3 g2! E3 e4! E5 b3 B4 b5 A3 a4! (25) __

DIAGRAM 290
Black to play (will lose)

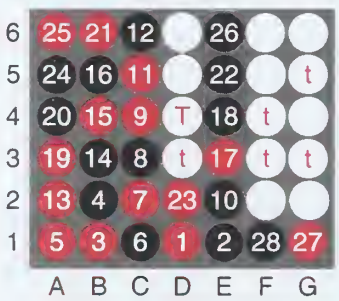


Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 C5! c6 A2 b3 B4 b5 E3 e4 A3! a4 B6 e5 (23) D2! (24) __

Diagram 291 shows a continuation from Diagram 290. Red must play F2; any other move gives up Red's sextuple threat.

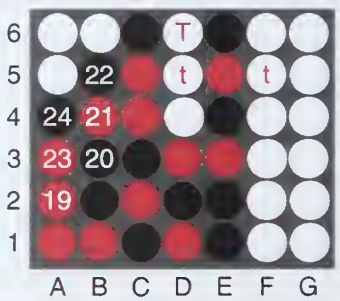
The solution to Problem 282 is as follows: Red must start with 19 A2. Red A2 forms a third minor odd-row threat (B3) to go with the two Red already has (D5, F5); moreover, Red is now certain to get an even-row threat at D4 or D6. One possible result is shown in Diagram 292.

DIAGRAM 291
Red to play, with only one move to win



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 C5! c6 A2 b3 B4 b5 E3 e4 A3! a4 B6 e5 D2! a5 A6 e6 G1 f1 (29) __

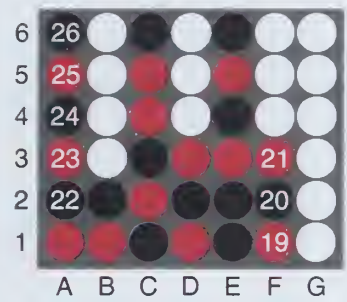
DIAGRAM 292
Red to play and win



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 C5! c6 E3 d2 D3! e4 E5! e6 (19) A2! b3 B4! b5 A3 a4 (25) __

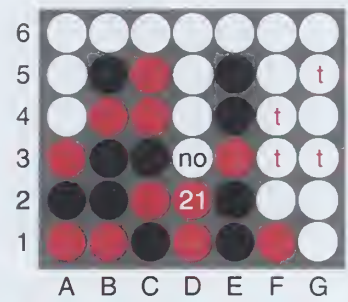
The move 19 A2 should be clear-cut, if only because every other possible move is clearly bad! Even 19 F1 isn't harmless; Black ends up getting a2 and eventually places Red in zugzwang, as shown in Diagram 293.

DIAGRAM 293
Red to play (will lose)



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 C5! c6 E3 d2 D3! e4 E5! e6 (19) F1?? f2! F3 a2! A3 a4! A5 a6 (27) __

DIAGRAM 294
Black to play (will lose)



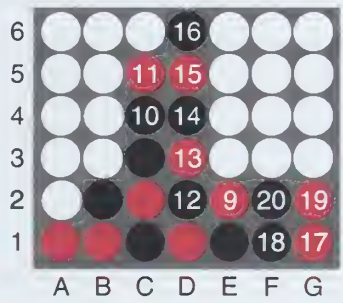
Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 C5! a2 A3 b3 B4! b5 E3 e4 F1 e5 (21) D2! (22) __

In Diagram 294, Red plays 21 D2 and wins with the combination of threats shown. Black can't take d3 because D4 is triple check.

Red can substitute 9 E2 in Diagram 286. This may result in Diagram 295, where Red can announce mate in three.

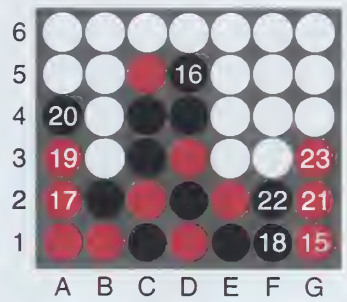
Black's plays in column F do Black no good, since in Diagram 295 Red is sure to find his or her one winning move (which happens to give very fast victory). As usual, Red has played very precisely to guarantee victory, but Diagrams 296 and 297 (and, earlier, Diagram 145) show some alternative ways to play.

DIAGRAM 295
Red to play, with only one move to win



Moves: D1! e1 B1 b2 A1! c1 C2 c3 (9) E2 c4 C5 d2 D3! d4 D5 d6 G1! f1 G2! f2 (21) __

DIAGRAM 296
Black to play (will lose)



Moves: D1! e1 B1 b2 A1! c1 C2 c3 E2 c4 C5 d2 D3! d4 (15) G1 d5 A2! f1 A3 a4 G2! f2 G3! (24) __

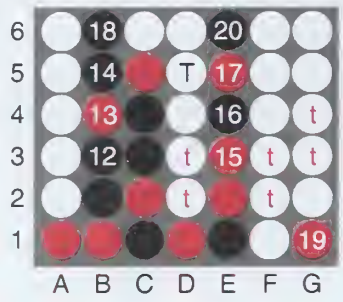
DIAGRAM 297
Black to play (will lose)



Moves: D1! e1 B1 b2 A1! c1 C2 c3 E2 c4 (11) G1 c5 C6! a2 E3 e4 E5! e6 G2 d2 G3 g4 D3! (24) __

DIAGRAM 298

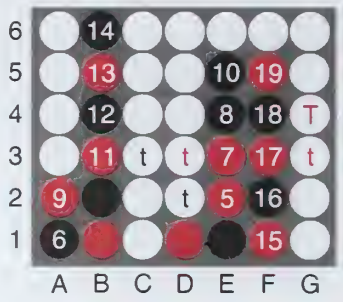
Red to play, with only one move to win



Moves: D1! e1 B1 b2 A1! c1 C2 c3 E2 c4 C5 (12) b3 B4! b5 E3 e4 E5 b6 G1 e6 (21) __

DIAGRAM 299

Black to play and draw



Moves: D1! e1 B1 b2 (5) E2? a1! E3 e4! A2 e5! B3 b4! B5 b6! F1 f2! F3! f4! F5! (20) __

If Black replaces 12 d2 with b3 in Diagram 295, Diagram 298 might follow. I have indicated Red's commanding minor threats there in the lower right side. Red must play 21 C6 next, or Black gets an adjacent threat pair.

Finally, Red may overlook the urgency of 3 A1 in these openings and play 3 E2, as shown in Diagram 299. In that diagram, Red has a triple threat in the lower right, but Black's b4-e1 diagonal undercuts it. (Red's diagonal threats A4-C2 are both even-row threats so are useless to Red, even for undercutting.) Observe the exquisite precision with which Black plays to achieve a draw. Although Black's threat c3-d2 is key to his or her draw, Black must not play d2.

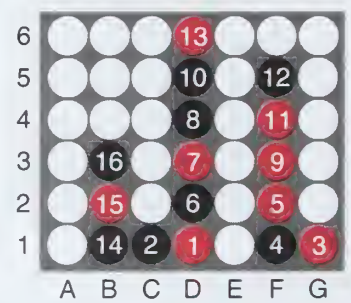
Third-Row Gambit

In chess, a *gambit* is a ploy in which one player sacrifices something hoping for compensation. In the *Third-Row Gambit*, Black lets Red establish an early major third-row threat but hopes to get compensation, usually in the fifth row.

Diagram 300 shows a typical Third-Row Gambit. Note that Red has played four disks to Black's two on the right side (columns E, F, and G), while Black has a similar two-disk advantage on the left side. Furthermore, Red dominates the third row, while Black dominates the fifth row. This lopsided disk distribution is typical of the gambit opening. Black lets Red make an easy third-row threat on the right side but develops a strong attack and is likely to get an odd-row counterthreat on the left side. In Diagram 300, Red has six losing moves! What is Red's only move to win? (Hint: Where does Black intend to build odd-row threats?)

DIAGRAM 300

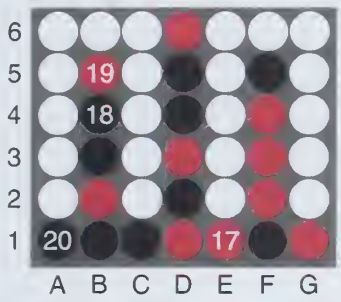
Red to play, with only one move to win



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! b3 (17) __

DIAGRAM 301

Red to play, with only one move to win



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! b3 (17) E1! b4 B5! a1 (21) __

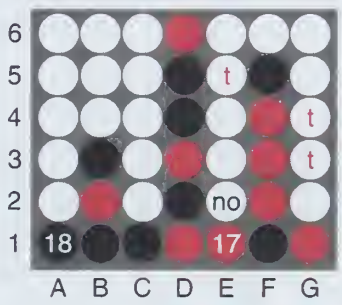
Red must play E1 now, as shown in Diagram 301, since an alternative such as F6 b4 E1 b5 would be devastating. Note from the exclamation points that nine of Red's ten moves in this game were uniquely forced if Red wishes to guarantee victory. Red's next move is also forced; where must Red play in Diagram 301? (This is a very difficult problem and is discussed at Diagram 370 later in this chapter.)

Instead of Diagram 301, I continue from Diagram 302, which may be easier for Red. (Note that by now, Red has a *three*-disk preponderance on the right side, and Black a similar preponderance on the left side.) As implied by the depicted threats, Red can play E2 now to get a routine triple threat, though to win Red will still have to play carefully and prevent a Black counterthreat on the left side. Note the value of E2 in this opening; if you focus your attention on just the third row, you may overlook that E2 forms a triple threat. (As we will see presently, in some variations of the Third-Row Gambit, D2 plays a role like E2 does here.)

Although Red can play E2 now and win, he or she may prefer to defer this move and work to counter Black's left-side attack. After all, Black can hardly play e2—it would give Red *two* major odd-row threats—so there may seem to be no rush for Red to play E2. But let's look at Diagram 303, where mistiming E2 turns out to be a mistake.

DIAGRAM 302

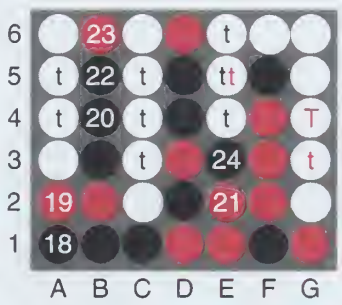
Red to play and win



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! b3 (17) E1! a1 (19) __

DIAGRAM 303

Red to play (will lose)



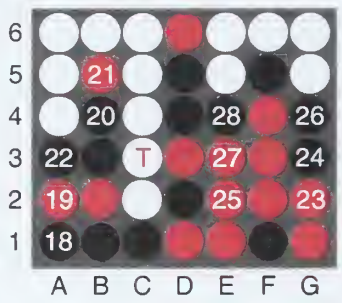
Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! b3 E1! (18) a1 A2 b4 E2?? b5 B6 e3! (25) __

In Diagram 303, Black has a plethora of minor odd-row and even-row threats on the left side. (Black's minor threats at e5 and e6 may come in handy but are not

essential.) Black will win despite Red's left-side triple threat. Black is not ready to play e4 yet (Red would play E5 for a quick column G win), but Red can't play E4 either. In Diagram 304, Red avoids the losing play 21 E2, plays up column G first, *then* plays up column E. This is successful Red play.

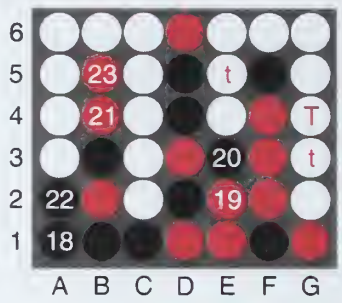
Diagram 305 shows another variation where Red can play E2 early and still succeed. (If Black substitutes 22 b5 for a2, Red must answer with 23 A2.)

DIAGRAM 304
Red to play and win



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! b3 E1! (18) a1 A2 b4 B5 a3 G2 g3 E2! g4 E3! e4 (29) __

DIAGRAM 305
Black to play (will lose)



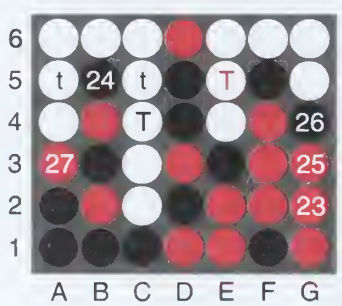
Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! b3 E1! (18) a1 E2 e3 B4! a2 B5! (24) __

Red's move 23 B5 was essential; otherwise, Black will obtain a triple threat that overcomes Red's triple threat, as shown in Diagram 306.

Black has several threats here; most simply, Black can play up column C to get c5 and a winning pair of major odd-row threats. However, the triple threat depicted in Diagram 306 is all Black needs for victory. Since three columns (A, C, and E) have odd-row threats and three is an odd number, you might think that Red has zugzwang control. But don't forget that a threat pair like Black's c4-c5 should be counted as *two* columns when computing zugzwang control.

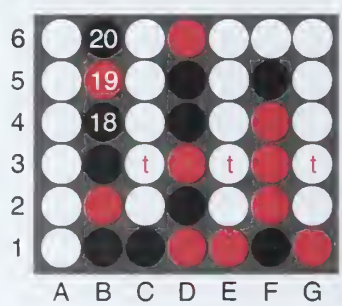
Diagram 307 shows a variation full of traps and complexity for each player.

DIAGRAM 306
Black to play and win



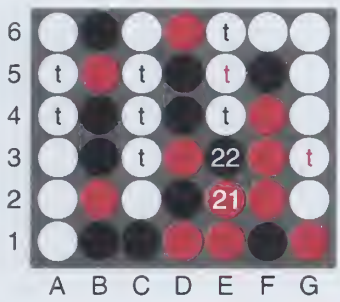
Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! b3 E1! a1 E2 e3 B4! a2 (23) G2?? b5 G3 g4! A3 (28) __

DIAGRAM 307
Red to play and win



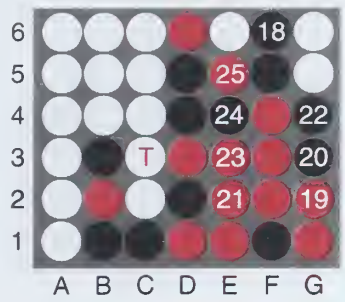
Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! b3 E1! (18) b4 B5! b6 (21) __

DIAGRAM 308
Red to play (will lose)



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! b3 E1! b4 B5! b6 (21) E2?? e3! (23) __

DIAGRAM 309
Black to play (will lose)



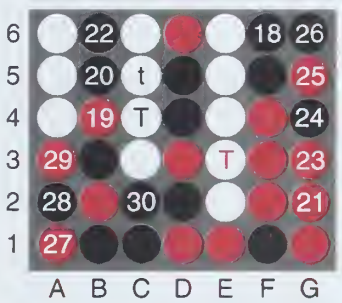
Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! b3 E1! (18) f6 G2! g3 E2! g4 E3! e4 E5 (26) __

Red has other threats (e.g., diagonals involving G3 and G4) besides the depicted third-row threats. Red must play 21 F6 or 21 G2 in Diagram 307. As in a previous variation, 21 E2 would be a mistake, leading to Diagram 308. Here Red has a right-side triple threat, but Black's plethora of threats in the upper left and center are equivalent to a winning quadruple threat.

There's still play left in Diagram 308, but Black will win if he or she is aware of his or her key threats, as shown. (Black's diagonal threat a2-b3-c4-d5 is useless since Red will take care to occupy A2 rather than A1.)

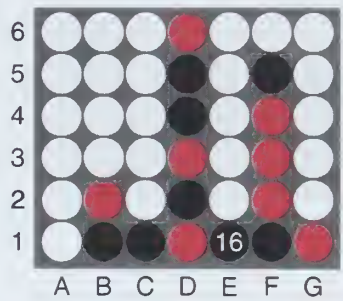
Diagram 309 shows how Red prevails against a different defense. With a simple tactical sequence, Red forces Black to give Red a major odd-row threat at either E5 or C3. (The move 19 G2 is the solution to Problem 244. After 19 G2, Red has various threats that might come into play depending on Black's responses.)

DIAGRAM 310
Red to play (will lose)



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! b3 E1! (18) f6 B4?? b5! G2 b6 G3 g4! G5 g6 A1 a2! A3 c2! (31) __

DIAGRAM 311
Red to play and win



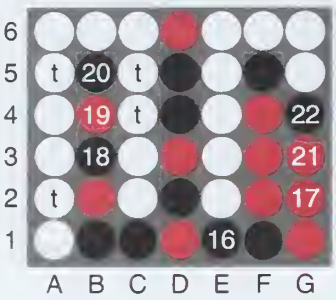
Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! (16) e1 (17) __

Less obvious is that 19 G2 is the *only* way for Red to win against this Black defense. Red might be tempted to seize a strong point with 19 B4, but Black would play as shown in Diagram 310, where Red is already in zugzwang.

Here's a variation of this opening where an expert might go wrong. Where should Red play in Diagram 311?

DIAGRAM 312

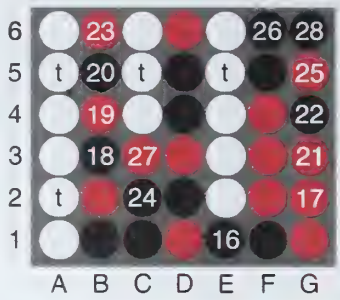
Red to play (will lose)



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! (16) e1 G2?? b3! B4 b5! G3 g4! (23) __

DIAGRAM 313

Red to play (will lose)



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! (16) e1 G2?? b3! B4 b5! G3 g4! B6 c2 G5 f6 C3 g6! (29) __

Red can quickly force Black to give Red a major odd-row threat, but Diagram 312 shows that to be a losing idea.

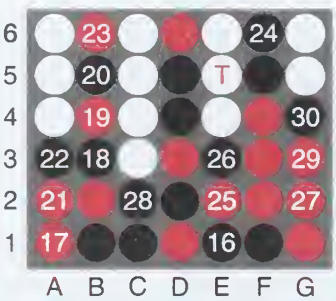
Although the threats have not become major, Black's adjacent threat pair in column C will win. Diagram 313 shows one variation.

Red is in zugzwang and will probably let Black have a winning double major odd-row threat in the fifth row. The fact that Red also has major odd-row threats in column E will be irrelevant. (If Red insists on never playing C4, Black will finish the game with an a2-b3-c4-d5 win.)

In Diagram 311, Red's only winning moves are 17 B3 or, as shown in Diagram 314, 17 A1. Red has a winning threat at E5, and Black's counterattack hasn't quite been successful. By the way, where do you think that Red should play next in Diagram 314?

DIAGRAM 314

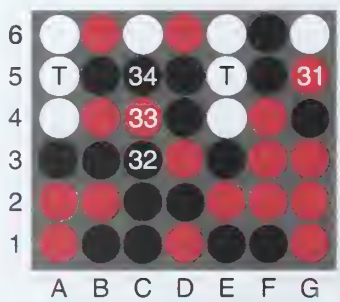
Red to play, with only one move to win



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! (16) e1 A1 b3 B4! b5 A2 a3 B6 f6 E2! e3 G2 c2 G3 g4 (31) __

DIAGRAM 315

Red to play (will lose)



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! e1 A1 b3 B4! b5 A2 a3 B6 f6 E2! e3 G2 c2 G3 g4 (31) G5?? c3! C4 c5! (35) __

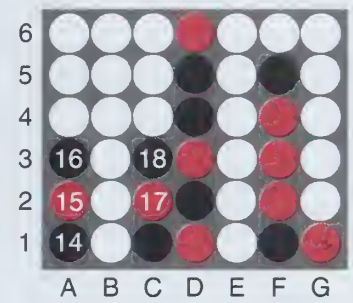
Red must play 31 C3. Otherwise, Black makes the push-up play at c3 and gets a winning double major threat, as shown in Diagram 315.

Diagram 316 shows yet another variation of this basic opening. Where should Red play move 19 now?

The only winning move for Red in Diagram 316 is 19 F6. If Black played there, Black would have a major threat at E5, but more important is that all of Red's other choices play into Black's hands. Diagram 317 shows one variation to indicate how easy it is for Black to win here if Red is careless.

DIAGRAM 316

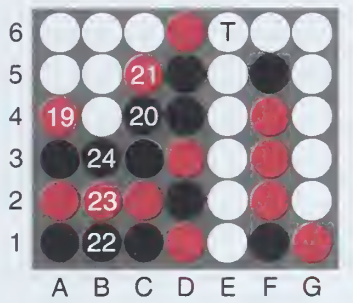
Red to play, with only one move to win



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! (14) a1 A2 a3 C2 c3 (19) __

DIAGRAM 317

Red to play (will lose)



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! a1 A2 a3 C2 c3 (19) A4?? c4 C5 b1 B2 b3! (25) __

In Diagram 317, Black threatens to play 26 b4 (double check) or 26 f6 (creating the adjacent threat pair e5-e6). Red can prevent only one of these moves.

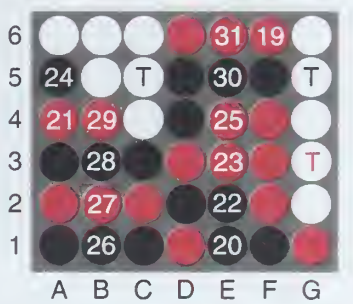
Even if Red plays 19 F6 correctly, Red will still have great trouble against an expert opponent. Suppose Black answers 19 F6 with 20 e1. Where should Red then play?

If the game continues as shown in Diagram 318, Black's two major odd-row threats defeat Red's single major threat. (If Red avoids 25 E4 and plays to capture E5, Black will get c5.) Red's 21 A4 turned out to be a subtle blunder. (You'd do well to study Black's play in this diagram. The move 30 b5 could have been substituted for e5.)

The proper Red play was 21 E2, but Red will then be faced with another quandary, shown in Diagram 319.

DIAGRAM 318

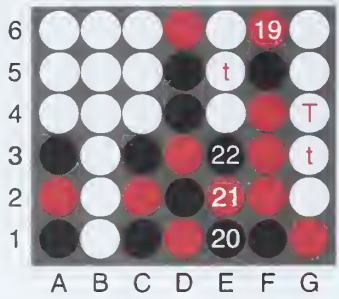
Black to play and win



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! a1 A2 a3 C2 c3 (19) F6! e1 A4?? e2! E3 a5! E4 b1! B2 b3! B4 e5 E6 (32) __

DIAGRAM 319

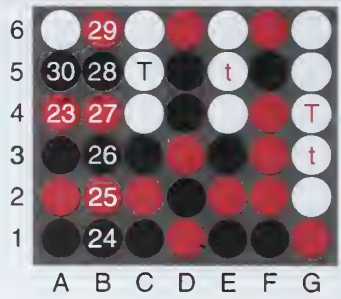
Red to play, with only one move to win



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! a1 A2 a3 C2 c3 (19) F6! e1 E2! e3 (23) __

DIAGRAM 320

Red to play and draw



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! a1 A2 a3 C2 c3 F6! e1 E2! e3 (23) A4? b1 B2! b3! B4! b5! B6 a5! (31) __

Diagram 320 is a possible continuation.

Black has achieved a major odd-row threat at c5. (If Red had played 29 A5 instead of B6, Black would take b6 with the same result.) Red will get a major odd-row threat along the E5-G3 diagonal, of course, but odd-row threat versus odd-row threat is a draw. Red can *always* win by playing *perfectly*, so where did Red go wrong in this variation?

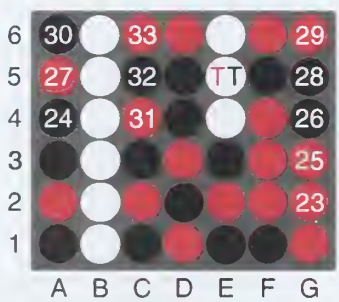
The answer is that it was Red's piece at A4 that allowed Black to play a5 and build a major threat. Red's 23 A4 seemed to have (a tiny bit of) offensive and defensive value, but it ended up helping Black.

If we look at the alternatives to 23 A4, we will see that almost all of them help Black! Red's only winning move in Diagram 319 is 23 G2, a move that has *absolutely no* offensive or defensive significance! (See Diagram 321 for the result.)

The guiding principle here is simple: Black seeks fifth-row horizontal threats, so both players should avoid the fourth row. If Red plays the (surprising) winning

DIAGRAM 321

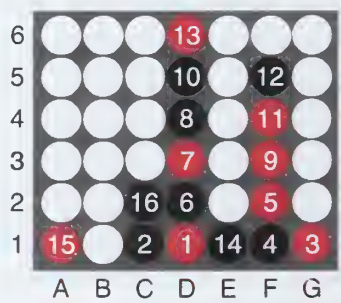
Black to play (will lose)



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! a1 A2 a3 C2 c3 F6! e1 E2! e3 (23) G2! a4 G3 g4 A5 g5 G6 a6 C4! c5 C6! (34) __

DIAGRAM 322

Red to play, with only one move to win



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! e1 A1 c2 (17) __

move 23 G2, the game plays out easily, but it will be Black who succumbs to zugzwang.

Black gets a major odd-row threat, but it's at the same cell as Red's winning threat. Black needs an odd-row threat in a *different column* to get the draw.

Here's another variation on this opening. Red also could have succeeded with 15 C2 or 15 E2, but where should Red play in Diagram 322?

A popular choice might be 17 C3, making a major odd-row threat at once. Diagram 323 is then a plausible continuation.

DIAGRAM 323

Black to play, with only one move to win



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! e1 A1 c2 (17) C3?? a2! C4 c5! A3 c6 F6 g2! G3 g4! G5 g6! A4 a5! A6 (32) __

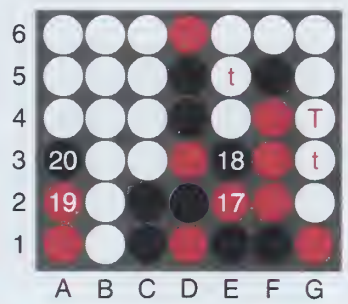
Black wins with two major odd-row threats. To achieve this, Black must play exactly as he or she did in Diagram 323 (the temporizing move 22 c6 could have been played at f6 or g2 also).

The move 17 C3 didn't work for Red. Although Black eventually relinquished the major threat at b2, it was essential to Black's victory, because it forced Red to play 19 C4 and give up 20 c5. (If Black got 20 c4, he or she would win quickly with the adjacent threat pair at b2-b3.)

Red has one way to prevent Black's major second-row threat, and that's how Red must play. Playing correctly, Red arrives at Diagram 324, with the depicted triple threat promising an eventual victory. However, Red still has the worry that Black will make a major odd-row threat on the left side. Where must Red play next in Diagram 324?

DIAGRAM 324

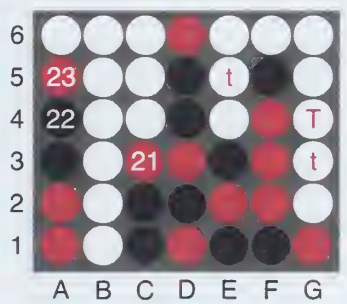
Red to play, with only one move to win



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! e1 A1 c2 (17) E2! e3 A2! a3 (21) __

DIAGRAM 325

Black to play (will lose)



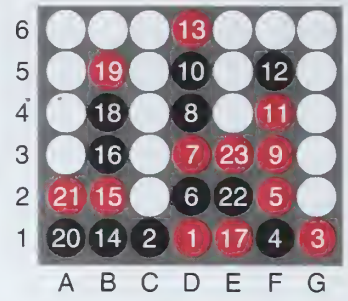
Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! e1 A1 c2 E2! e3 A2! a3 (21) C3! a4 A5! (24) __

Red must play 21 C3 (otherwise, Black can claim c5 with an easy push-up play). Red's 23 A5 is necessary as well. After this, Red's triple minor threat will prevail easily. (See Diagram 325.)

In Diagram 326, all of Red's plays (other than 3 G1) were uniquely forced. Red has played twelve disks, and only a single deviation was permitted if he or she is trying to win against an expert opponent!

DIAGRAM 326

Black to play (will lose)

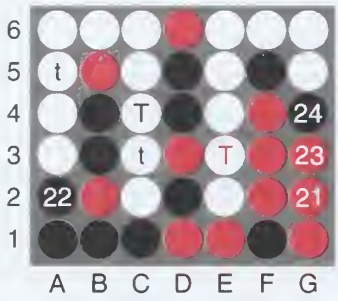


Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! b3 E1! b4 B5! a1 A2! e2 E3! (24) __

Let's pretend we're Black and want to punish Red if he or she deviates from the expert line just shown. Diagram 327 is a win for Black and shows why Red can't omit 21 A2. Diagrams 328, 329, 330, 332, and 333 show other variations where Red has deviated from the prescribed path. In each case, determine Black's only winning response.

DIAGRAM 327

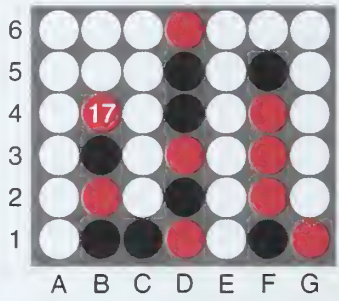
Red to play (will lose)



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! b3 E1! b4 B5! a1 (21) G2?? a2! G3 g4! (25) __

DIAGRAM 328

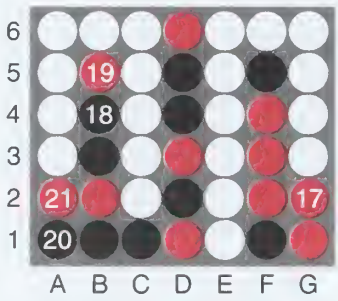
Black to play, with only one move to win



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! b3 (17) B4?? (18) __

DIAGRAM 329

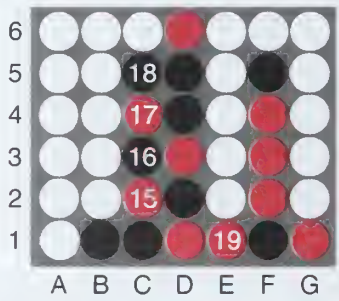
Black to play, with only one move to win



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! b3 (17) G2?? b4 B5 a1 A2 (22) __

DIAGRAM 330

Black to play, with only one move to win



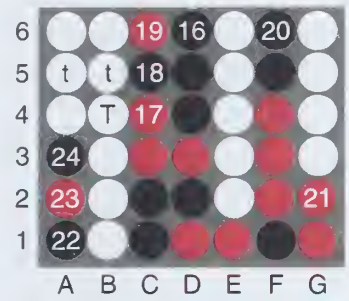
Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 (15) C2?? c3! C4 c5! E1 (20) __

Diagram 331 shows another variant.

The position shown in Diagram 332 is exactly the same as a common position (Diagram 192) from the 5-4 opening, *except* that Red has a very weak disk at G1 instead of a powerful disk at C4. Black is always delighted when he or she realizes this! (Similarly, Diagram 333 looks like a Candlesticks Opening in which Red has wasted a disk at G1.)

DIAGRAM 331

Red to play (will lose)

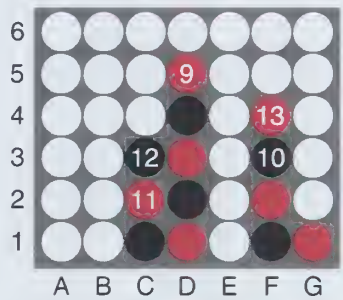


Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 E1?? c2 C3 (16) d6! C4 c5! C6 f6 G2 a1! A2 a3! (25) __

In Diagram 328, Black must play 18 b5, building minor odd-row threats in the fifth row. In Diagram 329, Black must play 22 e1 to build a major odd-row threat.

DIAGRAM 332

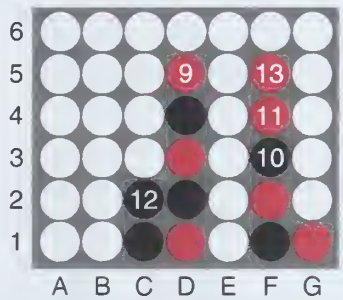
Black to play, with only one move to win



Moves: D1! c1 G1 f1 F2! d2 D3! d4 (9) D5?? f3! C2 c3 F4 (14) __

DIAGRAM 333

Black to play, with only one move to win

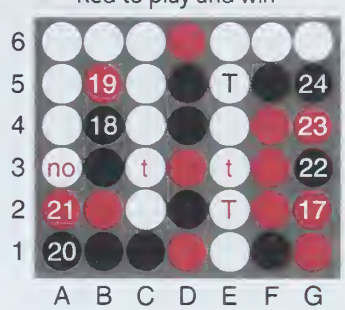


Moves: D1! c1 G1 f1 F2! d2 D3! d4 (9) D5?? f3! F4 c2 F5 (14) __

It's easy for Black to go wrong in Diagram 329 and play 22 g3 to kill Red's right-side threat, as shown in Diagram 334. There, Red has a winning triple threat, with Black's threat at e5 undercut by Red's major even-row threat at E2. Red must take care only not to play A3 (shown as "no"), since a Black disk at a4 would produce a Black major threat at c4 to overcut the Red minor threat at C3. (That is, Black would play up column C and capture c3.)

DIAGRAM 334

Red to play and win



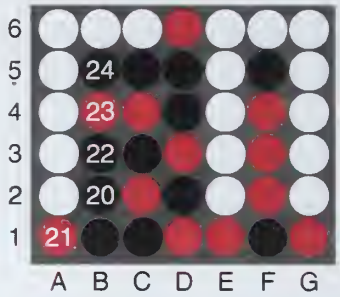
Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! b3 (17) G2?? b4 B5 a1 A2 g3?? G4! g5 (25) __

Did you solve Diagram 330? Black must adopt the forcing push-up play shown in Diagram 335.

In Diagram 333, Black must begin with the push-up play c3 C4 c5 (see Diagram

336.) After this, Black's second-row threats mean that Red's only chance is on the right side. Red has two minor odd-row threats but lacks the even-row threat he or she needs to go with them. Red may be trying to be a little tricky with the odd-looking move 17 F6; Black would certainly not play g2 if Red played 17 D6 but *might* overlook the danger in this diagram. (The play 18 g2 D6 g3 C6 loses for Black, though of course Black has no reason to play g2.)

DIAGRAM 335
Red to play (will lose)

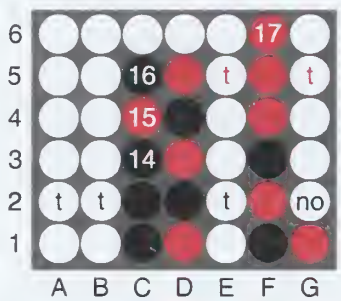


Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 C2?? c3! C4 c5! E1 (20) b2! A1 b3! B4 b5! (25) __

In Diagram 337, Black has more than one winning move; look for the quick mate in four. (See Diagram 372 for the solution.)

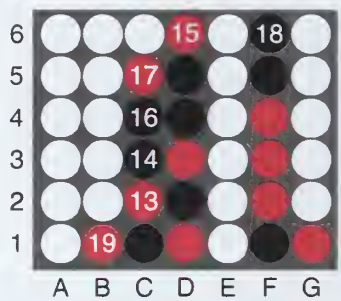
These variations shown after 1 D1 c1 G1 f1 F2 d2 require supersmart play by Red. To avoid these problems, Red will usually play 3 F1 instead of 3 G1. I now explore some of the opening variations in that case.

DIAGRAM 336
Black to play and win



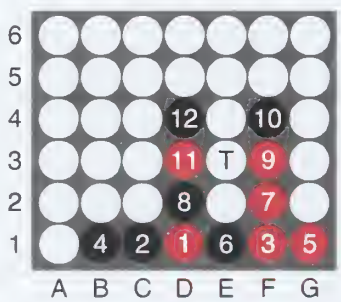
Moves: D1! c1 G1 f1 F2! d2 D3! d4 D5?? f3! F4 c2 F5 (14) c3! C4 c5! F6 (18) __

DIAGRAM 337
Black to play and win



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 (13) C2?? c3! D6 c4 C5 f6 B1 (20) __

DIAGRAM 338
Red to play and win



Moves: D1! c1 F1 b1 G1 e1 F2 d2 F3 f4 D3! d4 (13) __

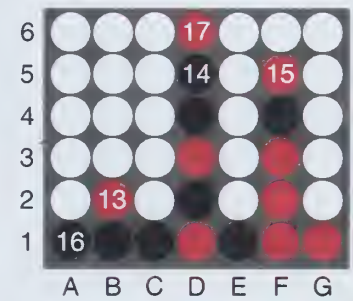
Diagram 338 shows an opening not uncommon among experts. Black has already formed a major odd-row threat at e3, so Red's position seems difficult. Not only will Red need to form a major odd-row threat, probably at E3 or G3, but also Red must prevent Black from constructing a second major odd-row threat on the left side. Where should Red play in Diagram 338?

D5 is often a key cell, and Red may be tempted to take it here, but this leads to a quick Black victory, as shown in Diagram 14 (see Chapter 2). Instead, Red must play 13 B2 or 13 C2. The game might then continue as shown in Diagram 339, with 17 D6 being a subtle blunder. Where should Black play, after 17 D6, to win?

In Diagram 340, Black plays perfectly at 16 a1, 18 a2, 20 b3, and 22 b5 and builds a left-side triple threat that, when combined with Black's major threat at e3, gives Black an easy win. (You may think of c4-c5 has an adjacent threat pair that, when played out, produces a major odd-row threat for Black at a5.)

DIAGRAM 339

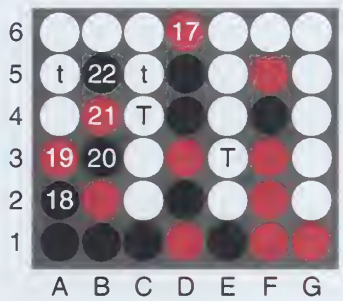
Black to play, with only one move to win



Moves: D1! c1 F1 b1 G1 e1 F2 d2 F3 f4 D3! d4 (13) B2 d5 F5! a1 D6?? (18) __

DIAGRAM 340

Red to play (will lose)

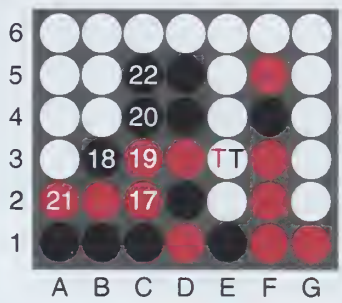


Moves: D1! c1 F1 b1 G1 e1 F2 d2 F3 f4 D3! d4 B2 d5 F5! a1 (17) D6?? a2! A3 b3! B4 b5! (23) __

In this game, the move 16 a1 may seem to have no offensive or defensive significance, and many players would fail to make it. In fact, as shown in Diagram 340, the value of 16 a1 comes from the follow-up 18 a2. Red can stop Black's win by playing 17 A2 or can instead win as shown in Diagram 341. (Red can also win with 19 B4 substituted for C3.)

DIAGRAM 341

Red to play, with only one move to win



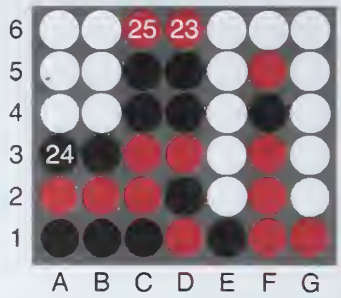
Moves: D1! c1 F1 b1 G1 e1 F2 d2 F3 f4 D3! d4 B2 d5 F5! a1 (17) C2 b3 C3 c4 A2! c5 (23) __

In Diagram 341, Black's disks outnumber Red's five to one in the upper half of the board, while Red has a corresponding majority in the lower right side. This is an elegant demonstration of the Third-Row Gambit. Where must Red play next?

Black has rich tactical possibilities on the left side, including the idea of playing b4-d6 with the winning adjacent threat pair a3-a4. Red can't react to this by playing B4; then Black would play b5 and get a routine endgame win with major odd-row threats at

both a5 and e3. Suppose instead that Red tries to thwart Black by playing 23 D6, as shown in Diagram 342. This gives the position in Problem 263, which Black solves by playing 24 a3. Red can't answer this at A4 or B4—Black would get a major fifth-row threat—so might as well choose to leave Black's powerful left side alone and make a useless move elsewhere.

DIAGRAM 342
Black to play and win

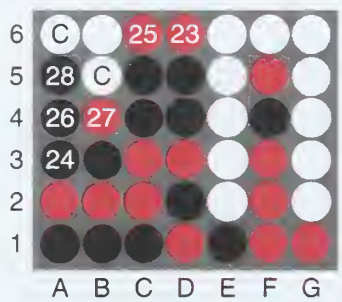


Moves: D1! c1 F1 b1 G1 e1 F2 d2 F3 f4 D3! d4 B2 d5 F5! a1 C2 b3 C3 c4 A2! c5 (23) D6?? a3! C6 (26) __

In Diagram 342, Black can first make some temporizing moves if he or she wishes but can then win the game only with an elegant forcing series (see Diagram 343). Black can also do the same thing in column B instead of column A.

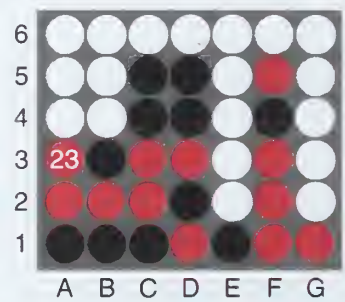
Return to Diagram 341 now, and determine what Red's only successful move is!

DIAGRAM 343
Red to play (will lose)



Moves: D1! c1 F1 b1 G1 e1 F2 d2 F3 f4 D3! d4 B2 d5 F5! a1 C2 b3 C3 c4 A2! c5 (23) D6?? a3! C6 a4 B4 a5 (29) __

DIAGRAM 344
Black to play (will lose)



Moves: D1! c1 F1 b1 G1 e1 F2 d2 F3 f4 D3! d4 B2 d5 F5! a1 C2 b3 C3 c4 A2! c5 (23) A3! (24) __

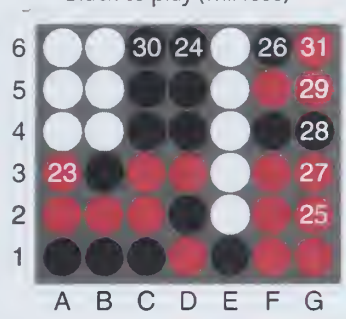
Diagram 344 shows the solution. The play 24 b4 A4 b5 A5 no longer works for Black, because A5 then completes a four-in-row for Red. Black has an amazing preponderance in the upper left of the board but can do nothing with it. The game might continue as shown in Diagram 345; Black is forced to play first on the right side and can accomplish nothing there. After a spurt of excitement, Red will end up with the routine win at E3.

I mentioned earlier that Red could have played 13 C2 in this opening. Let's look at that possibility. The most likely continuation is shown in Diagram 346.

In Diagram 346, Black may appear to have neutralized Red's major odd-row threat with one of his or her own at e5. Furthermore, Black's major threat at b4

DIAGRAM 345

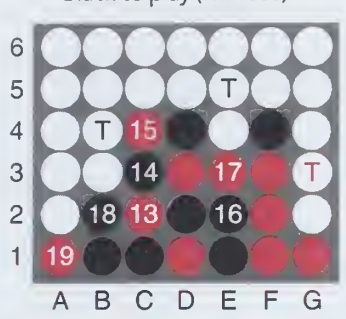
Black to play (will lose)



Moves: D1! c1 F1 b1 G1 e1 F2 d2 F3 f4 D3! d4 B2 d5 F5! a1 C2 b3 C3 c4 A2! c5 (23) A3! d6 G2 f6 G3 g4 G5 c6 G6! (32) __

DIAGRAM 346

Black to play (will lose)



Moves: D1! c1 F1 b1 G1 e1 F2 d2 F3 f4 D3! d4 (13) C2 c3 C4! e2 E3! b2 A1! (20) __

seems to curtail any Red chances on the left side. But this is one of those games that demonstrate the adage "First Player can win every time in CONNECT 4, but to do so he or she may have to play *very* well." In this case, Red will start by eliminating Black's threat at e5. It's Black's move now, but he or she can't preserve the threat because Red *has two different ways to eliminate this threat.* Do you see what they are?

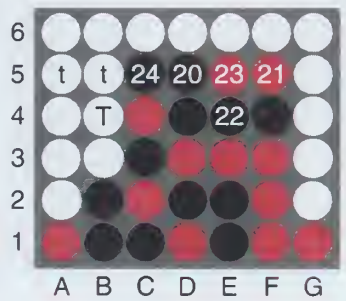
(By the way, instead of playing 18 b2 as shown in Diagram 346, Black might play 18 d5. See whether you can figure out the one sequence of forcing moves Red must then adopt to ensure that Black gets no left-side odd-row threat.)

One continuation is shown in Diagram 347. Red's 21 F5 allows Red to kill Black's e5 threat, but Black gets a triple threat in the upper left. (If Black had played 20 f5, Red would play 21 D5. Do you see why that also eliminates the e5 threat?)

Where must Red play move 25 in Diagram 347 to win?

DIAGRAM 347

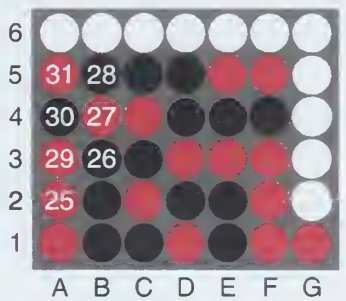
Red to play, with only one move to win



Moves: D1! c1 F1 b1 G1 e1 F2 d2 F3 f4 D3! d4 C2 c3 C4! e2 E3! b2 A1! (20) d5 F5! e4 E5! c5 (25) __

DIAGRAM 348

Black to play (will lose)

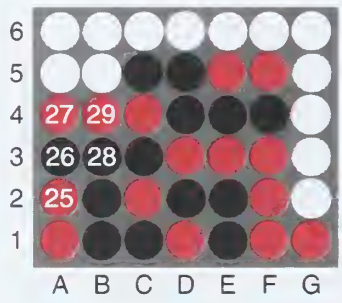


Moves: D1! c1 F1 b1 G1 e1 F2 d2 F3 f4 D3! d4 C2 c3 C4! e2 E3! b2 A1! d5 F5! e4 E5! c5 (25) A2! b3 B4! b5 A3! a4 A5! (32) __

Black threatens b3 B4 b5—a push-up play that would establish a major odd-row threat for Black at a5. That wouldn't be enough for Black to win, but it would be enough for a draw. The play 25 A2 is the only move Red can make to prevent that. In Diagram 348, I show the continuation where Black follows 25 A2 with a column B push-up play, but Red responds with a column A push-up play to destroy the Black threat. Black had nothing better to offer. Diagram 349 shows another variation with a similar result.

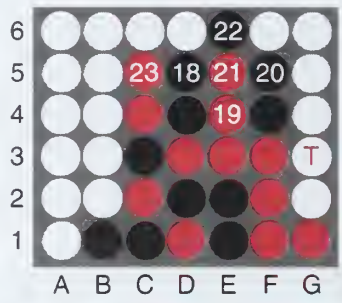
Earlier I mentioned the variation 18 d5. Diagram 350 shows how Red prevails then. (Red can substitute 23 A1 for C5.)

DIAGRAM 349
Black to play (will lose)



Moves: D1! c1 F1 b1 G1 e1 F2 d2 F3 f4 D3! d4 C2 c3 C4! e2 E3! b2 A1! d5 F5! e4 E5! c5 (25) A2! a3 A4! b3 B4! (30) __

DIAGRAM 350
Black to play (will lose)



Moves: D1! c1 F1 b1 G1 e1 F2 d2 F3 f4 D3! d4 C2 c3 C4! e2 E3! (18) d5 E4! f5 E5! e6 C5 (24) __

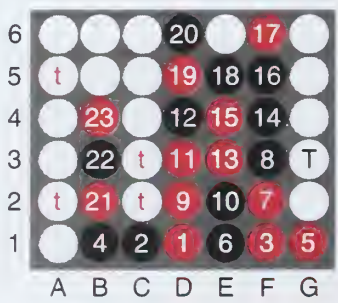
DcF Variations

There are far too many variants of the first-row opening to show them all. Although Black's 4 b1 seems weak in Diagram 351, it leads to a position where Red makes the subtle blunder 13 E3 and then needs precision just to draw. Red has an effective triple threat in this diagram, but Black has the counterthreat g3.

Red 5 E1 can be substituted for 5 G1 here. Black may then do best to play as in the Third-Row Gambit; two possible continuations are shown in Diagrams 352 and 353.

Diagrams 354 and 355 are variations on Diagram 139 (see Chapter 10). After Red errs with 11 D2, Black can win with either 12 e3

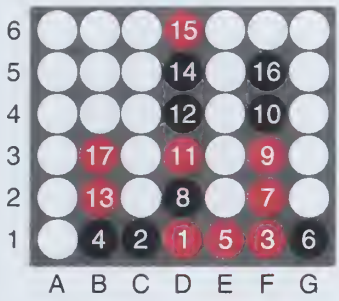
DIAGRAM 351
Black to play and draw



Moves: D1 c1 F1 b1 G1 e1 F2 f3 D2 e2 D3 d4 E3? f4! E4! f5! F6! e5! D5! d6 B2! b3 B4! (24) __

DIAGRAM 352

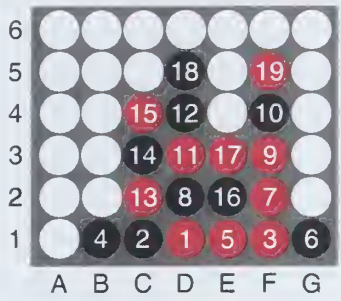
Black to play (will lose)



Moves: D1! c1 F1 b1 E1 g1 F2! d2 F3 f4 D3! d4 B2 d5 D6 f5 B3! (18) __

DIAGRAM 353

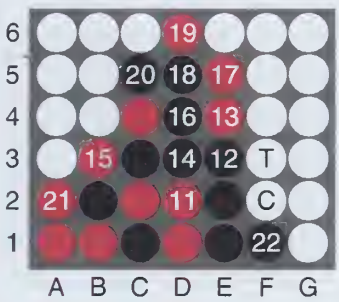
Black to play (will lose)



Moves: D1! c1 F1 b1 E1 g1 F2! d2 F3 f4 D3! d4 C2 c3 C4! e2 E3! d5 F5 (20) __

DIAGRAM 354

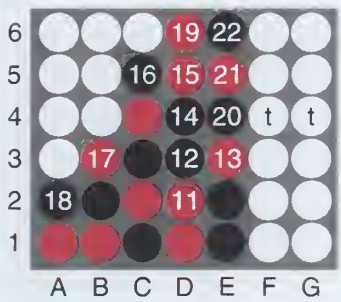
Red to play (will lose)



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 (11) D2?? e3 E4 d3! B3 d4! E5 d5! D6 c5 A2 f1 (23) __

DIAGRAM 355

Red to play (will lose)

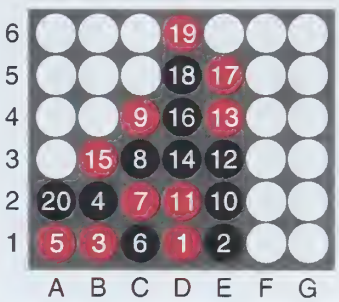


Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 (11) D2?? d3 E3 d4! D5 c5 B3 a2! D6 e4 E5 e6! (23) __

(Diagram 354) or 12 d3 (Diagram 355). (The former, in which Black's moves from 14 d3 through 18 d5 are the only winning line, is the solution to the mate in seven exercise mentioned in the discussion of Diagram 138.)

DIAGRAM 356

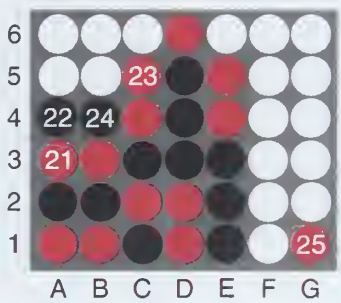
Red to play and draw



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 D2?? e3 E4 d3! B3 d4! E5 d5! D6 a2? (21) __

DIAGRAM 357

Black to play and draw



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 D2?? e3 E4 d3! B3 d4! E5 d5! D6 a2? (21) A3 a4! C5! b4! G1! (26) __

The DcF opening is, of course, the same as the DeB opening; so that you can practice viewing positions both ways, I show most problems and some further examples using this mirror-image form.

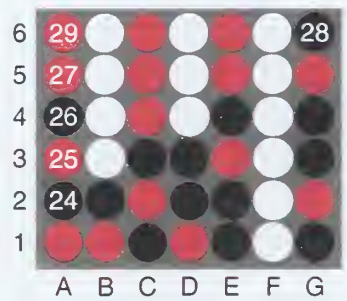
Diagram 356 shows a variation of Diagram 354 where Black makes a peculiar blunder. How can Red now salvage a draw? (There are two successful first moves.)

Diagram 357 shows one answer to Diagram 356. (Red can also start with 21 C5.)

The solution to this problem posed in Diagram 138 is as follows. If Black plays as shown in Diagram 358, he or she will soon be in zugzwang and forced to let Red get D5. If Black plays 24 d4, a series of forced moves follow, with Red finally playing D6 with a winning threat at A3. Instead, Black must temporize with 24 g6, as shown in Diagram 359.

DIAGRAM 358

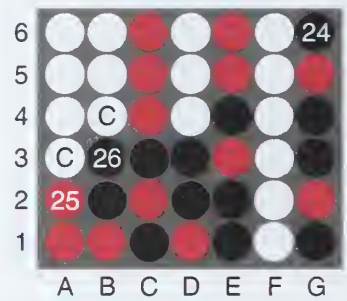
Black to play (will lose)



Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 E3?? e4 E5 d2 C5 g1 G2 d3! E6 g3 C6 g4! G5 (24) a2?? A3! a4 A5 g6 A6! (30) __

DIAGRAM 359

Red to play (will lose)



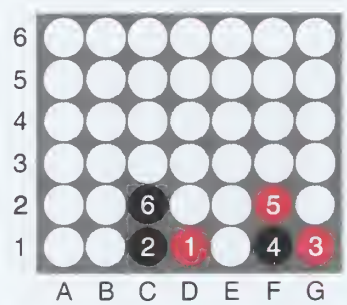
Moves: D1! e1 B1 b2 A1! c1 C2 c3 C4 e2 E3?? e4 E5 d2 C5 g1 G2 d3! E6 g3 C6 g4! G5 (24) g6! A2 b3 (27) __

DcG Variations

Red can defeat Black's 2 c1 by playing at 3 G1. When you get comfortable winning with 3 F1, try this alternate path to victory!

DIAGRAM 360

Red to play, with only one move to win



Moves: D1! c1 G1 f1 F2! c2 (7) __

DIAGRAM 361

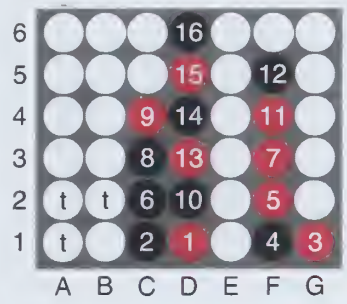
Black to play and draw



Moves: D1! c1 G1 f1 F2! c2 C3? c4! F3! g2! F4 f5! C5 c6! A1 a2 E1 a3 A4! a5! A6! f6! D2 d3! E2! (26) __

DIAGRAM 362

Red to play (will lose)



Moves: D1! c1 G1 f1 F2! c2 F3?? c3 C4 d2! F4 f5! D3 d4! D5 d6! (17) __

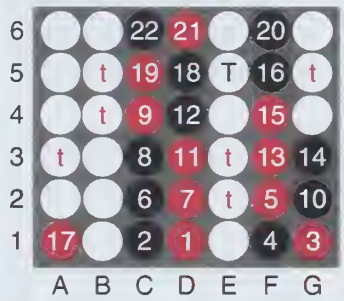
Diagram 360 is an opening where many Red players will go wrong with disk 7. A natural temptation is to occupy a third-row cell, but 7 C3 draws, as shown, for example, in Diagram 361. Black's 14 c6, killing Red's C6-F3 diagonal, was essential; note that 14 f6 would *not* fully kill the B2-F6 diagonal.

The other third-row play in Diagram 360 leads to a loss, as shown, for example, in Diagram 362.

Red should play 7 D2 instead of F3 in this opening but will need to continue skillfully. In Diagram 363, Red has solved many problems correctly but isn't out of the woods yet. Red has odd-row threats in all four remaining columns; in other words, Red has a (nonstandard) triple threat on the right side and another (sort of) triple threat on the left side. Black's major odd-row threat at e5 might lead to a drawn game against a triple threat involving column E (Black would play up column E so that Red's major threat ends up in column G) but perhaps not against a left-side triple threat, let alone two. However, the left-side triple threat is not properly formed (the threat B5 depends on E2, which in turn relies on E3), so the outcome will depend on timing.

DIAGRAM 363

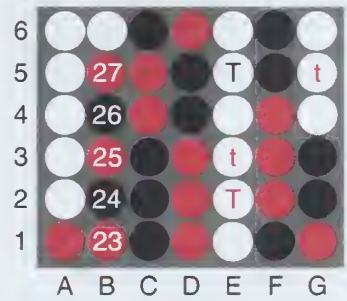
Red to play, with only one move to win



Moves: D1! c1 G1 f1 F2! c2 D2! c3 C4! g2 D3 d4 F3 g3 F4 f5 A1! d5 C5! f6 D6! c6 (23) __

DIAGRAM 364

Black to play (will lose)

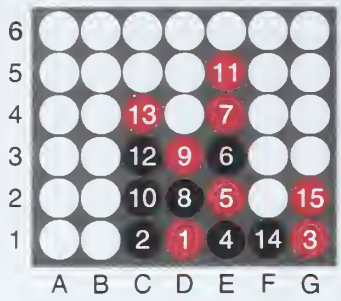


Moves: D1! c1 G1 f1 F2! c2 D2! c3 C4! g2 D3 d4 F3 g3 F4 f5 A1! d5 C5! f6 D6! c6 (23) B1! b2 B3! b4 B5! (28) __

Red plays 23 B1, of course, the only move that doesn't give up any of his or her threats. Play might continue as shown in Diagram 364, with Red's right-side triple threat obviously victorious. Black could have played in column E to activate his or her e5 counterthreat, but then Red would get E2 and an easy win with his or her *left-side* threats.

DIAGRAM 365

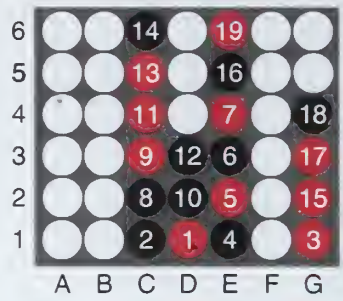
Black to play (will lose)



Moves: D1! c1 G1 e1 E2! e3 E4! d2 D3! c2 E5! c3 C4! f1 G2 (16) __

DIAGRAM 366

Black to play (will lose)



Moves: D1! c1 G1 e1 E2! e3 E4! c2 C3! d2 C4 d3 C5! c6 G2 e5 G3 g4 E6 (20) __

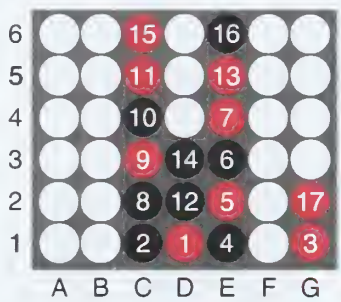
Diagrams 365 and 366 show additional variations of the first-row opening.

In Diagram 367, G2 is a better temporizing move than F1 because Red will get G3 or G4 before Black can exploit his or her tactical threats at f4-f3-g3.

In Diagram 368, Red's 21 D6 is key. If Black were permitted to occupy d6, he or she would get a triple threat at a3-b4-b5.

DIAGRAM 367

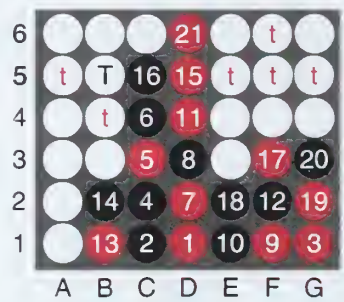
Black to play (will lose)



Moves: D1! c1 G1 e1 E2! e3 E4! c2 C3! c4 C5 d2 E5! d3 C6 e6 G2! (18) __

DIAGRAM 368

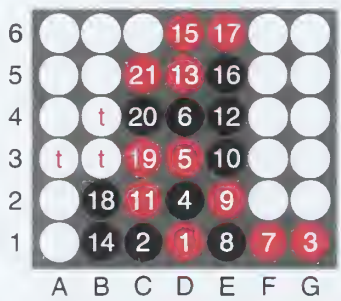
Black to play (will lose)



Moves: D1! c1 G1 c2 C3 c4 D2 d3 F1! e1 D4! f2 B1! b2 D5! c5 F3! e2 G2 g3 D6! (22) __

DIAGRAM 369

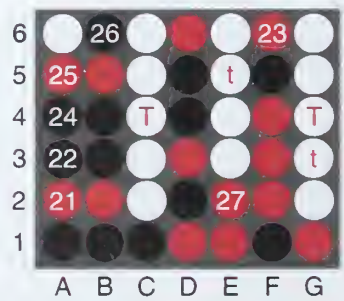
Black to play (will lose)



Moves: D1! c1 G1 d2 D3 d4 F1! e1 E2! e3 C2 e4 D5! b1 D6! e5 E6! b2 C3! c4 C5! (22) __

DIAGRAM 370

Black to play (will lose)



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! b3 E1! b4 B5! a1 (21) A2! a3 F6 a4 A5! b6 E2 (28) __

Diagram 369 is another interesting variant.

In Diagram 301, Red's winning move is 21 A2, as shown in Diagram 370. Red gets a triple threat on the right side and a major threat at C4 to undercut Black's potential threat at c5. Black's 24 a4 may seem to be a weak move, since Black's chances evaporate when his or her minor threat at c3 evaporates, but Black has no good alternative. If Black held a2 instead of Red, Black would already have a major threat at c4, forming a triple threat a5-c3-c4, and would even be threatening the push-up play a3 A4 a5 to get a major threat at c3.

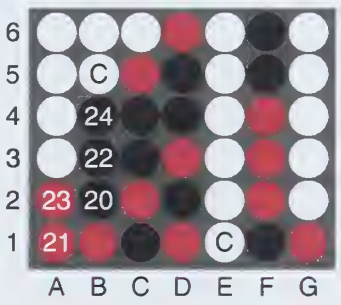
The "logical" move 21 E2 in Diagram 301 loses to e3, as shown in Diagram 371, which ends in an elegant flurry of forcing moves.

DIAGRAM 371
Red to play (will lose)



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 D6! b1 B2! b3 E1! b4 B5! a1 (21) E2?? e3! B6 c2 F6 g2 A2 a3! G3 g4! G5 g6! A4 a5! C3 c4! E4 e5! (39) __

DIAGRAM 372
Red to play (will lose)



Moves: D1! c1 G1 f1 F2! d2 D3! d4 F3! d5 F4! f5 C2?? c3! D6 c4 C5 f6 B1 (20) b2 A1 b3 A2 b4! (25) __

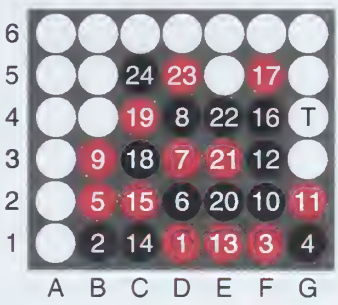
Diagram 372 gives the solution to the problem in Diagram 337.

DbF Variations

Diagram 373 shows an exquisite, instructive game. Early in the game, the players seem to have built threats on the second and third rows, but these evaporate. Black prevails by playing very precisely. Red's only blunder is 9 B3, a third-row move that many players would make.

In the opening shown in Diagram 374, Black plays an early 6 d2, letting Red take key cells in the second and third rows but hoping for some counterplay in the fifth row, as in the Third-Row Gambit. I'd definitely avoid the 6

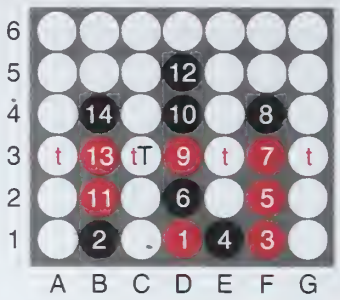
DIAGRAM 373
Red to play (will lose)



Moves: D1! b1 F1 g1 B2 d2 D3 d4 B3?? f2! G2 f3 E1 c1! C2 f4! F5 c3! C4 e2! E3 e4! D5 c5! (25) __

d2 play Black makes, but many players will play this way. (Red has *four* odd-threated columns, but the term *winning quadruple threat* is reserved for a special configuration.) Where should Red play next in Diagram 374?

DIAGRAM 374
Red to play, with only one move to win



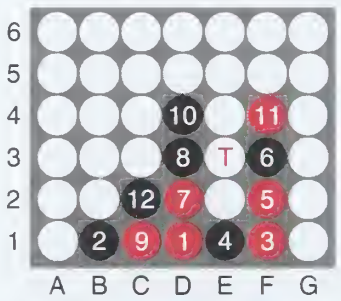
Moves: D1! b1 F1 e1 F2! d2 F3 f4
D3! d4 B2! d5 B3 b4 (15) __

Red must next play 15 C1 in Diagram 374 to kill Black's threat at e3. There's no mystery to this; Black already has one major odd-row threat; a second would give Black victory. That Red has four odd-threated columns instead of an ordinary triple threat is irrelevant. To win against two Black major odd-row threats, Red would need two major odd-row threats of his or her own, *and* one of those would have to be in a third column, separate from Black's threats.

In Diagram 375, Red has two winning plays and five losing plays. Only experts will get this right, but try anyway: Where should Red play next in this diagram? (If you solve this problem, maybe *you* should be writing this book!)

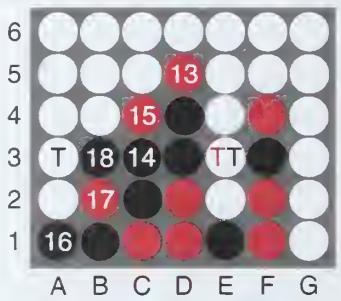
A popular choice might be 13 D5. If the game continues as shown in Diagram 376, Black gets two major odd-row threats to win. (Black's major threat at e4 doesn't count as part of an adjacent threat pair because Red's major threat at E3 undercuts e4.)

DIAGRAM 375
Red to play and win



Moves: D1! b1 F1 e1 F2! f3 D2!
d3 C1 d4 F4! c2 (13) __

DIAGRAM 376
Red to play (will lose)

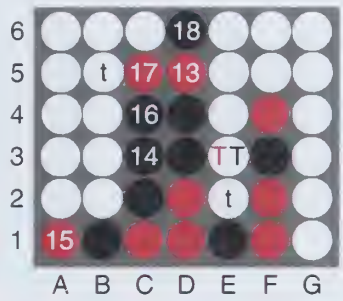


Moves: D1! b1 F1 e1 F2! f3 D2! d3 C1 d4
F4! c2 (13) D5?? c3! C4 a1 B2 b3! (19) __

Red can play 15 A1 to make the game a bit tougher for Black. Then play might continue as shown in Diagram 377, with Black's triple threat giving him or her a winning endgame. (Eventually Red will be forced to play E2 or B4.) Black's move 18 d6 is key (although it could have been delayed) to prevent Red from getting A3 and D6 and making B4 into an undercutting threat.

DIAGRAM 377

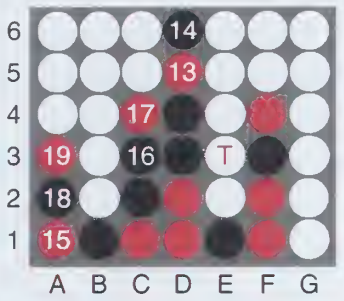
Red to play (will lose)



Moves: D1! b1 F1 e1 F2! f3 D2! d3 C1 d4 F4! c2 (13) D5?? c3! A1 c4 C5 d6 (19) __

DIAGRAM 378

Black to play (will lose)



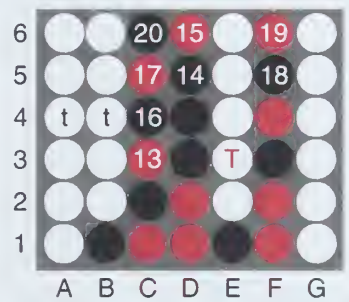
Moves: D1! b1 F1 e1 F2! f3 D2! d3 C1 d4 F4! c2 (13) D5?? d6?? A1! c3 C4! a2 A3 (20) __

The position shown in Diagram 377 has surprising subtlety. Black may calculate that there is no hurry to take 14 c3, but Diagram 378 indicates how Red then kills Black's counterthreat.

So 13 D5 loses for Red in Diagram 375. Another popular choice might be 13 C3. A likely continuation is shown in Diagram 379. (This game is not going to end in a tie. If Red can't win at E3, Black will win with the minor even-row threats shown in Diagram 379.) Both players have avoided columns A, B, E, and G in this game, but it is Red who finally succumbs to zugzwang. Red's 21 G1 would let Black play g2 and have no other purpose; 21 E2 gives up Red's only threat; 21 A1 would let Black build a major odd-row threat by playing a2. So Red tries 21 B2, but now it is the d5-a2 diagonal that gives victory for Black, as shown in Diagram 380.

DIAGRAM 379

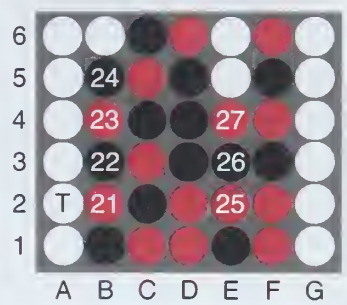
Red to play (will lose)



Moves: D1! b1 F1 e1 F2! f3 D2! d3 C1 d4 F4! c2 (13) C3?? d5! D6 c4! C5 f5 F6 c6! (21) __

DIAGRAM 380

Black to play and win



Moves: D1! b1 F1 e1 F2! f3 D2! d3 C1 d4 F4! c2 (13) C3?? d5! D6 c4! C5 f5 F6 c6! (21) B2 b3! B4 b5! E2 e3! E4 (28) __

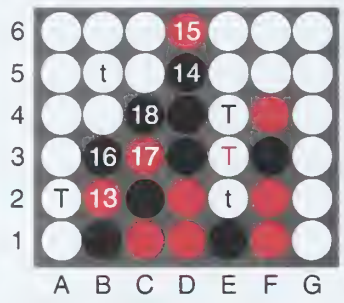
In Diagram 380, Black is left with only a single even-row threat, at a2—but one even-row threat is all it takes for Black to win, once Black has confounded Red's odd-row threats.

So 13 C3 also loses for Red in Diagram 375. Red's 13 B2 may also be plausible, but again careful Black play prevails, as shown in Diagram 381, where there are two odd-threated columns, so Black has zugzwang

control and wins at a2. (Red's G3 doesn't count as a threat, of course. Red takes G3 and gets a major odd-row threat in column E, but he or she *already has* a column E threat. Black's 16 b3 is a key hard-to-find move.)

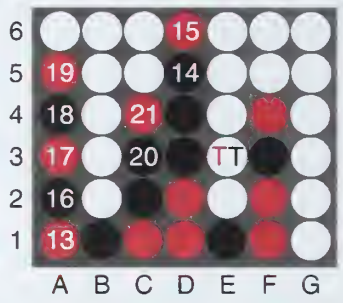
Finally, Red's winning move in Diagram 375 is the bizarre-looking play 13 A1, with a likely continuation shown in Diagram 382. Red must play as shown, except for a few deviations: 17 B2 would work instead of A3. (The useless move G1 could have been played by Red at almost any point since it forces the response g2.)

DIAGRAM 381
Red to play (will lose)



Moves: D1! b1 F1 e1 F2! f3 D2! d3 C1 d4 F4! c2 (13) B2?? d5 D6 b3! C3 c4! (19) __

DIAGRAM 382
Black to play (will lose)

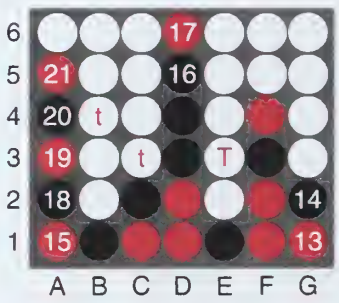


Moves: D1! b1 F1 e1 F2! f3 D2! d3 C1 d4 F4! c2 (13) A1 d5 D6! a2 A3 a4 A5 c3 C4! (22) __

The other winning move in Diagram 375 is, surprisingly, 13 G1. (The play G1 is of no value to Red, but it does force Black's response, after which Red gets on with the necessary moves on the left side.) If the game continues as shown in Diagram 383, Red has a winning threat at E3, along with the A5-D2 diagonal threats to undercut Black's b5-e2 undercutting diagonal.

The key to Diagram 375, therefore, is that Red must play in columns A and B to resolve the plethora of potential Black threats there. The play 13 A1 or 15 A1 is not good enough unless Red follows through. If Red wastes time with 19 F5 or 19 G3 in Diagram 383, Black can play either 20 a3 or 20 c3 and win easily.

DIAGRAM 383
Black to play (will lose)



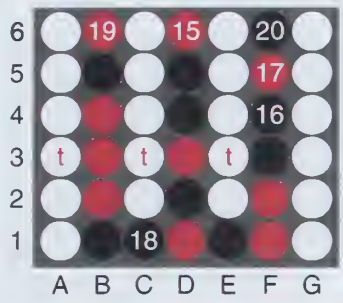
Moves: D1! b1 F1 e1 F2! f3 D2! d3 C1 d4 F4! c2 (13) G1 g2 A1! d5 D6! a2 A3 a4 A5! (22) __

DbB Variations

Red may play B2 when Black takes b1 early in the opening. After this move, there are far too many options to do a full study, but I review a few sample variations here.

DIAGRAM 384

Red to play, with only one move to win



Moves: D1! b1 B2 d2 D3 d4 B3 d5 B4 b5 F1 e1 F2 f3 (15) D6! f4 F5! c1 B6 f6 (21) __

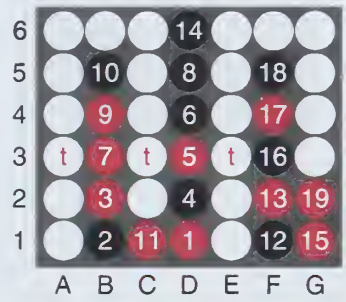
Diagram 384 shows a variation that has been played well. Red's 15 D6 is mandatory (the particular configurations of disks in columns B, D, and F make cells such as A1 and C1 almost worthless, and F4 would give up f5); D6 may lead to threats along the C3-D6 and D6-G3 diagonals.

In Diagram 384, Red has a triple minor threat and should win. (The three-column odd-odd-odd triple threats are less common than the two-column even-odd-odd variety, but don't overlook them.) Since 21 G1 g2 would give Black a quick win with the adjacent threat pair e3-e4, Red's only correct move is 21 A1, with victory following easily.

Diagram 385 shows another variation. Red 15 F3 might win easily, but Red explores a slightly more interesting win with 15 G1. Red plays 19 G2 so Black can't play there and get an overcutting threat.

DIAGRAM 385

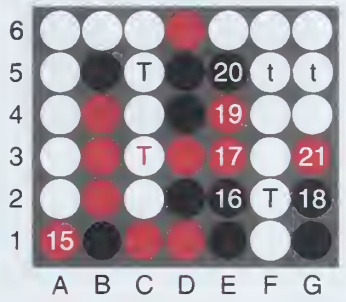
Black to play (will lose)



Moves: D1! b1 B2 d2 D3 d4 B3 d5 B4 b5 C1 f1 F2 d6 G1 f3 F4! f5 G2! (20) __

DIAGRAM 386

Black to play and win



Moves: D1! b1 B2 d2 D3 d4 B3 d5 B4 b5 C1 e1 D6! g1 (15) A1?? e2 E3 g2! E4 e5! G3 (22) __

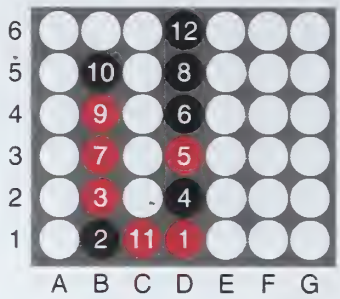
Problem 211 is also taken from this opening. There Red has a triple minor threat at A3-C3-E3 but must maintain column F possibilities as well and play 15 G2 before Black seeks an f2 threat. Otherwise, Black may get the winning position shown in Diagram 386.

The way to judge this endgame quickly is to recall that Black's triple threat f2-f5-g5 becomes equivalent to a major odd-row threat at f5. Then, with major odd-row threats in two columns, Black wins. From Diagram 386, Black will win automatically (as long as he or she doesn't do something foolish like playing f1 before he or she is forced to).

Diagram 387 shows another variation. Red will probably next take 13 F1, followed by 15 F2 or E2. Red can also win with 13 G1, answering 14 f1 with 15 F2.

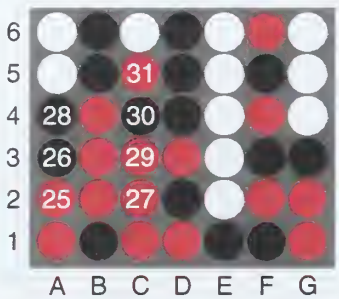
If Red plays 13 G1 in Diagram 387, the position shown in Problem 168 is likely to result, as Red blocks potential Black minor threats with 15 F2, 17 F4, and 19 G2 and temporizes with 21 F6, 23 A1. If Red solves Problem 168 correctly (starting with A2), the position in Diagram 388 will result. Although 31 C5 is not shown with an exclamation point (31 A5 also kills Black's fifth-row threat), the tempo-gaining power of Red's push-up play C1 C2 C3 C5 was essential.

DIAGRAM 387
Red to play and win



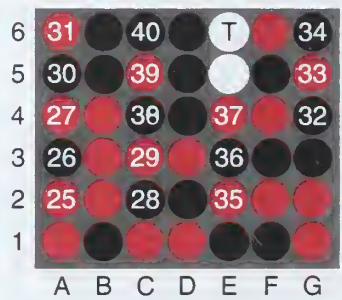
Moves: D1! b1 B2 d2 D3 d4 B3 d5 B4 b5 C1 d6 (13) __

DIAGRAM 388
Black to play (will lose)



Moves: D1! b1 B2 d2 D3 d4 B3 d5 B4 b5 C1 d6 G1 f1 F2! f3 F4! f5 G2! g3 F6 b6 A1 e1 (25) A2! a3 C2! a4 C3! c4 C5 (32) __

DIAGRAM 389
Red to play (will lose)



Moves: D1! b1 B2 d2 D3 d4 B3 d5 B4 b5 C1 d6 G1 f1 F2! f3 F4! f5 G2! g3 F6 b6 A1 e1 (25) A2! a3 A4?? c2! C3 a5! A6 g4! G5 g6! E2 e3! E4 c4! C5 c6! (41) __

If Red plays 25 A2 but then blunders with 27 A4, Diagram 389 is a likely result—pretty because of the utter precision Black must exercise to win.

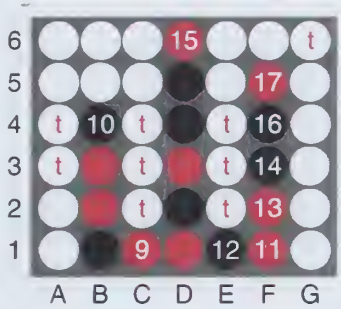
Diagrams 390 and 391 show two variations of a common opening when Red plays 9 C1 instead of B4 as in the above variations. (The moves 9 F1 and 9 G1 are also winning moves for Red; indeed, they may lead to easier play for Red.)

Diagram 392 shows another variation of the DbBdDd opening, with Red's 11 B3 a key possibly hard-to-find move. Black has nowhere to build a counterthreat, so Red will win with a triple minor threat.

If Diagram 392 makes this variation look too easy, see what happens when Red plays slightly differently, as shown in Diagram 393.

DIAGRAM 390

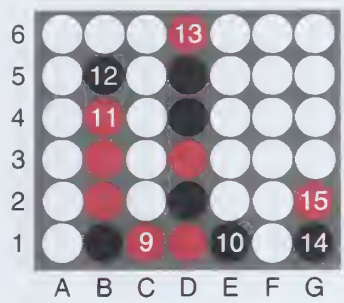
Black to play (will lose)



Moves: D1! b1 B2 d2 D3 d4 B3 d5 (9)
C1 b4 F1! e1 F2! f3 D6! f4 F5! (18) __

DIAGRAM 391

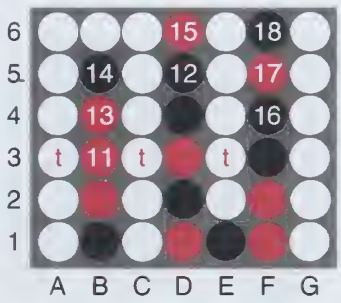
Black to play (will lose)



Moves: D1! b1 B2 d2 D3 d4 B3 d5 (9)
C1 e1 B4! b5 D6! g1 G2! (16) __

DIAGRAM 392

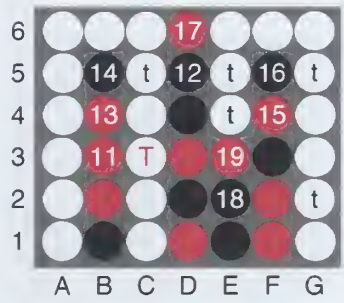
Red to play and win



Moves: D1! b1 B2 d2 D3 d4 F1 e1 F2! f3
(11) B3! d5 B4 b5 D6! f4 F5! f6 (19) __

DIAGRAM 393

Black to play and win



Moves: D1! b1 B2 d2 D3 d4 F1 e1 F2! f3 (11)
B3! d5 B4 b5 F4? f5! D6? e2! E3 (20) __

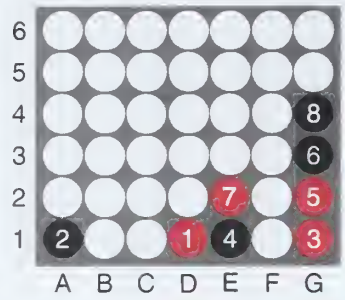
Now Black's plethora of minor threats will lead to Black victory. (Black's four threats in the columns E and G form a triple threat that neutralizes Red's C3. Black needs a fifth threat to win rather than just draw but already has it at c5.) A comparison of Diagrams 392 and 393 shows that the lesson is to remember the importance of the odd rows; Black's f5 disk in Diagram 393 works better than f4 in Diagram 392. (There are many similar situations where a player should prefer a play at D6 to one at F4; F4 lets the opposing player capture f5, while D6 gives up nothing.)

DaG Variations

In the opening D1 a1 G1, both a1 and G1 may seem like somewhat weak moves, but this "Swedish opening" (so called because it originated with some of Sweden's strong players) is frequently played, by arrangement, between good players, as leading to a severe challenge for Red. Diagram 394 on the next page is one way to continue.

DIAGRAM 394

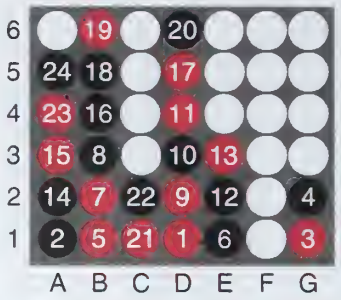
Red to play and win



Moves: D1! a1 G1 e1 G2! g3 E2! g4 (9) __

DIAGRAM 395

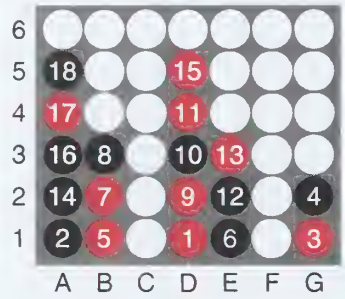
Red to play and draw



Moves: D1! a1 G1 g2 B1 e1 B2! b3 D2? d3! D4 e2! E3! a2! A3 b4! D5 b5! B6! d6 C1 c2 A4 a5 (25) __

DIAGRAM 396

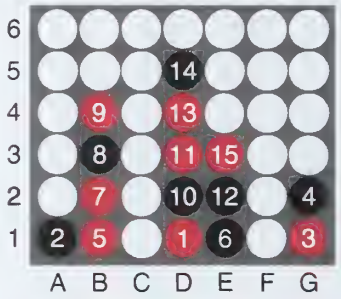
Red to play and draw



Moves: D1! a1 G1 g2 B1 e1 B2! b3 D2? d3! D4 e2! E3! a2! D5 a3! A4! a5! (19) __

DIAGRAM 397

Black to play (will lose)



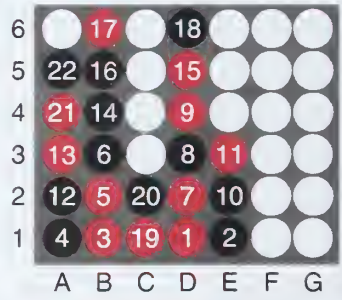
Moves: D1! a1 G1 g2 B1 e1 B2! b3 B4! d2 D3! e2 D4! d5 E3! (16) __

Diagrams 395 and 396 show variations in which Red errs with 9 D2. According to Swedish grand master "Eliten," he has yet to see a human player play correctly the first time the player encounters the position shown in Diagram 397.

It is interesting to contrast Diagram 395 with a variation of the DeB opening shown here as Diagram 398. Here, if Red next relinquishes his or her future threats on the D5-G2 diagonal by playing 23 G1?, Black will grab 24 g2! immediately, producing the position in Diagram 395.

DIAGRAM 398

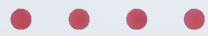
Red to play and win



Moves: D1! e1 B1 a1 B2! b3 D2 d3 D4 e2 E3 a2 A3! b4! D5! b5 B6! d6 C1 c2 A4 a5 (23) __

Chapter 14

CONNECT 4 PopOut



In 2009, Hasbro began marketing a new CONNECT 4 set with a "PopOut bar." With this set it is possible to poke a disk in the bottom row and pop it out, or remove it from the board. When the disk pops out, any other disks in the same column descend, under the force of gravity. For example, if you pop out the disk at B1, the disk originally at B2 will move to B1, the disk at B3 to B2, B4 to B3, and so on.

Hasbro includes rules for three games in its new set:

- Original CONNECT 4—the game I discuss in 12 of the first 13 chapters of this book.
- PopOut—a game similar to original CONNECT 4, but with a special pop-out move.
- Pop 10—a fun, fast-moving game in which *every* move is a pop-out move.

In each of the pop-out games (PopOut and Pop 10), you are allowed to pop out disks only of your own color. There is a PopOut bar with two positions: open for the pop-out games, closed for original CONNECT 4. (It is easy to change the PopOut bar position in the middle of a game; when playing PopOut, heavy-handed players might even prefer to leave the bar in the closed position except when actually making a pop-out move.)

PopOut

PopOut is a fun and exciting game that many players will prefer to the original CONNECT 4 game. Space constraints do not permit me to treat it in as much depth as I have done for the original CONNECT 4; here I give the rules, strategic principles, and a few illustrative games.

Rules of PopOut

The rules of PopOut are exactly the same as those of CONNECT 4, except for the pop-out move. At a player's turn, *instead* of dropping a disk of his or her color into the board at the top, he or she may pop a disk of his or her own color out at

the bottom, with remaining disks in that column each descending one row. If this pop-out move leads to a four-in-row, the game is won by the player whose color matches the four-in-row. There are three special rules to handle special cases associated with PopOut:

1. If the pop-out move leads to two (or more) four-in-rows, so that both Red and Black seem to have won, then *the player who made the pop-out move wins.*
2. When the board is completely full (with 42 disks), the player to move has his or her choice: play a pop-out move or declare the game to be drawn.
3. It may happen that a repeating series of pop-out moves and regular moves means a game never ends. (This is similar to a "perpetual check" in chess.) When this happens, either player can point out that the same game position has occurred at least three times and declare the game drawn.

While the first special rule will need to be applied quite frequently, it will be more seldom that the second and third special rules come into play. You might even guess that the repeating moves in the third special rule would happen only if both players were being deliberately silly, but the game in Diagram 404 (see the section "PopOut: Example Games") shows both players playing rather well and Red needing the repetition to salvage a draw.

Strategic Principles of PopOut

Same parity as CONNECT 4. In ordinary CONNECT 4, after Black takes turn 10, there will be ten disks on the board (five red, five black). The same applies in PopOut if there have been no pop-out moves. If Black plays a pop-out move at his or her turn 10, there will be eight disks on the board (five red, three black). Thus, in either case there will be an even number of disks after Black's move. Similarly, there will always be an odd number of disks on the board after any Red move, regardless of how many pop-out moves have been played. For this reason, the *fundamental strategic rule of CONNECT 4 still applies*: Red (First Player) tries to establish a winning odd-row threat; Black (Second Player) is more likely to win with an even-row threat.

Nonwinning pop-out moves are usually wrong. As I just mentioned, if Black plays a pop-out move at turn 10, he or she will be left with three black disks to Red's five red disks. This is a huge handicap, so pop-out moves must not be

made lightly. In most games of PopOut, play will proceed without any pop-out moves at all until Red or Black plays a pop-out move and wins at once. There are exceptions, however, and in the illustrative games I describe examples where a player needs *two* pop-out moves to win.

Draws are rare. Because the pop-out move gives players new ways to win, drawn games are uncommon in PopOut. This keeps the tension and excitement high! When play is going poorly in ordinary CONNECT 4, a player can abandon his or her own offense and try to secure a draw, but this tactic won't work in PopOut. Red must be especially alert for a possible Black pop-out move creating a second-row win.

Play in lower rows. In ordinary CONNECT 4, the third row is usually key, and it is often correct to grab the third-row cell whenever a second-row play by the opponent allows the possibility. Cells on the fourth and fifth rows are also often valuable. If you've mastered CONNECT 4, you've learned to play on the fourth or fifth row in many situations where a less-advanced player might think a first- or second-row play is more logical. This principle changes in PopOut. Since pop-out moves are possible *only* with first-row cells of *a player's own color*, occupying the first row becomes more desirable. Even when a player has no intention of ever popping that disk out, the fact that the player occupies that first-row cell prevents the opposing player from playing a pop-out move there. Furthermore, because of the possibility of a sudden win on the first or second row after a pop-out move, disks on the second row are more valuable in PopOut than in ordinary CONNECT 4.

Red should win. In ordinary CONNECT 4, Red must play almost perfectly to win against an expert Black player. In PopOut, Red can relax somewhat: His or her advantage is much stronger. The same move a player would make in ordinary CONNECT 4 will often work in PopOut; the player must just remember to lean toward a first- or second-row play in close situations where he or she would lean toward a play higher in the board in ordinary CONNECT 4. Black should try to occupy cells in the first, second, and third rows and should look for a chance to pop out a disk and win suddenly.

If you and your partner get so good at PopOut that Red wins almost every game, you may wish to handicap Red. Restrict Red to a single pop-out move per game (letting Black still have unlimited pop-outs).

PopOut: Example Games

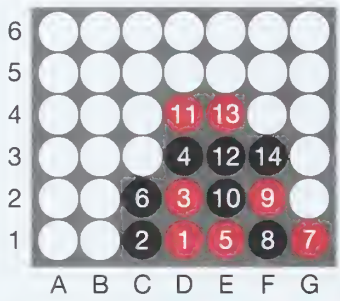
As far as I know, there has not yet been a complete computer analysis of PopOut. Red's strongest first move is 1 D1, but it probably is not the *only* winning move, so it is not shown in these diagrams with an exclamation point.

Diagram 399 shows a typical development. In ordinary CONNECT 4, 3 D2 would be a bad move, but the increased value of first- and second-row cells makes it fine here. In the diagram, Black may seem on the verge of victory, but Red pops out at E1 and gets four-in-row.

Diagram 400 follows the same pattern as a standard CONNECT 4 opening. Just as in ordinary CONNECT 4, Red's disk 7 can be played at E2 or B1. (First- and second-row moves are good in PopOut.) Unlike in ordinary CONNECT 4, Red's disk 3 and disk 5 at D3 and D5 were *not* essential: First-row moves would have worked also. For the same reason, Black's 4 d4 was a weak move.

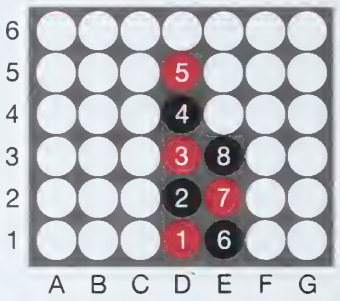
Now see whether you can guess where Red must play next in the diagram.

DIAGRAM 399 (PopOut rules)
Red to play, with only one move to win



Moves: (1) D1 c1 D2 d3 E1 c2 G1 f1 F2! e2 D4 e3 E4! f3 (15) __

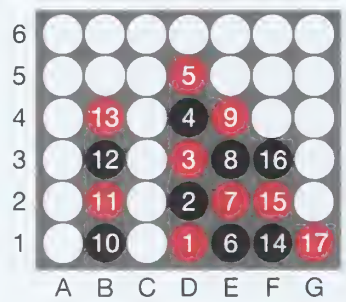
DIAGRAM 400 (PopOut rules)
Red to play and win



Moves: (1) D1 d2 D3 d4 D5 e1 E2 e3 (9) __

In Diagram 401, players continue with an ordinary CONNECT 4 opening, but then Black makes threatening plays in column F. (Black could have begun his or her columm F attack at disk 10 or disk 12.) Red's 17 G1 prevents an easy Black win (g1 followed by a pop-out move at f1), but Black still prevails. See whether you can find Black's next, winning move.

DIAGRAM 401 (PopOut rules)
Black to play, with only one move to win



Moves: (1) D1 d2 D3 d4 D5 e1 E2 e3 E4? b1 B2 b3 B4 f1! F2 f3! G1 (18) __

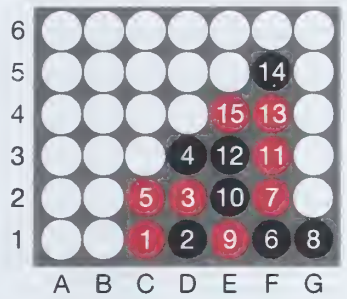
In the solution to Diagram 401, the game will continue 18 c1! F4 f5!, with Black winning on the following move with a column F pop-out move. Thus, Red's 9 E4, which wins in ordinary CONNECT 4, loses in PopOut. Contrarily, Red's winning moves in PopOut are moves that would *lose* in ordinary CONNECT 4!

In Diagram 400, Red must play 9 C1 or 9 G1 to win. Diagram 402 shows one possibility after 9 C1. In the diagram, Black threatens g1 or a pop-out move at f1, either of which produces four-in-row. But it's

Red's turn, so he or she just pops out at D1 to win!

First Player advantage in PopOut is so strong that Red can probably win even with 1 C1, as in the game shown in Diagram 403. Red threatens to pop out E1 with a winning four-in-row at C1-D2-E3-F4. (Black also gets four-in-row, at d1-e1-f1-g1, but under the rules Red wins since it was Red's move.) Black's plays at 2 d1 and 4 d3 would *seem* to be his or her strongest defense, but I didn't actually *prove* that.

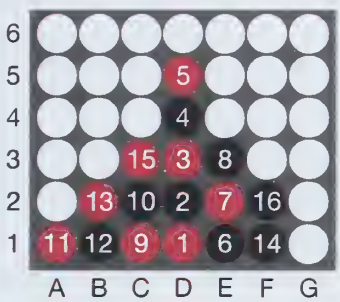
DIAGRAM 402 (PopOut rules)
Red to play, with only one move to win



Moves: (1) C1 d1 D2 d3 C2 f1 F2! g1 E1! e2 F3 e3 F4 f5 E4 (16) __

In Diagram 404, both players play *very* well (Red 7 D3 was mandatory), but Red makes one slip. Red's 9 E2 may seem logical (to block Black's second row), but a play at C3, D4, F3, or even G1 would have won. Instead, after 9 E2, the best play by both players leads to the continuation shown. In the diagram, Red is almost in zugzwang and will pop out again at G1 for lack of a better move. Black then has to pop out (again) at g1 to stop a quick Red win; Red will play G5 (or pop out at G1 again): draw by repetition of moves! If either player refuses to accept this draw and tries a deviation, that player will lose.

DIAGRAM 403 (PopOut rules)
Black to play (will lose)



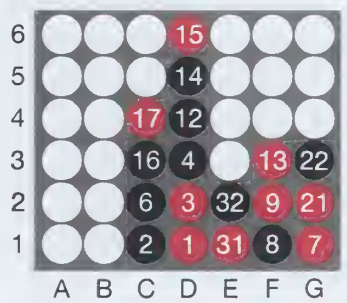
Moves: D1! a1 D2 d3 D4 b1 E1 a2 D5 c1 (11) G1 f1 F2 a3 A4! c2 G2 (18) __

DIAGRAM 404 (PopOut rules)
Red to play, with only one move to draw



Moves: (1) D1 c1 F1 e1 F2 d2 D3! c2 E2? e3! C3 c4! E4 a1 D4 d5 E5 a2 C5 a3 A4! a5 D6 a6 C6 e6! G1! g2! (29) G3! g4! G5 g6! Pop:G1! Pop:g1! G5 g6! (39) __

DIAGRAM 405 (PopOut rules)
Red to play, with only one move to draw

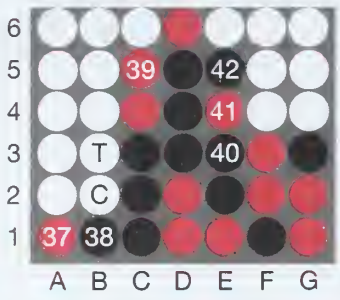


Moves: (1) D1 c1 D2 d3 E1 c2 G1 f1 F2! e2 E3 d4 F3? d5 D6! c3 C4! e4! Pop:E1! g2 G3! Pop:e1! (25) E3! e4! Pop:E1! Pop:e1! E3! e4! Pop:E1! Pop:e1! (37) __

Diagram 405 on the previous page is another example of draw by repetition. Moves 23–26 are the same as moves 27–30, and these moves will be recycled ad infinitum.

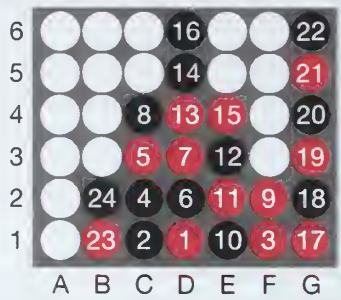
It isn't obvious that the repeated moves are forced in Diagram 405. If Red should deviate, one winning continuation for Black is shown in Diagram 406.

DIAGRAM 406 (PopOut rules)
Red to play (will lose)



Moves: D1 c1 D2 d3 E1 c2 G1 f1
F2! e2 E3 d4 F3? d5 D6! c3 C4! e4!
Pop:E1! g2 G3! Pop:e1! E3! e4! Pop:E1!
Pop:e1! E3! e4! Pop:E1! Pop:e1!
(37) A1? b1 C5 e3 E4 e5 (43) __

DIAGRAM 407 (PopOut rules)
Red to play, with only one move to win



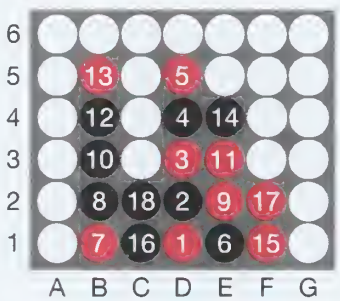
Moves: (1) D1 c1 F1 c2 C3 d2 D3 c4
F2 e1 E2? e3! D4 d5 E4 d6? G1! g2
G3! g4 G5! g6 B1! b2 (25) __

Sometimes a player will need *two* pop-out moves to win. Let's see some examples of that.

There were some weak moves leading up to Diagram 407, but from here Red wins easily. First Red pops out at D1 (preventing the winning Black pop-out move at e1). Black might answer at e5, but Red then pops out at G1 and wins at D2-E2-F2-G2.

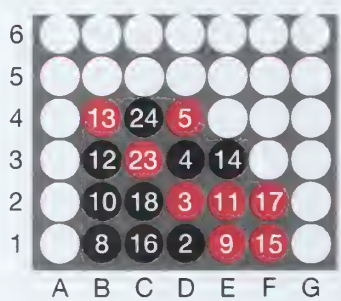
Diagram 408 is another example where Red may need *two* pop-out moves to win. (If Red were restricted to a single pop-out move, 7 B1 would not work: 7 E2 would be mandatory.)

DIAGRAM 408 (PopOut rules)
Red to play, with only one move to win



Moves: (1) D1 d2 D3 d4 D5 e1 B1 b2 E2!
b3 E3! b4 B5! e4 F1! c1 F2! c2 (19) __

DIAGRAM 409 (PopOut rules)
Red has won



Moves: (1) D1 d2 D3 d4 D5 e1 B1 b2
E2! b3 E3! b4 B5! e4 F1! c1 F2! c2
Pop:D1! Pop:e1 C3! c4 Pop:B1!

If you find the correct continuation in Diagram 408, you'll end up winning as shown in Diagram 409.

Pop 10

After completing a game, or whenever the board is full or almost full, you may enjoy playing a game of Pop 10. Alternate turns, popping out one of your own colored disks from the bottom row. If that disk was part of a four-in-row, score a point, drop the disk in anywhere at the top, and take another turn. The first player to reach ten points wins the game.

It is doubtful that serious players will study this game the way they do CONNECT 4 or PopOut, but it can be a fun change of pace. Here are some simple tactical ideas to get you started:

- Look for a way to score two or more points at your turn.
- If you can score only one point, it may be preferable to play in a way that denies the opposing player any points at his or her turn.
- When dropping a scoring disk at the top, take care in choosing which column. Make sure you have at least one disk in every column; otherwise, the opposing player may be able to score repeating points.

Glossary

?: Indicates a move that is a blunder.

??: Indicates a blunder that changes a win to a loss.

!: Indicates a move where every other choice would have been a blunder.

Adjacent Threat Pair: Two threats belonging to the same player in the same column in adjacent rows.

Black: The player who moves second. (Any color may be used, but in this book I assume Second Player uses black disks.)

Blue Cell: In rule variations where Red plays with a handicap, a cell that Red cannot use as part of a four-in-row.

Blunder: A move that, assuming perfect future play, doesn't produce the perfect result; for example, a move that leads to a draw when a win was available.

Candlesticks Opening: A sequence of initial moves that leads to two or three separated columns of disks, looking like candlesticks.

Cell: Any of the 42 places on the board where a disk can eventually be placed.

Check (or Immediate Threat): A cell that is available for immediate capture and whose occupation would form four-in-row.

Counterthreat: A threat that neutralizes an enemy threat that would otherwise win. The term is used specifically for a Black major odd-row threat in a separate column from Red's major odd-row threat.

Diagonal Threats: Cells whose occupation would form four-in-row along a diagonal.

Diagonal Undercutting Diagonal Undercut: A diagonal undercutting threat such that the threat being undercut is itself an undercutting diagonal threat. This is a very common situation; creating or destroying such an undercut is often a key tactical idea.

Disk: A game marker. Can also be called *piece*, *counter*, *checker*, or *token*.

Double Check: Two simultaneous checks by the same player.

Draw: A game that no one wins. The board fills up with 42 disks, but neither player gets four-in-row.

Endgame: The late stage of a game, usually when zugzwang is about to occur.

Even-Row Threat: A threat cell on the second, fourth, or sixth row.

Four-in-Row: Four disks of the same color in a continuous line horizontally, vertically, or diagonally.

Future J Shape: A configuration of Red disks, such as B1-C2-D3-D1-E2, with E3 empty. It is called *future J shape* because Red can convert it to a J shape whenever he or she chooses.

Horizontal Threat: Cell whose occupation would form four-in-row along a horizontal row.

J Shape: The common winning Red configuration comprising D1-D3-E2-E3 or its mirror image.

Key Threat: A threat that is essential to victory.

Major Threat: An empty cell whose occupation would complete four-in-row, but which is not available for immediate play.

Mate: Victory by completion of four-in-row.

Minor Threat: One of two (or three) empty cells in a partially completed four-in-row.

Mixed Threat: One of a pair of minor threats, of which one is odd and one is even; can occur only along a diagonal.

Odd-Row Threat: A major or minor threat cell on the third or fifth row.

Odd-Threated Column: Any column with an odd-row threat. Each odd-threated column implies an eventual reversal of zugzwang control. (A column with same-player threats in both the second and fifth rows may be treated as two odd-threated columns.)

Opening: The initial move sequence of a game, especially a standard, well-played sequence.

Overcut J Shape: A Red J shape together with three Black disks above it that negate Red's triple threat.

Overcutting Threat: A threat that neutralizes an enemy threat immediately below it. Normally the overcutting threat will be a major threat, and the threat under it will be a minor threat.

Plethora of Threats: A group of five or more threats that work together to have strong strategic value.

Pop-Out Move: A move in which a player removes a disk instead of adding one. This is available only with Hasbro's newer game sets.

Push-Up Play: A tactic in which a player occupies cells such as C1-C2-C3 to ensure his or her capture of C5.

Red: The player who moves first. (Any color may be used, but in this book I assume that First Player uses red disks.)

Row Parity: The attribute of a threat determined by whether it lies on an even-numbered or odd-numbered row.

Sextuple Threat: A particular configuration of six threats equivalent to a triple threat.

Strategic Threat: A cell that a player hopes eventually to occupy (or threaten to occupy) as part of a four-in-row but that is not an immediate check.

Tactical Sequence: A series of two or more forcing moves to attain an objective.

Tempo-Gaining Move: A play which forces the opponent to play an otherwise-useless disk to block a threat. The player who makes a tempo-gaining move gets to play, in effect, two good disks in a row.

Third-Row Gambit: An opening style in which Red succeeds in forming a third-row threat on one side of the board, but Black compensates with fifth-row threats on the opposite side.

Threat: A valuable empty cell. Threats are classified as Red or Black; strategic or tactical; major or minor; even-row or odd-row; horizontal, diagonal, or vertical. (A single cell may represent more than one threat.)

Transposition: Reaching the same position by a different ordering of the moves.

Triple Mixed Threat: Two (or more) minor odd-row threats in two separate columns, which are made equivalent in value to a major odd-row threat by virtue of one or more even-row threats in one of the two columns.

Triple Odd-Row Threat: Three minor odd-row threats in three separate columns. This is usually equivalent in value to a major odd-row threat.

Triple Threat: Either a triple mixed threat or a triple odd-row threat.

Undercutting Threat: A threat that neutralizes an opposing player's threat above it.

Winning Quadruple Threat: A particular configuration of four threats equivalent to either (1) two major odd-row threats or (2) an adjacent major threat pair.

Zugzwang: A position in which a player doesn't want to move and would prefer to pass if that were legal. Most CONNECT 4 games are won in the ending after the loser is placed in zugzwang and forced to play. Zugzwang cannot occur in most games of the tic-tac-toe family, but the "gravity rule" makes CONNECT 4 an exception.

Zugzwang Control: The dominant (or upper-hand) position in zugzwang. Normally Red has zugzwang control when there are an odd number of odd-threated columns; Black has it when there are an even number of such columns.

Answers to Problem Sets

Answers to Problem Set A (Easy)

Problem 1: Red wins with 15 C4; if Black played there, he or she would have an adjacent threat pair (see Diagram 353).

Problem 2: Black wins with 20 b5, dominating the upper board (see Diagram 310).

Problem 3: Black draws with 16 b5, or else Red gets an adjacent threat pair (see Diagram 213).

Problem 4: Red draws with 11 D3 (see Diagram 157).

Problem 5: Black begins with 12 e2.

Problem 6: Black begins with 10 c2.

Problem 7: Black wins with 14 c3, making counterthreats on both sides of the board (see Diagram 376).

Problem 8: Black wins with 22 e3; Black cannot let Red play there (see Diagram 308).

Problem 9: Red wins with 11 E2, before Black locks up the second row (see Diagram 230).

Problem 10: Black begins with 14 d2.

Problem 11: Black wins with 12 d4, a key cell (see Diagram 153).

Problem 12: Red draws with 23 B4, a strategic cell for either player (see Diagram 351).

Problem 13: Black begins with 12 d3.

Problem 14: Red wins with 17 G2, or else Black plays there (or f3) and wins quickly (related to Diagram 296).

Problem 15: Red wins with 17 F3, a strategic center cell (see Diagram 368).

Problem 16: Black wins with 8 c3, preventing a left-side Red J shape (see Diagram 122).

Problem 17: Red wins with 19 E3, stopping a quick Black win (see Diagram 170).

Problem 18: Black wins with 14 e3, with 16 d3 to follow (see Diagram 183).

Problem 19: Red wins with 11 D5, the key fifth-row cell (related to Diagram 229).

Problem 20: Black wins with 22 g4, preventing the Red F3 threat (see Diagram 217).

Problem 21: Red wins with 21 D6, forming a (partial) triple threat at A3-B4-B5; combined with Red's right-side triple threat, Black's e5 counterthreat is overwhelmed (see Diagram 363).

Problem 22: Red begins with 7 B3 or 7 C1, as shown in Diagram 410.

Problem 23: Black wins with 10 f2, using the second row to undercut Red's third row (related to Diagram 373).

Problem 24: Black draws with 18 e4, preventing Red's odd-row threat (see Diagram 171).

Problem 25: Black wins with 16 c3 to block Red's third row (see Diagram 330).

Problem 26: Black begins with 12 e2.

Problem 27: Black plays 12 e2, 14 e3, 16 d3, 18 b2.

Problem 28: Black begins with 14 g1.

Problem 29: Black begins with 12 f1.

Problem 30: Black wins with 14 c5, then 16 b1 (see Diagram 232).

Answers to Problem Set B (Easy)

Problem 31: Black begins with 12 e3.

Problem 32: Black draws with 12 e2, using the second row to undercut Red's third row (see Diagram 245).

Problem 33: Black wins with 16 c4, pointing toward several possible future threats (see Diagram 71).

Problem 34: Black begins with 12 e3.

Problem 35: Black begins with 12 e2.

Problem 36: Red wins with 13 G3 (related to Diagram 167).

Problem 37: Red draws with 9 F3 (when in doubt, play on the third row!) (see Diagram 361).

Problem 38: Black draws with 24 a4, preventing a Red odd-row threat (see Diagram 173).

Problem 39: Red begins with 11 F3 or 11 G1, as shown in Diagram 411.

Problem 40: Black begins with 12 d3.

Problem 41: Black draws with 12 b4, a strategic cell. If Red responds with B5, Black must answer at b6! (see Diagram 299).

Problem 42: Black wins with 14 d3, killing Red's third row (see Diagram 277).

Problem 43: Black draws with 12 f1, the proper firm response since Red F1 forces after 12 g1 (related to Diagram 241).

Problem 44: Black wins with 14 d5, a much better cell than D6, which Red must now take (see Diagram 379).

Problem 45: Red wins with 29 A3, robbing Black of his or her a5 threat (see Diagram 348).

Problem 46: Red wins with 31 C1, the last harmless cell. This produces the rarish case of a triple threat based on four odd-row threats. (Many of the prior moves in this diagram are blunders.)

Problem 47: Black plays 10 e2, 12 e3, 14 g3, 16 f4.

Problem 48: Black draws with 16 f5, followed by 18 e5 (see Diagram 351).

Problem 49: Red wins with 21 D5, preventing a Black adjacent threat pair (see Diagram 279).

Problem 50: Black begins with 10 d2.

Problem 51: Black begins with 12 c3.

Problem 52: Red plays G1, F2, E3 (mate in five rather than four because Black can delay the inevitable with 10 a3).

Problem 53: Red begins with 5 D5, a central fifth-row cell. (Of course, you can't see 19 moves ahead!)

Problem 54: Black wins with 14 c3, blocking Red's third row (see Diagram 337).

Problem 55: Black wins with 16 d4; follow with 18 d5 to destroy Red's future (see Diagram 354).

Problem 56: Red begins with 11 D2, as shown in Diagram 412.

Problem 57: Red wins with 19 A3; Red's future winning fifth-row threat hasn't materialized yet, but the first step is to stop Black's third-row threat (see Diagram 290).

Problem 58: Black wins with 20 b6, winning eventually on sixth row (see Diagram 164).

Problem 59: Black plays 12 e3, 14 d3 to form an adjacent threat pair.

Problem 60: Black wins with 14 f1 (see Diagram 122).

Answers to Problem Set C (Medium)

Problem 61: Black wins with 20 b2; Red builds threats easily if he or she gets B2 (related to Diagram 293).

Problem 62: Black draws with 16 f5, to make fifth-row threats (see Diagram 393).

Problem 63: Black wins with 20 c6, the only play that doesn't hurt Black (see Diagram 379).

Problem 64: Black draws with 10 b1 (any other move except c1 gives Red a quick win) (related to Diagram 148).

Problem 65: Red wins with 15 G4; if Black gets g4 and d4, he or she will lock up the board with even-row threats (see Diagram 227).

Problem 66: Red wins with 19 F6 (see Diagram 316).

Problem 67: Black wins with 16 f4, with 18 c3 to follow (related to Diagram 373).

Problem 68: Black draws with 10 f1, killing Red's offense (see Diagram 243).

Problem 69: Black wins with 30 e4; Black will get either e5 (and a strategic win) or e6 (and a quick win in column B). If Red is allowed to take G2 first, Red can answer e4 with double check (related to Diagram 248).

Problem 70: Black wins with 30 a6, eventually winning on the sixth row (related to Diagram 89).

Problem 71: Black wins with 16 f1, preventing Red adjacent threats C3-C4 (related to Diagram 235).

Problem 72: Red begins with 13 E1, as shown in Diagram 413.

Problem 73: Red wins with 21 E6, preventing Black adjacent threats at d5-d6 (related to Diagram 298).

Problem 74: Black wins with 16 d4, followed by d5 to kill the fifth row (see Diagram 227).

Problem 75: Black wins with 26 f3 followed by f5, a push-up play (related to Diagram 16).

Problem 76: Black plays 10 e3, 12 c3, eventually winning with an adjacent threat pair.

Problem 77: Black plays 10 b2, 12 b3, 14 d3, 16 c4.

Problem 78: Black draws with 10 c3 (similar to Diagram 81).

Problem 79: Black wins with 22 g4, preventing a Red odd-row threat (see Diagram 180).

Problem 80: Black wins with 28 a6, taking a neutral cell to preserve Black's threats at the upper right side (related to Diagram 122).

Problem 81: Red wins with 23 B6; either F5 or F6 will complete the triple threat, so defeating Black's e3 undercut is the urgent play (related to Diagram 210, which shows a losing variation.)

Problem 82: Red wins with 15 G2, completing the triple threat D5-F3-F2 (related to Diagram 269).

Problem 83: Red wins with 9 F1 forcing Black to play e1.

Problem 84: Black begins with 12 d3.

Problem 85: Black wins with 22 a1, the only move that doesn't hurt Black (see Diagram 331).

Problem 86: Red wins with 25 E2; Red is not worried about Black taking e3 because of Red's E5-G3 threat (see Diagram 314).

Problem 87: Red wins with 13 A2, blocking Black's second-row threat (see discussion at Diagram 110).

Problem 88: Black begins with 12 d3.

Problem 89: Red wins with 15 B4 (see Diagram 288).

Problem 90: Red wins with 19 A2; if Black plays there, he or she gets a key overcutting threat (related to Diagram 385).

Answers to Problem Set D (Medium)

Problem 91: Black draws with 8 a3, preventing a Red horizontal third-row threat.

Problem 92: Red wins with 25 A3, followed by A5, winning in the upper left (related to Diagram 126).

Problem 93: Black wins with 18 a2, killing any Red left-side chance (related to Diagram 386).

Problem 94: Red wins with 15 E1, blocking Black's possible threat at c3 (related to Diagram 374).

Problem 95: Red wins with 21 G6, making a major odd-row threat (related to Diagram 228).

Problem 96: Black wins with 22 a2, the only move that doesn't hurt Black (see Diagram 293).

Problem 97: Red wins with 23 G3; this threatens G4 with strong follow-up moves, so is better than the more passive D2 (related to Diagram 172).

Problem 98: Red wins with 21 E4, preventing a Black threat at f3 (related to Diagram 375).

Problem 99: Black wins with 8 d2, forcing Red; then Black answers C2 at d3 (see Diagram 81).

Problem 100: Red wins with 17 F5, where Black would love to play (see Diagram 392).

Problem 101: Red wins with 15 F5, a strategic cell for either player (see Diagram 341).

Problem 102: Black wins with 16 d6, blocking Red's A3-D6 diagonal (which would combine with the A4-D1 diagonal for an unusual triple threat) (related to Diagram 362).

Problem 103: Black draws with 16 b4 (see Diagram 88).

Problem 104: Red wins with 19 B1, to kill Black's undercut at e4 (related to Diagram 73).

Problem 105: Red wins with 13 E3, preventing Black e3-e5, which would give a c3 threat solely to Black (see Diagram 53).

Problem 106: Red wins with 19 B3, followed by E1 and eventual victory in the upper left (related to Diagram 239).

Problem 107: Red wins with 13 C3; Red has the winning triple threat C5-E3-E2 but must not let Black destroy it with the push-up play c3 c5 (related to Diagram 85).

Problem 108: Red wins with 21 C6, completing a triple threat (related to Diagram 194).

Problem 109: Black draws with 16 e4 (see Diagram 156).

Problem 110: Black wins with 14 b5; Red would get a triple threat if he or she captured B5 (related to Diagram 332).

Problem 111: Red wins with 13 E4, following with G2 if Black takes e5 (related to Diagram 218).

Problem 112: Black wins with 10 f3, blocking Red's third-row threat (see Diagram 332).

Problem 113: Black wins with 12 f5; Red would eventually get a triple threat if he or she played there (related to Diagram 219).

Problem 114: Red begins with 11 G1, as shown in Diagram 414.

Problem 115: Red wins with 11 C5, a good cell that gives up nothing (related to Diagram 365).

Problem 116: Red wins with 27 G4, the only safe cell (related to Diagram 54).

Problem 117: Black wins with 18 f5, preventing a Red threat at E4 (related to Diagram 244).

Problem 118: Red wins with 19 G2, not letting Black make an easy threat at e4 (related to Diagram 388).

Problem 119: Red wins with 17 B3, to give third-row threats on both sides (see Diagram 352).

Problem 120: Black wins with 22 g4, the only play that doesn't hurt Black (see Diagram 138).

Answers to Problem Set E (Medium)

Problem 121: Black draws with 32 f2, the only safe cell (see Diagram 76).

Problem 122: Red wins with 25 F5 for fifth-row threats (see Diagram 54).

Problem 123: Red wins with 19 E5, avoiding a Black c5 threat (related to Diagram 363).

Problem 124: Red begins with 9 E1, as shown in Diagram 415.

Problem 125: Black wins with 26 a6 (related to Diagram 59).

Problem 126: Red wins with 23 A4, completing the triple threat C2-C3-E3 (related to Diagram 334).

Problem 127: Red wins with 5 A1, which is generally more valuable than C1.

Problem 128: Black wins with 8 g1, preserving Black's second-row threats (related to Diagram 78).

Problem 129: Red wins with 17 C4, threatening both A2 and E2 (related to Diagram 239).

Problem 130: Red wins with 17 G1, to keep the right side in play (related to Diagram 148).

Problem 131: Red wins with 7 E4, a strategic center cell (see Diagram 367).

Problem 132: Red wins with 13 E2, which offers strong follow-up moves at C4, E3, or G4 (see Diagram 234).

Problem 133: Black begins with 12 c3.

Problem 134: Red begins with 13 F2, as shown in Diagram 416.

Problem 135: Black wins with 14 d2, to lock up the second row (see Diagram 275).

Problem 136: Red wins with 31 G6, the only play that doesn't hurt Red (see Diagram 345).

Problem 137: Black draws with 22 g3, to avoid Red adjacent threats at F3-F4 (see Diagram 100).

Problem 138: Red wins with 15 B5, forming a triple threat in the fifth row (related to Diagram 61).

Problem 139: Red wins with 15 A1, a forcing move that prevents Black's possible threat at b2 (related to Diagram 99).

Problem 140: Red begins with 13 B4, as shown in Diagram 417.

Problem 141: Red wins with 15 B2; Black would have strong threats if he or she plays there (see Diagram 300).

Problem 142: Red wins with 17 D2, preventing Black from locking out the left side with a second-row threat (related to Diagram 278).

Problem 143: Black wins with 18 a2; in the endgame, Red will block at B2 then C4, and Black will get c5 (see Diagram 323).

Problem 144: Red wins with 17 G2; if Black plays there, Red can't win in column F (see Diagram 282).

Problem 145: Red begins with 7 E1, as shown in Diagram 418.

Problem 146: Red wins with 27 F6, preventing a Black counterthreat (related to Diagram 38).

Problem 147: Red wins with 21 F5, robbing Black of an e5 threat (see Diagram 347).

Problem 148: Black wins with 28 a6, the only harmless cell left (related to Diagram 313).

Problem 149: Red wins with 19 C1; Black's c1 would kill Red's future threat at F5 (see Diagram 199).

Problem 150: Black draws with 24 g6, the only safe cell (related to Diagram 48).

Answers to Problem Set F (Hard)

Problem 151: Red wins with 11 A2, relevant to both second- and third-row horizontal threats (related to Diagram 234).

Problem 152: Black wins with 12 e1, preventing Red's F1-F2 (related to Diagram 32).

Problem 153: Black wins with 10 a1, aiming at the key a1-d4 diagonal (related to Diagram 244).

Problem 154: Black begins with 12 a4, eventually building an adjacent threat at b3-b4.

Problem 155: Black wins with 12 f5; after Red's forced reply, Black will take c3 (related to Diagram 51).

Problem 156: Red wins with 21 F3, defeating Black's f4-g5 diagonal and so preserving Red's F5 threat (related to Diagram 46).

Problem 157: Red wins with 19 A2; Red will follow with either A3 or A4 to win (related to Diagram 295).

Problem 158: Black begins with 12 c4, continuing as shown in Diagram 419.

Problem 159: Black draws with 14 c2, blocking Red's possible threat at D3 (see Diagram 171).

Problem 160: Black wins with 18 d3, building a triple threat at a3-b3-b4 (see Diagram 138).

Problem 161: Red wins with 13 B1, leading to mate in three moves (see Diagram 15).

Problem 162: Black wins with 8 g1, guaranteeing Black a major second-row threat (related to Diagram 39).

Problem 163: Black wins with 18 c5, with either b5 or b6 to yield an eventual odd-row threat (see Diagram 330).

Problem 164: Black draws with 22 f3, so the d2-g5 diagonal kills Red's right-side threats (related to Diagram 46).

Problem 165: Black wins with 18 c2 (related to Diagram 393).

Problem 166: Red wins with 15 E5, aiming at eventual D4 or D5 (see Diagram 278).

Problem 167: Red wins with 17 C5, making fifth-row threats (see Diagram 285).

Problem 168: Red wins with 25 A2, with a major odd-row threat then forced by disk 29 (see discussion at Diagram 388).

Problem 169: Red wins with 13 B1; if Black plays there, his or her e4 threat will end Red's chances in the upper right (see Diagram 368).

Problem 170: Black wins with 12 c3, blocking Red's third-row threat (see Diagram 221).

Problem 171: Black wins with 18 f6, preventing Red's play at E4 (related to Diagrams 192 and 84).

Problem 172: Red wins with 9 A2; left-side horizontal threats are urgent (related to Diagram 153).

Problem 173: Black draws with 14 d3, completing the undercut of Red's potential D5 threat and destroying other potential Red threats as well (related to Diagram 74).

Problem 174: Black begins with 12 d3; one continuation is shown in Diagram 420.

Problem 175: Red wins with 19 B2, the only way to guarantee left-side odd-row threats (see Diagram 95).

Problem 176: Black wins with 28 g3, the only safe cell (related to Diagram 371).

Problem 177: Black wins with 20 b2, seizing control of the left side (see Diagram 232).

Problem 178: Black draws with 12 f2, aiming at a counterthreat at e3 (related to Diagram 193).

Problem 179: Black begins with 10 e1, with various ways to eventually form an adjacent threat pair depending on how Red plays.

Problem 180: Red wins with 21 D2, building the diagonal threat d2-g5 (related to Diagram 291).

Answers to Problem Set G (Hard)

Problem 181: Red wins with 13 E5, killing Black's b2-e5 diagonal (see Diagram 269).

Problem 182: Black plays 10 e2, 12 c3, 14 d3, 16 c4.

Problem 183: Red draws with 11 F4, blocking the fourth row (related to Diagram 245).

Problem 184: Red wins with 23 D2; if Black plays there, he or she locks out the right side (see Diagram 290).

Problem 185: Red wins with 15 A1 (related to Diagram 214).

Problem 186: Red wins with 17 G4, with the winning triple threat E2-E5-F5 (related to Diagram 211).

Problem 187: Red wins with 11 D2, leading to mate in four moves (related to Diagram 15).

Problem 188: Black wins with 24 a2, the only harmless cell (related to Diagram 323).

Problem 189: Red wins with 13 E5, aiming at a fifth-row threat (related to Diagrams 151 and 227).

Problem 190: Red wins with 11 E5 (related to Diagram 287).

Problem 191: Red wins with 19 G1, racing up column G to establish a major odd-row threat (related to Diagram 105).

Problem 192: Red wins with 21 D6; Red's diagonal A3-D6 may undercut Black's b5-e2 diagonal (related to Diagram 52).

Problem 193: Black draws with 8 c4, the strategic central cell (see Diagram 361).

Problem 194: Black wins with 14 c3, to block Red's A5-D2 diagonal and thus preserve Black's undercut at b4-e2 (the e3 threat is irrelevant) (related to Diagram 51).

Problem 195: Black begins with 10 d3; one continuation is shown in Diagram 421.

Problem 196: Black wins with 22 e2 and answers 23 E3 at a5! This is a very subtle game (see Diagram 318).

Problem 197: Red wins with 17 D6, followed by C5 or G3 (Black cannot block both diagonals) (related to Diagram 110).

Problem 198: Black wins with 24 a6, the only harmless, temporizing move (related to Diagrams 358 and 359).

Problem 199: Black begins with 10 c2.

Problem 200: Black wins with 20 g1, sealing off Red's right-side threats (related to Diagram 155).

Problem 201: Red wins with 29 B2; Black is in zugzwang (related to Diagram 291).

Problem 202: Red wins with 19 C3, with C5 to follow (related to Diagram 154).

Problem 203: Black draws with 12 b2, answering B3 at b4 (see Diagram 262).

Problem 204: Black plays 12 d3 A3 e2 E3 c4 to form an adjacent threat pair.

Problem 205: Red wins with 13 B3, eventually developing threats in the upper part of column C (see Diagram 211).

Problem 206: Black draws with 14 c6 to kill Red's C6-F3 diagonal (see Diagram 361).

Problem 207: Black wins with 18 b3 (see Diagram 312).

Problem 208: Red begins with 13 C5, as shown in Diagram 422.

Problem 209: Red wins with 15 E3, followed by G4 to establish an odd-row threat (see Diagram 152).

Problem 210: Red wins with 13 D2 (related to Diagram 274).

Answers to Problem Set H (Expert)

Problem 211: Red wins with 15 G2; although Red has a left-side triple threat, he or she must maintain right-side possibilities as well. (Diagram 386 shows a result after Red errs here.)

Problem 212: Black draws with 6 g1, blocking the first row and aiming at g2 and the d4-g1 diagonal (related to Diagram 250).

Problem 213: Red wins with 17 F6, allowing a future Red play at E4 (related to Diagram 192).

Problem 214: Red wins with 15 A4, forcing Black to waste a turn at c2 and preventing a horizontal Black undercutting threat at 16 d4 (related to Diagram 256).

Problem 215: Black wins with 22 e6 (related to Diagram 149).

Problem 216: Black wins with 12 c4, preventing Red from making a threat at E2 (related to Diagram 42).

Problem 217: Black wins with 16 a4, for a fourth-row horizontal threat (related to Diagram 153).

Problem 218: Red wins with 21 B6, the only safe move (related to Diagram 249).

Problem 219: Black wins with 16 f3, building useful even-row minor threats (related to Diagram 381).

Problem 220: Red wins with 17 F1, claiming a threat at C4 and keeping the play at G2 in reserve (see Diagrams 195 and 199).

Problem 221: Black draws with 12 d3, blocking two important diagonals (related to Diagram 256).

Problem 222: Black wins with 24 e6, placing Red in zugzwang (related to Diagram 143).

Problem 223: Red wins with 15 B2 to stop any Black second-row threat (see discussion at Diagram 387).

Problem 224: Black wins with 28 g6, the only harmless cell left (related to Diagram 180).

Problem 225: Red wins with 13 D6 (related to Diagram 337).

Problem 226: Red wins with 19 B6, producing triple threats in both the upper left and upper right (related to Diagram 241).

Problem 227: Black draws with 20 f4, blocking a future Red threat at D6-G3 (see Diagram 261).

Problem 228: Black draws with 30 b2, before Red constructs a major threat at B3 (related to Diagram 146).

Problem 229: Red wins with 17 D6, getting a sextuple threat on the left side (related to Diagram 264).

Problem 230: Red wins with 19 E1, after which either E2 or E3 will give Red a major odd-row threat (see Diagram 148).

Problem 231: Red wins with 15 D2, since Black d3 would give Red D4 and an easy win (related to Diagram 280).

Problem 232: Red wins with 19 G2, blocking a second-row threat (related to Diagram 324).

Problem 233: Red wins with 17 G4, forcing e2 E3 and also preventing any Black fourth-row horizontal threat (related to Diagram 259).

Problem 234: Red wins with 21 B4, preventing a Black fourth-row horizontal threat (related to Diagram 283).

Problem 235: Red wins with 27 G2, temporizing (see Diagram 145).

Problem 236: Red wins with 11 F4, making a triple threat (related to Diagram 271).

Problem 237: Red wins with 9 A1, directed at the A1-D4 diagonal (related to Diagram 234).

Problem 238: Red wins with 9 G3; G5 would be a very useful cell for Red, so he or she stops Black from capturing it in a push-up play (related to Diagram 17).

Problem 239: Red wins with 19 A2, preventing the Black diagonal threat d5-a2 (related to Diagram 179).

Problem 240: Red wins with 15 A2, blocking Black's second-row threat (related to Diagram 391).

Answers to Problem Set I (Expert)

Problem 241: Black wins with 22 d6, eventually establishing a sixth-row horizontal threat (related to Diagram 163).

Problem 242: Black wins with 18 d6, preventing Red diagonal threats at D6-G3 (see Diagram 51).

Problem 243: Red wins with 17 E2, followed by A2 after e3 to destroy Black's second-row horizontal threat (see Diagram 322).

Problem 244: Red wins with 19 G2; Red has a plethora of threats, including the B5-E2 diagonal and the E column; G2 is the way to activate the right side tactically (see Diagram 309).

Problem 245: Black draws with 22 b6 (related to Diagram 361).

Problem 246: Black wins with 16 d6 (related to Diagram 331).

Problem 247: Red wins with 21 C2, forming the triple threat A3-A4-C5 (related to Diagram 319).

Problem 248: Black draws with 20 f1, preventing a Red threat at C4 (see Diagram 215).

Problem 249: Black wins with 10 b4, blocking Red's A5-D2 diagonal; Red's B2-E5 diagonal is less urgent because Black can undercut it (see Diagram 271).

Problem 250: Black wins with 16 e5, building a major threat at d4 (see Diagram 141).

Problem 251: Red draws with 29 F4, followed by B6 (F4 stops a quick Black win, and also forces Black to spend his turn preventing Red's winning F5 follow-up, so Red gets B6 as well) (see Diagram 126).

Problem 252: Red wins with 15 D6, the D6-G3 diagonal forming part of a sextuple threat (related to Diagram 369).

Problem 253: Black wins with 14 d2, making even-row threats (see Diagram 289).

Problem 254: Red wins with 17 C1, interfering with Black's possible threat at e3 (related to Diagram 300).

Problem 255: Red wins with 17 A4, to deny Black a fourth-row horizontal threat (related to Diagram 226).

Problem 256: Red wins with 17 A2, a temporizing move better than B1 (related to Diagram 367).

Problem 257: Red wins with 19 A1 (related to Diagram 55).

Problem 258: Red wins with 11 C4, building a threat at E2 (related to Diagram 112).

Problem 259: Red wins with 21 E1, aiming at threats on the upper left side (Red's threat at B3 is partly undercut by Black's b2-e5 diagonal) (related to Diagram 198).

Problem 260: Black draws with 16 b2, eliminating Red's second-row tactical threats (related to Diagram 299).

Problem 261: Red wins with 23 G5, depriving Black of a potential threat at f5 (related to Diagram 325).

Problem 262: Red wins with 21 D6, preventing the Black triple threat f5-f4-g3 (related to Diagram 368).

Problem 263: Black wins with 24 a3, forcing the construction of a major threat at a5 or b5 (see Diagram 342).

Problem 264: Red wins with 19 D3, blocking some possible Black left-side threats; victory will come at D5 eventually (related to Diagram 285).

Problem 265: Red wins with 19 A4, avoiding a future Black fourth-row threat (related to Diagram 184).

Problem 266: Red wins with 21 G5, interfering with Black's possible threats at e5-f5 (related to Diagram 383).

Problem 267: Red wins with 21 G1, racing to capture E6 eventually (see Diagram 208).

Problem 268: Black draws with 18 b2, threatening push-up plays in both columns B and D (related to Diagram 261).

Problem 269: Red wins with 21 G1, the only safe move (related to Diagram 384).

Problem 270: Red draws with 23 C6; Red can't win so just stops Black from getting a winning sixth-row threat (related to Diagram 18).

Answers to Problem Set J (Challenger)

Problem 271: Red wins with 25 G2, later capturing G5 with a push-up play (related to Diagram 347).

Problem 272: Black wins with 22 d6, the only way to get an even-row threat (related to Diagram 69).

Problem 273: Red draws with 9 E2; this move loses forcing value if D2 is played first (related to Diagram 82).

Problem 274: Red draws with 23 F6, blocking Black threats e5 and e6 (related to Diagram 216).

Problem 275: Red wins with 11 F1, with B2 or C2 to follow for a left-side win (related to Diagram 231).

Problem 276: Red wins with 21 A3, blocking Black's third-row threat (see Diagram 187).

Problem 277: Red wins with 17 D6; Red already has a triple threat, but d6 would give Black a triple threat of his or her own (related to Diagram 272).

Problem 278: Red draws with 13 E1, blocking Black's b4 threat and preserving a possible future Red counterthreat at B5 (related to Diagram 256).

Problem 279: Red wins with 15 E6, getting a triple threat after B3 (or after B4 and B5 or B6) (see Diagram 210).

Problem 280: Red wins with 29 A2, so that Black c4 can be answered at C5, with a quick win to follow (related to Diagram 248).

Problem 281: Black draws with 22 f3, followed by f5 to block the fifth row (related to Diagram 124).

Problem 282: Red wins with 19 A2, aiming at a B3 threat to undercut Black's threat at b4 (see Diagram 292).

Problem 283: Red wins with 13 G2, preventing a Black second-row threat (see Diagram 288).

Problem 284: Black wins with 18 g2, after which either f3 or f4 gives Black a key major threat (related to Diagram 339).

Problem 285: Red wins with 19 E2, destroying Black's second-row threat (related to Diagram 58).

Problem 286: Red draws with 19 A4 (see Diagram 361).

Problem 287: Red wins with 27 F6, since a Black major threat at e5 would override Red's minor threat there (related to Diagram 204).

Problem 288: Red wins with 23 A2; Red already has a triple threat (A3-A4-C5) on the left side, but the tactical possibility of double check after Red A3 and C4 means that A2 gives Black no time to construct a counterthreat (related to Diagrams 319–21).

Problem 289: Red wins with 23 G3; Black has strong tactical possibilities in the upper right, which Red can destroy only with counterplay involving a column G vertical threat (related to Diagram 341).

Problem 290: Red wins with 23 F1 (related to Diagram 363).

Problem 291: Black draws with 24 d2; if Red plays there, he or she gets a winning, albeit unusual, triple threat (related to Diagram 176).

Problem 292: Red wins with 15 A1, interfering with Black's potential threat along the a1-d4 diagonal. B2 would fail since it lets Black get b3, which works well with either c3 or c4 (this position is from the opening shown in Diagram 383).

Problem 293: Red wins with 15 D6; B4 is also a useful cell but would let Black take b5 (related to Diagram 384).

Problem 294: Black wins with 24 a5, building the left-side triple threat b4-b5-c5; by itself the triple threat would be enough only to draw, but Black has other potential threats, for example, on the b3-e6 diagonal (see Diagram 318).

Problem 295: Red wins with 17 F5, with the plethora of threats shown in Diagram 390; B5 is less valuable because Red's disk at F1 renders it redundant for the C4-E2 diagonal.

Problem 296: Red draws with 15 A1, blocking the d4 diagonal and aiming at A1-A2-A3 (related to Diagram 157).

Problem 297: Red wins with 11 F3 to make minor third-row threats (see discussion at Diagram 392).

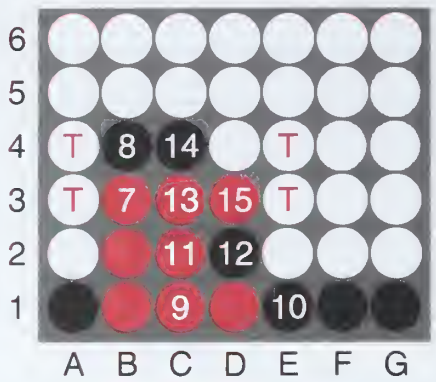
Problem 298: Black wins with 18 a2, avoiding a Red undercutting threat along the A2-D5 diagonal (see Diagram 160).

Problem 299: Red draws with 15 D2, to block the second row and aim at diagonal threat A5-D2 (related to Diagram 140).

Problem 300: Red wins with 15 C2, destroying Black's second-row horizontal threat (see Diagram 55).

DIAGRAM 410

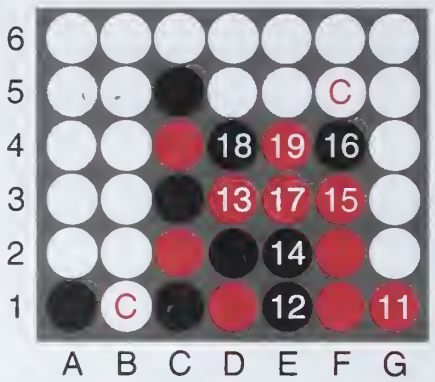
Black to play (will lose)



Moves: D1! f1 B1 a1 B2 g1 (7) B3 b4 C1 e1 C2 d2 C3! c4 D3! (16) __

DIAGRAM 411

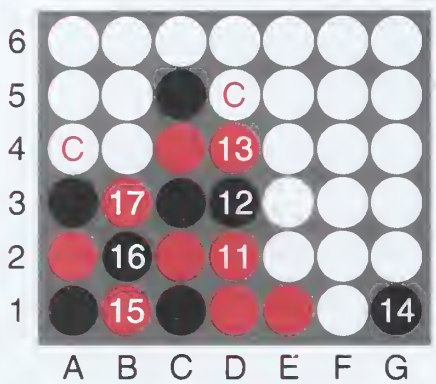
Black to play (will lose)



Moves: D1! c1 C2? c3! C4 c5? F1? a1? F2? d2? (11) G1 e1 D3 e2 F3 f4 E3! d4 E4 (20) __

DIAGRAM 412

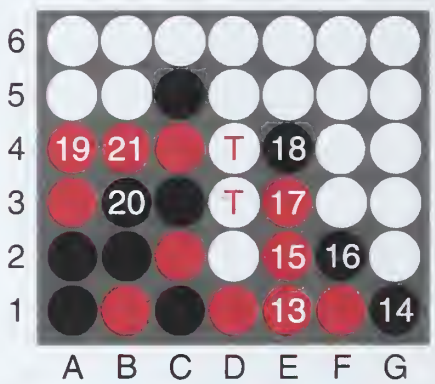
Black to play (will lose)



Moves: D1! a1 E1 c1 C2 c3 C4 c5 A2 a3 (11) D2 d3 D4 g1 B1 b2 B3! (18) __

DIAGRAM 413

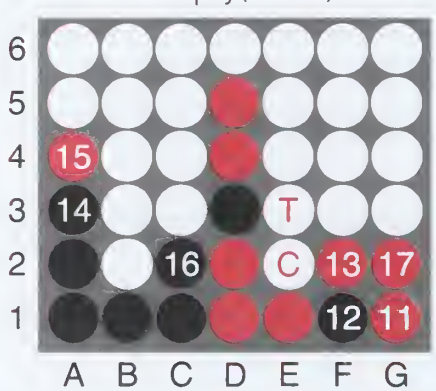
Black to play (will lose)



Moves: D1! c1 C2? c3! C4 c5? F1? a1? B1? a2? A3 b2 (13) E1 g1 E2 f2 E3 e4 A4 b3 B4 (22) __

DIAGRAM 414

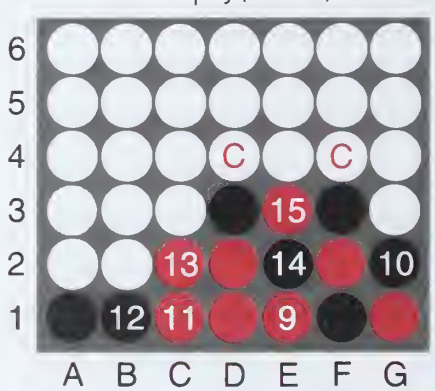
Black to play (will lose)



Moves: D1! a1 D2 d3 D4 b1 E1 a2 D5 c1 (11) G1 f1 F2 a3 A4! c2 G2 (18) __

DIAGRAM 415

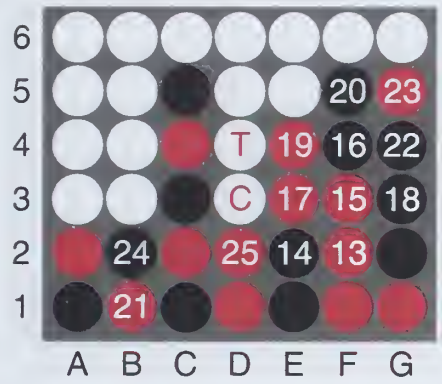
Black to play (will lose)



Moves: D1! a1 D2 d3 G1 f1 F2 f3 (9) E1 g2 C1 b1 C2 e2 E3 (16) __

DIAGRAM 416

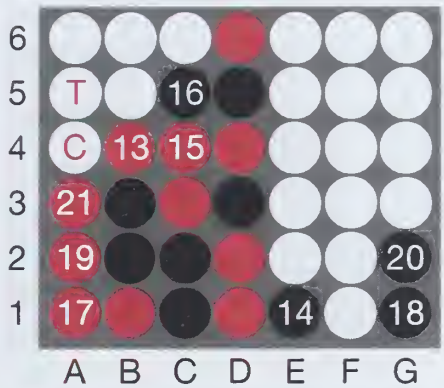
Black to play (will lose)



Moves: D1! c1 C2? c3! C4 c5? F1? a1? A2? e1? G1 g2 (13) F2 e2 F3 f4 E3! g3 E4 f5 B1 g4 G5! b2 D2 (26) __

DIAGRAM 417

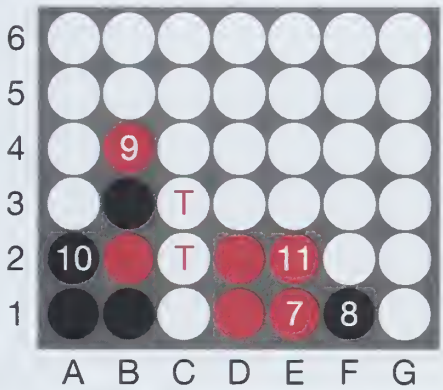
Black to play (will lose)



Moves: D1! c1 D2? d3! D4! c2 C3 d5! B1 b2! D6 b3? (13) B4! e1 C4! c5 A1 g1 A2 g2 A3 (22) __

DIAGRAM 418

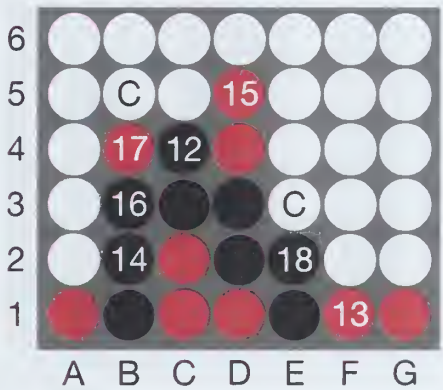
Black to play (will lose)



Moves: D1! b1 B2 b3 D2 a1 (7) E1 f1 B4 a2 E2 (12) __

DIAGRAM 419

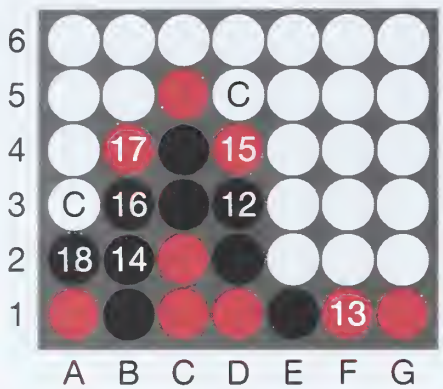
Red to play (will lose)



Moves: D1! d2 C1?? e1 A1 b1! C2 c3 G1 d3 D4 (12) c4 F1 b2 D5 b3 B4 e2 (19) __

DIAGRAM 420

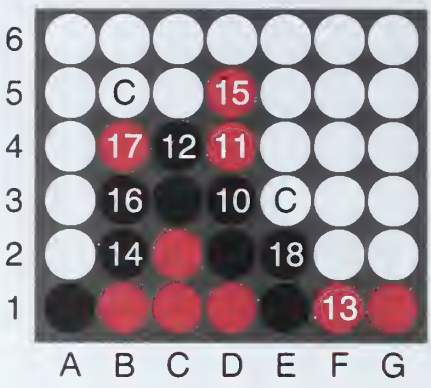
Red to play (will lose)



Moves: D1! d2 C1?? e1 A1 b1! C2 c3 G1 c4 C5 (12) d3 F1 b2 D4 b3 B4 a2 (19) __

DIAGRAM 421

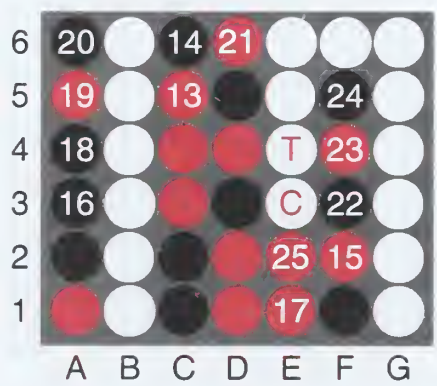
Red to play (will lose)



Moves: D1! d2 C1?? e1 C2 c3 B1 a1! G1 (10) d3 D4 c4 F1 b2 D5 b3 B4 e2 (19) __

DIAGRAM 422

Black to play (will lose)



Moves: D1! c1 D2? d3! D4! c2 C3 d5! A1? a2? C4? f1?? (13) C5 c6 F2 a3 E1 a4 A5! a6 D6 f3 F4 f5 E2 (26) _

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